INVESTIGATION OF MOTOR VEHICLE FIRES



FOURTH EDITION

Lee S. Cole

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Fourth Edition

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He is the author of a large number of training bulletins, articles and books on vehicle identification, theft, fraud and fire investigation published by Lee Books.

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By Lee S. Cole

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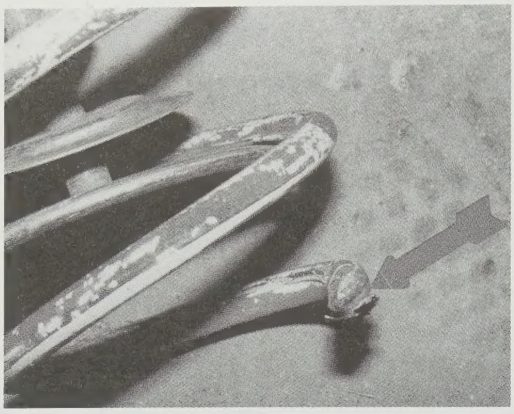
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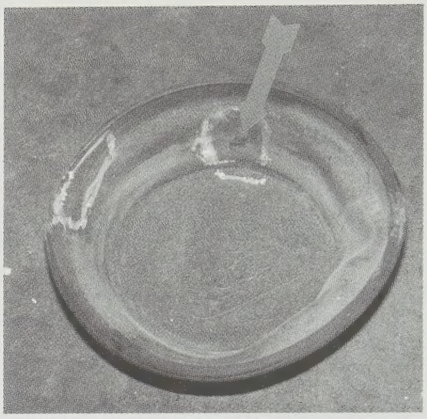
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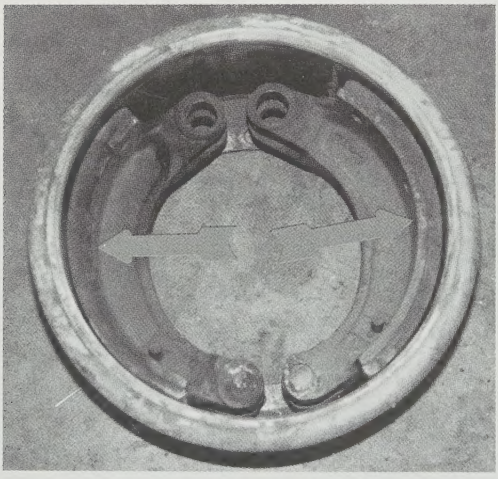
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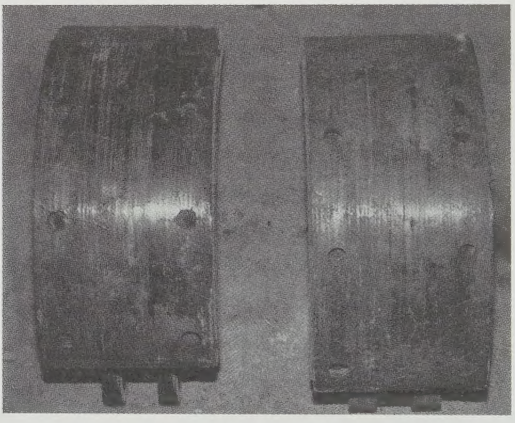
A sequence of events that caused a major bus fire.This suspension spring broke at arrow...



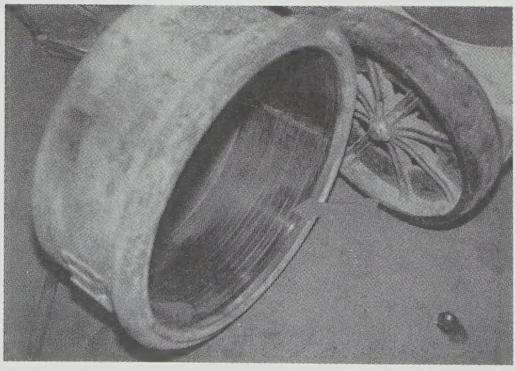
...and punctured a hole in the suspension airbag.



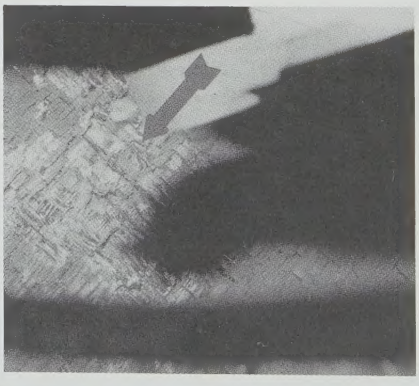
This caused the emergency brake shoes to clamp down on the drive shaft.



The shoes quickly overheated...



...as did the brake drum.



Heat was transmitted to the plywood underside of the floor and a fire broke out.

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Foreword

This book is an updated and supplemented version of the original "Investigation of Motor Vehicle Fires", first published in 1980 and revised in 1985 and 1992. The information in the original books was based on material available at that time and our experience in the investigation of these fires. A fourth edition of this book is necessary because of the use of additional light weight materials, the trend toward down sizing, the increased use of plastics and the higher operating temperatures of engines and exhaust systems. This has materially effected flammability and fire behavior. We have also gained a great deal more experience in vehicle fire investigation since the last edition and this also needs to be imparted to the investigative and prevention community.

No book of this type could possibly be the work of only one person. It is an accumulation of material, personal observations and experience collected over forty years. It is meant to serve as a guide in investigating motor vehicle fires. As in the case of most guides it tends to generalize in some areas. Your own organization's procedures and policies are the only sources that should be looked to for specific instructions. Nonetheless, I believe the information in this book is the most useful collected anywhere.

In compiling this material I have consulted numerous books, pamphlets, texts and individuals on the subject of fire investigation, vehicle nomenclature and construction. If I were to enumerate all this material I daresay there would be no room for anything else in this book.

I am indebted to all who have researched and written on the subject of vehicle fires. My thanks to you all.

Lee S. Cole

Introduction

Motor vehicle fires have been occurring for as long as people have been building automobiles. I'm sure the earlier points of origin of these fires were fairly easy to spot. The fuel line broke and sent a shower of gasoline over a hot engine, the electrical spark malfunctioned or a dragging brake line caused a fire by frictional heat. Since cars were simple in their design and construction the determination of fire cause was much easier for the investigator. By way of comparison just take a look at the engine compartment of a 1970 model auto and then raise the hood of a new car. The difference is mind boggling! Yet, bear in mind that the basic function of both systems you have observed is to furnish propulsion for the automobile. All the extra hoses, belts, motors and other paraphernalia filling the space are for emission controls, power steering, power brakes, air conditioning and all the other luxuries and necessities that are now a part of our motoring lives. All these extra appurtenances can make fire cause determination much more difficult in the modern motor vehicle.

In addition to determining the cause of an accidental vehicle fire there is the growing problem of arson. Many cases of "accidental" fires are, in fact, deliberate instances of fire setting. Unless the investigator knows the signs of intentional and accidental vehicle fires, he or she is lost.

I can recall many years ago looking at a burned Hudson and being advised that the car "burst into flames" after the driver saw "some sparks under the dash". This version was accepted without question and I'm sure some insurance company paid for this "accidental" fire. From the experience gained over the last forty years, I feel now that the fire was anything but accidental. It's a matter of observation and experience.

The purpose of this book is to give you some ideas on determining vehicle fire causes, both accidental and deliberate. It is only a guide and will not give you miraculous solutions. I do hope, however, that it will give you some workable ideas and a sound basis for your investigations.

Looking at Vehicle Fires

Why should we want to investigate a vehicle fire? Certainly not every case is one of arson and even if there is good suspicion that it is, how do we establish this fact? Arson is a most difficult crime to establish and prove. Probably it is the *most* difficult. In many cases the evidence is gone with the fire - or so it certainly seems. And as if this weren't trouble enough, bear in mind one other fact — *all fires are legally presumed to be accidental.* Just in case you missed it, read that sentence again. In other words, in order to establish arson we must rule out every chance of the fire being accidental! By doing this we establish the "corpus", but more about this in later chapters.

If not to establish arson why else investigate vehicle fires? We want to ascertain the cause so that responsibility may be established and proper redress made in products liability cases. We want to establish cause so that future engineering can reduce the chances of this hazard re-occurring. We want to establish cause since the primary purpose of any investigator is to determine what happened.

In order to conduct a vehicle fire investigation, we must be familiar with the parts of a vehicle most often involved in fires.

The Engine

This is the heart of the system. An engine converts heat energy to motion. As soon as you start the engine this process begins. Inside the engine block there is a rotating oddly shaped rod called a crankshaft. Attached to this device by connecting rods are pistons. The number of pistons is determined by the number of cylinders. As the crankshaft turns each piston is pulled, one at a time, to the bottom of the cylinder. When this happens a vacuum is created and the intake valve opens, sucking in a mixture of air and fuel. The piston then returns to the top of the cylinder, compressing the mixture. At this precise moment a spark plug fires, igniting the compressed mixture. This causes an explosion forcing the piston down, turning the crankshaft and supplying power to the transmission and from there to the drive wheels. This is a simple explanation of the operation of the internal combustion engine. How does the gasoline engine differ from the diesel? The diesel engine is fired by the use of heat generated from the compression stroke. It requires no spark plugs (although it does have "glow plugs" to facilitate

starting). Since it does not have an "ignition system" created by an electric impulse to create the heat required to fire the mixture, a diesel engine requires a 22-1 compression ratio as opposed to about 12-1 for a gasoline engine. This is why a much heavier block is required for a diesel engine. Some engines are mounted "In Line" in the engine compartment with their axis perpendicular to the axle. Power is transmitted to the rear wheels via the transmission and drive shaft. Other engines are "transverse" mounted with their axis parallel to the axles. Most of these engines are "front wheel drive" and power is supplied through the trans-axle (or transmission) directly to the front wheels.

A few engines are rear mounted, or "pusher" types. Some of these are "in line" and some are "transverse" mounted. In either event power is supplied directly to the rear wheels via the trans-axle.

A necessary component of the engine is the transmission. Without it the wheels would have to turn at whatever speed the crankshaft turned, much as a light aircraft engine turns the propellor shaft. The transmission allows the engine to run at one speed while the wheels are turning at another speed. This is accomplished by means of a series of internal gears and shafts. The torque produced by the engine is multiplied and allows the engine to run at an efficient speed.

Some transmissions are called "manual" meaning that the proper gear is selected by the driver by manually operating the gear lever. Others are "automatic" and the driver has only to place the gear selector in "drive" position and the transmission automatically progresses from low to high gear. This is done by the circulation of fluid between the engine driven "impeller" and the "turbine" which supplies power to the output shaft. This whole unit is called a "torque converter". The output shaft then transmits power to the internal gears and shafts inside the transmission housing where the proper speed and power selection is made. Transmission fluid and its flammability will be an important consideration in future chapters.

The Fuel System

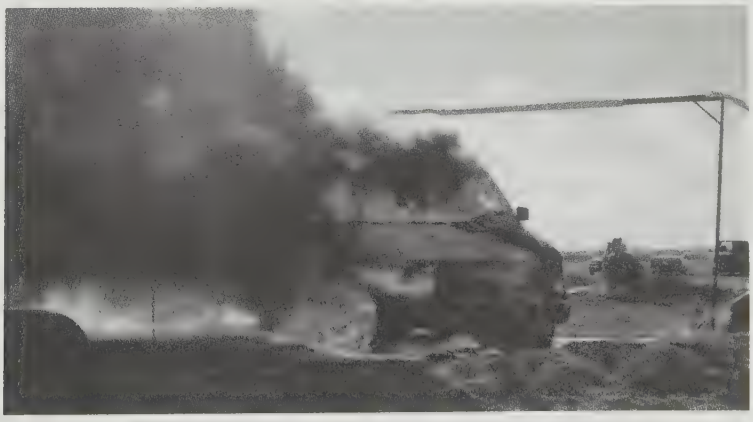
Since the engine is the heart of the motor vehicle the fuel system is both its life blood and digestive system. Gasoline is stored in the car's fuel tank. It is drawn from this storage unit through a filter system and into the carburetor or throttle body or direct injectors mounted on the engine top. The carburetor mixes the gasoline with the air and allows an atomized mixture to be drawn into the cylinder through the intake manifold. If the vehicle has fuel injection instead of a carburetor the fuel and air mixture is electronically or mechanically metered and

fed directly into the cylinders. Throttle body injection makes use of one or two injectors mounted above the intake manifold. A spray of fuel mixed with air is fed directly into the throttle throat. The carburetor is fast going the way of the dinosaur on the road to extinction. The reason for this is the strict requirement of air to fuel ratio in the modern motor vehicle. This requirement is dictated by the emissions control systems now in effect and the carburetor cannot handle this. The ratio for complete gasoline combustion is 14.7 to 1. In other words, 14.7 pounds of air mixed with 1 pound of gasoline will produce the most complete combustion with the least amount of Hydro-Carbon (HC) and Carbon Monoxide (CO) emissions. This is called the Stoichiometric air-fuel ratio.

In order for gasoline to burn as an energy producing fuel it must be vaporized. Remember, a liquid does not burn, it is the vapor produced by the liquid that burns. Escape of this vapor in the presence of an ignition source creates the hazard that can produce the vehicle fire.

Gasoline vapors are about three to four times heavier than air. They will travel to the lowest level and will accumulate until a source of ignition is reached. Garage fires involving vehicles may occur when the pilot light of the water heater ignites the vapors that have "sunk" to the lower level.

Vapors from all petroleum products are heavier than air and will settle to the ground when released. Like liquids they tend to run toward the lowest point. This may be near or remote from the point of greatest damage. In a fire scene, caution must be exercised in making origin determinations.



Expelled fuel from the tank will create an inferno such as this.

The Electrical System

The center of this system is the battery. The 12 Volt battery has six compartments containing seven to fifteen plates. These plates are composed of lead and are submerged in a solution of sulpheric acid and water. This solution causes a flow of electrical current. This flow is augmented and supplemented by the alternator. The alternator is a form of generator. Older vehicles used the generator which produced direct current (DC) for the electrical system when the engine operated belt turned the generator shaft.

The alternator can produce more electrical current than did the generator and is smaller and lighter. Alternators produce alternating current (AC) which must be converted to direct current (DC). This is done by the use of rectifiers made of silicon which allow current to ;pass in one way only. If this were not done electricity would flow back and forth in polarity, hence its name "alternating current". The alternator can run at much higher speeds than can the generator. It is driven by a belt from the engine crankshaft and creates an electrical current by rotating soft copper wire coils over an iron core. The magnetic field resulting from this produces voltage and powers the accessories and charges the battery. The voltage regulator controls the electrical output of the generator or alternator. It prevents battery overcharging by decreasing output of the alternator/generator.

On this system operate the headlights, turn signals, interior lights, horn, gauges and all the other electrical accessories. In addition we have the system that starts the engine and maintains its heartbeat. The starter motor is directly connected to the battery and is engaged by turning the ignition key. This activates the starter solenoid and a gearing system turns the engine flywheel.

When electricity leaves the battery it flows through the ignition coil and builds up voltage which then goes to the distributor. This process boosts the voltage from the battery to 20,000-40,000 volts required to fire the spark plugs and create the engine's ignition. This is accomplished by the passing of current through two coils of wire wound around an iron core. This process boosts the voltage in the same manner as a transformer. It is enclosed in an oil filled case and is commonly called "the coil". Leading from the coil to the distributor is a black high tension wire. The voltage carried on this cable is "distributed" to the spark plugs by means of a rotor which directs the charge from the coil to the towers inside this distributor cap. The "points" inside control the pulses of electricity through the coil. When the rotor inside the distributor cap turns it supplies an impulse to each of the high tension wires leading to the spark plugs causing them to fire one at a time.

The "condensor" protects the points from burning and builds up high voltage for the spark plugs. A faulty condensor can cause the points to malfunction in a matter of seconds. Wiring from the battery to the coil and distributor is called "primary" wiring. From the coil to the distributor and to the spark plugs wiring is called "secondary".

Most 1970-1990 vehicles have electronic ignition systems frequently called "breakerless" systems. These systems differ slightly from conventional ignition systems. There are no basic differences in the two systems in the secondary wiring. The big difference occurs in the primary wiring. Instead of a distributor cam, points and condensor, the electronic system has an armature, a pickup coil and an electronic control module. When the ignition is turned on current flows from the battery to the coil primary. Primary current is turned on and off by the armature as it revolves near the pickup coil. The pickup coil induces a voltage which signals the electronic module to turn off the coil primary current. When the current is off, the magnetic field. in the coil collapses and a high voltage impulse is induced in the secondary wiring of the coil. The system now operates on the secondary circuit in the same manner as a conventional ignition system.

In 1984, distributorless ignitions were introduced. The distributor is replaced by a module, coil pack, and a crankshaft sensor. The crankshaft sensor is now the trigger which controls the on and off of the primary ignition current. There is one coil for each two cylinders. The coil fires both spark plugs on each crankshaft rotation; one cylinder is on the exhaust stroke and the other cylinder is on the compression stroke. This system eliminates the rotor, distributor cap, and distributor moving parts and reduces emissions by firing each spark plug on its exhaust stroke. In 1996, approximately 90% of vehicles sold in the USA used distributorless ignitions.

The Exhaust System

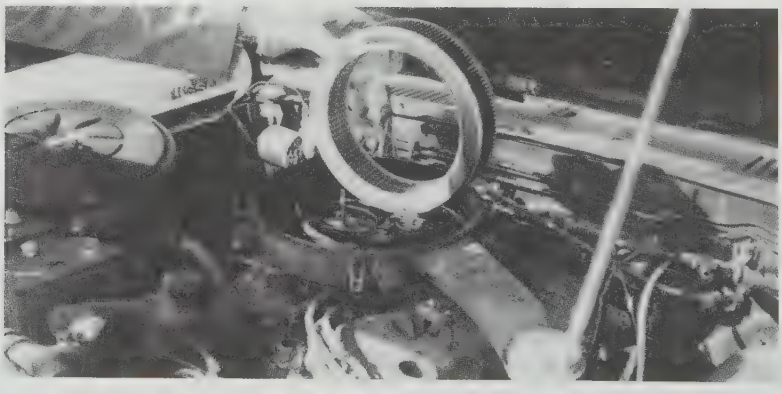
This is the part of the car that takes the waste products from the car's engine through the tail pipe. When the engine's exhaust valve opens, spent gases enter the exhaust manifold. From there the gases enter the exhaust pipe, the muffler and out the rear through the tail pipe. Some of the fire problems confronting investigators are related to the Emissions Control Systems that are directly linked to the vehicle's exhaust. These control systems have been in effect for over a decade and are:

1. Exhaust Gas Recirculation (EGR). This system uses exhaust gas to lower peak combustion temperature and control the formation of NOX. Exhaust gases

are introduced into the combustion chamber and take the place of a certain amount of air-fuel mixture. When the spark ignites the fuel mixture, it is diluted and the resulting explosion is not as hot. Unfortunately, the engine does not put out as much power.

The EGR valve is mounted on the intake manifold so that when it opens, gases are allowed to go into the intake manifold. The valve is vacuum operated and opens when the throttle opens. A temperature control is used allowing a single modulator to open at low temperatures. At high temperatures a second opens and allows air to enter the vacuum system. The vacuum is weakened and the opening of the valve reduced.

2. Open Positive Crankcase Ventilation (PCV). This system uses intake manifold vacuum to draw fumes from the crankcase into the manifold. These fumes are burned in the combustion chambers and go out in the engine exhaust. A tube from the air cleaner goes to the rocker cover, sometimes called the "valve cover". Air goes through this and into the crankcase. From there blow by gases go through the hose into the PCV valve and into the manifold. Flame arrestors are inserted into the hose connections and are designed to prevent a backfire and explosion. Sometimes these devices plug up and cause the system to fail. PCV valves will close if there is an intake manifold backfire. As long as the PCV valve is in place the flame cannot travel along the hose and cause crankcase explosion.



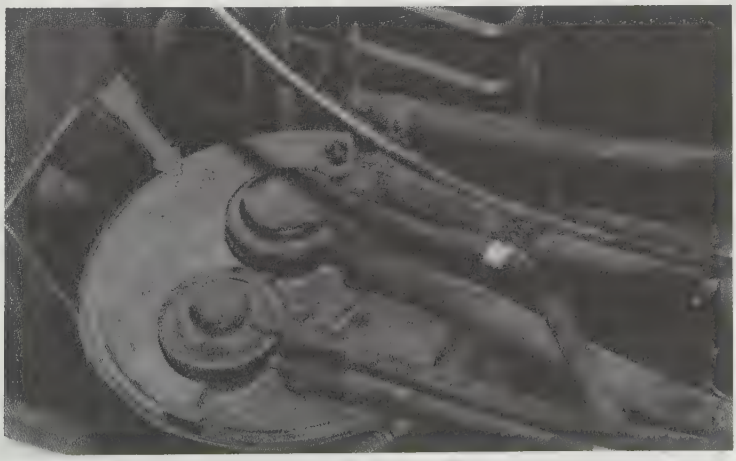
A condition known as "blow by" forced oil mixed with crankcase vapors into the air cleaner and down the heat tube where exhaust manifold heat ignited it.

3. Closed Positive Crankcase Ventilation (PCV). The PCV valve and the hose from the crankcase to the intake manifold are the same on both the open and closed systems. On the closed system fresh air enters the air cleaner and from there into

a hose and into the crankcase. Usually the fresh air enters a PCV filter inside the air cleaner and then goes through a hose to the rocker cover. This system does not pass fumes from the crankcase into the air at high throttle opening or if the PCV valve is clogged.

4. Air Pump Exhaust Control (AIR). The air pump is driven by a belt at the front of the engine and pumps air into each of the exhaust ports. This extra air under pressure causes a fire in the exhaust manifold that oxidizes the carbon monoxide into carbon dioxide and burns the hydro-carbons into carbon dioxide and water. Stainless steel nozzles send the air into the port as near as possible to the exhaust valve. Between these nozzles and the air pump is a check valve. The valve prevents the exhaust gases from flowing back into the pump and hoses. If the check valve fails, burned hoses and pump can be the result. Some V-8 and V-6 engines use two check valves (one per cylinder bank) while others use only one. In addition to the check valve there is a diverter valve and it uses a diaphragm operated from the intake manifold vacuum. In deceleration the diverter shuts off air to the nozzles and sends it to the air cleaner. Later models of these valves exhaust the air through a bronze or cotton filter directly into the atmosphere. Failure of the diverter valve can result in backfiring and a fire hazard. Air is also injected directly into the catalytic converter to enhance its action.

5. Vapor Control System. This system is entirely separate from the exhaust and crankcase emissions controls. The system takes care of any gasoline vapors that escape from the fuel tank or the carburetor. The fuel tank is sealed except for a line that goes to a charcoal filled canister. Vapors move into this line and are stored in the canister until the engine is started. These vapors are then sucked into the



The vapor storage canister is part of the emissions control system. It holds fuel vapor that would escape into the atmosphere until it can be burned in the engine.



It is generally located in the engine compartment.

intake manifold and burned along with the regular mixture. In order to keep liquid fuel from entering the canister a separator on top of the tank sends all the liquid back to the tank. Over production of vapor in the tank can result in overloading the canister and a resulting over rich condition in the intake system. More about this problem and its relation to fires in subsequent chapters.

The carburetor float bowl also connects to the vapor canister so that vapor from the float bowl will not escape. If the vehicle has fuel injection this connection is eliminated.

6. Catalytic Converters. Since 1975 this system has been used on American cars and most imported vehicles. This is nothing more than a container that holds a chemical substance through which the exhaust gases are passed en route to the muffler. The reaction of the chemical substance on the hydro-carbons and carbon monoxides in the exhaust converts them into water and carbon dioxide by promoting additional burning of the pollutants. Many cars are equipped with air pump systems in conjunction with the converter that provide a fresh air source to enhance the conversion process. In most modern vehicles an oxygen sensor is mounted in the exhaust stream upstream of the catalytic converter. Information on the amount of oxygen in the exhaust is transmitted from the sensor to the vehicles computer and this data is used to make adjustments in the fuel/air mixture.

Since leaded gasoline will contaminate the catalytic agents all converter equipped vehicles are designed to operate on unleaded fuel. The reaction that takes place inside the catalytic converter is of a combustion type and raises the

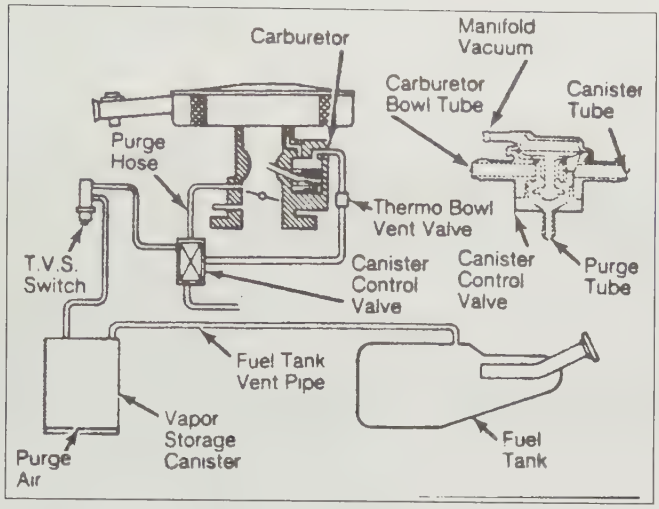


Figure 1. Evaporative emission system for 3.0L, 3.8L & 5.0L carbureted engines. Other models are similar

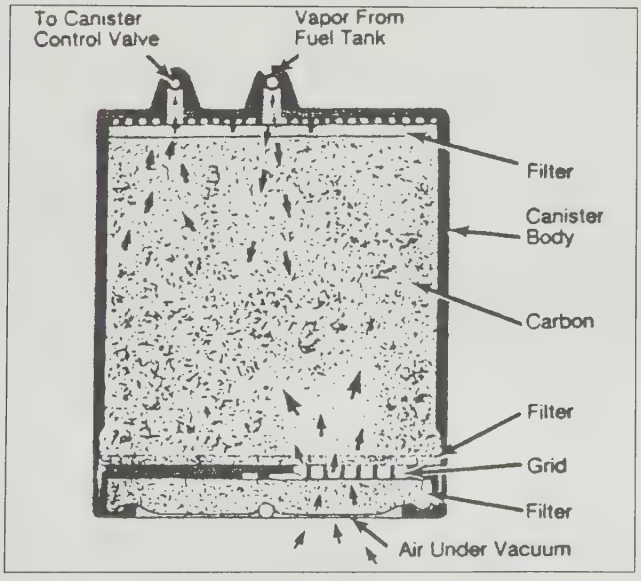
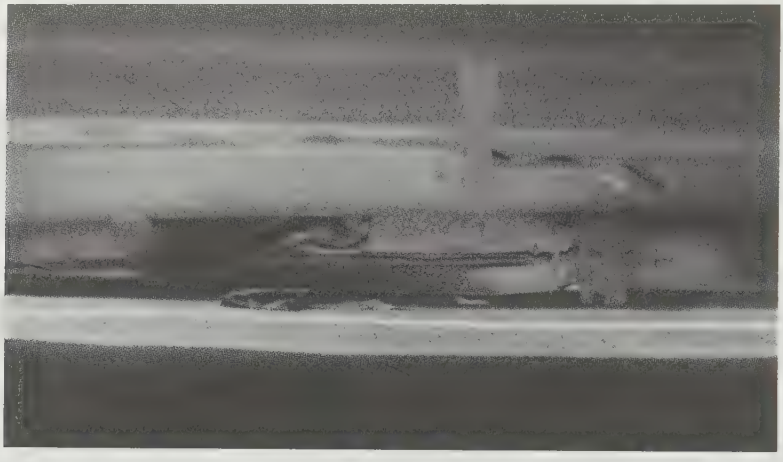


Figure 2. 2-Tube open bottom canister.

heat level of the converter exterior by several hundred degrees. Special heat shields must be used to protect the underside of the vehicle's body, both from the converter and adjacent exhaust surfaces.



Catalytic converters are mounted under the vehicle.



This catalytic converter and its heat shield have been removed.

There are, of course, many other elements and materials in the modern motor vehicle that can contribute to or be the cause of a fire hazard. Plastics, rubber, paper, cloth, magnesium, all are used in the construction of a vehicle and the flammability of these materials is well established under the right conditions. Always bear in mind the necessary elements for a fire:

1. Oxygen or other oxidizing agent.
2. Fuel.
3. Sufficient temperatures to establish combustion.
4. Molecular chain reaction to sustain combustion.

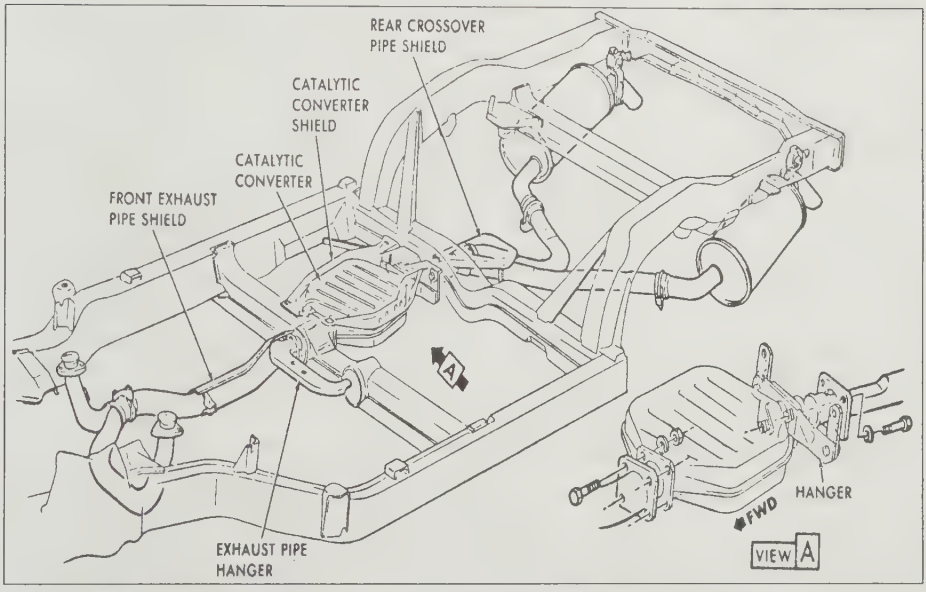


Figure 3 shows catalytic converter location in exhaust system of a Corvette.

In order for us to understand fully the product we are dealing with and its possible reaction to heat and flame it is necessary that we know its composition. See **table 1** to give you an idea as to the materials used and the trend toward overall vehicle weight reduction for the period 1975-1985.

As downsizing continued the total weight of the average American sedan has shrunk. Most of this reduction has been due to efforts to improve fuel economy.

TABLE 1

MATERIALS USAGE IN NEW CARS 1975-1985						
(Dry Weight — lbs.)						
Model Year	**1975**		**1979**		**1985**	
Total Weight	**3,970**		**3,330**		**2,400**	
Material Mix	**%**	**lbs.**	**%**	**lbs.**	**%**	**lbs.**
High Strength Steel	2.7	106	4.6	154	12.5	300
Plain Carbon Steel	58.3	2,315	55.1	1,833	44.0	1,056
Iron	15.8	626	14.7	490	9.0	216
Aluminum	2.2	86	3.9	129	6.5	156
Copper	0.9	37	0.8	28	1.0	24
Lead	0.7	29	0.7	24	1.0	24
Zinc	1.3	53	0.8	27	0.5	12
Glass	2.4	94	2.5	84	3.0	72
Rubber	4.0	160	4.3	143	4.5	180
Other Plastics	4.2	168	5.8	191	10.5	252
Other	7.5	297	6.8	227	7.5	180

Source: U.S. D.O.T., The U.S. Automobile Industry

A materials list for a 1990 vehicle given in pounds shows the following:

Materials List for 1990 Vehicle	
Plastic	260
Wrought aluminum	41
Cast aluminum	107
High-strength steel	354
Cold-rolled steel	433
Other steel	747
Cast iron	341
Glass	67
Rubber	119
Sound deadener	53
Copper	22
Lead	26
Zinc	11
Minor materials	55
Total materials	2,636

source: Ford Motor Co.

In continuing this trend the following table shows material content in North American produced vehicles:

Material Content for North American-Produced Passenger Cars
(weight in kilograms)

Material	1990	1995
Total steel	691	600
Total cast iron	172	136
Total plastic	100	136
Total aluminum	66	77
Total all other	65	61
Total rubber	59	54
Glass	36	36
Powdered metals	12	13
Total zinc	11	11
Copper	10	9
Total Vehicle	1,222	1,133

**To convert kilograms to pounds remember that 1 pound equals .453592 kilograms.*

Material Composition for North American Produced Passenger Cars by Component in the mid to late 1990's

Component	Steel	Aluminum	Plastic
Outer hood	70%	2%	28%
Inner door	69	0	31
Bumper fascia	1	6	93
Suspension control arms	94	6	0
Suspension springs	63	0	37

Where and how did it start?

When we start the investigation of a motor vehicle fire our most important consideration must be to determine the point of origin. When you stand back and survey a burned car you might think this is something almost impossible to determine. It may be difficult but it can be done in most cases.

Start your examination by determining where the burning was most intense. If your observation reveals the vehicle completely burned and the time element is very short, it might be well to consider an incendiary origin of the fire — but more about that later and we must never be hasty.

Suppose we determine the vehicle's interior is burned out with the fire diminishing into the trunk and engine compartment. What could cause a fire of this type? Burning material left on the seats? A deliberate fire? There are many possibilities but the choices begin to narrow when we see diminished burning in the engine compartment.

If the burning is most intense in the engine compartment we must again consider several possibilities. Was the fire the result of the fuel system malfunction or a problem within the electrical system? Did a problem in the emissions control system contribute to the fire? Did the transmission erupt and spill fluid on the exhaust manifold?

If the most intense burning occurred under the vehicle what caused this? Was it the catalytic converter? Did friction in the wheel bearings lead to the fire? Was tire malfunction the culprit? There are obviously many points to consider in determining origin but the main things to look for are:

1. Area of most intense burning, and
2. Systems and materials within that area that will support ignition and sustain continued burning.

When observing a burned vehicle your initial task will be to isolate the general fire cause. Be careful to distinguish between damage caused by primary and secondary causes.

Suppose our observation leads us to believe the fire's point of origin was the engine compartment. In this case, we must look for indications of failures in the fuel, emissions, electrical systems, or escape of operating fluids. If it was the interior body of the car we must look for smoldering materials or electrical shorts. If the main fire seems to have been centered under the vehicle, look for frictional causes or heat transmission from the catalytic converter or exhaust surfaces. Was the fire centered toward the rear of the car? We can again suspect the fuel, electrical or exhaust systems. If it wasn't , we must find what was being carried in the trunk that might cause fire generated by the causes we have just enumerated. We would certainly consider the possibility of arson, but before we think about this, let's examine the origins of most fires considered as "accidental".

Electrical Fires

Electrical circuits in the modern day automobile are well fused, but fires originating from this cause are not uncommon. A short in the wiring from the power source to the fused circuitry will result in sufficient heat to generate melting, separation and burning of the insulation. So will a wire carrying a higher current then it is designed for. If this heated material comes in contact with any other combustible material a fire can be the result.

Fuses are replaceable conductors in little glass cylinders or plastic clips. The conducting part of the fuse is made of metal with a low melting point. If the current passing through the fuse exceeds its capacity the conduction material will melt and stop the flow of electricity. There are also circuit breakers. These are switches that open the electrical circuit when the current exceeds a certain specified level. After a cooling period the contacts close again and the circuit is restored. Circuit breakers are generally used to protect electric motors that are used intermittently. You will find them on auxiliaries such as power windows, tailgates, etc. Circuit breakers can fail, sticking in closed position and allowing current overload to continue until a fire point is reached.

In recent years automobile manufacturers have added an additional safeguard against electrical fires. This is a device called a "Fusible Link". A fusible link is a section of wiring about four gauges smaller than the regular wiring. If an overload occurs the link will burn out before the regular wiring is damaged. In some cases melting fusible links have contacted nearby plastic components and a fire has resulted. Such a situation caused a General Motors recall in the late 1980's when a melting fusible link contacted the plastic support bracket for the windshield washer fluid.

Wires subjected to extreme shorting show evidence of heat. When a wire develops sufficient heat to burn in two it generally exhibits a "bead" on each end. Sometimes this heat causes the wire to melt with a rounded end on each portion of the separation. When either of these is found it can be concluded that the heat generated by this short was capable of starting fire. Wires burned in two from external heat generally show jagged ends.

Remember, the finding of beaded wiring does not necessarily mean that this is the cause of the fire. Beaded, or shorted, wiring can be the result of a fire as well as the cause. Let us say that during the progress of a fire the insulation is burned or melted from an energized wire. When this wire makes contact with another wire or metal surface a short will result and the wire will become beaded or welded to metal at contact point. This condition will be the result of the fire, not the cause.

The solution of this problem is one of the most challenging in vehicle fire investigations. There are numerous theories regarding the establishment of which came first, the short or the fire. In a study prepared by Robert N. Anderson, Ph. D. the following theory is advanced in the abstract: "Fires caused by electrical shorting are difficult to identify with certainty from fire pattern analysis alone. Surface analysis techniques, such as electron spectroscopy for chemical analysis (ESCA) and auger electron spectroscopy (AES) can be used to determine the history of an electrical short circuit by characterizing the environment in which the electrical arc residues were formed.

In a fire that destroys insulation, the surface of the arc bead formed from the resultant electrical short circuit would show the evidence of combustion products. If the electrical short circuit occurred before the fire had started then arc bead analysis would show the absence of combustion products. (Anderson, R.N. "Surface Analysis of Electrical Arc Residues in Fire Investigation," Journal of Forensic Sciences, JFSCA Vol. 34, No.3)

This method is often referred to as the "auger analysis" and is cited by fire investigators as a procedure for determining whether or not electrical arcing was the cause or result of the fire.

Dr. Anderson goes on to advise "other information such as fire pattern analysis, must be used to assess the importance of the arc". In other words, the arc or short should be considered on the basis of where and in what circumstances it is found. If we find a bundle of shorted wiring under the dash or in the engine compartment of a thoroughly burned out vehicle it may have no definable meaning. If, however,

we find such a condition and fire damage extending outward in a diminishing pattern we clearly have a strong indication that an electrical short has occurred.

A contrary view is expressed by Dr. David G. Howitt, Professor of Material Science, University of California, Davis. In an article titled "The Chemical Composition of Copper Arc Beads; A Red Herring for the Fire Investigator" appearing in the March 1998 issue of *Fire & Arson Investigator* he states "...it is apparent that the concentration of oxygen and carbon measured by auger analysis, accurately reflect the local composition of the beads. The problem is however, that since neither the extent of oxide formation on the surface of an arc bead...nor the amount of fire debris that contaminates, can be seriously considered to reflect a snapshot of conditions at the time of the arc, it would seem that the determination of the arc bead composition is simply a rather expensive waste of time... The similarity between the conposition variations which are found on a single arc bead to those of different beads is a strong indication that even under controlled laboratory conditions there is probably no information that can be gleaned from the surface of an arc bead using auger analysis. The idea that such information would pertain universally to the circumstances that existed at thetime of the arc is absolute nonsense because there is no change in the properties of copper that accompany melting."

Controversy continues on this subject and in the abstract to his paper "Which Came First...the Arcing or the Fire" which appeared in the March 1996 issue of *Fire & Arson Investigator*, Dr. Anderson writes,"The appreciable solubility of gases in metals at elevated temperatures has led to the determination of electrically caused fires by analysis of the chemical elements picked up during the formation of electrical arc residues. The presence or absence of elements of combustion products in the arc residue has proven effective in identifying whether a fire is of electrical origin."

And so it goes. There are a number of well established authorities on both sides of this question. The investigator must use these theories as only one of many factors in coming to a conclusion.

Just how much heat is generated at an electrical short? Remember that copper melts at 1980°F and when we see wiring with beaded ends we know that the temperature had to reach almost 2000°F to create this melted condition. Nearby insulation will be heated since it is plastic and will begin to give off a flammable gas which will ignite. Though the original fire will be small a cycle of heating and release of flammable gas is established and the fire will grow. This is especially



This extensive damage in a BMW's dash —



was caused by this wire shorting when insulation wore away by contact with the radio

true of automotive wiring that has become coated with dust and roadfilm that frequently contains oil.

Automobile wiring consists of "primary" and "secondary" wiring. Primary wiring is that which leads from the battery to the ignition switch and on to the coil. This is "hot" line at all times. Secondary wiring is that which runs from the coil to the distributor or coil packs and onto the spark plugs.

"Fusible Links" are generally found in the primary system. Following is a fusible link guide on some American manufactured vehicles. The chart is indicative of what may be expected in checking for these links. It should not be regarded as absolute since new systems are being added regularly.

Guide to Fusible Links

Make	Location	Wire Color
AMC		
	Battery terminal of starter to main wire harness	Red
	Battery terminal of horn relay to main wire harness	Pink
	Accessory terminal of ignition switch to wire harness	Brown
	Battery terminal of starter relay to heated rear window relay	Red
	B-3 terminal of ignition switch to circuit breaker	Red
	I-3 terminal of ignition switch to circuit breaker	Yellow
	I-3 terminal of ignition switch (single wire at switch splits into two feed wires)	Yellow
	I-3 terminal of ignition switch to throttle stop solenoid	Yellow
	Engine compartment harness at AV terminal of dash connector	Dark Green
	SOL terminal of ignition switch to wire harness	Dark Green
Chrysler		
	Rear Wheel Drive Vehicles	
	Rear of battery positive terminal in circuit leading to #10 cavity (hazard flasher circuit) of bulkhead disconnect	Orange
	Between alternator and wiring harness main splice R6	Black or red
	Four links at rear of left front wheelhouse between wiring harness main splice R6 and bulkhead disconnect cavities 4, 38 (to ignition switch), 6 (to heated rear window), and 40 (to headlamp switch)	Gray *(4,38, 40)* Dark Blue*(6)*
	Front Wheel Drive Vehicles	
	Rear of battery positive terminal in circuit leading to #12 cavity (hazard flasher circuit) of bulkhead disconnect	Pink
	At left side shield between warness main splice A1 & alternator	Red*(65 amp alternator)* Dark Blue *(60 amp alternator)*
	Four links at left side shield between wiring harness main splice A1 & bulkhead disconnect cavities 4,37 (to ignition switch), 6 (to headlamp switch, and 35 (to heated rear window)	Gray*(4, 6, 36)* Orange *(37)*

Guide to Fusible Links - continued

Make	Location	Wire Color
Ford **Lincoln** **Mercury**		
	Between Starter relay and alternator.	OEM links are Green or Black
	Looped outside wire harness between starter relay and alternator	Links are colorcoded according to wire gauge as follows:
	Twin links—one between starter relay and alternator, and one between starter relay and vehicle equipment harness	Blue—20 ga. Red—18 ga. Yellow—17 ga. Orange—16 ga. Green—14 ga.
	Triple links—one between starter relay and alternator, and two between starter relay and vehcile equipment harness	
	Four links—one between starter relay and alternator, and three between starter relay and vehicle equipment harness	
General Motors		
	At starter solenoid battery terminal	Black (except H&X-body)
	To cooling fan relay, X-body	Red (H&X-body)
Cadillac		
	One, two, or three links at starter solenoid battery terminal (depends upon model year/body style)	Black or Dark Green

Chevrolet

Single link at solenoid battery terminal	Brown
Twin links at starter solenoid battery terminal	Red *(gas vehicles)*
	Black *(diesel vehicles)*
To cooling fan relay, X-body	Red
Molded splice at horn relay	Black
Molded splice in voltage-regulator #3 terminal wire	Orange
Molded splice in ammeter circuit (both sides of meter)	Orange

Oldsmobile

Single or twin links at starter solenoid battery terminal	Black
	or Brown
	(all except H& X-body)
	Red *(H&X-body)*
To cooling fan relay, X-body	Red

Pontiac

Positive battery cable pigtail lead (requires entire cable replacement)	Brown
At horn relay	Black
Molded splice in circuit at junction block and horn relay (some single and some twin)	Orange
Single or twin links at starter solenoid battery terminal	Black
	(all except H& X-body)
	Red *(H&X-body)*
To cooling fan relay, X-body	Red

*Not all links are used on any one vehicle, unless otherwise specified. Colors and locations subject to change at manufacturer's discretion. Additional links may be added as a running change during the model year and not appear on vehicle wiring diagram for that year.

Generally speaking the plastic material covering the wiring harness will operate in a temperature environment of 190-200° Fahrenheit. While engine heat and extreme outside temperature can cause a softening of this material it would cause no damage. When heat generated by shorting or from other outside sources exceeds the maximum temperature this material will begin to melt or burn. When other fuel is in close proximity a fire can result. When a hot smoldering wire comes in contact with sound deadening material (fiberglass firewall or dash pads or plastic ducting material) a fire is very likely to follow. Once this fire is

BELDON AUTOMOTIVE PRIMARY WINDING SAFETY GUIDE

Total Approx. Circuit Amperes		Total Circuit Watts		Total Candle-power		Wire Gauge (For Length in Feet)											
6V	**12V**	**6V**	**12V**	**6V**	**12V**	**3'**	**5**	**7'**	**10'**	**15'**	**20'**	**25'**	**30'**	**40'**	**50'**	**75'**	**100'**
0.5	1.0	3	12	3	6	18	18	18	18	18	18	18	18	18	18	18	18
.75	1.5			5	10	18	18	18	18	18	18	18	18	18	18	18	18
1.0	2	6	24	8	16	18	18	18	18	18	18	18	18	18	18	16	16
1.5	3			12	24	18	18	18	18	18	18	18	18	18	18	14	14
2.0	4	12	48	15	30	18	18	18	18	18	18	18	18	16	16	12	12
2.5	5			20	40	18	18	18	18	18	18	18	18	16	14	12	12
3.0	6	18	72	25	50	18	18	18	18	18	18	16	16	16	14	12	10
3.5	7			30	60	18	18	18	18	18	18	16	16	14	14	10	10
4.0	8	24	96	35	70	18	18	18	18	18	16	16	16	14	12	10	10
5.0	10	30	120	40	80	18	18	18	18	16	16	16	14	12	12	10	10
5.5	11			45	90	18	18	18	18	16	16	14	14	12	12	10	8
6.0	12	36	144	50	100	18	18	18	18	16	16	14	14	12	12	10	8
7.5	15			60	120	18	18	18	18	14	14	12	12	12	10	8	8
9.0	18	54	216	70	140	18	18	16	16	14	14	12	12	10	10	8	8
10	20	60	240	80	160	18	18	16	16	14	12	10	10	10	10	8	6
11	22	66	264	90	180	18	18	16	16	12	12	10	10	10	8	6	6
12	24	72	288	100	200	18	18	16	16	12	12	10	10	10	8	6	6
15	30					18	16	16	14	10	10	10	10	10	8	6	6
20	40					18	16	14	12	10	10	8	8	6	5	4	2
25	50					16	14	12	12	10	10	8	8	6	6	2	2
50	100					12	12	10	10	6	6	4	4	4	2	1	0
75	150					10	10	8	8	4	4	2	2	2	1	00	00
100	200					10	8	8	6	4	4	2	2	1	0	4/0	4/0

* 18AWG indicated above this line could be 20 AWG electrically — 18 AWG is recommended for mechanical strength.

communicated to other materials such as flex fuel lines, foam rubber, etc. the vehicle will likely be totally destroyed. In most cases, however, the destruction will not be that complete since fire fighting equipment is generally nearby. On examining the auto's engine compartment or interior you will be struck by the heavy pungent odor of burned plastic insulation. You should begin tracing the wiring system from the battery, searching for the evidence of heavily burned and melted insulation, exposed wiring and signs of "fusing" of the wiring against bare metal.

Some vehicles are now using two batteries instead of the usually found single 12 volt battery. Recently we examined a late model BMW that had a severe short occur when a wire from which the insulation had rubbed off made contact with the firewall. The battery in the trunk compartment was checked and it had been disconnected by the fire department. When we pushed at the fused wiring with a screw driver, a shower of sparks flew. That is when we found there was a second 12 volt battery also in the trunk compartment on the opposite side.

Loose connections in wiring can result in a heat build up with fire resulting. In tracing wiring check to see that all connecting points are tight. A faulty connection will impede the flow of electricity and result in an increase of resistance. With heightened resistance comes heat. This is generally evidenced by melting of copper or brass connectors. Wiring that is subjected to constant binding pressure, such as a too tight clamp can cause insulation to break down and a short to occur. Wiring that rubs against sharp metal edges will cut the insulation and the grounding of the wire against the metal. When this occurs in the primary wiring system a fire is almost certain.

Alternators can be sources of vehicle fires. Plug type connectors are used in some of these units and can become loose or develop high resistance after being removed and reconnected. Corrosion can also produce the same condition. Occasionally when alternators are changed a 75 AMP will be substituted for a 65 AMP unit. The additional output may cause melting and a good chance of fire.

Another cause of alternator fires can be bearing failure. This creates intense heat which is communicated to the varnish covering of the copper windings and wiring insulation. Ignition of vapors occurs and a fire capable of spreading to surrounding material is the result.

Tests have shown that vehicle wiring harnesses can be damaged by fluids that enter at unsealed points and by a "wicking" action permeate the entire system.

When the water combines with dirt corrosion results and the resistance in the wiring changes, making relays, switches, splices and connectors unreliable and overloading the system. This causes hot spots to develop and the burning or melting of insulation.

An infrequently encountered electrical fire cause is the breaking of the ground straps. As you know, an electrical circuit flows from the positive side to the negative, or ground. In a motor vehicle the frame and body make excellent grounds for the circuit. On some occasions we have observed cases where the ground strap leading from the engine block to the frame has separated. When this happens the circuit is interrupted and the electricity seeks another path to ground. Sometimes the path chosen will be the throttle cable, emergency brake cable or some other metal line. High resistance is created and the chosen line will glow red hot. If it contacts anything flammable a fire results. On one such case the selected path was the metal fuel line which then grounded against the oil dipstick. The fuel line melted and allowed fuel to escape under pressure. The heat from the contact ignited the vapors and an engine compartment fire quickly followed.

In late 1990 we observed our first fire that appeared to have originated in the steering column near the ignition switch of a Ford product. The fire occurred while the driver was attending an automatic teller machine and since it was discovered almost immediately the damage was not severe. No exact cause could be determined other than concluding there had been a "short in the ignition switch."

Following this case a number of similar fires were observed in Ford passenger and light truck vehicles. We published an article on this problem in our newsletter Vehicle Safety & Security Report and received a number of responses from other investigators reporting similar cases in late 1980's and early 1990's Fords.

We conducted a number of investigations and extensive research into these problems and in 1995 issued special reports which are excepted below:

"The seriousness of this problem was brought home to us during an investigation into the cause and origin of a fire in the front passenger compartment of a Ford Aerostar in the Sacramento area. The fire in the vehicle was discovered by a neighbor at about 2:00 AM who reported it to the fire department. The Aerostar had been parked by the owner in the driveway at 10:00 PM. It usually was parked in the garage but on this occasion the owner's son had come home early and parked his car in the garage. Had it not been for this the Aerostar would have

burned in the garage with the fire undetected until it became intense enough to break through the walls. This could very well have resulted in death or serious injury to occupants of the house.

Our research involved studying new ignition switches, used ignition switches, partially burned ignition switches and heavily burned ignition switches. The failure of all these switches was due to a short between the battery terminal and chassis ground terminal inside the ignition switch.

This switch design was used from 1986 through 1991 or 1992, depending upon the application. A new design which relocates the chassis ground terminal inside the switch and also approximately doubles its square surface area was available in 1992 and replaced the faulty switch.

The switch which was most helpful in locating the problem was switch #E6EC-11572-AA. It was taken from a vehicle where the fire was extinguished before the housing of the switch was destroyed. Another switch with the exact same number was obtained from a used vehicle in a salvage yard to use as a comparison. The damaged switch had suffered melting point damage to the chassis ground terminal due to a short to the battery terminal located approximately 3/32 inches away. The sliding contact leg, used in this part of the switch, is convex and measures approximately 6/64 inches across the top when new but measured 9/64 inches on a worn and failed switch. The contact on the slider measures larger as more wear is present due to it's convex design. It measures approximately 12/64 at its widest point. Only a small plastic divider approximately 2/32 inches wide separates the two problem terminals. It should be noted that the chassis ground terminal on the non-damaged switch had excessive lateral movement as well as up and down movement. Any failure of the small plastic divider (each time the vehicle is started, a contact on the movable slider which is spring loaded rubs against the plastic divider) or any tracking due to sparks arcing (this is normal inside of an ignition switch) along with the widening of the movable slide contact due to wear, allows a direct short between the battery and chassis ground terminals inside the ignition switch. This is a direct short and causes heat and fire.

The actual position of the ignition switch when this short occurs is between the "run" and "start" positions, although with tracking it could be any position including "off" or "lock" or "run".

There are two three-legged contacts and the smaller of the two has one rear leg which contacts terminal A2 for accessories and the other rear leg contacts the

battery during the "run" position. When the key does not fully return from start to run, it provides a perfect position to form an arc between the battery and chassis ground terminal through the leg of the small moveable slider. This also explains why some fires occur while the vehicle is running.

The new designed switch has a suffix of BB and has been designed with three changes. Contact P2 has been moved up to a position between battery terminal and chassis ground, increasing the distance from approximately 3/32 inches to 12/32 inches between the two problem terminals. The chassis ground terminal surface area has been approximately doubled and the contact leg from the small moveable slide is now the front leg on the new design; on the old design it was the right rear leg. On the new switch, the contact point for the P1 terminal is the left rear leg; on the old design the contact point for the P1 terminal was the front leg. The third change is for terminal P2; the new switch uses the right rear leg while the old switch used the left rear leg. The slider assembly stayed the same. The new design appears to solve the problem. By increasing the distance between the two problem terminals by more than 1/4 inch (8/32), the moveable slider cannot contact both battery and chassis ground at the same time."

This problem resulted in a massive recall involving the following years, makes and models:

7.9 million Ford vehicles recalled in USA
1988-90 Escort
1988 EXP
1988-92 and early build 1993 Mustang
1988-92 and early build 1993 Tempo/Topaz
1988-92 and early build 1993 Thunderbird/Cougar
1988-92 Crown Victoria/Grand Marquis
1988-89 Lincoln Town Car
1988-91 Aerostar
1988-91 Bronco/F-series light trucks

651,000 Ford vehicles recalled in Canada
1988 Escort
1988 EXP
1988,1992 and early build 1993 Mustang
1988-92 and early build 1993 Tempo/Topaz
1988, 1989, 1992 and early build 1993 Thunderbird/Cougar
1988 Lincoln Town Car

1988 Aerostar
1988, 1989 and 1991 Bronco/F-series light trucks

248,000 Ford vehicles previously recalled in Canada
1989-90 Escort
1989-90 Mustang
1990-91 Thunderbird/Cougar
1988-89 Crown Victoria/Grand Marquis
1989 Lincoln Town Car
1989-91 Aerostar
1990 Bronco/F-series light trucks

•Total Ford vehicles in this recall: 8.7 million (7.9 million in USA)

While it is rare, a battery explosion can be the source of a vehicle fire. One type of such an explosion is caused by an excessive charge rate which causes hydrogen gas to be released. If there is insufficient ventilation, a spark can ignite this gas.

In 1978 the National Highway Traffic Safety Administration issued a report titled "Battery Explosion Tests and Labeling". While this report is over 20 years old it provides excellent background information on the question of battery explosions. The following information from this report appeared in *The Vehicle Fire Reporter*, Volume 2, Summer 1997 by Lee S. Cole & Associates, Inc.

Properties of the Gas Mixture

Hydrogen and oxygen gases are natural constituents of the air we breathe. Hydrogen exists in a concentration of approximately 0.01% and oxygen approximately 21% by volume in air. As for gases generated within a storage battery, electrolysis converts or separates each water molecule into two parts hydrogen and one part oxygen. Thus, under steady state conditions such as continuous charging over a period of hours, the composition of gas in the battery ullage space above the electrolyte may approach 66-2/3% hydrogen and 33-1/3% oxygen.

Even after charging, the mixture in the ullage space may remain hydrogen-rich for a period of several hours because it is thoroughly mixed with oxygen and vapor and because restriction to the escape of hydrogen is inherent in the small cell vent hole, normally 0.06 inch diameter or less. The vent hole must be of minimum size so as to retain as much liqud as possible within the battery.

Once the hydrogen escapes the vent hole, however, its remarkable diffusion properties allow it to disperse with extreme rapidity into the surrounding air. Some researchers have found that highly concentrated hydrogen escaping from a cell vent hole of a battery undergoing severe high current charging dispersed to a concentration of less than 4% within approximately one to two inches immediately above the vent.

Combustion Characteristics of Hydrogen

When mixed with oxygen or air and heated to an ignition temperature of approximately 1074°F, hydrogen gas may be extremely flammable or explosive across a broad range of concentrations. Many texts list the flammability range of hydrogen as being between approximately 4% and 74% by volume in air at standard temperature and pressure. Other data indicate that the upper limit of the flammability range may extend to approximately 95% in oxygen. Within the broad flammability range, hydrogen gas may explode or detonate within a somewhat narrower band of concentrations thought to be from approximatley 15% to 90% oxygen.

Hydrogen, of course, is an extremely efficient fuel, releasing approximately 62,000 BTU/lb. of heat energy when ignited, compared to only about 20,000 BTU/lb. for gasoline. When burned, hydrogen exhibits an almost invisible blue flame and is entirely smokeless. The only product of combustion is water vapor.

The detonation of an explosive hydrogen-oxygen mixture may be characterized as follows. Upon being ignited by any heat source which exceeds the 1074°F, auto-ignition temperature of hydrogen, a flame front expands rapidly in approximately spherical shape from the point of ignition.

Rapid expansion of the flame front compresses the unburned mixture ahead of the front, creating a substantial shock wave. Researchers have estimated that the speed of the shock wave emanating from the ignition of a stoichiometric or ideal mixture of hydrogen and oxygen may approach 3000 meters per second (Or 1.86 miles per second.)

As previously discussed, storage batteries produce hydrogen and oxygen gases as natural byproducts of normal operation. These gases are generated and may exist in the battery's ullage space above the electrolyte in approximately stoichiometric or ideally combustible proportions. If

ignited by a high temperature heat source, such as a spark or flame, the gas mixture may detonate.

The shock wave created through compression of the unburned gas by the rapidly expanding flame front may impact the inside walls or cover of the battery, causing rupture of the container. Experimentation has shown that the rupture pressure pulse created by the ignition of a stoichiometic mixture of hydrogen and oxygen in the ullage space of a typical 12-volt battery may exceed 300 pounds per square inch with a time duration of approximately 1.3 milliseconds. Thus, the explosive forces created within the battery may be several times greater than any conventional container can withstand. The speed of the reaction precludes the possible use of rupture discs or other devices designed to release or redirect a relatively-gradual increase in overpressure.

Once rupture of the case or cover of the battery occurs, gas in adjoining cells may be ignited. Observations of the remnants of many exploded batteries have indicated that virtually simultaneous explosions may have occurred in two, three, or more cells as a result of the ignition of only one cell. However, the total heat energy released by these explosions is very low. With the electrolyte level at or below the top surface of the plates, the ullage space of a typical 12-volt storage battery may approximate a volume of no more than 9 cubic inches. Assuming a stoichiometric mixture of hydrogen and oxygen, the quantity of hydrogen gas contained in the cell is computed to be 0.00002 lb. which, if ignited, can release only approximately 1.2 BTU. Even in the event of nearly simultaneous explosion of the gases contained in all six cells of a typical 12-volt battery, the total heat released can only be approximately 7.25 BTU, the amount of heat required to raise the temperature of one pound of water 7.25°F.

The forces generated by a hydrogen-oxygen explosion within a battery are directly proportional to the quantity of fuel oxidizer available. Therefore, the forces can be limited by reducing the volume of the ullage space to an absolute minimum. However, sufficient space is required to maintain an adequate reserve of electrolyte and to allow for the separation of entrained liquid from the gases being vented from the battery. Researchers have found that even with the ullage space reduced to as little as 3 cubic inches, the explosive pressure pulse generated by detonation was sufficient to rupture the container.

Characteristics of Battery Explosions

Based on numerous tests and observations of exploded storage batteries, certain common characteristics are clearly evident.

Of paramount importance is the fact that storage battery explosions of significant intensity can only be produced through ignition of an explosive mixture of hydrogen and oxygen or air within the battery. Although rupture of a totally sealed battery can occur through a pressure buildup of internally generated gases, the rupture will occur at relatively low pressures resulting only in separation of the case-to-cover seal or the dislodging of vent caps and a gradual release of the overpressure. There is no scientific evidence to support the theory that substantial fragmentation of the container or violent spraying of acid occurs through gradual internal pressure buildup. Further, ignition of the internally confined gas mixture is often difficult to achieve. Not only must the ignition source exceed the auto-ignition temperature of hydrogen, but the gas mixture must exist in explosive concentrations at the moment of ignition and at the site of ignition.

Because the quantity and concentration of ignitable gas can vary greatly, so too does the intensity of the resultant detonation. Minor detonations may manifest themselves as a mere "popping" sound, reminiscent of the removal of the cork from a champagne bottle, accompanied by ejection of a vent cap, slight splashing of battery electrolyte through a filler hole, or the occurrence of a slit or crack in the battery case or cover.

More severe detonations may cause substantial fragmentation of the battery cover and upper portions of the case walls and cell partitions, and violent splashing or spraying of electrolyte. The noise levels associated with these detonations have been likened to the discharge of a small bore rifle or shotgun. Although detonations may occur in several adjoining cells as a result of the ignition of a single cell, the rapidity of the reaction precludes the observer from distinguishing separate explosions.

In the course of a severe detonation, vent caps, electrolyte, and fragments of the battery case and cover may be thrown in generally upward and outward trajectories. Initial velocities of such debris have been measured at 80-100 feet per second. However, individual fragments seldom weigh more than a few ounces, thus substantially limiting their kinetic energy.

Even in the event of a severe detonation, the combined mass of the plates, separators, and electrolyte prevents any significant damage to the lower portion of the battery. Fragmentation is ordinarily limited to the cover, case walls, and partitions in the immediate vicinity of the ullage space(s) in which the explosion(s) occurred. This superficial damage to the containing structure often will not prevent the battery from being used as a power source until replacement can be made.

Storage battery explosions may be further characterized by the absence of smoke, visible flame or flash, or residual charring or melting within the area of the detonation. The ignition of a stoichiometric mixture of hydrogen and oxygen results in complete combustion, leaving nothing but residual water. The speed of the flame front, the low visibility characteristic of the hydrogen flame, and the remarkably small quantities of gas involved in the reaction combine to minimize heat transfer and self-illumination.

Accidental Ignition Sources

As suggested above, the search for the cause of a battery explosion begins with a search for a source of ignition. The preponderance of accident and test data indicate that most explosions are caused by the introduction of an ignition source such as a spark or flame from outside the battery. The ignition source must be placed in a flammable zone of venting hydrogen, which occurs within approximately one to two inches of the point of exit of the gas from the battery's ullage space. Upon ignition, the external flame may propagate to the interior of the cell causing detonation of the contained hydrogen-rich mixture.

While open flames in the form of lighted matches, cigarettes, cigars, pipes, cigarette lighters, etc. can easily be introduced within the flammable zone of venting hydrogen, some distinctions must be made with respect to the behavior of sparks near the battery. For example, a tiny blue electric arc may occasionally be observed at the terminal of the battery during routine disconnection of battery cables or booster cables. Although the blue arc exceeds the ignition temperature of hydrogen, the location of the arc is generally outside the flammable zone immediately above the cell vent opening and thus cannot ignite the gas escaping therefrom.

On the other hand, reversed or cross-connected booster cables between

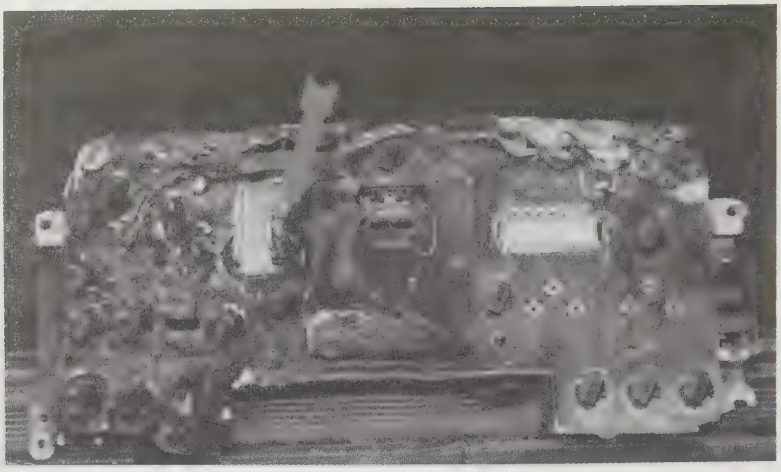
two 12-volt batteries create the electrical equivalent of a 24-volt, direct short circuit. Very high currents, attempting to bridge the comparatively high-resistance area of contact between the booster cable clamp and battery terminal, cause intense localized heating. Resulting melting and boiling of the locally heated, mated metallic surfaces causes white-hot, molten metal particles to be ejected from the area of contact. This phenomenon is often described as a "shower of sparks". Although such data suggests the possiblity of internally generating sparks as ignition sources, tests have shown that sparks intentionally created beneath the surface of the electrolyte cannot propagate to the battery's ullage space. Within the engine compartment of a vehicle, a number of possible ignition sources may exist; however, in virtually all instances, the spark and flame sources are far removed from the battery cell vent flammable zone. Although other suspected ignition sources including static electricity have been investigated, none have been found to bear technical credibility.

From the foregoing information it appears that battery explosions accompanied by fire are unlikely but not impossible. The following information was obtained from a former battery shop operator:

"The number one cause for battery explosions is the 'loose battery terminal' which causes a spark for the gas...Sometimes if the battery is not properly mounted, or the wrong size battery is installed, it can rub a hole in the side. This empties the acid from one cell. This causes high resistance and forces the alternator to run all out. Sometimes an arc will be produced between the cell plates and cause an explosion." While this information may not be based on scientific data it should not be ignored. A recent investigation revealed a situation exactly as that described concerning an emptied battery cell and a severely overheated alternator.

The fad of "lowriding," which was popular in some areas, can also lead to a battery explosion hazard. The power source for the hydraulic lifters used to raise and lower the "low rider" is a series of 12 volt batteries installed in the trunk. Accumulated gas from these batteries in the closed confine of the trunk could explode with dire consequences.

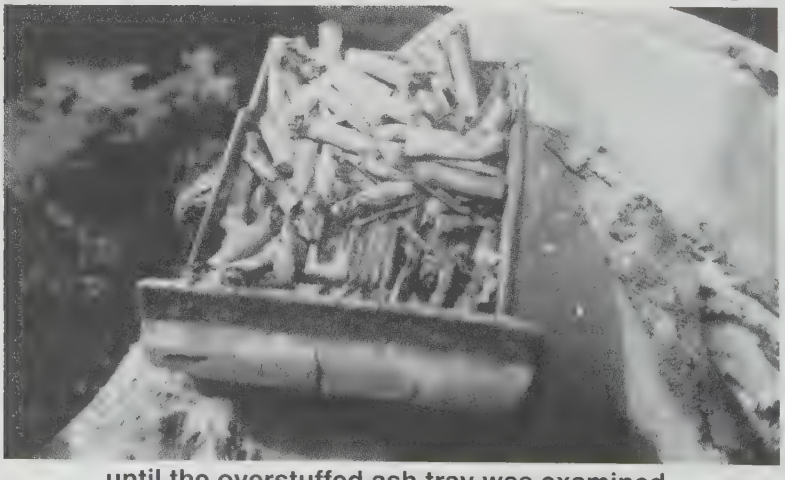
In checking for the source of an electrical fire don't forget "add ons" by the car owner such as aftermarket alarms, stereo systems, air conditioner units, etc. Improper installation of these units can result in shorted wiring and a fire hazard.



This photograph shows the plastic backside of a dash from a late model Ford pickup. A poor electrical contact at the arrow caused overheating and the plastic melted and ignited.



This dash fire was at first thought to be electrical in origin...



until the overstuffed ash tray was examined.

We have conducted investigations into several fires where the alternator was identified as being the most probable cause. On one of these cases the driver reported that she smelled smoke while driving her 1991 Aerostar. She allowed the vehicle to coast downhill and pulled into the parking lot of a video store. When she stopped she could see there was a fire under the hood. The fire department was called and the fire extinguished. Our inspection of the vehicle revealed the wiring at the plug connector to be melted and the plastic burned. This appeared to have occurred as a result of a high resistance overload. this was a result of an in-line resistance to the load. Burn patterns in the engine compartment indicated the fire spread from this area.

Another alternator problem resulting in fire has been observed when the alternator had been replaced. A 1991 Ford Aerostar caught fire while the owner was making floral deliveries. An inspection revealed the area of origin was once again the alternator. After a preliminary investigation we determined that the alterator had been replaced. Following is a portion of our report of investigation:

> The alternator and wiring from the alternator of the 1991 Ford Aerostar were evaluated. The regulator is mounted on the back of the alternator making it an integrated designed system. A three wire harness with female blade terminals plugs into the regulator. A second three wire harness with female blade terminals plugs directly into the alternator. The highest heat area and damage occurred at the harness connection which connects directly to the alternator. Both male terminals of the alternator connection which are connected to battery (B+) were completely melted. The third terminal at this connection, which connects to the stator terminal of the harness, was still present but heavily damaged.
>
> Two different size wires are used for the battery (B+) terminals of the harness from the factory. Vehicles equipped with a 4.0L engine use two 12 gauge wires and vehicles equipped with a 3.0L engine use two 14 gauge wires. This vehicle was equipped with 3.0L engine.
>
> Our original investigation report shows that the alternator and wiring harness were replaced. The 3.0L engine is equipped with a 60-65 amp alternator and uses the 14 gauge wiring harness. The replacement alternator #75A-E79FBA-C217C is a 75 amp alternator and should not have been used to replace the 60-65 amp alternator. A bulletin also warns of the problem due to the design of the connectors (slide-on) and the possibility of fire.

It is my professional opinion that the cause of this fire is most likely due to the wrong replacement of the alternator or the problem with the slide-on connectors at the alternator (B+) terminals.

The bulletin referred to follows:

1. Before installing this alternator carefully inspect the connector from the wiring harness. (See illustration below.) If it is corroded, deformed or in generally poor condition, it must be replaced. Failure to replace the connector could result in a melting condition that may or may not cause a fire.

2. The gel supplied with this alternator is of a special type. It prevents corrosion, promotes better conductivity and is necessary for the correct operation of the alternator with the wiring harness. Failure to apply gel, as shown below, may cause a melting condition, which may or may not cause a fire at once or several months in the future.

3. Do not substitute a 75 amp alternator for a 65 amp alternator. The wiring harness is designed to fit the alternator. The additional output may cause melting which in turn may or may not cause a fire. If you are uncertain, consult with the store from which the alternator was purchased. In addition, if the wiring harness is very hot or starts smoking when the alternator is near 70 amps output, this is an unsafe condition.

4. Do not run this alternator at full output for more than 10 seconds. Alternators and particularly these Ford alternators, are designed to operate at 75% of the maximum. Operating these alternators at full output for more than 10 seconds may cause melting of the connector, which in turn may or may not cause a fire.

5. The full output of this alternator is delivered through the two large spade terminals. If the connector does not fit snugly, excessive heat will build up and the connector may melt. A careful installer must take the time to be certain the connector fits with no loose play.

Mr. Keith Fowler of Western Canadian Mechanical Investigations in Edmonton, Alberta, Canada advised his experience relating to a certain group of alternator fires:

On the subject of alternators, General Motors has had problems with the front bearing. I have investigated several fires in GM trucks including the S series where the front bearing of the alternator suddenly seizes. As the

newer model trucks use a serpentine belt which is capable of transmitting large amounts of horsepower compared to a conventional V-belt and as they also use an automatic tensioner the belt does not burn or loosen off. The heat generated by the bearing seizing results in the aluminum housing expanding. When the housing expands the complete bearing will start to turn in the housing. This generates heat from friction which is absorbed by the housing. Eventually the heat melts the electrical insulators and an electrical short occurs followed by fire. As the alternator's supply is not protected by any type of fuse or fusible link the supply wire will quickly overheat. In older vehicles which used a conventional belt, the belt would simply start to slip and loosen off.

I have investigated several vehicles which were total burns and the only way to find the problem is to recover the alternator remains and cut open the front bearing race and examine the races for evidence of damage. In total burns, often the alternator's aluminum housing is totally consumed.

In cases of a relatively small burn, examination of the backside of the alternator fan will show a rotary pattern of melted aluminum sprayed out along the backside of the fan. Oddly enough, very few of the vehicle operators reported hearing any unusual noises prior to the fire occurring, one would expect them to hear a bearing noise but the failure may be very sudden. Almost all operators reported the lights dimming and a discharge condition on the voltage gauge just prior to the fire. There does not seem to be any pattern to these failures and I suspect GM had a bad batch of bearings from a supplier for a period of time. The year range is from late 1990 to 1994. GM has been paying insurers for the fires but has not issued a recall as they say this is a limited problem. When I first discovered the problem I checked with local GM dealers who reported they have seen many cases of bearing failures in these alternators, some reported having cars towed in with the front end of the alternator melted away.

A second report was sent to us by Ken Feder of Forensic Automotive and was authored by Paul Adams, Master Technician of Forensic Automotive:

In recent investigations conducted by our firm, a trend has surfaced concerning General Motor's Blazers, Jimmys, S-10, S-15, and Astro vans equipped with 4.3L V-6 engines.

We have found a number of cases involving alternator front bearing failures causing the alternator cooling fan blades to contact the front

mounting frame and separate from the fan hub, becoming high velocity projectiles cutting and damaging fuel lines or power steering lines. The fluid or fuel in these lines is easily ignited by heat from the exhaust manifold.

All of the vehicles in which this scenario has been seen are higher mileage, usually in excess of 60,000 miles. Statements from the vehicle owners at the time of the fire do not indicate any unusual noises heard prior to ignition. This seems to be unlikely based upon the amount of damage to the front end alternator frames, although we have not conducted any sound testing in this area.

The bearings that have been inspected show the inner race seized to the armature shaft and the outer race grinding on the aluminum front housing frame of the alternator. The outer housing damage is always in a downward pattern accelerated by the tension of the serpentine drive belt. The serpentine belt is under constant tension by a spring-loaded mechanical tensioner. The routing of the serpentine belt causes the majority of the tension to be between the 12:00 and 1:00 o'clock positions when viewed from the front of the vehicle.

The National Highway Traffic Safety Administration opened a defect investigation on July 29, 1994. The ODI Resume includes many more models than the few listed. The preliminary analysis was upgraded to an Engineering Analysis in July 1994. The analysis was closed July 31, 1996 stating: "More importantly, ODI believes the majority of the alternator fires are associated with low mileage failures, and since most of the subject vehicles have exceeded the mileage at which the failures arise, the majority of the fires attributed to alternator failures should have occurred by now.

This is in direct conflict with findings compiled when investigating fires in these vehicles. It should be noted that our findings have been limited to the vehicles listed. No vehicles with 5.0 liter and 5.7 liter engines have been examined. Our investigators are finding bearing failures on both original equipment alternators and after-market remanufactured units.

The extent of damage to the vehicle varies from confined to the alternator and either fuel or power steering lines to complete destruction of the engine compartment and exterior paint damage.

Fuel System Fires

The motor vehicle's fuel system is the most common source blamed in vehicle fires. This blame is well deserved in some cases and is a handy "scape goat" in some others. Let's think a little about gasoline, the most common fuel for the motor vehicle. Remember that under the proper circumstances gasoline can produce an explosion more powerful than dynamite. It vaporizes very easily and these invisible vapors are heavier than air. In closed structures, such as a garage, they will spread along the floor to a water heater, an electric motor or a car's engine and produce an explosion and fire of disastrous results. Many fires of this type result from siphoning gas in the garage, washing off car parts with gasoline or similar activities. I have worked on a number of investigations where fatalities resulted from the water heater igniting fumes accumulated in garages where car thieves used gasoline to wipe fingerprints from stripped vehicles.

The gasoline shortages that occur occasionally can produce another hazard, that of the worried traveler who carries a can of gasoline in the trunk so that he will not be stranded without fuel. In case of ignition of the vapor from these cans the driver's concern for being stranded will be academic. The explosion would be almost powerful enough to put driver and car in orbit!

Let's review for a minute. Fuel is stored in the tank and is pulled or pushed from this receptacle through lines by the fuel pump and into the injector system or carburetor where it is mixed with air, drawn into the manifold and cylinders where it is ignited. Fuel lines are for the most part metal but in places where they must bend at acute angles or are subjected to excess vibrations they are composed of that we call "flex lines". These lines are rubber or plastic or metal fiber composition. During the process of extraction from the tank to deposit in the intake system there are numerous connections of the main line and its "flex lines". These are generally made with either clamps, screw type threads, crimped fittings or snap type fittings. At any of these connections loose fittings can result in leaks. The farther away from the engine compartment this occurs the less the danger of fire. This is fairly obvious since ignition possibilities decrease as the volatile fuel is removed from sources of heat and electrical sparking. Some manufacturers now use plastic fuel lines throughout the fuel delivery and return system. This plastic is "Nylon 12" and will melt when exposed to temperatures above 360°F. This has been the cause of many engine compartment and underbody fires when exhaust surfaces have become superheated.

Some of the most common engine compartment fuel fires in older vehicles result from float sticking, frozen needle valve or improperly fitted needle valves.

Any of these conditions allow gasoline to escape through the anti peculator hole in the top of the carburetor. In the case of fuel injected vehicles a leaking high pressure fuel line is the most frequently observed cause. A poorly fitted connection from the fuel line to the carburetor or a clogged filter can produce the same result. Once gasoline has begun to flow and collect on the top of the hot manifold it begins to give off vapors. The ignition of these vapors can be only a matter of time. If you suspect a carburetor related fire look for burning and melt down of the carburetor itself. If the carburetor shows only smoke and light heat damage chances are the fire didn't start there. A backfire in the carburetor can result in a large fire in the engine compartment. A backfire blows fuel all over the engine and any ignition source can set off a flash fire. The air cleaner filter is an excellent spark arrestor, but this element can become rapidly saturated with gasoline oil and road film. Since it is porous it will continue to hold this flammable mixture and will burn once flame is applied.

On some rear engine mounted vehicles the fuel line connection is not securely clamped at the carburetor inlet. Vibration can result in the connection being broken and the spilling of gasoline in the engine compartment. Ignition occurs immediately and hundreds of these fires have occurred.

Leaks at the fuel pump can also produce fires. Check for evidence of intensive burning around the pump and the connecting flex line. Once the flex line becomes disconnected at the pump, this can also result in a fire hazard. A "soft plug" fitting in the inlet of certain carburetors has been known to leak and cause engine compartment fires. Any leaking gasket, fitting, connection or cracked fuel line is a potential fire. A broken engine mount can allow the block to twist enough to rupture fuel lines and allow fuel to escape.

Fuel injection systems are now the common mode of fuel delivery today. There are two basic methods of fuel injection systems in use in passenger cars, light trucks and vans today:

Electronic Fuel Injection (EFI)

In this system fuel flows in short spurts as the injectors turn off and on sending fuel into the intake system. An electronic control unit determines the length of time the injector is held open. This unit has memory capability and monitors other systems of the vehicles operation and uses this information to compare data against a pre-programmed memory system.The electronic control unit then signals the injectors to adjust or change the air/fuel ratio for various speeds, loads and operating parameters including the oxygen content in the exhaust stream.

Continuous Injection System (CIS)

This system supplies fuel to the injectors in a constant, continuous flow. This flow is controlled by a distribution unit that feeds fuel to the engine directly proportional to the airflow requirement. CIS is basically mechanical and makes use of a limited electronics system.

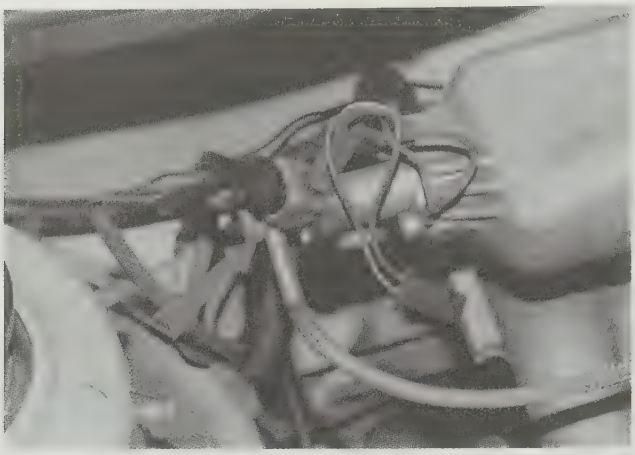
Fuel injection system can be: "port injection" or "multi port" where there is one or more injectors per cylinder and fuel is supplied by individual lines or a fuel rail or: "throttle body injection" where one or two injectors are mounted in the throttle body and deliver the fuel directly into the intake throat. This system also is controlled by a computer and is thought of as a compromise between the port injection system and the carburetor. It is more efficient than the carburetor but does not deliver fuel equally to all cylinders. It is mainly used on smaller, less expensive vehicles.

The aim of fuel system in todays vehicles is to produce a constant air/fuel ratio of 14.7:1. This balance as we have already observed is called "stoichiometric" and is the ratio where the most efficient burning of fuel is achieved and the least amount of pollutants produced. The carburetor can not attain this fine balance.

A large number of engine compartment fires have been caused by breakdowns inthe fuel injection system. The fuel lines are subjected to high pressure, varying from 28-78 PSI. The basic reason for this higher pressure in the injection unit is that fuel is constantly injected and must be atomized by the injector nozzles.

Any leak caused by a pinhole in the fuel injector line, a break in the line or a loose fitting fuel rail will result in a fine spray of fuel in the engine compartment. One small spark is all that is required to set off an intense fire. One manufacturer had to recall hundreds of its vehicles because the battery was positioned over the fuel line at the fuel pressure regulator. As water accumulated on the battery top it dripped down on the fuel line. The caustic material caused tiny holes in the fabric-rubber line. This resulted in a high pressure fuel leak and a fire.

Another factor contributing to fires in the fuel injection system has been the "cold start valve". This is an injector that sends fuel directly into the intake manifold. A backfire in this area can cause a sufficient fire to melt the injector. When this occurs fuel under pressure escapes and a good sized fire results. Leaking diaphragms inside the warm up regulator have also been blamed for fuel injector fires as have loosened connections between the fuel accumulator and the fuel filter. In one model of an imported vehicle, numerous fires have been



A leak at the cold start valve can be a fire cause in some fuel injected vehicles.

observed caused by the separation of the line to the cold start valve. Separation of spring lock fuel rail fittings caused Ford Motor Company to make a massive recall of vehicles in the late 1980's.

As stated earlier, smog control devices have contributed to the vehicle fire hazard. In the A.I.R. system a clogged or inoperative check valve can allow hot exhaust gases to flow back into the air pump and hoses, causing them to burn. The air pump parts are manufactured from a hard plastic material and can burn. An inoperative diverter valve will allow backfire at deceleration and can result in an engine compartment fire.

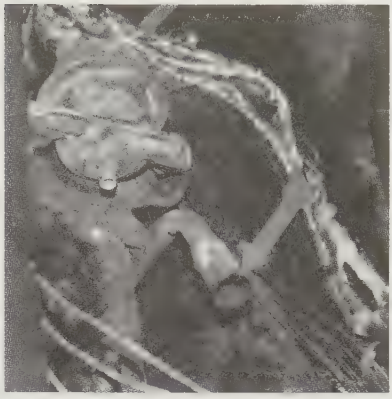
The Exhaust Gas Recirculation (EGR) valve can be another fire hazard. It allows a small amount of exhaust gas to enter the combustion chamber where it replaces a like amount of air and fuel mixture. When the ignition process takes place the fire isn't as hot because there isn't as much to burn. The EGR valve has a small hole in it that allows the recirculation of the exhaust gases. If the valve becomes clogged or deteriorates it can result in a backfire or an increase in the temperature of the catalytic converter. In either event a fire can result.

The metal tube for the EGR valve becomes extremely hot during operation since it serves as a conduit for exhaust gases. On occasion plastic pieces from the hood inner liner or other materials can fall onto or make contact with the EGR tube. When exposed to this hot surface for some period of time a fire can result.

The closed tank *vapor control system* has been another source of vehicle fires. A fire in the air cleaner can result in communicating the fire through the



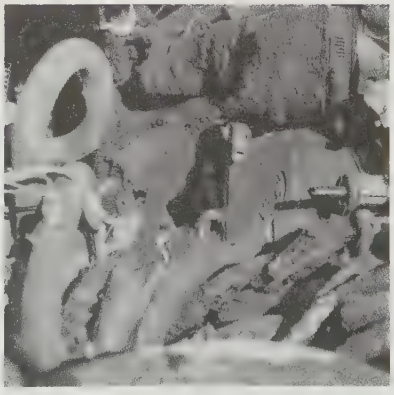
These spring lock fittings caused fire problems in Ford products when they parted...



as one did in this case.



A major recall resulted and retaining clips were added.



But the plastic fuel lines continued to melt and burn when heat was encountered from another source.

connecting hoses and into the vapor storage canister. An intense fire can be the result.

As we have outlined before, the vapor control system is a part of the vehicles emission control system. In the old days gasoline vapors from the fuel tank were allowed to escape through a vent pipe and fuel cap. Likewise any vapors from fuel in the carburetor bowl were allowed to escape into the atmosphere when the vehicle was standing. Neither of these is acceptable now under stringent emissions control regulations. Vehicles for the last two decades are equipped with a fuel vapor emission control system that requires the fuel tank to be sealed and the escaping vapors flow through a recovery line to a charcoal canister located in the engine compartment. Vapor from a carburetor float bowl also flows to the canister via a similar line. The charcoal particles in the canister trap the vapor and hold it. When the engine is started air flows through the canister and the stored vapor is sucked into the intake system by engine vacuum and becomes part of the overall air-gasoline mixture.

Occasionally one can be misled in investigating an engine compartment fire since the destruction in the area of the canister will be so complete as to suggest origin at that point. Canister fires are generally secondary but the burning of the stored vapor can create a fire of great intensity since the canister body is plastic in manycases.

We have observed fires caused by the presence of gasoline in the vapor canister being forced through the line to the intake manifold. The resulting flooding has caused a vapor concentration in the engine compartment and ignition occurred with an electrical spark. This can be caused by one of two things:

a. Overfilling of the fuel tank forcing gasoline into the vapor line and into the canister.

b. The addition of an extra fuel tank (especially on motorhomes, vans and pickup trucks) without allowing for the increased amount of vapor and fuel flow into the canister.

On several occasions I have seen vapor canisters literally dripping gasoline. Even without the overflowing of the line into the engine compartment the danger caused by such a condition in the canister can be readily seen when a parked vehicle inside a closed garage is considered. A case in point occurred in Reno, Nevada where a home was destroyed because of fuel leaking from the vapor

storage canister. The tank had been over filled and heat expansion caused fuel to flow into the vapor line. Collected vapors were ignited by a hot water heater.

In late 1997 the Environmental Protection Agency (EPA) ordered that 40% of U.S. manufactured vehicles be equipped with a second vapor recovery canister called ORVR (on-board refueling vapor recovery canister) for the production year 1998 and that by 2006 100% compliance would be required on all new cars and trucks. The National Highway Traffic Safety Administration (NHTSA) and the EPA have been at odds over this second canister as a safety hazard. A 1991 NHTSA study of crashes found 27% more fires in vehicles with evaporative canisters than in vehicles without them. There is no reference made to the year model of these vehicles and so this statistic may not be meaningful since canisters have been in use for almost two decades.

The purpose of the second canister (ORVR) is to capture gasoline vapors that escape during the refueling process. The location of this canister will be in the rear area of the vehicle where many believe it will be more susceptible to collision damage. As we have stated in previous reports on this subject:The vapor storage canister is not so much a hazard when it comes to causing a fire. It does, however, greatly increase the intensity of a vehicle fire when it is exposed to flame or heat impingement from a fire that has started from other causes.

Contrary to popular belief fuel tank explosions are rare. This is true of metal tanks as well as the plastic ones now in use. When vehicle fires occur and spread to the fuel tank area, the gasoline inside the tank will even boil and the fumes from the tank will ignite and burn, but the tank will not explode. What does occur is that the fire will be intense enough to burn through a plastic tank or destroy the filler neck connection. When this happens the fuel inside the tank is dumped under the vehicle and the fire rapidly accelerates. This generally occurs in the latter stage of a vehicle fire.

Fuel tanks in older vehicles made use of a vent pipe that allowed air to escape while fuel was used. This allowed the equalization of pressure inside and outside the tank. This system was changed with the introduction of the Vapor Emission Control System we discussed earlier. A special filler cap is now used with a two way relief valve that operates on pressure and vacuum. When gasoline is withdrawn from the tank a slight vacuum develops causing the valve to open and admit air. If interior pressure builds up the valve opens to relieve it. Many of these caps have a "rollover" check valve which closes if the car overturns. The use of a properly fitted filler cap designed for use on the right vehicle is vital since the

wrong one can cause the collapse of the fuel tank from atmospheric pressure. Some vehicle maintenance advisors recommend the complete draining of the fuel tank yearly to get rid of sediments. Fuel tanks on older cars were equipped with drain plugs that made this a fairly easy task. Drain plugs are a rarity now.

In fire tests conducted on vehicles equipped with Vapor Emission Control Systems a new phenomenon has been observed. In two out of four cases the plastic threads on the gas filler cap melted and the cap blew due to internal pressure in the tank. This was immediately followed by a gush of flaming gasoline that initially spread several feet from the filler neck. This could certainly be a hazard to fire fighters. It could also lead the fire cause investigator to a faulty conclusion since it would be possible to find traces of fuel on the ground some distance from the vehicle and lead him to believe that flammable liquid had been poured around the vehicle.

Tanks can be split by collisions and the rapid dumping and ignition of fuel can take place with explosive force. This is still not an explosion of the tank, however.

In SAE paper #97004 titled "Fuel Tank Combustible Mixtures — New Lessons Learned," authors Floyd A. Wyczalek of F.W. Lilly, Inc. and Chung M. Suh of NGK Spark Plug Manufacturing state on page 667 the following:

Explanation of Diagram

Figure 8 shows lean flash point and rich limit temperature conditions for an automotive gasoline fuel tank, as a comparison. For an automotive fuel tank at normal ambient temperatures and above about 1% full, air/fuel vapor mixture in the right hand zone is too rich to ignite. However, as the fuel level drops under 1% full, the mixture crosses into the flammable zone bounded by the rich flammability limit because this vertical curve is also asymptotic to the horizontal temperature axis. Consequently, an automotive tank must be virtually empty, significantly under 1% full, before there is any possibility that the vapor can ignite in the range of normal ambient temperatures.

This study confirms what we have observed before — that while an in-tank gasoline vapor explosion in a motor vehicle fuel tank is not impossible, it is very unlikely.

Before we go on with the broad subject of fuel fires it might be well for us to examine the various types of fuels used to power motor vehicles. We will not be considering diesel type fuels here since we will be covering this in the chapters

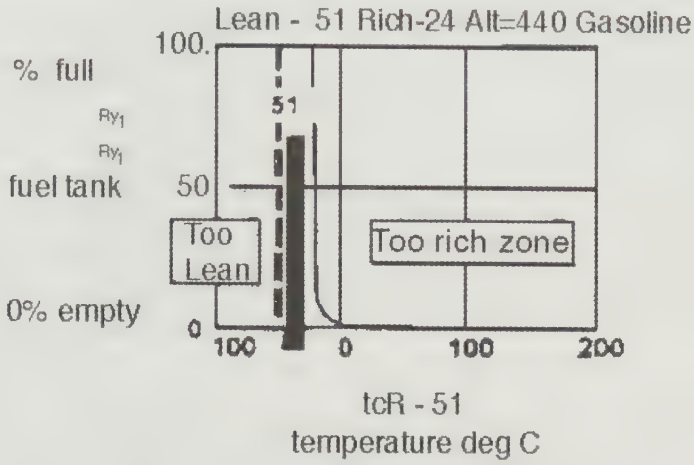


Figure 8: Automotive gasoline flash point and rich flammability boundaries as itank full v. fuel temperature •C
lean flash point=-51•C (-60•F)
rich limit=-24•C (-11F)
For gasoline, spontaneous ignition temperature is AIT=440•C
(Zabetakis 1962 Bureau of Mines circular 8137)

on Trucks, Buses, Heavy Equipment and Boat Fires.

Gasoline is still the most common fuel used in spite of the great emphasis on the development and use of what are referred to as "alternate fuels". Gasoline is a liquid hydrocarbon and burns intensely with a bright orange flame producing quantities of black smoke. Modern gasoline is a blend of hydrocarbons and their isomers ranging from pentane to nonane. An isomer is a revised line up of carbon atoms that differ widely in structure. Most of them are saturated and contain 4-12 hydrocarbons per molecule. Lighter, more volatile components are needed for fast starting and warm up in colder weather. but a different blend is needed to prevent high evaporation losses and vapor lock problems in higher temperatures. This, plus the increasing demands for a cleaner burning fuel, makes the gasoline production process one of constant change. Different grades of gasoline, with and without additives, must be produced for the various tasks to be performed on land, sea and air applications.

One of the major problems encountered in the use of gasoline in the internal combustion engine has been that of "knock", a rapid detonation occurring toward

the end of combustion. The octane scale was developed for measuring anti knock characteristics. "Iso-octane", a highly branched hydrocarbon was assigned the value of 100 while heptane was assigned 0. The percentage of the iso-octane in the blend is termed the octane number. Most octane numbers encountered range between 84 and 100.

The heat of combustion of gasoline is 16,000-21,000 BTU's per pound. "Heat of combustion" is the amount of heat released during a fuels complete oxidation. A BTU (British Thermal Unit) is the amount of heat required to raise the temperature of one pound of water one degree Fahrenheit. Its metric equivalent is the calorie which is the amount of energy required to raise one gram of water one degree celcius. 1 BTU = 242 Calories.

As a comparison one pound of paper or cardboard in complete oxidation produces 6000 BTU's. One gallon of gasoline weighs 6 pounds.

Gasoline vapor is 3-4 times heavier than air and will sink to the lowest level it can find. Gasoline is lighter than water and will float on the surface. The following properties relate to gasoline: (see table on next page).

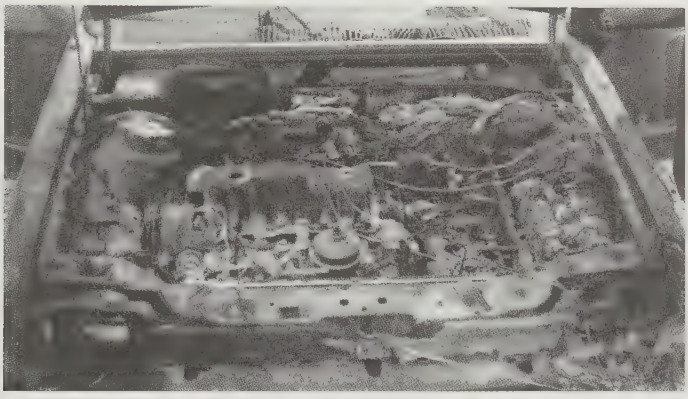
- One gallon of gasoline will produce 30 cubic feet of vapor at room temperature (70 F).
- One gallon of gasoline will raise 230 cubic feet of air to the lower flammable limit (1.4%).
- One ounce of gasoline produces 0.234375 cubic feet of vapor.
- Gasoline vapor will not ordinarily rise to a greater depth than 3 feet unless confined.



In an experimental fire a fuel leak was created in this Ford product.



Ignition of vapors occurred when an electrical spark was created.



Even though the fire was extinguished a few minutes later this damage occurred.

There has been much discussion over the auto ignition temperature of gasoline and the likelihood of ignition taking place in the presence of a heated engine compartment surface. This subject is also discussed in other sections of this book. NFPA 921, 1995 edition, section 15-2.1, page 119 makes the following observation:

"Whether a given fuel can actually be ignited depends on the properties of the fuel, its physical state, the nature of the ignition source, and other variables. Flash point is of little or no significance when a fuel is released in spray form. Ignition on hot external surfaces may require temperatures of 200°C (400°F) above published ignition temperatures."

In an article prepared for the British Fire Prevention Journal, Mr. John Butler, M.SC. Eng, M.I. Mech E., DIP Eng., M.I. Fire E., of the Dublin Institute of Technology and Dr. Trevor Orr, PhD., Trinity College, Dublin University made

Gasoline Properties

Flash Point	-36° to -45°F	-38°C to -43°C	varies according to octane	
Ignition Temperature	536°F* to 853°F ***	280°C to 456°C	varies according to octane	*60 octane **100 octane
Flammable Limits (% by volume in air)	upper = 7.6% lower=1.4%			
Specific Gravity (water =1.0)	0.8			
Vapor Densidty (air=1.0)	3 to 4			
Boiling Point	100°F to 400°F	38°C to 204°C		

these observations:

Petrol leaks seem to be largely responsible for the fact that some 60% of car fires occur in the engine compartment. These leaks are usually due to the flexible fuel lines becoming porous or due to the lines loosening at the spigots. Though the source of ignition is usually electrical, it can also in theory be due to auto-ignition or spontaneous combustion.

Characteristics of Petrol

Petrol is widely recognized as being highly flammable. Its vapours are denser than air and will lodge in low spots in the event of a petrol spill. Its flammability limits in air are approximately 2% to 9%. Petrol contains some 230 different hydrocarbons. These are mainly liquids but there are also a few gases which are soluble in liquid hydrocarbons. The mix of hydrocarbons forming any one type of petrol is not constant and may change with the type of process used to distill, reform or crack the heavier fractions of the crude oil.

Though the process may influence the octane number of a petrol in general, additives are included to provide cheaper and simpler control of the higher octane numbers. Petrols are formulated to suit winter and summer conditions in particular regions, high mountain regions, etc.

The boiling point of petrol ranges from about 25°C to approximately 210°C[1] and the boiling range is continuous between these points. The initial boiling point (IBP) determines the ease of starting of the engine and the end boiling point determines the possible build-up of carbon in the engine. The important temperatures are the values at which the first stream of tiny bubbles (IBP) rises through the sample of petrol, and the temperature at which 10%, 30%, 50% and 90% of the initial volume distills over.

In general, it can be expected that in the case of premium grade petrol, the 10% distillation will occur at a temperature of approximately 60°C, the 50% occurs at a temperature of 110°C and the 90% occurs at a temperature of approximately 170°C.

In engine compartment temperature measurement tests carried out on a Ford Cortina, we registered temperatures of around 280°C during average urban travel — with a further rise of some 30°C in the exhaust manifold temperature when the car is momentarilly stopped at traffic lights.

The exhaust manifold temperature would probably be higher in the case of a transverse engine, and higher still if a turbo charger is fitted. The position of the exhaust manifold in front of or behind the engine and the load on the engine will influence the final equilibrium temperature of the exhaust manifold. There have been claims of temperatures as high as 500°C being noted.

In general, outbreaks of fires in the engine compartment of cars increase with the age of the car. These fires are largely due to deterioration of the flexible fuel lines.

If the incidence of fuel related fires in three year old cars is X, a rule of thumb says that when the same car models are five years old the incidence of fuel related fires is 2X and at 7 years old the incidence is 3X if the flexible fuel lines have not been replaced during this time.

Because of provisions to allow for engine vibrations, changes in the working temperatures of the engine etc., it is not practical to run the complete length of the fuel line from the petrol tank to the carburetor in metallic tubing.

The short flexible parts of the fuel line are generally run in rubber or plastic tubing which may or may not be reinforced with a metallic canvas braid. In certain circumstances hydrocarbons can attack the flexible materials and degrade them to the point that they become porous. The hydrocarbons may also cause environmental cracking of the material.

I. Porous

If the material is porous, the leakage is greater when that section is pressurized with petrol — i.e. when the engine is operating. When the engine stops, the leakage through the porous pipe progressively decreases as the pressure within the pipe falls. For this reason, you may smell petrol in the engine compartment for a short period after the engine is switched off — the smell will be less evident or absent when the engine is cold. Porosity of the tube is usually accompanied by a swelling of the material forming the tube. Even in the absence of a smell of petrol, if swelling is noticed, it is advisable to replace the tubing.

II. Environmental Cracking

If you suspect there is environmental cracking of the tube material, this can be checked by flattening the tube betweeen your fingers. If environmental cracking is present, you may then see crevices appear which usually run parallel to the longitudinal axis of the tube. If the tube is allowed to return to its normal shape, it becomes extremely difficult to see the cracks. Obviously it is advisable to replace such degraded tubing.

Other Sources of Leaks

There may be other sources of petrol leaks in engine compartments. For example, a defective carburetor float may allow petrol to overflow from the float chamber, A petrol tube may become loose on a spigot, allowing petrol to leak at the joint. Or the tube may fall off, allowing petrol to flow freely until the fall of petrol in the carburetor causes the engine to stop.

In the case of petrol-methanol fuel, if water separates and collects in some part of the carburetor, corrosion pitting of the carburetor can develop at the fuel-water interface — thus allowing the fuel to leak out of the carburetor.

Dangers Besides Petrol

If the vapour of a flammable liquid and the oxygen in the surrounding air forms a vapour-air mixture that is within the flammable range for that

particular liquid, ignition can occur if the temperature of a body or surface in the immediate vicinity of the mixture exceeds the auto-igntion temperature for that flammable liquid.

People accept that petrol is more easily ignited by a pilot flame than would a liquid such as diesel. But this is not necessarily the position in the case of spontaneous combustion. Textbooks give values ranging from 280° to 460°C being the approximate auto-ignition temperature for various petrols, approximately 250°to 270°C[2,3] as being auto-ignition temperature of diesel oil.

In effect, this is stating that the auto-ignition temperature of diesel is lower than the lowest auto-ignition temperature given in the books for petrol and as mentioned earlier temperatures in excess of these values have been measured in engine compartments.
Hydraulic and brake fluids are also flammable fluids and it is now recognized that, given certain condition, the vapours of these fluids can ignite and cause outbreaks of fire in engine compartments.

Any condition that may cause fluids such as hydraulic fluid, brake fluid, diesel oil, petroleum lubricating oil to change from a liquid form to a vapour or spray in the engine compartment is a potential fire hazard.

References:
1. Williams, D.A. et al Liquid Fuels, Pergard Press, Oxford
2. Meyer, Eugene Chemistry of Hazardous Materials, Prentice-Hall Inc., Englewood Cliffs, NJ
3. Brame, J.S. et al Fuel—Solid, Liquid and Gases, 5th edition, Edward Arnold, London

Liquified Petroleum Gas (LPG) is about two times heavier than air. It will settle in lower elevations and if ignited will explode. LPG can be liquified too by moderate pressure and transported and stored in a liquid state. It consists mainly of propane and butane. These are distillates of gasoline.

LPG Systems - liquified petroleum gas systems are used in some passenger cars, light trucks and motorhomes. Occasionally a fire will involve one of these systems and it is incumbent on the fire investigator to be conversant with them. LPG molecules are related to gasoline molecules but they are smaller. Because of this LPG is a vapor at ordinary temperature. At high pressure it is a liquid and

PROPERTIES OF BUTANE & PROPANE		
	Propane	**Butane**
Flash Point (of liquid)	Below -100°F (-73°C)	-76°F (-60°C)
Ignition Temperature	871°F (466°C)	761°F (405°C)
Flammable Limits	2.2% to 9.5%	1.9% to 8.5%
Specific Gravity of Liquid at 60°F	0.509	0.582
Vapor Density	1.6	2.0
Boiling Point	-44°F (-42°C)	31°F (-0.6°C)
Water Solubility	No	Slight
Volume of gas per lb. of liquid (at 60°F & atmospheric pressure)	8.5 cubic feet	6.5 cubic feet
Vapor pressure at 70°F (21°C)	110 psig	19 psig
Vapor pressure at 100°F (38°C)	192 psig	59 psig
Critical Temperature	206°F (97°C)	306°F (152°C)
Critical Pressure	617 psig	551 psig
Weight per gallon of liquid	4.24 lbs.	4.84 lbs.
Flame temperature (approx.)	3497°F (1925°C)	3443°F (1895°C)

when this pressure is relieved it turns to a gas. It must be stored as a liquid in a pressurized tank. There are two basic types of LPG, propane and butane. Propane is the one most used as a vehicle fuel.

In a propane fuel system LPG is stored in a pressure tight tank at high pressure. The fuel is forced through a filter, a high pressure regulator and a vaporizer. The high pressure regulator reduces pressure and the fuel begins to turn to vapor. It then enters the vaporizer which has an inner tank surrounded by a water jacket through which engine coolant circulates. The coolant adds heat to the fuel so that it is completely vaporized. The gas then passes through a low pressure regulator where the pressure is further reduced. It then enters the carburetor where it is mixed with air.

Many of these LPG systems are "add ons" since only a few are produced at the assembly plant as part of a new vehicle. We have investigated several of these fires that resulted from leaks in the system. Many of the leaks occurred in the long flex line leading from the tank to the pressure regulator.

Flash Point	Ignition Temperature	Flammable Limits (% by volume in air) Upper - Lower	Specific Gravity (water= 1.0)	Vapor Density (Air=1.0)	Boiling Point
52°F 11°C	867°F 464°C	7.3 — 36.0	0.8	1.1	147°F 64°C

Methanol is commonly called "wood alcohol" and was originally prepared by heating wood in the absence of air. It is mainly derived now by applying heat, pressure and oxygen to natural gas. It can also be prepared from coal, a fuel that the United States has in great abundance. Methanol is a simple fuel. It breaks apart easily and burns more completely than petroleum fuels. It has the following properties:

The desirable properties encouraging the use of methanol are that the basic fuel source is plentiful (especially coal) and would reduce enormously the U.S. dependence on outside sources. The amount of carbon dioxide produced in the burning of methanol is 7-16% less than gasoline and it does not release unburned hydrocarbons into the atmosphere because of its more complete combustion process. Particulate, or "soot" is not produced by methanol, making it a desirable substitute for petroleum based diesel fuel. Methanol has a high octane rating and can be used in engines with higher compression ratios. It has been used in racing cars for a number of years.

Unfortunately, Methanol has some bad points, too. It is extremely toxic if ingested and reportedly can enter the body through the skin. It has no odor and burns with a flame that is not visible. It can form an explosive mixture in the tank, note the wide band of its explosive limits, which is unlike gasoline with a narrow band of 1.4 - 7.6% by volume in air. This is the reason that gasoline tank explosions are such a rarity, television programs and movies notwithstanding.

The energy density of methanol is about one half that of gasoline, requiring a fuel tank twice as large or a cruising range of half as much as in today's cars. Because it does not vaporize as readily as gasoline (note the difference in flash

points — gasoline is -45° F while Methanol is plus 52°F) it makes cold starting more difficult. Methanol can be corrosive and requires fuel tanks, lines and distribution systems to be made of special anti-corrosive materials.

Of great concern to environmentalists is the fact that the combustion of methanol produces 3-5 times as much formaldehyde as a gasoline vehicle. Formaldehyde is suspected of being a producer of ozone - a chief ingredient of smog. The production of methanol, especially when it is made from coal, produces twice as much carbon dioxide as in gasoline production.

In spite of these drawbacks Methanol still seems to be a fuel of the future. A blend called "M85" is now being used. It consists of 85% Methanol and 15% gasoline. "M100" will be 100% methanol and will come into use about the year 2000. The present aim is the production of 500,000 alternative fuel vehicles a year by the turn of the century. Governmental agencies such as Environmental Protection Agency (EPA) express a great deal of optimism that the problems with Methanol will be overcome or greatly lessened as we enter the new millenium. Auto manufacturers and petroleum companies don't seem to share this optimism.

Ethanol is commonly known as "grain alcohol". It is produced by fermenting crops such as corn, wheat, rice or sugar cane. It has been in use for well over twenty years in a gasoline blend called "Gasohol". The Renewable Fuels Association, based in Washington, D.C., estimates that between 8.5 and 9 billion gallons of gasohol were sold in the United States during 1989. The most common form of Gasohol is a blend consisting of 90% gasoline and 10% ethanol. If the blend is more than 10% the mixture must be labeled as Gasohol. If it is 10% or under it does not have to be so designated. Its chief use now is as an octane booster and has found much greater use since the elimination of lead in gasoline. Ethanol is not as toxic as Methanol and does not require as great handling care. Ethanol is not as corrosive as methanol and fuel delivery systems will not be required to have as high degree of corrosion inhibitors. Like Methanol its air to fuel mix ratio is 6.5 to 1. Gasoline is 14.7 to 1 and from this one can readily observe that one tank of Ethanol will produce about $^1/_2$ the energy as a tank full of gasoline. Either larger tanks or a shorter driving range would have to be expected.

The chemical formula for Ethanol (Ethyl Alcohol) is C_2H_5OH (Source "Flammable Hazardous materials", 2nd edition by James Meidl, MacMillan Publishing Co.) It has the following features:

Like Methanol, Ethanol is soluble in water.

Flash Point	Ignition Temperature	Flammable Limits (% by volume in air) Upper - Lower	Specific Gravity (water= 1.0)	Vapor Density (Air=1.0)	Boiling Point
55°F 13°C	793°F 423°C	19.0 — 4.3	0.8	1.6	173°F 78°C

Since Ethanol can be made from grain it would seem that this fuel would be our best answer to the problem of dependency on foreign countries for fuel. We have large grain surpluses and much land that could produce additional grain now lies unused under subsidy programs. The problem is cost and that it is not as simple as it may seem.

In 1975, Brazil decided to end its dependence on imported petroleum. Since sugar cane was an abundant product a program of ethanol production and vehicle conversion was embarked on. Vehicles produced in Brazilian factories were designed for ethanol use and after a period of almost 15 years Brazil saved 200,000 barrels of gasoline a day by using ethanol. To encourage cane production Brazilian farmers were given low interest loans to cover over 80% of investments in distilleries on their property. To aid in this activity Brazil received a $300 million loan from the World Bank and the country seemed well on the way to a total replacement of gasoline by ethanol. Of Brazil's 13 million motor vehicles, more than 4 million were fueled entirely by ethanol.

All this sounds wonderful but it was anything but. The Brazilian government subsidized the ethanol program. An artificially low price of $1.27 per gallon was created. In the meantime gasoline sold for $1.70 per gallon. When the ethanol project was begun oil was selling for $40.00 per barrel. Soon it was half that. The actual cost of producing a barrel of ethanol is about $45.00. It doesn't take a genius to figure out that one barrel of ethanol is costing over twice as much as gasoline and it will only produce one half the energy. This program generated a $2.7 billion annual deficit for the Brazilian government, according to a World Bank study.

While air pollution was reduced in many of Brazil's larger cities it increased in other areas. All sugarcane fields are burned after harvest to protect workers from the sharp leaves of the cane. Great clouds of smoke and pollution are produced and some ecologists are worrying this may contribute to the "Green-house Effect". It is little wonder that Brazilian car manufacturer Joao Gurgel was

quoted as saying "Alcohol is the AIDS of Brazil and we don't know how to cure it. Sometimes I don't sleep because I know that our country is going to be destroyed by this problem". The Brazilian government is presently ordering manufacturers to reduce the production of alcohol burning engines and says that it cannot afford to continue its alcohol subsidy program. It would be good for our government to take a long look at what happened in Brazil before embarking on a total program of alcohol fuel.

One other thing that has been occurring since the use of an ethanol-gasoline mix to boost octane has been the increase in the Reid Vapor Pressure (RVP) in the fuel tank and system. This is caused by the higher volatility of the ethanol-gasoline mix and this produced, under hot weather and hard driving conditions, the phenomena of "fuel spurting". This has caused a large number of vehicle fires and deaths in the United States. It has occurred mostly in those states allowing a 10% or more ethanol mix in gasoline. "Fuel spurting" usually occurs when the fuel cap is removed and pressurized fuel gushes out. Any nearby ignition source will cause an instant fire and in some cases the hapless operator is turned into a human torch.

We will now consider "Natural Gas" as a vehicle fuel. According to the Brooklyn Union Gas Company tests have shown that natural gas produces 85% less carbon monoxide, 25% less nitrogen oxide, 23% less carbon dioxide and 13% less hydrocarbons than gasoline.

The bad feature of *natural gas* is that it takes up to 1,000 times the space of equivalent gasoline and therefore must be compressed. Even then *compressed natural gas* (CNG) at 3,000 PSi only produces 29,000 BTU's per gallon as opposed to 110-126,000 BTU's produced by a gallon of gasoline. The energy density of CNG is less than 25% of gasoline.

In order to use CNG as a motor vehicle fuel a number of natural gas cylinders would have to be stored in the car or truck. Each cylinder would hold the equivalent of 3.5 gallons of gasoline and would be filled by connecting a tank fuel nozzle to a "quick connect" valve in the car. The vehicle's tanks would then be filled. Fuel from the tank would be fed to a pressure regulator which would reduce the pressure from the storage cylinder to the gas/air mixer. At this junction the mixer blends the gas with air to obtain optimal combustion. The mixture would then be fed through the throttle body and fired in the usual manner.

The American Gas Association says the savings of CNG over gasoline are 75

cents per gallon. Another feature of CNG is less wear on engine parts with fewer oil and spark plugs changes required.

The big drawbacks are space, the tanks would take up a great deal of room, and a lack of a nationwide refueling structure. CNG would be most useful in fixed base operations such as delivery and bus services. Refueling could be taken care of at return to base intervals where a supply of CNG would be maintained. Currently some United Parcel delivery vehicles in Brooklyn, New York area are being fueled by CNG. This is a study project that is designed to provide UPS and the Automotive Industry with data for evaluation of long term use of CNG. In all, there are reported to be about 30,000 vehicles in the U.S. operating on CNG. The Associated Press reports that CNG powered vehicles in Canada have logged 500,000 miles without an overhaul because a gaseous fuel does not cause the wear of a liquid fuel. About 8,000 vehicles in British Columbia are operating on CNG. In Saskatchewan, Canada CNG is being used on 3 diesel tractors converted to 100% CNG, or a dual fuel capability, on five farms. The experiment will last for two years. J.G. Davies, University of Saskatchewan and M.T. Sulatisky, Saskatchewan Research Council, reported in *Automotive Engineering* that "The feasibility of operating two wheel drive agricultural tractors on CNG has been demonstrated. These results are important because of the potential for substituting a cleaner, less expensive, more plentiful fuel for diesel fuel..."

What can we expect in the way of hazards in fires in vehicles fueled by CNG? The American Gas Association says "Natural gas tanks have proven to be much safer than other fuel tanks. Tests have shown they can even withstand bullets and dynamite...almost fifty years of experience worldwide has resulted in an unmatched safety record for natural gas vehicles". Natural gas is 65% as heavy as air. This means it will rise when released and will be rapidly diffused if escaping into the open. If contained it will gather in the highest elevation of the container. It has an ignition temperature of 1,100-1,200° F (593.3 - 648.9° C). The flammable limits are 4% lower to 14% upper. It is odorless and must have a detectable odor added, generally at the distribution point. Typical cylinder pressure for CNG would be 2,000 PSi. These cylinders are designed with pressure relief valves so that rupture would not occur in the cylinder walls. A hazardous situation would no doubt occur if the cylinder were broken open on impact and once the gas dispersed enough to reach the explosive limits.

Mercedes Benz and BMW have been experimenting with hydrogen powered vehicles. Mercedes Benz has a hydrogen powered research vehicle using a 2.3 liter four cylinder engine putting out 105HP. The engine has a hydrogen distributor instead of a carburetor and an intake manifold water injector. BMW

has produced a 735L with a 3.5 liter engine that operates on hydrogen. The vehicle's fuel tank is insulated with 70 layers of aluminum foil and fiberglass matting to keep the liquid hydrogen at minus 487°F. Chilled hydrogen is injected directly into the combustion chamber. The vehicle has a 186 mile cruising range. Liquid hydrogen has an energy density of 34,000 BTU's per gallon. Gasoline has a density of 110-126,000 BTU's for the same quantity.

Liquid Hydrogen has a wide flammability range, 4.1 -74.8% according to "Hazardous Materials" (by Isman and Carlson, 1980, Glencoe Publishing Company). The authors state, "To give you an idea of what can happen in a release, hydrogen gas when mixed with sufficient air to yield 4% of the lower flammable limit, forms a combustible hydrogen-air mixture with a volume 21,000 times that of the original liquid. When hydrogen was released to the atmosphere, ignition occurred 6.2% of the time; in release to enclosures, ignition occurred every time (100%)." Until dangers in the use are overcome, hydrogen's use as a future fuel for motor vehicles is limited.

The automobile powered by electricity has been the subject of speculative search since the beginnings of its existence. As early as 1899, Hautier was building a car powered by electricity produced by a series of storage batteries. The problem then, and for many years thereafter was limited range, low speed and the bulk of the batteries. Experiments continued from time to time throughout the 20's, 30's and 40's. Some success was reported in reducing battery weight and size, but range remained short and batteries required frequent overnight charging. With the exception of a few short haul delivery services the electric powered automobile remained a rarity.

With the growing concern over environmental pollution generated by motor vehicles, interest in the electric powered car has been rejuvenated.

In 1989, the U.S. Congress appropriated $14.1 million for research and development of an electric car. European Community Transport Commissioner Karel Van Miert told a seminar of electric transport experts in Belgium "We aim to introduce several million electric driven cars by the end of this century". Peugeot S.A. of France has developed the model 205 and expected to sell 10,000 of them by 1992. Peugeot 205's performance was listed as 62 MPH top speed, range of 62 miles and acceleration to 31 MPH in 11.6 seconds. The vehicles use 12 six volt batteries mounted under the hood. The battery powered 205 weighs 660 pounds more than the standard gasoline fueled 205.

BMW has developed a front wheel drive electric powered version of the "3" series. The vehicle has a 584 pound liquid sodium battery mounted in the rear. Power from this battery drives a front mounted Asea Brown Boveri direct current motor developing 145 volts at 2,400 RPM's. It has a 2 speed automatic transmission. First gear is used only on steep grades. Top speed is 62 MPH and a cruising range of 93 miles.

A representative of General Motors announced in early 1990 that his company has built a prototype electric car called the "Impact". The vehicle is a two seater, described as a "concept car" and capable of doing 0-60 MPH in 8 seconds according to the GM representative. The *Impact* takes advantage of lighter weight construction and improved battery technology. The range was announced as 120 miles. Obviously this vehicle is in early stages of development, but the announced results are impressive. Electric cars may answer part of our future transportation needs in a pollution reduced world.

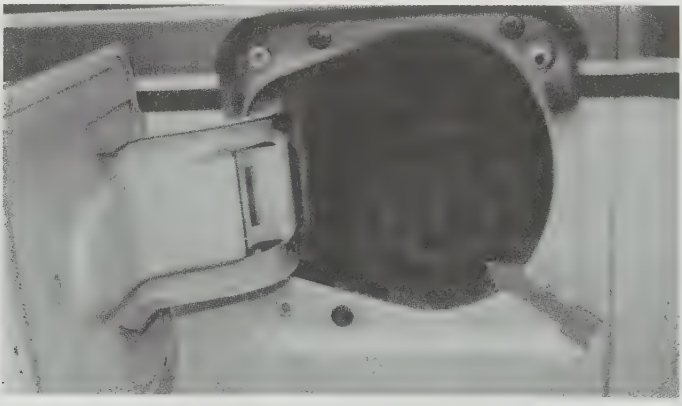
Emmissions Control & Exhaust Systems

Sometime back a California Highway Patrolman observed a strange sight along the roadside. A man was hopping about wildly with his pants legs on fire and — to put it in an old fashioned term — he was "exposing his person". It developed that the unfortunate man had spent a few hours drinking beer in a bar. As he drove home he had the urge to urinate and pulled off the road to relieve himself. As he got rid of the beer he had consumed earlier he failed to note the fire spreading through the dry grass under his car until he became painfully aware of it! Fortunately for him, the Highway Patrolman was able to put out the blaze with his fire extinguisher since the quantity of beer consumed by the driver was not equal to the task of quenching the fire.

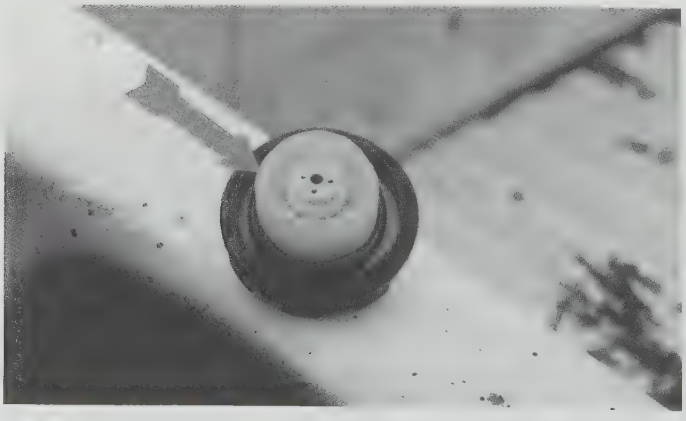
He had done what many have done with disastrous consequences. He had parked his catalytic converter equipped car over dry grass and a blaze resulted. Temperatures inside the catalytic converter can reach 1700° F under normal conditions with outside temperatures of around 700° F. Special heat shields are used to protect the underbody and components from this extreme heat. An overrich fuel mixture, failure of AIR and EGR systems or other smog control devices can result in overheating of the converter. This heat can be transmitted to the vehicle's interior by radiation or conduction. There have even been cases of explosions of these devices or mufflers due to super heating. A few earlier models of some imported vehicles placed the catalytic converter under the exhaust manifold. The resulting heat damaged the plastic fuel line and fires resulted. In one such case, the resulting fire in the vehicle communicated to a structure and

an apartment building was destroyed.

Catalytic converter problems have caused the recall of a large number of vehicles by manufacturers. One of the problems was the placing of the unit too close to the underside of the vehicle. The heat shields proved to be inadequate and a number of interior fires resulted. This can also happen in cases involving the exhaust system alone. A few years ago we were required to assist in some experiments dealing with the possibility that the exhaust system caused an interior fire under the vehicles back seat. We were surprised to learn that during an uphill pull over a distance of about five miles the exhaust system in the vicinity of the muffler reached a temperature of 625° F. Since the ignition temperature of paper is about 450° F it is easy to see that any such material in direct contact with this system would result in a fire. Occasionally imperfect assembly or repair work results in the placing of the exhaust system too close to the vehicle's underbody. This has resulted in the igniting of sound deadening material and carpeting in the vehicle's rear interior. Our experiment showed, however, that with normal design



Most fuel caps are plastic.



When subjected to high heat of a vehicle fire, the plastic threads will deform.

clearance on the properly assembled vehicle the radiated heat from the exhaust system will not reach sufficient temperature to ignite the vehicles interior. As we will see in the chapter "Special Surveys & Research" exhaust system temperatures are climbing in todays cars.

Before we leave our consideration of the fuel system we should briefly discuss the electric fuel pump. These pumps are generally mounted in or near the fuel tank. They are designated as either being "in-line", i.e. they are mounted in the fuel line itself, generally toward the rear, although they may even be located in the engine compartment, or "in-tank", i.e. located inside the fuel tank so that it "pushes" the fuel through the entire system. In some vehicles you will find both "in-tank" and "in-line" pumps mounted. The main advantages of the electric fuel pump over the mechanical one is that the fuel system is filled with fuel as soon as the ignition is switched on and since it does not rely on engine RPM's the engine will never be fuel starved. All fuel injection systems use electric fuel pumps.

Of particular interest to the fire investigator is the fact that some of these pumps will continue to function as long as the ignition is turned on, whether the engine is running or not. This is particularly true of some older vehicles. This is of significance in an engine compartment fire where the panicky driver forgets to turn off the ignition in his hasty exit. The pump will stay on and "add fuel to the fire", as the saying goes. Most newer model electric fuel pumps automatically shut off when the oil pressure drops to zero. This will result in the pump shutting down regardless of whether the ignition is still on when the engine stops. This still does not solve the problem that exists when a fire begins in the engine compartment while the engine is running. In these cases the engine continues to run and the pump will operate until a fuel line is burned away or the fire reaches such a state that the oxygen is exhausted. In experiments we have seen. this takes about 5 minutes.

Many late model cars and trucks are equipped with inertia switches that turn off the fuel pump on impact. This is designed to prevent the post collision fires.

Vehicle fires can be "fed" by gasoline gushing out of ruptures in the system because fuel tank vapor pressure or gravity causes gasoline to flow even after a fuel pump ceases to function. Stilson Corporation consulting engineer Tom Green advised NHTSA that the firm's research has documented these fuel system hazards that he described as "fuel siphoning". Green's comments to the agency discuss problems and solutions to failed fuel lines and connectors, fuel filler necks and caps, inertia switches, fuel siphoning and the state of crash testing.

The fuel siphoning hazard, which Green described as fuel flowing out of severed or burned fuel lines, results from gravity and fuel tank pressure. He stated that testing shows fuel will flow out of severed lines that are located significantly higher than the fuel tank even with only a small amount of vapor pressure in the tank. Green provided NHTSA with Ford and GM documents that show those companies recognized fuel siphoning hazards as early as 1988.

OTHER CAUSES

Automatic Transmission and Power Steering Fluids

The fluid used in the power steering unit and the automatic transmission is about the same. It is a light weight oil and when under pressure and heated it is very flammable. In tests conducted with a vehicle's wheels blocked and the gear selector placed in"drive" an explosion occurred when pressure blew out the seal and the underside of the vehicle immediately burst into flames. The atomized fluid, under heavy pressure, ignited on exposure to the heated exhaust system. In other cases pressure inside the transmission has forced the dipstick up from the filler tube, allowing the fluid to erupt over the tube. The fluid ignited as soon as it hit the exhaust manifold.

I once investigated a fire death case where an inebriated man was burned to death when he passed out while he had the wheels of his car turned as hard to the right as he could. He had placed the gear selector in neutral but continued to press his foot on the accelerator pedal, causing the engine to race and the pressure to build up in the power steering unit. Finally, a seal blew and the fluid ignited when it made contact with the exhaust. The intense fire that followed inside the garage resulted in the death of the driver and destruction of the property. The ignition temperature of this fluid is 500° - 700°F while the flash point is 385°F.

A few years back General Motors produced a series of transverse mounted engine vehicles whose power steering return lines were misrouted and could make contact with the exhaust manifold. A hole was burned in the line and fluid poured on the exhaust. Since the line was not under high pressure the fire that resulted was not large and when discovered was relatively easy to extinguish. Most of these fires did not occur until after the vehicle was parked. Ford Motor Company also had a similar problem.

Some have expressed doubt over the cause and origin of vehicle fires that occur after the engine has been shut down. Remember that within 2-3 minutes after shut down there is a rise of about 85°F (30°C) in the exhaust manifold. This is caused by diminished air flow, lack of coolant circulation and the stationary condition of the vehicle. This rise in temperature can be just enough to push any oil or fuel vapors over their ignition point.

In some cases we have found leaking oil from the engine was the cause of fire. A broken gasket or a separation of the silicone sealer used in place of the gasket can allow oil to drip down on a hot exhaust manifold. Unlike gasoline, which will evaporate as it hits hot metal (evaporation is a cooling process) oil will burn under these circumstances even though it has a high ignition point of about 700°F. It generally produces a smoky, smelly fire of short duration that does relatively little damage. Nonetheless, it can spread to other components and a large fire may result.



A broken power steering hose allowed fluid to spray on the hot exhaust manifold and a fire resulted.

Turbo Charger

In years past they called this system a "Supercharger""and it was generally used on racing vehicles. It originally made use of a chain or belt driven pumps that forced more air and fuel into the cylinders. The system frequently used now is called "Turbo Charging" and in place of the belt or chain drive the exhaust gases are used. "Superchargers" are now making a comeback and some late model vehicles have them. The purpose of the turbo charger or super charger is to increase the engines power output. This is of particular importance in the continuing cycle of overall vehicle weight reduction. By use of the "turbo" a small 4 cylinder engine can increase its power by as much as 40%.

Under normal circumstances the amount of power an engine can deliver is

limited by the amount of fuel and air delivered to its cylinders. In order to increase this a small turbine wheel is mounted in the exhaust system where the flow of gases cause it to spin rapidly and drive a compressor in the intake manifold. The compressor forces the air and fuel mixture into each cylinder causing the engine to run faster. So that this action-reaction chain won't cause the engine to tear itself apart a safety valve called a "waste gate" diverts the exhaust gases around the turbine when intake manifold pressure reaches a certain predetermined level. Under most driving conditions the turbo does not operate but only engages when extra power boost is needed. The turbine may turn as many as 100,000 RPM's and because of this more frequent oil and filter changes may be required. Of particular interest to the fire investigator is the fact that the turbocharger operates at extremely high temperatures and can be the ignition point for any fuel leak or other flammable material in its proximity. I investigated a very interesting turbo related fire in a Subaru 4 wheel drive station wagon.

For some time prior to the fire. the owner had smelled an odor of something burning and had called this to the attention of the dealership where he had purchased the car. They advised him that it was probably the catalytic converter burning the "newness" off and that it was nothing to worry about. On one occasion he observed thin wisps of smoke coming from under the hood and fender lines of the car. On examination I saw that there were burn patterns on the top of the hood indicating heat and fire concentration over the right front firewall area and extending toward the right front of the vehicle. The sound deadening material under the hood had fallen down on top of the engine. It too was burned completely through on the right side. The turbo is mounted over the right side of the engine and is covered by an alloy sheet metal. This metal is curved to fit over the exhaust pipe and turbo unit. Over this unit is another metal cover of light steel and connected to two arms extending from the left inner fender panel to the right inner fender panel. The two metal covers are separated by some insulating material and over the second cover is the mounting position for the spare tire and wheel. This complete unit showed extreme heat distress and the spare tire was burned on the right side; the fire beginning on its downward side. Examination of the lower heat shield revealed that it normally would be fastened to the upper cover with three rivets. The head of the middle rivet was missing and the other two rivets had apparently never been installed. This condition allowed the lower shield to "droop" and make direct contact with the exhaust and turbo. The spread of this heat by conduction reached the spare tire and over an extended period of time caused it to "bake" and allowed fine cracks to develop with some volatizing of the process oils in the tire's construction. According to experiments conducted in 1982 by Processor Michael E. Kurz, Professor of Chemistry, Illinois State

University, the elastomer in a tire will begin to burn at 635°F. Burning is facilitated when cracks are present.

By placing the spare tire location in the engine compartment the risk of elevated temperature is greatly increased in even normal conditions since the Subaru owners manual warns, "When you take out the spare tire just after driving. wear gloves as it could be hot". The failure to close up the heat shield over the exhaust and turbo insured a great increase in the tire's temperature.

It was my conclusion that the damage to this vehicle was caused by failure to install proper equipment during manufacture and this led to ignition of the spare tire in the engine compartment.

Friction

This well known fire cause can be the culprit in car fires. A dragging brake lining, soaked with brake fluid can produce a fire. Anything that rubs or scrapes together can produce heat and sparks. In either event fire can be the result, especially where other flammables are present such as gasoline, oil and other volatile fluids. Just remember that the smallest spark is hot enough to ignite vapors. Recently I checked the remains of a burned motorhome. The fire obviously originated in the rear of the vehicle and spread toward the front. Examination revealed that an auxiliary fuel tank had been added at the rear of the vehicle. A metal strap securing this tank to the frame had broken, allowing the right side of the tank to fall to the roadway and drag, creating a huge amount of sparks. Pressure from the fallen tank twisted the fuel lines, causing them to break and allowed gasoline to pour into the shower of sparks. The result was inevitable.

Tires

There are three basic types of tires used by passenger vehicles: bias, belted bias and radial. The bias ply tire has been in use since the 1920's and the cords in the plies cross each other at an angle called the "bias angle". Cords are arranged in two or more plies, always even numbered. These tires tend to run hotter than the other two types.

The belted bias tire had cords in a criss cross pattern but also has two or more layers of fabric, or "belts" under the tread. This tire features increased strength and stiffness in the tread. The body cords are rayon, nylon or polyester. The belt cords are fiber glass, rayon or steel. This tire runs cooler and gives more mileage than the bias ply tire.

The radial tire carries a lettered number that has an "R" in it such as "ER7814". The cords in the body run at right angle to the center line and may be arranged in one to three plies. Over this radial section is added a belt of up to four plies whose cords run at an angle of 15° to the centerline. In some radials the belts are made of steel, some of fiberglass or rayon. The radial tire has flexible sidewalls but with great stiffness and strength in the tread area. It runs cool and provides a very long wear. (Source: Tires — Their Selection and Care, National Bureau of Standards NBS C1.52.)

A motor vehicle tire can burn with sufficient intensity to communicate the fire to the passenger compartment. Likewise, a fire inside the engine or passenger compartments of a vehicle can communicate to the tires. In a study conducted by Professor Michael E. Kurz, Illinois State University, in 1983 he used the *Thermal Gravimetric Analysis* (TGA's) of compounded tire stocks to give an indication of how tires might behave at elevated temperatures present in fire situations. His conclusions were that "Tires require fairly high temperatures (650-700°F) before the bulk of the tire will volatilize to flammable fragments. Use of an appropriate amount of an accelerant capable of heating the tire to this temperature would be required to ignite the tires. Once ignited copious heat would be evolved to sustain virtually (95%) combustion of the tires with proper air present."

Tire fires are more frequently encountered in trucks, buses and heavy equipment. Friction with the ground surface produces heat. This heat is greatest in the shoulder of the tire since it does not have the cooling benefits provided by the tread design. When these tires are operated at too high speed or under stress of frequent stop and go driving the internal heat of the tire increases and is unable to escape. Eventually the ignition temperature is reached and the tire catches fire. This usually does not occur until the vehicle stops and there is no longer an airstream over the wheel.

Another type of tire fire encountered in the operation of trucks, buses and heavy equipment occurs when one tire on a set of dual wheels deflates and the load must be carried by the companion tire. Friction caused by the additional weight brings on a rapid increase in temperature. In addition the flat tire rubs against its companion and additional heat is created in a short period of time. While the vehicle is in motion there is no outbreak of fire even though ignition temperature has been reached. Once the vehicle is halted the fire follows very quickly. The following is from an investigative report concerning a transit bus fire:

"The major fire damage was to the right rear side of the vehicle with fire diminishing toward the front and opposite side. Examination of the engine compartment did not reveal any evidence of failure or malfunction in the fuel, electrical or transmission systems and they were eliminated as fire causes. There was considerable fire and heat damage in the right rear wheel well. The fiberglass top structure of the well was burned through at its top rear. The right rear interior of the bus showed the same condition with fire and heat damage diminishing toward the left side. The electrical wiring adjacent to the right rear door was examined for any evidence of arcing or similar malfunction. None was found. The emergency brake drum on the drive shaft was examined through the burned floorboard at the rear of the bus. If the vehicle had been operated with this brake engaged, the drum and shoes could be expected to show evidence of excessive heat and wear. None was found.

Both right rear dual tires and wheels showed exposure to considerable heat and fire. The tire on the inside dual exhibited indications of break up of the casing often associated with under inflation. There were two deep cuts in the tread with some penetration indicated on the inside of the tire. Such a cut would suggest that the tire had run over some sharp object that penetrated it, allowing the air to escape and the tire to deflate.

Conclusion as to cause of fire:

From examination of this bus, it is my conclusion that the fire originated in the right rear wheel well, burning through the fiberglass upper structure and entering the interior of the bus. The origin of the fire was the overheating of the right inner dual tire due to its continued operation after deflation. This caused overheating and a fire resulted due to friction heat.

Recommendations:

Consideration should be given to a program warning drivers of the hazards of not inspecting all tires after the bus has run over some object. It might also be well to instruct drivers in the 'feel' of a bus being driven with one of the rear duals deflated."

In studies conducted a number of years ago it was determined that fires have resulted when tread temperature has reached 400°F and there has been a break in the tire. Re-ignition is encountered in tire fires also. One experiment recorded a temperature of 440°F when a truck tire split and ignited. Twenty-five minutes later the tire re-ignited even though the tread temperature had lowered to 150°F.

Smoldering Material on Upholstery (experiments)

Vehicles of the 1960's and 1970's used a cotton filler and foam rubber seats covered with various materials, such as nylon and plastic. Tests allowed burning cigarettes to smolder as long as the vehicle remained closed. Cigarettes used in these tests were placed on the seat cushions and were generally consumed in about twenty minutes. In one hour and fifteen minutes a front door was opened and a smoldering area ignited with a match. The ensuing fire communicated to the entire interior of the car in a matter of seconds. In another test the doors were not opened for one and one-half hours. When the doors were opened flames were observed in the areas where the cigarettes had been placed. The seats and the entire passenger compartment soon became involved.

From this it was concluded that smoldering fires, such as associated with cigarettes, could occur. It required thirty minutes or more for a three inch circle burn pattern and up to an hour for a charred burn fourteen inches in diameter. Experiments conducted recently indicate that the total vehicle fire caused by a carelessly discarded cigarette or similar object may be a thing of the past. In not one instance was a fire produced of sufficient energy to cause more than superficial damage. Compliance with Federal Safety Standard #302 has undoubtedly led to this improved situation since it relates directly to the flammability standard of a motor vehicles interior. While this is the good news, the bad news is this: If the smoldering cigarette or other such object makes contact with paper and a fire develops from that a completely gutted car can result. This of course depends on whether or not there is sufficient oxygen inside the vehicle for the fire to continue. If the vehicle had its doors and windows closed the fire will extinguish itself generally within five to eight minutes.

Plastics

The growth of the use of plastics in the production of the modern automobile has greatly increased. In 1975 plastics comprised 150-160 lbs. of the average production vehicle. By 1980 this had risen to 203 lbs. per car or 6% of the total weight based on a 3,300 lbs. vehicle. Estimates for 1985-1990 production show the use of plastics rising from 240-300 lbs, or 13% of a total vehicle with a weight of 2700 lbs in 1985 and 2250 lbs in 1990. Clearly the total weight of the automobile is declining while the use of plastic is increasing. (Sources: *U.S. Automotive Industry in the 1980's, 1981 U.S. Industrial Outlook, U.S. Bureau of Industrial Economics, 1981 General Motors Public Interest Report and Information Handbook - Materials, 1982 edition, Motor Vehicle Manufacturers Association of the United States, Inc.)*

In addition to this we have seen a corresponding increase in the use of lighter metals such as aluminum and a decrease in steel and iron. This also effects the vehicle fire investigator's problems but since we are dealing with plastics in this chapter we will defer comment on these other materials.

This move toward weight reduction has been made largely to accommodate fuel economy standards required in the last few years. Increased miles per gallon fuel efficiency can best be reached by decreasing the overall weight of the motor vehicle. In 1968 the Plymouth 4 door sedan with 116 inch wheelbase and 318 cubic inch V-8 engine weighed in at 3,415 lbs. By 1973 the Fury III model had stretched out to a 120 inch wheel base and curb weight had increased to 4,014 lbs, of which 125 lbs, or 3.1% of the total weight was plastic. In 1981 the weight of the Plymouth Reliant 4 door sedan with 99.6 inch wheelbase and 2.2 liter 4 cylinder engine was 2337 lbs, at the curb and the plastic content was 174 lbs. (Source : *Materials in the Automobile, Chrysler Corporation Cars* and *Information Handbook - Materials*, Motor Vehicle Manufacturers Association of the U.S. Inc.) The same trend was matched by other vehicle manufacturers. Some vehicles make extensive use of plastic body panels. The following chart shows the growth in the use of plastic in vehicles from 1977 to 1990:

AUTOMOTIVE INGREDIENTS					
Pounds of material per U.S. passenger vehicle					
Material	**1990**	**1989**	**1987**	**1985**	**1977**
Steel	1,717	1,728	1,775	1,781	2,202
Iron	454	459	480	468	540
Plastic	229	225	222	211	168
Glass	87	85	86	85	86
Source: American Metal Market Newspaper					

In Society of Automotive Engineers (SAE) paper #970663, "Recent Developments in Shredder Downstream Separation Processes and Recycling Options of Automotive Shredder Residue," V. Sendijarevic, B. Pokorski, D. Klempner and

K.C. Frisch, University of Detroit Mercy observed, "The overall amount of plastics and polymer composites in the U.S. automobiles increased from 5.0% in 1978 to 7.7% in 1992. In the same time period the total weight of the average passenger car decreased from 1572 to 1411 kg."

In September 1997 Chrysler Corporation announced plans to test the possibility of making a car body encased in plastic. In early 1999 the company, now Daimler-Chrysler Corporation, produced a concept vehicle featuring a fiberglass body. This vehicle is a sports utility vehicle (SUV) called the Jeep Commander. If it goes into full production, polyethylene terephthalete would be a likely substitute for the composite material. The ability of this plastic to sustain side impact made it a choice over fiberglass according to Larry Oswald, executive engineer for body panels at Daimler-Chrysler Liberty and Technical Affairs Development Center in Madison Heights, Michigan. The vehicle is based on the 1999 Jeep Grand Cherokee, but is seven inches wider. A heavier power train has been installed weighing 2100 pounds. The entire vehicle weighs in at 5,000 pounds, keeping it in range with the average SUV even with the much heavier power train. The use of plastic could save as much as 50% body weight as compared to the use of steel, according to researchers at Daimler-Chrysler.

Ford Motor Company is considering introducing the plastic bodied electric powered Think City Car into the U.S. after the year 2000. Ford acquired a 51% interest in Pivco AS, an Oslo, Norway auto company, producer of the Think. The vehicle will be sold in select markets as a vehicle to be used to travel between commuter stations and home or in retirement communities. *(Source: Automotive News, January 18, 1999)*

Without going into more detailed elaboration it can readily be observed that plastics are, and will continue to be, a major component factor in the modern motor vehicle.

The question here is what, if any, problem does the use of this material present to the vehicle fire investigator? As a general rule, plastic material is difficult initially to set on fire but once ignited can burn with great intensity. In addition. some plastics give off toxic fumes during the burning process and increase the hazard of the fire. The examination of this debris by the fire investigator can also present a dangerous health condition. This will be discussed in the chapter "Arson".

Just what is plastic anyway? According to the Encyclopedia Britannica it is

TYPES OF PLASTIC IN THE MODERN VEHICLE

Area	Application	Material	Lbs. per car
Interior	Crash pad	Urethane, ABS, PVC	5
	Headrest pad	Urethane, PVC	4
	Trim, glove box	Polypropylene, PVC, ABS	18
	Seating foam	Urethane	24
	Seat belts	Nylon	3
	Upholstery	PVC, nylon, polyester	18
	Instrument panels	ABS, SAN, urethane	8
	Headliners	Sytrene, PVC	6
	Carpeting	Nylon	9
	Package shelf	ABS	3
Exterior	"Soft" bumpers	Urethane	20
	Fascia panels	Urethane	22
	Fascia retainers	G.R. polypropylene	7
	Fender liners	Polypropylene	12
	Front end panels	G.R. polyester & polycarbonate	14
	Wheel covers	Polyphenylene oxide	8
	Fender extension	G.R. polyester, nylon	3
	Grille	ABS	4
	Lamp housing (rear)	Polypropylene, ABS	4
	Styled roof	PVC	6
	Bumper sight shield	EPDM-rubber, urethane	12
	Window louvers	Polybutylene terephthalate	2
Under-the-hood	Ducts	Polypropylene	4
	Battery case	Polypropylene	2
	Fan shroud	Polypropylene	3
	Heater & air cond.	Polyester, polypropylene	9
	Electrical housing & wiring	Phenolic, PVC, silicone	11
	Electronic ignition comp.	Polybutylene terephthalate, phenolic	1
	Master brake reservoir	Nylon, HDPE	1
	Battery trays	G.R. polypropylene	1

**Other than soft interior components, many applications are not on every car; weights given are averages for fully formulated filled plastic parts and includes the weight of the resin, fillers, & reinforcements.*
Source: Society of the Plastics Industry, Inc.

"synthetic materials that are capable of being formed into usable products by heating, milling, molding and similar processes". The definition also expands to "resins in their molded form (although not all plastics are resins and not all resins can be or necessarily are molded; they can be cast or converted to coatings, self supporting films or fibers.)". The Society of the Plastics Industry defines plastics this way "any one of a large and varied group of materials consisting wholly or in part of combinations of carbon with oxygen, hydrogen, nitrogen and other organic or inorganic elements which, while solid in the finished state, at some stage in its manufacture is made liquid and thus capable of being formed into various shapes, most usually through the application, either singly or together, of heat and pressure".

Generally speaking, plastics are made up from a long chain of hydrocarbons linked together in various ways. Their chemical compounds are quite complicated and the setting out of their various chemical structures schematically would only serve to confuse both the writer (who can barely recall his high school chemistry!) and the reader. Suffice it to say that there are two extremes on the

U.S. AUTOMOTIVE PLASTIC CONSUMPTION

	Lbs. per Car		Total Resin Usage	
	1979 (1)	Forecast	1979 (2)	Forecast (3)
ABS	20	42	200	500
Acrylic	4	8	40	100
Cellulosic	1	2	10	20
Nylon	6	17	60	200
Phenolic	4	3	40	40
Polyethylene	11	14	110	170
Polypropylene	44	53	440	630
Polyurethane	46	48	460	580
Polyvinyl Chloride	27	38	270	460
Unsaturated Polyester	20	50	200	600
Other (4) (5)	11	25	110	300
Total	**194**	**300**	**1,940**	**3,600**

plastic flammability scale. Tetrafluoropolyethylene, better known by its registered trade name Teflon® is not combustible and will show heat damage only at very high temperatures. On the low end of our scale is Nitrocellulose. This material, under its old name of Xyloidin when combined with camphor and a material to make it flexible, produced the first usable plastic which became known by its brand name Celluloid. It was, and is, extremely flammable. (See table on prior page.)

The past and projected use of plastic in motor vehicle production is illustrated by the previous chart on page 86.

(1) Excludes acrylic paints, elastomeric absorbers, seals, tires, sealants, & polyvinyl butyral glass interlayer.
(2) Based on 10 million unit production.
(3) Based on 12 million automotive units built.
(4) Data includes acetal, polycarbonate, polyphenylene oxide, polysulfone, & thermoplastic polyester.
(5) Projected data includes thermoplastic olefin elastomers, acetals, polyphenylene oxide, polycarbonate, thermoplastic polyesters, etc.
(Source: The Society of Plastics Industry, Inc., "Facts & Figures of the Plastics Industry.")

The self ignition temperatures of plastics are generally from 840°F (450°C) to 1100°F (600°C) From this it can be readily determined that this material is not a primary cause of vehicle fires, but is a material that adds considerably to the fire load of a motor vehicle.

How does a plastic burn? The burning of a solid polymer or polymer composition goes through four stages:

1) Heating
2) Decomposition
3) Ignition
4) Combustion

During the initial heating a thermoplastic material will begin to soften or melt and begin to flow. A thermoset plastic generally will not melt below the decomposition point.

At the later stage of heating decomposition occurs and volatile gases or chemically degraded polymer fragments are released. Many polymers decom-

pose into monomers which are flammable at elevated temperatures.

Flaming Characteristics of Small Samples of Common Polymers

Polymer	Flame Color	Odor	Other Features
Nylon	Blue with yellow tip	Resembles burning grass	Tends to self extinguish, clear mel
Polytetra-fluoroethylene	Yellow	None	Chars very slowly; difficult to burn
Polyvinyl chloride	Yellow w/base green	Acrid	Tends to self extinguish, acidic fumes
Urea-formaldehyde	Pale Yellow	Fishy formaldehyde	Very difficult to igni
Butyl rubber	Yellow, smoky	Sweet	Burns
Nitrile rubber	Yellow, smoky	Sickly sweet	Burns
Polyurethane	Yellow, blue baseAcrid		Burns readily
Silicone rubber	Bright yellow-white	None	Burns with white fumes; white residu
Styrene-butadiene rubber	Yellow, very smoky	Styrene	Burns
Alkyd plastic	Yellow, smoky	Pungent	Burns readily
Polyester (contains styrene)	Yellow, very smoky	Styrene	Burns
Polyethylene	Yellow, blue base	Resembles burning candle wax	Burns readily; clear when molten
Polymethyl Methacrylate	Yellow	Characteristic	Burns
Polypropylene	Yellow, blue base	Resembles burning candle wax	Burns readily; clear when molten
Polystyrene	Yellow, blue base	Styrene	Burns readily

As the temperature of the plastic increases the flammable gases released reach an appropriate ratio of oxygen to air and ignition takes place. If heat is maintained and a supply of gaseous decomposition vapor is produced the burning process will continue. That is to say, it will continue as long as an oxygen supply is maintained. Combustion now continues in either a nonflaming or flame propagation state. This depends on the make up of the plastic and the oxygen index.

Here are some characteristics of common plastics on exposure to a small flame source:

(Source: "Kirks Fire Investigation", 4th Edition, John DeHaan, Brady,

Prentice Hall Division Publishing)

ABS plastics were first developed over 45 years ago and are used in a variety of items including battery cases, fittings and wheels. They are slow burning and do not drip.

Acrylics basically combine strength with clarity and light transmission. They are used for skylights, tail lights, paints and in a form of fiber. They will ignite readily but will not drip while burning.

Cellulosics are among the toughest of plastics and are widely used. As a general rule their maximum service temperature is relatively low and are readily burnable.

Nylon is an extremely useful plastic and is found in carpeting, belts, gears, washers, bristles, molding, etc. It is difficult to ignite but will melt and drip. Besides flammability considerations another very important aspect of the exposure of plastics to flame is their tendency to produce toxic gases. The following is an excerpt from the testimony of a consulting chemist in a case involving a dash fire caused by shorted wiring:

"Polyvinyl chloride...when heated gives off materials like hydrogen chloride, hydrochloric acid, chlorine and phosgene. The latter two are very toxic substances." His testimony continues "Many plastic materials are difficult initially to set on fire, much more difficult than fibrous materials such as cotton based materials or cellulose but once these other materials begin to burn then they can ignite the plastic materials which burn very intensely".

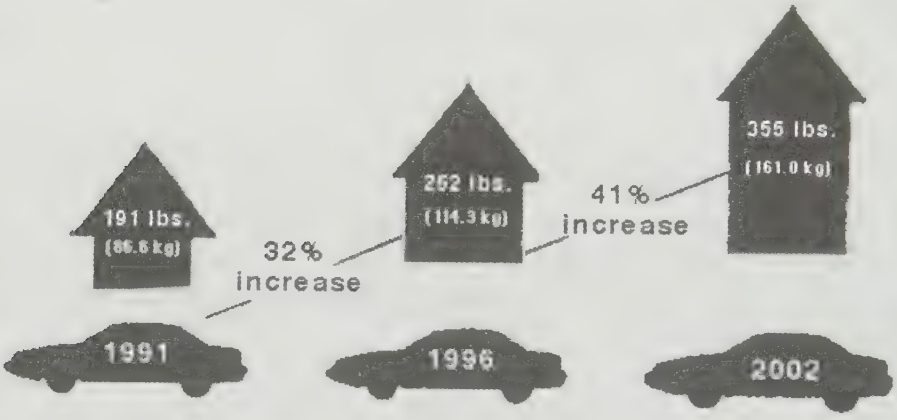


Arrow shows plastic covering and exposed foam sound deadening material in new vehicle's engine compartment. Its close proximity to the exhaust manifold could easily result in a fire.

(Pragoff vs. Putnam Buick et al).

Many of the synthetics used in cars today will reach a melting point before they actually burn. When flame is present they will continue to burn. The foam substance in use in many new vehicle seats is called "Self Extinguishing Foam" (SEF). What this means is that the substance will not support a flame but will melt instead.

The increased use of plastics in the production of motor vehicles continues to present a challenge to the motor vehicle fire investigator.



Average Amount of Aluminum in North American Passenger Cars

Aluminum Alloy

Some truck cabs are made from aluminum and magnesium alloys. A fire in the cab can result in almost complete destruction. The investigator in such a case should seek out the area of greatest fire and heat damage in helping to determine the cause and origin.

By the mid 1980's in the production of American motor vehicles the content of aluminum had greatly increased. The use of aluminum radiators results in a 20-40% reduction in weight compared to copper and brass. The average U.S. built car contains over 250 lbs. of aluminum according to the Amcast Automotive. The use of aluminum in 1990 represented 4.4% of the vehicles total weight against

Aluminum Casting Alloys	Intake manifolds, Oil filter blocks Transmission cases, Cylinder heads, Cast oil pans, Valve covers, Alternators, Water pump housings, Note: Alloy 319 is more in Ford products; 356, a purer, heat treated alloy is more common in GM vehicles.	319 alloy: 960-1120°F 356 alloy: 1035-1135°F (melting ranges of most common alloys).
Aluminum Die Casting	Carburetors, Fuel pump body Alternator bracket, A/C Compressors	380 alloy: 1000-1100°F (melting range)

2.0% in 1970. The accompanying chart by Amcast illustrates actual and projected usage.

Aluminum alloy has been used for years in the production of wheels, in intake manifolds, water pump housings and other engine accessories. Aluminum alloys have melting points between 950°-1150°F. If the alloy is a zinc based diecast this temperature is reduced to 715°-730°F. Following are some cast aluminum usages and their melting points.

Copper & Aluminum Alloying

While examining the remains of a burned motorhome or RV trailer you may have come across copper tubing bearing small holes with light colored edges and wondered what significance this had on the cause, origin or progression of the fire. We know that copper has a high melting temperature of 1980°F and you may make the assumption that something extremely hot came in contact with the tubing or possibly that the tubing failed due to some inclusion in the copper.

Neither of these assumptions would be correct. Aluminum is used in the construction of many motorhomes and RV trailers. Aluminum melts at 1200°F, far below that of copper. Copper and aluminum, however, when mixed form a combination with a melting point of 1020°F. This is below the melting point of aluminum and far below that of copper. When you observe these holes in copper tubing you can assume that the fire was hot enough to cause aluminum to melt and drip onto the copper. The resulting contact would cause the tubing to melt at 1020°F and not at 1980°F.

This same condition has been observed in vehicles where copper wiring has been exposed to melting aluminum. In these cases the wiring is completely severed by the melting process. Examine the ends very carefully for aluminum deposits.

Magnesium Alloy

Some vehicles use a magnesium alloy for their transmission housings and engine blocks. This alloy is about 80% magnesium and 20% aluminum. The melting point is around 1200-1400°F. If you check the engine compartment and find only the crankshaft and pistons from the engine, don't worry, the block hasn't been removed — it was aluminum-magnesium alloy and melted!

Magnesium	Motorcycle carburetors Volkswagen transmissions & engine cases, increasing use in future.	AZ 81 alloy: 914-1132°F (melting range)

Magnesium, pound for pound, is stronger than aluminum or steel and is about one third lighter than aluminum. Since it has a lower melting temperature than aluminum it is easier to use in casting dies since it flows better in molten form. Ford Motor Company is reportedly planning to increase their use of magnesium to include intake manifolds, oil pans, instrument panel braces, wheels and transfer cases. Ford plans to boost its magnesium use from three pounds to 12 pounds by 2003. (Source: *Automotive News*, January 25, 1999)

Here are some current uses of magnesium along with melting temperatures:

Safety Glass

American windshield glass is laminated and has a sheet of plastic in between. At temperatures between 250-300°F this plastic changes to a liquid and begins to bubble. As the temperature increases the material chars and pressure builds up until the glass breaks out. Recently there have been suggestions that an inner plastic sheet be added to the windshield to protect against glass splinters. These inner liners have been used on some European vehicles since 1978. Tempered glass which is used in windows other than the windshield flows at about the same temperature as laminated glass, that is about 1300°F. In most cases this tempered glass will break into small particles before it begins to flow. This is especially true in cases where the windows are up and interior pressure has built up. Any evidence of flowing or "icicling" of glass would indicate a temperature of over 1300°F. Black "charring" would indicate a temperature of between 800-900°F.

This generally indicates a slower build up of heat. Some general observations of glass are:

Glass (tempered)	Side windows, Rear window Moon roof	450°F: Loses temper 500-600°F: May shatter thermally shocked.
Glass	Windshields. Note: Glass is a liquid even at room temp., but is so viscous it behaves as a solid. Windshields have a layered construction sandwiching a tough plastic film between two layers of glass.	1000°F: Sags 1300°F: Loses shape rapidly. 2500°F: Drips, completely molten behaves like a non-viscous liquid

Ignition Key Material

In some cases where the vehicle was supposedly stolen and then burned the location of the ignition key becomes of vital importance. A complete search of the front floorboard area should be made for the ignition key since it is generally made of a medium leaded brass with nickel plate. The melting point of this material is about 1700°F. and it can survive a vehicle fire. Sometimes it drops out of the keyway in the earlier fire stage and falls to the floor of the car where it becomes imbedded in molted glass and pot metal. It should always be searched for. Its discovery can lead to the establishment of the truth or falsity of a theft claim.

Air Bags

The behaviour of air bags during the course of a vehicle fire and the possibility of being a causative factor in such fires is of major importance to the vehicle fire investigator. Some general information along these lines is contained in Department of Transportation information bulletin HS 807 579, revised August 1990 (we have excerpted some portions of this bulletin that are pertinent to our study):

The air bag is designed to supplement the protection offered by safety belts. In a frontal impact of sufficient severity (comparable to a collision into a solid wall at 10-14 MPH or above), sensors in the vehicle detect the sudden acceleration and trigger the inflator module. This causes the solid chemical propellant sealed inside the inflator, principally sodium azide, to undergo a rapid checmical reaction. This reaction produces primarily nitrogen gas, the same gas that makes up 80 percent of the air we breathe. The gas inflates a woven nylon bag packed inside the steering wheel hub or the instrument panel for the front seat passenger. The bag inflates in less than one-twentieth of a second, splitting open its protective cover, and inflating in front of the occupant. As the occupant contacts the bag, the nitrogen gas is vented through openings in the back of the bag which helps to cushion forward movement.

Is smoke produced during deployment?
There are three kinds of "smoke". First, many people mistake the corn starch or talcum powder used to lubricated the bag as smoke. These substances should not be a problem for rescue workers or accident victims. Second, a sealant which is used to prolong the life of the air bag system can smoke when the air bag is deployed. This smoke dissipates rapidly and should not be a cause for concern. Lastly, during deployment, small particles from inside some bag systems are vented into the passenger compartment. These airborne particles look like smoke, and some are deposited as a powdery residue on and around the bag.

Is the air bag hot?
The bag itself will not be hot. Some components within the air bag module will be hot for a short time, but they are relatively inaccessible and should pose no threat to rescue personnel or crash victims. However, personal contact with the steering wheel hub should be avoided for at least 15 minutes after deployment.

Is the sodium azide canister likely to explode during a car fire?
No. The air bag is designed to inflate normally in the event that a vehicle fire causes the canister to be heated above 300°F. Consequently, it is possible that the air bag will deploy in a car fire, but there should be no fragmentation of the inflator.

If there is a fire in an air bag car, can water be used to extinguish it?
Yes. Any effective fire-fighting medium, including water, may be used to extinguish a fire in an air bag-equipped car...

It should be noted that deployment of the air bag during the cause of a vehicle fire is not an uncommon occurrance. When this happens it is generally accompanied with a loud report and a fireball caused by the ignition of the sodium azide and rapidly expanding gases.

In the December 1996 issue of *Fire Engineering*, an article by Leigh T. Hollins reports that the Cedar Hammock Fire Department responded to a vehicle fire call. The car was a 1994 Ford Aspire and the passenger compartment was fully involved. After the fire was extinguished it was determined that the inflator assembly of the air bag exploded with such force that the top of the steel assembly was ripped in half, propelling it through the air and landing fifteen (15) feet in front of the car. A bowl sized impression was left in the roof of the vehicle.

Two rivets, each about 1/4" wide by 2" long, pierced the steel roof. Interestingly, this controverts information provided by the Department of Transportation and the major vehicle manufacturers in *Emergency Rescue Guidelines for Air Bag Equipped Cars, DOT-NHTSA, DOT HS 807 579 revised August 1990.*

Coolant Fires

There has been a great deal of speculation as to engine coolant being a fire cause in the engine compartment of motor vehicles. One does not normally think in terms of this liquid being a flammable since it is most frequently used in a 50% ratio to 60-40% mixture with water. It's use as both an engine coolant and anti-freeze is well known under a number of brand names.

"Coolant" or "anti-freeze" is an ethylene glycol with a flashpoint of 270°F, an ignition temperature of 790°F, a boiling point of 370°F and flammable limits of 2.6-12.5% by volume in air. It has a specific gravity of 1.0 and is water soluble. Will it burn? Obviously the answer is yes. Will it burn when mixed with 50-60% water? No, at least not while it is in that state. Does it remain in that state when extreme heat is applied? We conducted a test using engine coolant mixed with 50% water. An open flame from a propane torch was applied to the mixture. After a matter of 35 seconds the solution caught fire after the water had boiled away. The experiment was conducted again and again with the same results. The inescapable conclusion is that coolant/anti-freeze will burn after a sufficient

quantity of water has boiled away. A blown hose on an overheated engine is quite likely to cause an engine compartment fire.

Some recent tests conducted by vehicle manufacturers show the following with relation to coolant/anti-freeze:

	Flash Point	Ignition Temperature on Hot Surface
50/50%	270°F	720°-1143°F
100%	232°F	750°-1100°F

Freon & Refrigerants

"Freon" is a brand name of the DuPont Company. It is a halogenated hydrocarbon containing fluorine. It is used in air conditioning systems of older motor vehicles. It is considered to be non-flammable in its variations except for Freon 21 and Freon 113. These are considered to be "weakly flammable" with ignition temperatures of 1026°F and 1256°F respectively. In spite of this there have been occasions where engine compartment fires have started as a result of a leak in the Freon line. Before this possibility is dismissed remember this: a light oil is used to lubricate the seals in the refrigerant line. This light oil drips on the hot surfaces of the exhaust manifold and a fire results. The Freon doesn't burn, but the lubricant in the line does.

As of this writing, there are no known fire problems with the now required refrigerant R134a. Some substitutes contain propane and do not meet EPA requirements. One of these is HC12a and it has the following properties:

Ignition temperature		1490°F
Upper Limits		8.5%
Lower Limits		1.9%
Vapor Density		1.7%
Specific Gravity		0.552
NFPA Rating:	Health	2
	Flammability	4

Foreign Objects in the Engine Compartment

Sometimes the cause of a vehicle fire can be so obvious that it is overlooked. This seems to be a contradiction in terms, but is true nevertheless. Over the years we have seen a number of cases where the primary cause of the fire has been the introduction of a foreign object into the engine compartment during repair or service. On one occasion, we discovered that a mechanic's wrench left in the

engine compartment had contacted the positive pole of the battery and the resulting short started a fire.

A more common occurrence is the forgotten shop towel in the engine compartment. This happens sometimes when a towel is used to soak up the oil spilled when the filter is changed. In some transverse mounted engines this filter is in the immediate vicinity of the exhaust manifold and tube. The forgotten oil saturated cloth is then baked by the hot metal and a fire frequently is the result.

In early January 1999 we were asked to make an investigation into the cause and origin of a fire that damaged a 1997 Ford Escort. Excerpts from our investigation report follows:

> Examination of the exterior of the vehicle showed a localized burn pattern to the center of the hood. No other burn patterns were noted on the exterior of the vehicle. No collision damage was noted on the exterior of the vehicle. Inspection of the interior of the vehicle showed no fire damage. Examination of the interior fuse box showed no blown fuses. Examination of the engine compartment showed a localized fire primarily in the area of the coil pack. This corresponded to the burn pattern found on the hood of the vehicle. Inspection of the coil pack revealed what appeared to be the burned remains of a cotton shop towel. Inspection of the remains indicated this to be a shop towel as used in the dealership and the auto repair industry. I found a portion of the towel adjacent to the exhaust manifold next to the coil pack. In addition, I found remains of a towel wedged between the metal exhaust heat shroud and the plastic fan shroud at the front of the engine compartment. The plastic fan shroud was heat damaged and partially melted. Below this area I found additional melting of plastic and fire damage to a rubber hose from drop down of fire debris which caused additional heat damage.
>
> Examination of the fuel system showed no evidence of a fuel leak. No evidence of electrical shorting was found in the engine compartment. Inspection of the engine fluids (coolant, power steering, engine oil, and transmission fluid) showed all to be within normal operating range.
>
> My opinion is that the most probable cause of this fire is a shop towel contacting a hot surface, most likely the exhaust manifold or heat shroud, where it ignited and spread the fire to adjacent combustibles.

Crankcase Vapors

In older vehicles it is not uncommon for fuel to drain down into the crankcase and oil pan. This occurs when the gasoline leaks down past the piston rings. There have been reports of backfires causing an explosion of this gasoline/oil mixture in the crankcase, damaging the oil pan.

From time to time, piston rods fail causing them to strike the block with such force that a hole is created. When this happens the sudden expulsion of this gasoline/oil vapor from the crankcase is frequently ignited by the nearby exhaust manifold.

Exhaust System

Weld failures, cracked flanges, cracked tubes, loose connections and other leak sources in the vehicle's exhaust system can lead to a motor vehicle fire. Exhaust temperatures of 640°F have been noted during normal vehicle usage and temperatures of 1200°F (650°C) have been noted in experiments conducted in vehicle test burns. When hot exhaust gases escape from system leaks, any nearby combustible material can be ignited. A break in the exhaust tube over the rear axle of a vehicle communicated sufficient heat to the underbody to ignite the carpeting in the vehicle's interior. In another case, a failed weld between the exhaust pipe and muffler allowed the escape of hot exhaust gases with sufficient elevated temperatures to melt the body sheet metal. The same thing can occur with a cracked or broken catalytic converter. Grassland fires have resulted from super heated pellets failling from a deteriorated catalytic converter.

ARSON

Volumes have been written on *arson* and they all agree on one thing — it is difficult to detect and even harder to prove. Most of the writing in the field of arson relates to the burning of buildings and similar structures. Over the years little attention has been paid to the burning of motor vehicles even though suspicious fires occur in vehicles with great frequency. With the exception of the booklet titled, "Manual for the Investigation of Automobile Fire" published by the NATB (now a part of the National Insurance Crime Bureau) there is relatively little reference material on motor vehicle fires.

Arson is defined as "The crime of deliberately setting fire to another's building, property, etc. or to one's own, especially for a criminal or fraudulent purpose". This is the definition given in the D*oubleday Dictionary* and would seem to be one that we all could agree on. One of the main problems with arson (or suspected arson) is that we must get over that hurdle called "Corpus Delicti". This means, simply, the "body of a wrong" and requires that in any criminal case we must prove that a crime, in fact, occurred. This is easy in crimes against the person and in most property crimes. In murder cases we have the deceased. In robbery we have a frightened or injured victim, deprived of his property and, in many cases, witnesses to the event. In burglary, car theft and similar offenses we have probable evidence that property belonging to the victim has been taken. No problem in these cases with proving the "Corpus Delicti".

In fire cases it is a different story. Remember what we said in the early pages of this book? "All fires are presumed to be accidental." This presumption is of particular significance in motor vehicle fires when the modern auto contains so many seeds that can lead to its own destruction.

In order to successfully prosecute an arson case, whether it be vehicle or any other type of property, we must prove three elements:

1. That a fire occurred.
2. That the fire was the result of a criminal act.
3. That the defendant was responsible for the criminal act.

I think you'll admit that these elements are logical and in proper order. We

don't have a lot of problems with element number one. In any vehicle fire case we have proof that the burning took place.

With respect to element number two we do have a problem. This is where we must establish the "Corpus Delicti". The fact that there was a fire isn't enough. Proof that the fire was intentionally set is essential to establishing "Corpus Delicti".

Arson is not the type of crime that one usually commits in front of a host of witnesses. It is generally an act of secrecy and so the proof is difficult. The Corpus may be established by circumstantial evidence alone but never on the defendant's confession or admission alone. This is an important point to bear in mind. A "naked confession", one that is devoid of corroboration, is not admissible, nor should it be.

Acts and admissions are admissible in establishing the Corpus if there is sufficient corroboration. Conduct of the suspect; such as not summoning aid at a nearby home and instead casually walking considerable distance down the road to call the fire department can establish the Corpus.

An incident or fact or circumstance which would arouse the ordinary person's suspicion in connection with the cause of a fire is generally admissible to show the fire was of an incendiary origin. An example of this would be where a suspect was seen carrying a gasoline can to a vehicle. Shortly thereafter the vehicle burst into flames and there is a strong odor of gasoline at the scene.

"Similar Acts" can also establish the Corpus by showing that on prior occasions the suspect has had a history of fire involvement. This is particularly important in cases involving insured property where the suspect has "suffered" several other fire losses where insurance was collected. The introduction of this type of evidence can show:

1. Motive.
2. Intent.
3. Absence of mistake or accident.
4. Scheme or design of the suspect.

Another method of establishing the Corpus is that of qualified expert testimony establishing the fact the fire could not have started in the manner and nature claimed by the suspect.

All fires fall into one of three categories — accidental, providential or incendiary. There are at least seven motives for incendiary fire and they are:

1. Fraud.
2. To hide other crime.
3. Jealousy.
4. Revenge.
5. Thrill (including pyromania).
6. Riots and vandalism.
7. Terrorism

In most cases involving a suspicious vehicle fire the motivating factor is fraud. There can be other factors as well and they should not be ignored. Certainly many stripped vehicles are burned in order to get rid of fingerprints and other evidence that the suspect believes will lead investigating authorities to him. The same thing is true when a vehicle is involved in some other type of crime such as robbery, murder, etc. and the suspect believes that prints and any other evidence left behind will lead to his identity. In these cases, as well as the others listed as cause for vehicle fires, the investigator's first function is to establish that the burning was caused by human hands and not by accident.

Discovering evidence of the origin and cause of fire is a primary function of the fire investigator. This is of enormous importance and it should be remembered that it is equally important to establish what could not have caused the fire as well as to what did cause it. Proof of cause and origin must be established beyond any reasonable doubt and the fire investigator should establish that nothing other than his conclusions might have occurred.

On being asked to investigate a vehicle fire here are a few things for the investigator to do, both before and during the early stages of his investigation:

1. Determine insurance coverage, name of the company, type of policy, mortgage, inclusive dates of policy coverage, make and model of the vehicle and its identification numbers.

2. Ascertain date of purchase, name of dealer, amount of down payment, amount of mortgage, number of monthly payments, name and address of mortgagee, balance due at time of loss and the number of delinquent payments, if any. Ask the lienholder if a request was received for a contract payoff balance and obtain the explanation for any delinquent payments.

3. Visit the scene of loss and carefully inspect it for any evidence of the origin of the fire. Photograph the vehicle if it remains where it burned. Record the location, distance from nearest crossroad, distance from nearest house and the county in which the loss took place. Describe the terrain, trees, ditches and similar information and record the type of road on which the loss occurred. Make a sketch of the scene, noting any evidence found.

4. Search the vicinity for cans, bottles or other containers. Look for tools, tires or other accessories. The owner may have hidden them and made plans to pick them up at a later date.

5. Check the scene for footprints or related signs. Locate tire marks of the burned vehicle as well as imprints of tires of any other automobiles. If a getaway vehicle was parked on a soft surface and an imprint of the tires measured, it may be possible to ascertain the year, make and model of the vehicle used. Exact measurements made of the wheel base and front and rear tire treads can be checked against existing publications which classify vehicles by such data. Evidence that another car was at the scene may indicate someone picked up the owner after the burning. Casts made of the imprint of tire treads may be made and used later for comparison with a suspect's tires.

6. If containers, gas caps, hoses, burned matches, paper or other physical evidence is found at the scene, it should be carefully preserved since it may be possible to trace it to the owner. Photograph all items where they are found before removal and take care not to destroy fingerprints when handling them. All evidence gathered should be labeled with the date, time and location where found and marked with the initials of the investigator so proper identification can be made later.

7. If the salvage remains at the scene, inspect the tires. Usually the portion of a tire in contact with the ground will not burn. Since many claimants contend their tires were new or only slightly used at the time of loss, save these portions as evidence — they will prove condition of the tires prior to the fire. Soil samples from beneath the vehicle and wheels should be preserved in airtight containers for laboratory processing (as soon as possible) — they may reveal that ignitable liquids had been poured over the tires and other parts of the vehicle before ignition. Be aware, however, that this gasoline could have come from tank and filler neck failure as the fire progressed under the rear of the vehicle.

8. Investigation at the scene of a fire frequently discloses witnesses who can give additional information. Contact the nearest residents to the scene in order to fix the time and date. Ask them what they may have heard or seen. Was the owner known to them? Was he alone? If not alone, who was with him? Did they recognize the burned car? Was any other car there? Did they know the owner of this car? Did they go to the scene while the car was burning? What portion of the car was burning? What was the color of the smoke? Was any attempt made to put the fire out? Did the owner stay at the scene? If so, how long? Did he leave the scene? If so, how did he travel? Did the owner say how the fire started? What did he do? Were any remarks made about insurance? What other persons were present when they arrived at the scene? If any information is obtained from witnesses, statement summaries should be taken as soon as possible.

9. Attempt to locate the first person at the scene or the person who discovered the burning car. Obtain his statement.

10. In many instances the burned vehicle may have been removed from the scene prior to the visit of the investigator. In such cases it is recommended that the tow truck driver or persons towing the vehicle be interviewed. Facts contrary to those claimed by the owner may be developed by this interview. The tow truck driver may be able to furnish information about the condition of the car. Missing or displaced items may be accounted for in this manner.

11. Interview any law enforcement or fire official who may have made an investigation at the time of discovery of the fire. If the officer was at the scene while the vehicle was burning, the owner may have been questioned and a different version of the burning given to him or to her. Facts may have been developed which will be of assistance.

12. Check fire and police records for reports made to these agencies. These records will enable you to determine the action taken from the time the loss was first reported. If a fire truck was dispatched to the scene, interview the firemen as to what they found on arrival. What was used by the fire department to extinguish the flames? Was the owner present? Did he tell them how the fire started? It's good to have a form to use when interviewing the owner of the burned vehicle. The following "Statement Outline" can be useful.

Statement Outline

Day_____ Date_____ Time_____ CI.#_____
Policy_____
Name(full)_____ Birth Date_____
Spouse (full)_____ Birth Date_____
Home Address_____ Telephone_____
Business Address_____ Telephone_____
Occupation_____ Driver's Lic. #_____ Date Iss._____
Year_____ Make_____ Model_____ Color_____
VIN_____ License #_____ State_____
How registered_____
Legal Owner_____
If no Legal Owner, do you have Title in your possession?_____
If not, where is Title?_____
Monthly Payments_____ To Whom_____
Date Purchased_____ Purchase Price_____
Bal. of Loan_____ Payments Current?_____
Next Payment Due?_____
Name & Address of Seller_____
Was it a rebuilt wreck_____ If so, who rebuilt_____
Was it a recovered theft?_____
Have you rented a car?_____ From whom?_____
Make/Year/Model_____
Any luggage or valuables in car? (describe)_____
_____
Where were they located?_____
Any special equipment? (describe)_____
Stereo: Make_____ Value_____ Date. Pur._____
Mag/Chrome Wheels_____ Make_____ Value_____
Date. Pur. & from whom_____
Tires Make_____ Value_____
Date. Pur. & from whom_____
Any old damage to vehicle? (describe)_____
_____
Type Engine_____ Condition_____ Mileage_____
Type Transmission_____ Condition_____
Where car normally serviced?_____
When last serviced?_____

Any mechanical defects? (explain)_____
When was vehicle last in repair shop and for what?_____
What repair shop?_____
Was vehicle being offered for sale?_____Why?_____
In your own words briefly describe how the fire occurred and what actions you took_____

_____
Were there any witnesses?_____
Names and Addresses_____

_____

_____
How did you get home?_____
Please describe your actions and any activity with the vehicle for the 12 hours just prior to the fire_____

_____

_____

This form may be helpful or perhaps you would like to make your own outline for obtaining information. One of our investigators uses the following format for this purpose:

Auto Fire Evaluation
Interviewing the Insured
by Don Stasiowski

1. When did the fire occur?

2. Where did the fire occur?

3. Who was operating the vehicle?

4. Last person to operate the vehicle?

5. What were the driving conditions?

6. Vehicle mileage?

7. What alerted the driver to a problem?

8. Where was the smoke/fire first noticed?

9. What was the color of the smoke?

10. Any problems with the vehicle?

11. Any recent repairs?

In case the fire was preceded by a report of theft you will want to secure the details of this incident. The following guide may be helpful:

Report of Vehicle Theft

Policy No.
Name of Insured_____ Phone No. (Hm/Wk)_____
Address_____ City_____ State_____
Employer_____ Location _____
Driver's Lic. No._____ Marital Status_____
Date of Theft_____ Time_____ am/pm_____
Vehicle locked?_____ Keys in car?_____ Amt of claim_____
Specific location where vehicle was taken_____
Name & address of person who left auto at this location_____
Driver's Lic. No._____
Name & address of others who were present_____
_____
_____
When was theft discovered? Date_____ Time_____ am/pm
By whom?_____
Was theft reported to police?_____ Date_____ Time_____ am/pm
By whom?_____
Police Department & Case No._____
If not reported, why?_____
Date car purchased_____ New/Used_____
Purchase price $_____ Trade-in car _____
Allowance_____
Selling Dealer/Individual_____
Address_____
How was car paid for? Cash_____ Check No._____ Bank_____
If financed, name & address of Finance Co._____
_____
Acct. No._____ Bal. due_____
Loan Terms_____ Months @ $_____
Are keys in your possession? Yes/No Ignition Key No._____
Trunk Key No._____
Do you have any other auto insurance? If yes, Policy No._____
Name & address of Ins. Co._____
Make of vehicle_____ Year_____ Model_____
Body type_____ Color_____
Complete Vehicle Identification No. (VIN)_____ Lic. No._____

Some states require a recitation similar to the following to appear on this form: The Insurance Code makes it a felony to present a false or fraudulent claim to an insurer and provides as follows: "It is unlawful to: (a) Present or cause to be presented any false or fraudulent claim for payment of a loss under a contract of insurance. (b) Prepare, make or subscribe any writing, with intent to present or use the same, or to allow it to be presented or used in support of any such claim. Every person who violates any provision of this section is punishable by imprisonment in the State prison not exceeding three years, or by a fine not exceeding three thousand dollars ($3,000.00) or by both."

_____

Your Signature

Always remember the rule that must be observed and it is —*absolute attention to minute details*! The trained investigator sees a multitude of points as he inspects — points easily overlooked by the layman. Remember to look for little things and to associate irregularities with possible causes.

Before we go into the actual inspection of the burned vehicle, referred to as the "salvage", I would like to give you a few words of warning. Vehicle fire investigation really began to take shape as a specialized subject in the mid 1930's when the National Automobile Theft Bureau was handed the task of checking into suspicious car fires by the insurance industry. Early efforts were by trial and error but eventually aided by experiments, a detailed summary was put together listing the commonly observed factors in a suspicious fire and their relation to the cause of the fire. One must remember that the construction, size, and materials used in the present day automobile are quite different from its 60 year old forebearer. Unfortunately, theories that had validity in the 1940's, 1950's, 1960's and 1970's are still sometimes advanced today long after they are no longer valid. This leads to still often quoted phrases such as "they will not burn without help" and "The modern day vehicle is virtually fire proof". As one who sees 2-3 burned vehicles per week, let me assure you that neither of these statements are true. Cars do burn accidentally as completely as any I have observed intentionally set on fire. The most important factor is determining how much time did it take for the vehicle to arrive at such a state of destruction.

I always tell students in my classes on motor vehicle fire investigations that one must consider a suspicious fire case as a whole pie and that you cannot remove a slice and say "here it is, I have the proof!" The whole pie includes motive and opportunity as well as facts gathered at the scene and salvage examination. There are fire investigators who will tell you flatly after examining a burned vehicle that it was deliberately set on fire without once talking to the operator of the car or any witnesses. Such conclusions can be drawn only if the deliberate fire had reached only the preliminary state and an incendiary device found. Even if we find the presence of ignitable liquids in the vehicle's interior we must have a statement from the driver indicating that he was not carrying any such material in the passenger compartment for our find to have evidentiary impact. In short, do not look for easy answers, rarely will they be found. The following suggestions for salvage examination serve only as a guide. None of the indications we list should be considered as absolute. Avoid jumping to conclusions and remember the whole pie.

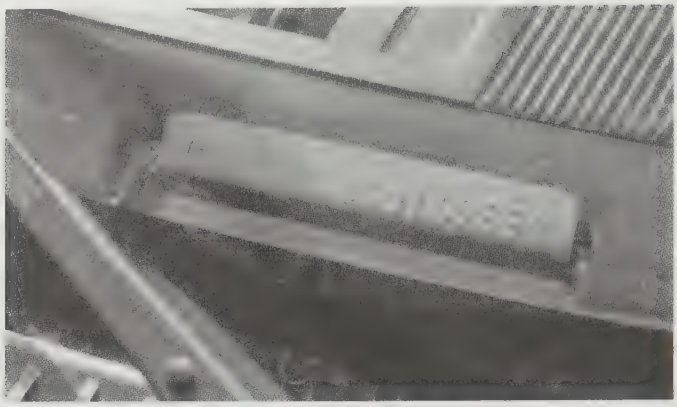
It is best to inspect the burned automobile before contact is made with the owner. This inspection should be made as soon after the fire as possible in order to prevent destruction or loss of evidence.

Inspections of the salvage and search of the scene should begin from the outside in. It is recommended that the examination be made in the following phases:

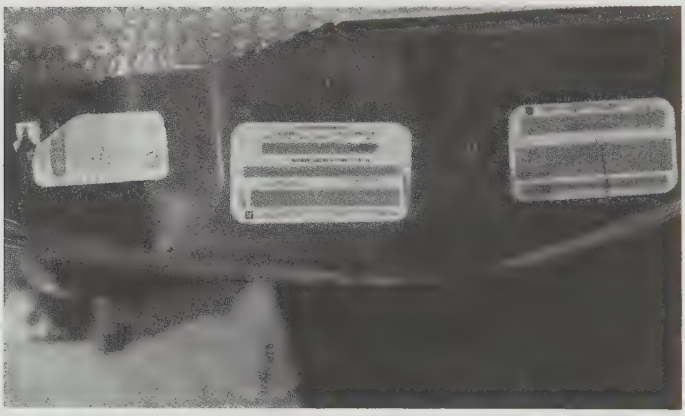
1. Exterior — The items listed below can all be inspected without seriously disturbing the salvage.

A. Vehicle Identification Number
B. Top
C. Glass
D. Windows
E. Doors
F. Paint
G. Hood
H. Trunk
I. Tires
J. Wheels
K. Fuel tank
L. Misc. accessories
M. Prior collision damage

Start by identifying the vehicle through the Vehicle Identification Number (VIN). Make certain it is the vehicle registered to the owner and described in the insurance policy.



Any vehicle examination begins with the establishment of its identity. The Vehicle Identification Number (VIN) plate will confirm this.



The Federal Standards Decals and service stickers can supply additional information.

Sagging or loss of temper in the spanned sheet metal may be indicative of a very hot fire in the interior of the car. Foam rubber and many plastics can generate enough heat to sag the top panel. Remember, a hot, fast fire can indicate an incendiary origin but it is not conclusive and determination of the time element is vital to the investigation.

Has the windshield glass melted and flowed in "Icicle" patterns? If so, this indicates a temperature of over 1200°F. But remember that this is not proof of the use of an accelerant and should be only noted as an observed condition.

Ascertain whether the windows were up or down during the fire. To support combustion, the fire must have air. If the fire melted or broke out the windows their position can be determined by inspecting the devices used to raise or lower



Fire may damage the VIN plate but in most cases it can still be read.

the windows. Sometimes the mechanism is so destroyed that this cannot be accurately ascertained. This is frequently the case with power windows.

Determine if the doors were open or closed during the fire. If a door was closed, combustion is usually complete to the edge of the door. An open door usually is not completely consumed unless wind carried the fire from the open door causing complete combustion of flammables.

Check the hood for signs of burn patterns. A carburetor or fuel injection fire will generally burn the paint on the hood in a pattern over the unit. Ignitable liquids poured on the hood or roof top will generally be outlined with a soot ring. The paint will frequently be intact or "checkered". Burning will be heaviest where the liquid drained into cracks.

When the spare tire burns in place, the metal strands of the tire's construction will remain on the wheel. At the time of examination of the spare tire, note whether items normally found in the trunk are in place, such as the jack, tire tools, etc. By examining the remaining portions of the burned tires you can determine their state of wear, and sometimes their make. Retain these portions as evidence. Examine the wheels and tires for evidence as to whether or not they were exchanged. The color of the wheel may not match the vehicle or perhaps all the lug nuts were not replaced.

Is the gas cap missing? Was it removed before the fire or was it blown off? Inspection of the filler neck flanges can help determine whether or not it was blown off if it was a metal cap. If the cap and threads are plastic the cap can be lost due to heat and pressure. The mixture of oxygen and fuel in a gasoline tank

must be exact before any explosion can take place. As noted before, it is extremely rare when one does and this generally occurs when the tank is thought to be empty. Welding accidents account for most such explosions.

Check the body carefully for any sign of a prior collision. The discovery of any such damage can assist in establishing a motive for the fire. Perhaps the fire was set to cover up a "hit and run" accident.

After examining the exterior thoroughly, be certain you have missed nothing before going on to the next phase of the examination. Make notes of what you have observed and preserve any pieces of evidence. Photograph any article before you remove it.

2. Interior

A. Trim	D. Any container
B. Keys	E. Contents
C. Accessories	F. any evidence of an accelerant

What did this car look like before it burned? What did it have in it? What kind of upholstery? Did the owner claim his car was stolen and he still has the keys? Any evidence of cans or melted glass jars?

Be sure to check the floor under the ignition lock and attempt to locate a key in the debris. The keyway is made of white metal and melts easily permitting keys to drop to the floor. Their location may be damaging evidence if the owner claims the vehicle was parked, locked and stolen prior to the fire.

As with the exterior, it must be ascertained if accessories inside the vehicle are intact. Are the radio and tape deck in place or were they removed prior to the fire? How about a cellular telephone? If one was installed there should be remains.

Throughout the interior inspection be alert for containers. If a glass container was used it is possible that it melted and you may find the debris in the remains. Sometimes people claim clothing, fishing tackle, tools, etc., were in the vehicle when it burned. Undisturbed ashes should disclose noncombustible items such as metal hooks, rings, buttons, etc. If a highway flare was used for ignition you may find silver/gray ash in the debris.

The ashes and debris from the salvage should not be just dumped on the ground. Work the ashes and debris out of the vehicle into a carton or similar

container. A box with a wire mesh bottom is excellent for this. Following the inspection this container can be placed in the vehicle. It may be that you will wish to recheck the ashes to prove or disprove information that comes to light during the investigation.

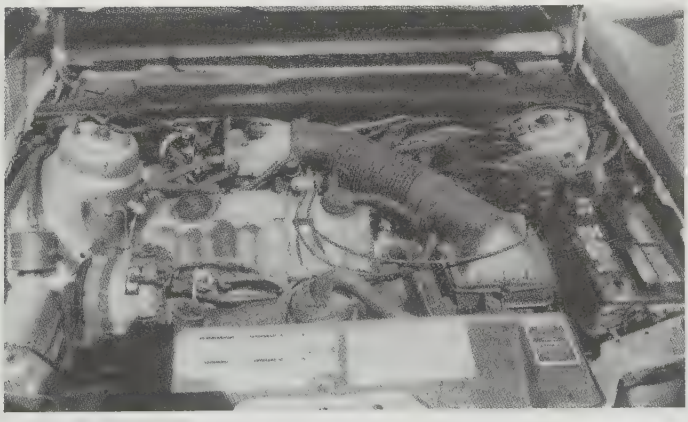
Search carefully for evidence of ignitable liquids inside the car. Automobile floors have low spots and excess liquid ignitables will often run into these areas and a search may produce gasoline or other flammable remains. If found, place in an airtight container, dated and initialed. If an excessive amount of ignitable liquid is used, it may run through openings in the floor and burn underneath causing soot to form on the underside of the vehicle. One of the best places to check for ignitable liquids is the floor matting material. Liquid can soak into the base and be present days after the fire.

When dirt has been thrown on the car, allegedly to put the fire out, search under the dirt to see at what stage of the fire it was thrown. If sand has been thrown in the car, check to see if it has burned.

3. Mechanical Condition of the Automobile

A. Engine
B. Transmission
C. Rear end
D. Other

The mechanical section of the inspection may produce the motive. This phase can prove to be the most important part of the examination.



The engine compartment of the modern automobile can be a confusing place.

Make an overall inspection of the engine compartment. Are engine accessories missing? Alternator? Carburetor or fuel injection unit? Are head bolts loose?

Is the manifold loose? Has the owner been tampering with the engine? Any unburned matches left on the block or in the engine compartment? Matches sometimes remain in evidence in the engine compartment after an extensive fire. A broken connecting rod could be in evidence from an exterior inspection. In order to make sure, have the oil pan dropped. The rod and main bearings should be inspected to determine if the engine was on the verge of "letting go". Check the compression and, if necessary, pull the head and examine the block and pistons for any evidence of mechanical failure. If possible, operate the engine to determine its general condition prior to the fire. Check the oil for evidence of water which could indicate a possible crack inside the block. Oil samples should be submitted to a competent lab for examination.

Possible places for an accidental fire in the engine compartment are the fuel pump, carburetor, or fuel injector and fuel lines, air cleaner, smog controls and the wiring. Evidence of fire elsewhere may indicate use of an ignitable liquid.

Check the transmission, driveshaft and differential by jacking up the rear of the vehicle and turning the wheels. This may show breaks or other mechanical damage that existed prior to the fire.

Check the muffler, manifold and tail pipes for breaks or leaks. Don't forget to closely check the catalytic converter. Overheating of this unit or the muffler is generally evidenced by a bluish discoloration of the exterior skin.

4. Fuel System

A. Fuel lines
B. Fuel tank
C. Carburetor or fuel injection unit
D. Any fuel system tampering

Since gasoline is the most common accelerant used in automobile fires, inspection should be made for signs of tampering with the fuel system.

Inspect the gasoline tank and ascertain if the filler cap was in place at the time of the fire. Trace the fuel line all the way from the tank to the fuel pump, engine compartment and fuel inlet.

If the vehicle is equipped with a gasoline tank drain plug, examine the plug to see if it was removed to drain gasoline from the tank. Pliers are the most common tool used to remove the plug and many times the owner does not bother to replace it. If the drain plug is missing and collar is still intact, the plug has been

removed. When the plug is melted out the collar usually goes with it. These plugs are found only on older cars or in large trucks, buses and heavy equipment. Plastic fuel tanks are increasing. Fire in the vicinity of these tanks will cause them to melt and dump fuel. This will also occur if the fuel tank splits along its seams or the filler neck separates.

Carefully examine the fuel line at the fuel pump and the carburetor or fuel injection unit. Owners have been known to remove or loosen lines at these points to allow gasoline to run over the engine for interior saturation. Pliers may have been used and the fittings may not have been reconnected or tightened.

If the fuel pump is melted there should be evidence of fire in its immediate vicinity. If the fuel pump was disconnected to allow gasoline to run out and then set on fire, there may be carbon deposits inside the gas line at the fuel pump. This, of course, would not apply if plastic fuel lines are used.

5. Electrical System

A. Battery
B. Wiring
C. "Shorted" Wiring
D. Evidence of ignition by pass

Automobile electrical systems are better than in prior years and engineers have reduced the possibility of fire as the result of short circuits. We should bear in mind, however, that such fires can occur and if allowed to spread unchecked can result in the vehicle's destruction.

If the wires have not been completely destroyed, evidence of shorting can be located. As we have pointed out earlier, a short may melt and separate the strands of wire and cause small beads or knots of melted wire to form on the ends.

Ascertain whether or not the ignition system has been circumvented if the vehicle was supposedly stolen. Don't forget to look for the key.

Determine whether the battery was in place during the fire. If the battery is intact and unburned, check to see if it still has power.

Here is a form that will aid you in the salvage examination:

Vehicle Inspection

Date _____ By _____

Damage from fire:

A) Exterior _____ B) Engine Comp. _____ C) Interior _____

Exterior:

Collision and/or vandalism damage: _____

_____

Tires, Rims & Hub Caps: (Inspect for loose nuts, mismatches, etc.)

	Type	Condition	Remarks
Lt. Frt.	_____	_____	_____
Lt. RR	_____	_____	_____
Rt. RR.	_____	_____	_____
Rt. Frt.	_____	_____	_____

Doors: (Open, closed, locked, unlocked, electric locks, door jamb, stickers, etc.)

Lt. Frt. _____

Lt. RR _____

Rt. RR _____

Rt. Frt. _____

Windows: (Stained, melted, shattered, up, down, electric, etc.)

Front Windshield _____

Rear Windshield _____

Driver's side windows (front & rear) _____

Passenger's side windows (front & rear) _____

Vent Windows _____

Gas Cap & Fuel Filler Spout: (Conventional or locking) _____

_____

Fuel Tank: (Damaged, etc.) _____

_____

Antenna: _____

Mirrors: _____

License Plates: (Front & rear) _____

_____

Vehicle Identification Number (VIN): (Read off vehicle) _____

Paint: (condition & etc)_____

_____

Roof Lines:_____

_____

Hood:_____

_____

Trunk: (burn pattern, contents, spare tire, car jack, personal property, etc.)_____

_____

_____

Engine Compartment:_____

Condition and/or damages:_____

_____

_____

Fire Wall: (Condition & description)_____

_____

Fluid Levels: (Level & Condition)

Oil:_____

Transmission:_____

Radiator:_____

Power Steering:_____

Brake Fluid:_____

Rear End:_____

Battery:_____

Electrical: (Insulation, beading, etc.) (Note: Inspect for loose wiring that could have been used to ignite fire.)_____

_____

Carburetor: (Melted, air filter, carb. cover, linkage, gas line connections, gas filters, fuel pump, etc.)_____

_____

Fuel rail or inlets (Fuel injection):_____

_____

Interior:

Vandalism and/or theft damage:_____

_____

_____

Keys: (Single, a ring, spare set, etc.)_____

_____

Vehicle Inspection

Ignition: (Intact, dislodged, melted, on floor, etc.)_____

_____

Locked Steering:_____

_____

Odometer Reading:_____

_____

Wiring:_____

_____

Fuse Box:_____

_____

Accessories: (Stolen and/or missing) (i.e. Stereos, speakers, equalizers, boosters, car heaters, gauges, C.B.s, etc.)_____

_____

Glove Compartment: (Open, closed, contents, etc.)_____

_____

Seats: (Covers, materials, distended, loose items)

Front: (Buckets or bench)_____

_____

Rear:_____

_____

Flooring: Floor Mats:_____

Carpet: _____

Carpet Underlay: _____

Misc. contents in vehicle: (Items under seats, on seats, papers, tools, containers, tapes, etc.)_____

_____

Exhibits & Evidence:

	Date	Time	Location
#1			
#2			
#3			
#4			
#5			
#6			
#7			

Photographs:_____

Concluding Remarks:_____

_____

_____

Sketch of Scene:

Laboratory Methods

When charred debris is furnished to a laboratory and the circumstances are described to an analyst, he has some idea of what to look for in the debris.

The first thing is a separation of the flammable residue from the charred material. If hydrocarbons or organic materials are suspected, the debris is covered with water and heated with a condenser apparatus and a device called a Dean & Stark trap. This trap catches hydrocarbons which condense and allows water to recycle. The distillation should be performed for a minimum of 72 hours and is often left as long as two weeks.

The residues separated are put through a gas chromatograph. This instrument separates a hydrocarbon mixture like gasoline into its many parts and a determination can be made as to nature and quantities of these constituents. The gas chromatogram of the residue serves as a "fingerprint" which can be compared to like parts from cleaning fluids, lighter fluid, gasolines, kerosenes, diesel oils, etc.

Infrared spectrophotometry (IR), differential thermal analysis (DTA) and thermogravimetric analysis (TGA) are other techniques used for identifying various organic materials. IR is used to determine the actual types of molecular bonds in the material, which, like the gas chromatograph, furnishes a different type "fingerprint". The DTA and TGA furnish information about the oxidation characteristics of materials. A combination of analytical instrumentation tools can be used to determine residues of organic materials. Instrumentation and wet chemistry techniques are used to determine the presence of reaction products which result from use of materials such as strong inorganic oxidants.

Before concluding the subject of salvage examination it would be well to be reminded of the fact that we have new hazards to consider that did not exist in the past. The following is from the February 1991 issue of our newsletter *Vehicle Safety & Security Report*:

"Examining burned vehicles has never been a clean job, (because of fire debris, grease, oil etc.), but now because of materials being used a real health hazard can be encountered. The following information is from an internal memo circulated by an organization warning their personnel about breathing fire debris ash when conducting an examination:

'He was inspecting a burned vehicle and sifting through the ashes for evidence. When he got home that night, he experienced what he thought was a heart attack. He had numbness to his arms and a very erratic heartbeat. He was in intensive care for three days and in the hospital for several weeks.

His illness was finally diagnosed as poisoning due to breathing the ash of various materials used in vehicle interiors. Apparently, some plastics when burned form a residue which acts much like a nerve poison causing paralysis and loss of muscle control.' The wearing of a mask during examination was recommended.

In addition to this, I would recommend that you consider wearing heavy gloves while removing debris, etc. from a vehicle's interior. It is very easy to cut yourself while doing this and a severe infection can result. I recently examined a partially burned ambulance that had been used in treating AIDS patients. Fortunately, I had been warned of this beforehand and was extremely careful to protect myself from sharp material. Sometimes, we are not that fortunate in being forewarned."

Since this article appeared a number of similar cases were reported and some agencies are requiring the use of masks and self contained breathing apparatus' in making debris examinations. This hazard seems to be greatest in burned motorhome examinations. This is due to the presence of of the compound CCA (chromium, copper, arsenate) which is used as a wood preservative in the fire debris. Also suspect in these cases are burned remains of ABS and molded polyurethane plastics. Whatever the cause, it is something that should be borne in mind when examining a burned vehicle.

There are many commercial kits available to assist the fire investigator in "field" situations. Following is a description of one such unit:

An *arson investigation kit* is available to aid the investigator in determining if hydrocarbon accelerants are present at suspected arson sites. The kit contains an electronic nose unit employing patented microcircuitry systems specifically designed for arson investigation and gas detection. When these detectors are

exposed to even the smallest amount of accelerants or gas, they will annunciate by beeping. The more rapid the beep rate becomes, the higher the concentration of accelerant or gas.

Cheaper, less sophisticated "combustible vapor detectors" are also available on the market for around $200.00.

When using any of these machines one should be careful and not be misled. Reactions may be had from petroleum base used in the production of such items as plastics etc. Look on these machines as investigative tools that give you a strong indication as to where to search and remove samples for laboratory examination. They should not be considered as the final word on the presence or absence of accelerants.

Establishing a motive

In order to establish a motive for the owner to cause the destruction of his vehicle we must examine his current and past relationship with the car. In most cases the deliberate burning of a vehicle by the owner is an afterthought as opposed to a preconceived notion at the time he originally took possession of it. In most cases he was happy with the car originally. It's just that something — or some things — came along to change all this. What could it be? Motives come in all forms. Let's examine some factors that can give us a strong indication of *motive*.

1. Secure purchase details, such as date, cost, trade-in, down payment, balance of mortgage, payments due, payments past due, name of salesman, etc.

2. General condition of the car at the time of the fire, including any defects, mileage, extra or unusual equipment, etc. Did the owner have any trouble with the car? Were any repairs made? If so, what, when and where? It has been established that motives in vehicle fire cases generally fall under three main classifications: dissatisfaction with the car, domestic difficulties and financial problems:

   A. Dissatisfaction with the car includes:
   1. Mechanical trouble.
   2. Car is a "gas hog".
   3. Car is a "lemon" or a source of problems.
   4. Car depreciated due to hard use.
   5. Owner believes that the insurance company will pay more for the car than

it is worth.
6. Car not suited to the needs of the owner.
7. Owner has no further need for the car.

B. Domestic difficulties may include:
1. Pending divorce or separation.
2. Husband or wife did not want car purchased.
3. Excessive drinking or gambling.

C. Financial problems may include:
1. Owner becomes unemployed or income is reduced.
2. Gambling debts.
3. Business reverses.
4. The dealer refuses to take car back when the owner finds he cannot pay for it.
5. The finance company demands the owner pay up or surrender the car, in which case the owner knows he will get nothing for his equity.
6. Owner of the car lives beyond his income.
7. Unable to sell his equity in the car and desires to liquidate mortgage.
8. Fears adverse effect repossession will have on his credit rating.

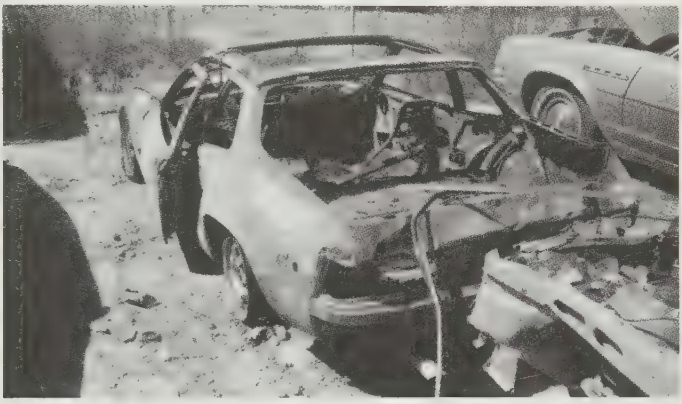
The investigation of arson is a very difficult and specialized undertaking requiring a knowledge of detailed methods and procedures. The success or failure of an arson investigation frequently depends upon the thoroughness of the investigator. Although evidence may be destroyed, some indication of its presence may be found through the application of scientific techniques. The investigator should note, safeguard, and evaluate all evidence of arson. He should utilize photographs, fingerprints, testimony of witnesses and the results of chemical analysis of material found at the fire.

Always remember that:

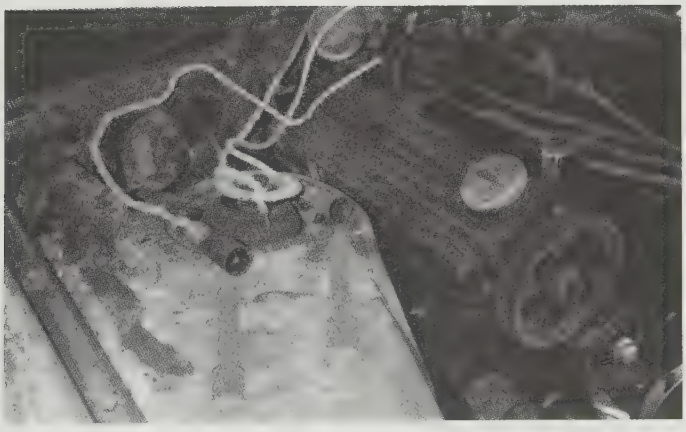
a. The assumption that the fire is of accidental origin must be overcome before the prosecution can establish arson. This fact must be established by the evidence.

b. The investigator must collect all available direct evidence, circumstantial evidence, admissible testimony and any other pertinent facts. The successful completion of such an investigation requires the application of many techniques and taxes the ability of the most skillful investigator. It

is the most challenging of any criminal investigation.



A clever arsonist burned this Porsche 924...



...by creating a fuel leak and disconnecting these spark plugs.



1. One ounce of brake fluid is poured into paper cup containing dry pool chlorine.



2. After 45 seconds...



3. The chemical reaction results in a violent fire.



4. That rapidly spreads inside the interior.

Special Surveys & Research

It is extremely difficult to get a precise "handle" on the amount of loss directly attributed to motor vehicle fires. The National Fire Protection Association (NFPA) reported that in 1982 there were 946,500 structural fires in the United States. This was a reduction of 8% over the prior year and 129,000 of these fires were identified as incendiary. In 1980 mobile property, i.e. motor vehicles, aircraft, boats, mobile homes etc. comprised 24% of the total number of fires reported. Of this 24%, a little over 90% were motor vehicles. From this we could reasonably conclude that there were in excess of 200,000 vehicle fires in the United States annually in this time frame. In 1978, according to the NFPA, the third and tenth ranking causes of fire deaths in the U.S. involved ignition of spilled fuel following motor vehicle accidents and accounted for 5% of all fire deaths. In 1983, according to the same information source, fires in vehicles ranked next to residential fires as being the greatest cause of deaths by fires in the U.S. By the mid 1980's fire departments in the United States responded to 454,500 vehicle fires (*NFPA Fire Protection Handbook*).

The NFPA Fire Analysis and Research Division published a report in 1997 titled "U.S. Vehicle Fire Trends and Patterns Through 1994." This report covered the years 1980-1994. The total number of vehicle fire s for this period was 6,691,000. The largest number in any one year was 477,500 in 1988. The smallest number was 405,000 in 1992. The last year reported on, 1994, listed 422,000 vehicle fires. The 1990-1994 annual average was 422,500 with 651 deaths and 2,940 in juries. $606.2 million in direct property damage was lost annually during this period. The annual average of passenger road vehicles by material first ignited reveals:

Fuel	104,500 (33.3%)
Electrical wire or cable insulation	75,150 (24.0%)

"Ignition factor" reveals:

Part failure, leak or break	67,210 (21.4%)
Short circuit or ground fault	53,000 (16.9%)
Incendiary or suspicious	52,620 (16.8%)

Perhaps one of the most reliable sources of information regarding motor vehicle fire losses is the experience of one insurance company in California and Nevada during a ten month period from January — October in the mid 1980's.

Total number of fire losses = 1,448
Average cost per loss = $3,186.50
Total losses paid = $4,614,050.90

Of particular interest to the fire investigator is the breakdown of fire causes:

Arson	49
Engine Compartment (fuel related)	320
Electrical	149
Catalytic Convertor	16
Power Steering	6
Propane/Butane	8
Oil	6
Muffler/Exhaust	10
Transmission	5
Garage Fire	48
Cigarette/Smoldering Material	16
Fire in Pickup Bed	3
Fueling	4
Collision	3
Brush Fire	3
Tire Fire	1
Paper in Motor Home/Van	5
Refrigerator in Motor Home	6
Air Conditioning Compressor	2
Welding Accident	2
Friction	6
Rag in Engine Compartment	1
Ether spray on engine	1
Unknown	778

Note that the "Cause Unknown" category comprises almost 54% of the total number of fires reported. Granted, in some cases additional investigation did in fact establish or suggest a cause it is unlikely that this occurred in no more than 10% of these cases. This would still leave the "Cause Unknown" factor at well over 40%. Only 83 vehicles were of the year models between 1982-84. The vast

majority were in the 1976-79 year models which suggests that vehicles become more fire prone as their age increases and flex lines, plastics etc. become more brittle and cracked due to extended heat and vibration.

Over half of the vehicles involved in these 1,448 fires were declared total losses by the insurance company.

In 1988 we produced the publication "A Survey of Vehicle Fire Causes". In this book we attempted to present more precise information on vehicle fire causes related to vehicle types and year models. With respect to existing data we stated:

"What little that has been written has mostly been concerned with crash related fires. Even here the data appears to be far from complete. In their excellent paper 'A Perspective on Automobile Crash Fires' by Charles Y. Warner, Michael B. James and Ronald L. Woolley, Collision Safety Engineering, SAE publication #8050092, state 'The relatively rare occurrence of injury or fatality in fuel fed fires has received considerable attention in automotive safety rule making and products liability litigation...there are no reliable field data which confirm a need for further injury reducing effect related to FMVSS 301. (Federal Motor Vehicle Safety Standard #301) NHTSA (National Highway Traffic Safety Agency) has acknowledged this by removing crash fire rule making from its priorities plan...'

FMVSS 301 set the standards for control of motor vehicle crash caused fires by preventing crash induced escape of fuel. As one can see, the main thrust of inquiry into vehicle fire causation has been from the standpoint of those cases where a traffic accident has resulted in the disruption of the vehicle's fuel system and an ignition source touched off vapor. The concern caused by these events can hardly be diminished when the plight of injured occupants in such cases is considered. Stories of such tragedies have far reaching affect and create lasting impressions. Add to this the almost daily diet of television and movie dramas where the "crash and burn" scene has become almost obligatory and there is little wonder that the general public believes such events to be commonplace. In spite of this, one of the conclusions reached in the Warner, James, and Woolley study 'A Perspective on Automobile Crash Fires' was that 'Less than one half of one per cent of all automobile accidents involve fire. Post collision fuel fed fires account for only one to one and a half percent of all occupant fatalities.'

I do not believe that enough research has been done in the area of those vehicle fires resulting from causes other than collision. As previous studies have shown, collision caused fires while certainly life threatening, do not present sufficient



This fire was caused by a fuel leak at the carburetor inlet.



Such fires can be caused by deteriorated "O" rings.

numbers to constitute a clear and present danger."

Of interest to the vehicle researcher is FMVSS #302 since it specifically deals with flammability of interior materials in passenger cars, multi purpose passenger vehicles, trucks and buses. The preamble of the standard states:

"...the occurrence of thousands of fires per year that begin in vehicle interiors provide ample justification for a safety standard on flammability of interior materials. Although the qualities of interior materials cannot by themselves make occupants safe from the hazards of fuel-fed fires, it is important, when fires occur in the interior of the vehicle from such sources as matches, cigarettes, or short circuits in interior wiring, that there be sufficient time for the driver to stop the vehicle, and if necessary for occupants to leave it, before injury occurs."

The standard arrived at for this requirement was a burn rate of four inches per minute measure in a horizontal test. If material stops burning before 60 seconds have elapsed and the burn does not exceed 2 inches the standard is considered to have been met. This standard was amended in 1975 to clarify materials for compliance to the basic requirement.

The requirements of this standard are obviously aimed at preventing or minimizing the small smoldering fire resulting from a dropped cigarette, match or something of this nature. Fire penetrating the interior from engine compartment blaze could hardly be expected to confine itself to the four inch per minute spread factor requirement in FMVSS #302. Fires originating in the plastic loaded dash of a post 1980 model car can reach a temperature of 1650°F in ten minutes time. ("Vehicle Fire Experiments", Aetna Life & Casualty Company, John Barracato.)

To add to this, consider the fact that Standard FMVSS #302 was not tested for compliance for the previous five years, according to a Government Accounting Office report issued in December 1986. In short, federally mandated safety standards as they apply to non collision related vehicle fires are without any direct relationship to the vast majority of vehicle fires.

What are the major causes of vehicle fires? Fuel leaks comprised 40% of the 233 cases surveyed in the preparation of this study. A breakdown of cases within this category showed the following:

Carburetor or carburetor inlet leaks	45
Fuel Injection - either injector or injector line	15
Fuel line leaks	34

The fuel tank itself was only involved as a causation factor in 2 cases and was not figured in the 40% total of the fuel leak category. Of these 2 cases one involved a tank ruptured during a collision and one fire resulting from an overflow.

Continuing with the breakdown of fire causes:

Catalytic converter	3
Overheated plastic in engine compartment	2
Electrical short	42
Transmission fluid	5
Oil leak	3
Overheated engine	3
Hydraulic line	2
Power steering fluid	5
Vapor canister	4
Propane leak	6
Turbo charger heat or malfunction	2

The total of these cases will not add up to 233 since I did not list a number of minor causes and arson.

You will note that the second largest causation factor was electrical shorting comprising about 18% of the total. I am sure this will cause some surprise since there has been a commonly expressed belief that due to fusing, fusible links and circuit breakers, an electrical fire is next to impossible. This, as will be seen, is not the case.

In a study prepared for publication in the British journal "Fire Prevention" Mr. John A. Butler of the College of Technology, Dublin, Ireland states that 90% of engine compartment fires were caused by faulty carburetors and "Fires due to leaking fuel are more likely to occur when the vehicle has stopped moving — the cooling effect of the moving air stream has stopped, allowing the temperature of the leaking fuel vapor to rise to the auto ignition temperature". This is an interesting observation and is another factor not completely agreed on by fire experts. I happen to agree with Mr. Butler, with whom I have had the privilege of consulting and meeting with on several occasions. Recently I measured the

exhaust manifold temperature on a three year old passenger vehicle after it had been operated at freeway conditions for one and one half hours. Five minutes after the ignition had been turned off the temperature was 558°F. The ambient temperature was 90°F. Another vehicle registered 480°F after five minutes had elapsed after operation. The ambient temperature was 70°F. With the auto ignition temperature of gasoline between 495-536°F. (depending on octane), it is not difficult to understand the engine compartment fire that occurs after the vehicle has stopped and there is "no known source of ignition" as it has sometimes been phrased.

Exact ignition sources are sometimes hard to determine in vehicle fire cases. There have been occasions where investigations have come to a standstill because of this. This should not happen and the investigator should note all possible sources of ignition and proceed with the inquiry. This was a major concern during the lengthy investigation into the loss of the Boeing 747 known as TWA flight 800. Finally a spokesperson for the National Transportation Safety Board (NTSB) was quoted as saying, "Whether we identify a specific ignition source is almost irrelevant."

Mr. Butler also states that his study of car fires in Ireland show that about 35% originate in the passenger compartment due to "smoking materials and defects in electrical wiring". This would seem to be higher than our experience and would tend to show that FMVSS #302 has been complied with sufficiently in the U.S. and Canada to reduce the rate of interior fires.

The age of vehicles involved in fires is always of interest since older vehicles may be expected to be more fire prone due to conditions brought on by heat, vibration, wear etc.

In reviewing material for preparation of this book we determined that the average age of the vehicle suffering a fire loss was seven years. In our book "A Survey of Vehicle Fire Causes" the vehicle year and number involved in fires were:

1987 — 5	1980 — 14
1986 — 29	1979 — 15
1985 — 24	1978 — 17
1984 — 25	1977 — 15
1983 — 16	1976 — 7
1982 — 20	1975 — 7
1981 — 15	Older than 1975 — 24

In 1990, the average age of automobiles in use in the U.S. was 7.8 years (*MVMA Motor Vehicle Facts & Figures, 1991*)

Using the experience of our own company, Lee S. Cole & Associates, Inc., we surveyed our records for a four month period in 1998-99 and compiled the list of the following:

Vehicle Year, Make & Model	**Cause of Fire**
1992 Beaver Motorhome	Undetermined
1993 Freightliner	Arson
1992 BMW 750iL	Shorted brake light switch
1990 Volkswagen Corrado	Oil supply line to supercharger detached
1997 Harley Davidson Motorcycle	Transmission expulsion
1985 Ford Pickup	Wiring at front headlight and turn indicator shorted
1994 Cruise Master Motorhome	Transmission expulsion
1994 Saturn	Fractured rod knocked hole in block — oil and gas ignited by exhaust manifold
1997 Chevrolet Camaro	Fuel tube retaining clip failed
1987 Caterpillar	Fuel leak on exhaust pipe
1992 Tioga Motorhome	Transmission fluid expulsion
1987 Ford pickup w/lift bed	Manifold melted plastic insulation on power cable for lift
1994 Mercury Sable	Wiring loom was misrouted, melted and ignited by manifold
1997 Ford Ranger	Insulation wore away from wiring adjacent to fuel tank. Arced at frame and insulation burned.
1989 Chevrolet Astro Van	Alternator bearing failed
1977 Pace Arrow Motorhome	Propane leak at refrigerator
1989 Volvo 760	Wiring harness insulation melted and charred due to high resistance in foglight circuit.
1997 Dodge Pickup	Expulsion of transmission fluid
1998 Ford Taurus	Power steering leaked on exhaust manifold.

Vehicle Year, Make & Model	Cause of Fire
1986 Ford Bronco	Spring lock coupling on fuel line separated
1985 Lincoln Town Car	Undetermined
1996 Ford Pickup	Plastic shroud contacted right rear of engine
1987 Ford Thunderbird	Alternator/power steering belt was slipping and generated sufficient heat to ignite
1990 Chevrolet Corvette	Fire at battery — cause not determined
1991 Ford Explorer	Arson
1992 Ford Aerostar	Fuel leak at right rear injector
1996 Plymouth Grand Voyager	Battery explosion and fire
1990 Jeep Cherokee	Leak in hgh pressure hose from ABS system sprayed on exhaust manifold
1994 Jaguar XJ-12	Malfunction in distributor caused plastic shroud to burn
1987 Nissan 200SX	Overfilled tank caused canister to overflow
1998 Chevrolet Blazer	Arson
1997 Chevrolet Tahoe	Arson
1985 Cadillac Limousine	Arson
1992 Ford Pickup	Transmission fluid expulsion
1997 Ford Escort	Shop towel left in engine compartment
1987 Dodge Caravan	Power steering fluid leaked on exhaust manifold
1993 Ford Mustang	Short circuit between battery wire to alternator and alternator housing
1988 Jeep Pickup	Left rear axle bearing overheated and ignited differential oil
1997 Ford 4x4 Super Cab	Undetermined
1987 Toyota Cressida	Transmission fluid cooler hose split & sprayed on catalytic converter
1993 Toyota Corolla	Broken rod knocked hole in block. Oil and vapor ignited by exhaust tube.

Vehicle Year, Make & Model	**Cause of Fire**
1990 Cadillac El Dorado	Leaking fuel from #7 injector. Ignition by disconnected spark plug wire. Occurred at dealership during repair.
1995 Acura	Arson

Over the years a number of experimental burnings of vehicles have been carried out to determine their behavior and characteristics during the progress of fire. Most of these tests have been along the lines of comparison between the deliberately burned vehicle and the accidental fire. These tests have been handicapped by the unavailability of late model vehicles to give the experiments validity and adequate equipment for accurate recording of data. This is no doubt one of the primary reasons for the persistence of the inaccurate information that we spoke of earlier.

The most significant experiments during the last 10 years have been conducted in the provinces of Saskatchewan, Alberta, and Manitoba, Canada. In 1977 the first series of these experiments was conducted at Regina and involved the burning of 6 late model vehicles. The stated purpose was "to provide reliable documentation on the flammability of late model vehicles and the manner and time in which the fire spreads to each compartment of the vehicle. This was done: (1) to assist fire fighters, insurance and police investigators in determining (a) locations and source of ignition of fires (b) whether ignition was accidental or deliberate. (2) To assist fire fighters (a) in arresting fire with minimum damage, (b) by providing information on time available to rescue occupants." You will note that temperature is given in Celcius as well as Fahrenheit. In order to obtain a Fahrenheit reading from Celcius multiply by 1.8 and add 32.

These experiments reached the following conclusions:

"Motor vehicle fires have three phases.

Phase 1 From time of ignition to approximately 200°C (392°F). This phase has 2 stages:

(1) Initial ignition with open flame which normally lasts 2-3 minutes.
(2) Smoldering with steady rise in temperature until conditions become critical in terms of temperature and oxygen supply for open flame

combustion. Fires caused by electrical ignition may not have stage one but initiate at stage two.

Phase 2 The second phase is reached only after the critical stage in phase one. The temperature rises very rapidly from approximately 200°C. (392°F) to approximately 900°C (1652°F) in about ten minutes in the vicinity of the roof interior. The second phase continues until most of the combustible material in the top two thirds of the passenger compartment is consumed. At this stage the temperature at the roof area would probably start dropping.

The second phase lasts anywhere from twenty minutes to one hour. During this phase the intense heat is capable of melting tempered glass. The outer panel paint also blisters and the white metal items melt. The temperatures reached in this phase are above the flash point for almost all interior combustible material. The engine compartment and the passenger compartment could go through the second phase at different times. Ignition of combustible material in the trunk may occur. The ignition of the tire in the trunk generates enough heat to cause expansion of the fuel tank due to the increased vapor pressure. Any leakage of gasoline vapor from the gas cap or filler pipe would burn in a 'blow torch' fashion. The tires may also be ignited in this phase.

Phase 3 The third phase of the fire is the cooling phase, that is where the temperatures vary from a high of 200°C (392°F) to ambient temperature. The combustion present in this phase is occurring from material on the floor where oxygen supply was not available in the second phase. All rubber material and insulation on wires was previously burned. The iron parts, steering wheel, control handles and linkages in the doors were a dark gray in color. Tires, if ignited in phase two, will continue to burn in this phase."

The general conclusions reached in these experiments were that the time span of burning was shortened by the use of accelerants, the source of ignition of fires which do not progress beyond the first phase can probably be identified through examination of remains and if the second phase is nearly completed before the fire is extinguished it is very difficult to detect whether or not an accelerant has been used.

The report continues that "...it was our experience that by three quarters of the

way through the second phase the remaining evidence in all cases is the same in nature. At this point, apart from recording the physical evidence of the articles consumed in the fire, there is very little a fire investigator could do to (a) locate the initial point of fire, (b) trace the path or spread pattern of fire or (c) collect samples for lab examination. The change in the property of iron and the deformation of steel in car seats and other panels is of the same nature in all cases. The material used in the car interior today acts the same way as excessive use of an accelerant. Both generate relatively high amounts of heat in a short time. The temperatures reached are more than sufficient to cause the glass to melt and flow." (Source - "A study of six Automotive Fires of Known Ignition Source" prepared for the Saskatchewan Government Insurance Office, Regina Saskatchewan by A. Patel, SGIO Auto Experimental Center, D.K. Lischynski, University of Saskachewan Transportation Center, S/Sgt. Anderson, RCMP, "F" Division, R. Hrynchuk, RCMP Crime Lab, Chemist, B. Jones, Saskatoon Fire Department, Investigator.

Needless to say, this report was greeted in some circles as heresy since it flew in the face of what had been conceived as positive proof of the use of accelerants such as flowing glass, annealed seat springs and sagged top. This was not, however, particularly surprising to many of us who had been observing for some time in experiments that the "set" fire seemed to be no more destructive than the "accidental fire" when both had been allowed to run their complete course. The main consideration, and this is extremely important, is how long did it take for the vehicle to reach this stage of destruction.

In 1983 in cooperation with the Santa Clara County Fire Association (California) a number of vehicles were burned experimentally. In one comparative test a 1972 Chevrolet Impala 4 door sedan was set on fire after pouring one gallon of gasoline over the front and rear seats. An engine compartment fire was started in a 1972 Ford LTD 2 door sedan by simulating a back fire in the carburetor and air cleaner. After burning for 21 minutes the Impala was totally involved in flame with both rear tires having blown out and all glass burned or melted out. After burning for the same period of time the fire in the LTD had breached the firewall and entered the passenger compartment where the windshield glass began to crack and the fire communicated to the dash and the seats. The elapsed time of burning was the same as that of the Impala, 21 minutes. If allowed to continue the fire would have produced the same overall condition in both vehicles but the time is the matter of greatest importance.

Additional experiments were conducted in Canada and a subsequent report

prepared. The principal editor of this report was Mr. R.J. Hrynchuk, M.Sc., Forensic Sciences, Laboratory Specialist Royal Canadian mounted Police Crime Detection Laboratory Edmonton, Alberta, Canada. Now at the RCMP lab in Winninpeg, Canada. It has been my good fortune to confer with Mr. Hrynchuk on several occasions regarding the findings and conclusions drawn from these experiments. He and his colleagues have done a tremendous service to the field of vehicle fire investigation. Unfortunately it will not be possible to reproduce the complete report of these experimental burnings here since it is over 70 pages in length. We will, however reproduce excerpts from this report that was originally prepared in January 1983 but is just now beginning to circulate.

A report prepared for the International Association of Arson Investigators Alberta Chapter, Edmonton, Alberta by K. Allen, Insurance Crime Prevention Bureau, K. Bett, The Co-operators Insurance Service, D.L. Danchuk, Fire Prevention Branch, Alberta Labour, A. Devins, Automotive Investigators. 202151 Holdings Ltd., R.J. Hrynchuk, (Editor), R.C.M.P. Crime Detection Laboratory, G.W. Kilert, Edmonton Fire Department, D.K. Lischynski, Saskatchewan Government Insurance, D.K. Matheson, Edmonton Fire Department, Bill Mitchell, Insurance Crime Prevention Bureau, A.P. Patel, Saskatchewan Government Insurance, D.R. Phillips, President, International Association of Arson Investigators, Alberta Chapter, J.V. Schick, Saskatchewan Government Insurance, J. Tash, Alberta Motor Association, D.W. Wolowshuk, Saskatchewan Government Insurance

"In May, 1982, four test vehicles were started on fire using different methods of ignition. These vehicle fires were intended to simulate noncollision related fires that totally destroy a vehicle. The progression of each fire was observed and the temperatures reached at different locations in the vehicles were monitored and recorded. A videotape record of each test fire was made. The dynamics of vehicle fires in these real fire situations were analyzed. Four late model vehicles were used in these tests because they fairly represented the year range of vehicles currently found on the road, and because they represented the modern manufacturing technology and construction materials currently being used. Each vehicle was allowed to burn to completion. The physical characteristics of the burned remains of each vehicle were then examined and related to the methods of ignition. This included an inspection of the mechanical condition of the motor and the drive train of each vehicle before, and after the tests.

Reasons for doing these tests:

The major purpose of these tests was to study and compare vehicle fires that are

caused by different ignition sources, that are noncollision related, and that totally destroy the vehicle. For each method of ignition, the rate of fire spread, the temperatures reached, and the path the fire takes through the vehicle could be observed and recorded. This was also the main purpose for the vehicle fire study conducted by Patel and co-workers in 1977. In their tests, vehicles in the 1970-74 year range were used. It is well known that vehicles manufactured in the 1980's are increasingly smaller in size, and a much higher proportion of plastic (combustible) components in order to reduce the overall weight of the vehicle. Therefore, it becomes desirable to repeat these test fires on new vehicles that would reflect these changes in construction, and that would reflect the types of vehicles currently on the road. Any data collected could then be compared to data from the work of Patel et al.

Choice of ignition attempts:

The possible causes of vehicle fires that have been suggested include accidental methods such as mechanical failures, electrical malfunctions, smoldering cigarettes, and intentional methods such as the use of a flammable liquid. The use of a flammable liquid as a fire accelerant is the most obvious method and it is well known that it is an effective method. However, very little work has been done in reproducing the so-called accidental methods and in determining whether they are indeed reasonable ignition sources. If these accidental methods are successful, it is important to know what differences if any, exist when compared to a vehicle fire started using a flammable liquid. Therefore, tests on three of the four vehicles in this study involve reproducing some accidental methods of ignition. Attempts were made to produce a fire in a vehicle due to an electrical failure, a fuel leak on an engine, an overheated catalytic converter, an overheated wheel bearing, and an overheated transmission. For comparison purposes, the fourth vehicle was set on fire using a flammable liquid. Some of these methods can be viewed as low energy ignition sources (smoldering cigarettes) and others as high energy ignition sources.

Finally, these test fires involved vehicles that, once ignited, were allowed to burn to completion. The reason for this is that most vehicle fires occur in circumstances where the fires are not likely to be noticed and subsequently extinguished in a short space of time. Vehicle fires that occur in urban areas are probably extinguished soon after they start.

Test Vehicles and Methods

1. Test vehicles and preparations:
The following vehicles were donated for use in these test fires:

Vehicle #1: 1980 GMC Sierra Grande, blue/white.
VIN: TCL44A1539122; Odometer 22375km

This was a truck with an eight cylinder engine and automatic transmission. The engine was in working order. The front end had severe prior collision damage, which was partially repaired to facilitate operation and running of the engine and transmission. The automatic transmission cooler was damaged in the collision and was bypassed at the radiator. The truck box was empty except for some concrete blocks, a sheet of plywood, some lumber and a pair of truck mirrors.

Vehicle #2: 1980 Pontiac Sunbird, yellow
VIN: 2M27VA7563235; Odometer not available

This was a two door car with a four cylinder engine and standard transmission. The engine was in working order. The left quarter panel and door were damaged and the glass in this damaged door was missing. This vehicle was equipped with a catalytic converter.

Vehicle #3: 1978 Chevrolet Impala, blue
VIN: 1L47U81130237; Odometer 101,957km

This was a two door car with an eight cylinder engine and automatic transmission. The engine was in working order and the vehicle could be driven. This vehicle had prior collision damage to the right front fender and wheel area. This vehicle was also equipped with a catalytic converter.

Vehicle #4: 1976 Ford Granada, red/white
VIN: 6W83H277377; Odometer 69552m

This was a four door car with an eight cylinder engine and automatic transmission. The engine was in working order and the vehicle could be driven. This vehicle was virtually undamaged prior to the tests.

Vehicle #5: 1975 Ford Country Squire Station Wagon
VIN: 5P76S149295; Odometer 82237m

This was a four door station wagon with an eight cylinder engine and automatic transmission. The engine was in working order and the vehicle could be driven. This vehicle was virtually undamaged prior to the tests.

Only the first four vehicles above were completely burned in this study because of time constraints in the course of the tests. Tests on the fifth vehicle were limited only to attempts to produce a fire from smoldering cigarettes and cigars.

1. 1980 GMC Sierra Grand

The first ignition attempt was to produce an engine fire due to a leaking fuel line. A hacksaw cut was made in the steel fuel delivery line approximately 15 cm from the carburetor. A high tension wire was detached at the spark plug to ignite the gasoline. The engine was started and the resulting fire was confined to the engine compartment.

The second ignition attempt was to produce a fire by shorting the wires either on the firewall or under the dash. The fuses in the wiring system all functioned to prevent an overload. A direct short between an unfused power line to ground finally produced ignition under the dash. The resulting fire destroyed the entire truck.

2. 1980 Pontiac Sunbird

The first ignition attempt was to overheat the catalytic convertor by running a rich fuel mixture to the engine. Dry grass and straw in contact with converter and the muffler ignited but no fire was produced in the vehicle.

The second ignition attempt was to produce an engine fire due to a leaking fuel line. This time, a threaded fitting to the carburetor was backed off until a fuel leak occurred. The engine was allowed to run for several minutes with the gasoline leaking on the hot manifold but no ignition occurred. The insulation on a high tension spark plug wire was partially cut to finally ignite the gasoline. The engine was restarted and the resulting fire destroyed the entire vehicle.

3. 1978 Chevrolet Impala

The first ignition attempt was to damage the right rear wheel bearing, raise the rear end of the vehicle then to run the engine at a high speed in order to overheat this wheel bearing and its surroundings. No fire in the vehicle was produced.
The second ignition attempt was to overload and overheat the transmission by

doing a prolonged severe stall test. The engine compartment caught on fire and the entire vehicle was destroyed.

4. 1976 Ford Granada

The first ignition attempt was to cause a fire by igniting the vehicle with a mixture of gasoline and fuel oil. Initially, the fuel was poured on the exterior of the vehicle only and ignited. No sustained fire in the vehicle was produced. Then, approximately two liters of fuel were poured on the left front and rear, floor and seat areas in the passenger compartment. It was ignited by a match through a trailer of fuel on the ground leading to the vehicle. The resulting fire destroyed the entire vehicle.

5. 1975 Ford Country Squire

The first ignition attempt involved placing several lit cigarettes and cigars directly on the seat coverings and between the bench and the back of the seat. The windows in the vehicle were down. Only minor, localized damage to the seat padding and covering were observed after about one hour. No open flames were obtained. No further tests were performed on this vehicle.

Phase 2. Physical characteristics and mechanical condition:

1. 1980 GMC Sierra Grande

The fire, as a result of the first attempt, occurred at the source of the fuel leak. The fire damage was to the electrical insulation, vacuum and coolant hoses in the proximity of the right front quadrant of the engine compartment. Beyond the immediate fire location, no other damage was found to have occurred. A specific inspection was done of the headlight wiring harness. The insulation was found to be burned. However, the lighting systems, heater, radio, warning lights, indicator lights and the windshield wipers were all operable.

Fire cause and repairability: The damage due to the fire was economically repairable and the cause of the fire would have been determined in this case.

In the course of the second ignition attempt both fusible links, one at the firewall and one at the starter motor, had failed independently and had to be bypassed. Continued ignition attempts failed and the observed damage was to the wiring in the immediate area of the dead shorts. As no fire occurred, the only damage was to the wiring system involved in the attempt. A dead short, between a power line in the instrument panel wiring harness and the brake pedal, finally overheated the

insulation and produced a fire that started in the dash. After the fire was over, an inspection of the vehicle showed the following:

Engine compartment: All plastic items, such as radiator overflow tank, windshield washer tank, etc., had been completely burned away. The rubber items, such as radiator hoses, heat hoses, etc. had only the area under their respect clamps remaining; this area being charred and brittle, but still intact. The insulation for the wiring harnesses and the plastic junction blocks had been completely burned away. The spark plug high tension wires were also consumed by the fire, but had left a distinctive white ash trail, indicating their positions prior to the fire. All the light metal, or pot metal parts, such as the carburetor, fuel pump, alternator housing, etc. had been distorted and/or melted.

Cab interior: The floor panel carpeting and underlay, after the removal of a substantial layer of debris, were found to be charred and blackened, but basically still intact. The bench style seat had its padding and covering completely burned away, leaving the seat frame undistorted. The padding support springs retained their shape but had lost tension. The other combustible materials in the cab, such as the headliner, trim panels, dash coverings, and other plastic items were completely destroyed. The windshield had melted and left glass icicles hanging on the dash. The left and right side windows shattered during the early stages of the fires. Some of their remains that had fallen into the cab had partially melted. The checkered fracturing of the tempered glass was still visible. One section of the sliding rear window was not present prior to the fire, and the right door vent glass was still intact after the fire.

Sheet metal body parts: The cab inner roof panel, along with the dash panel, had sagged. Further, the door outer panels, cab back panel, and box side panels were displaying signs of warping and distortion. The front hood and fender panels were badly damaged prior to the fire. As a result, any further damage from heat exposure was not taken into account.

Fire cause and repairability: The extent of damage rendered the vehicle unrepairable. The cause of the fire, being a dead short from the instrument panel wiring harness to the brake pedal was still evident after fire.

2. 1980 Pontiac Sunbird

After the first ignition attempt, an inspection of the area inside the vehicle and directly above the catalytic converter revealed that the only damage was to the

floor panel insulation which was made from a tar-like substance. This layer had begun to melt at the floor surface.

Fire cause and repairability: The minor amount of damage that occurred was cosmetic and repairable. The heat source was evident.

After the fire due to the second ignition attempt was over, an inspection of the vehicle showed the following:

Engine compartment: The extent and nature of damage in this area was found to be similar to that observed in the cab of the first vehicle.

Trunk compartment: All items of a combustible nature were completely burned. The tension in the trunk lid hinge springs was relaxed.

Sheet metal and body parts: The roof had sagged and the balance of the sheet metal body parts were displaying signs of warping and distortion, similar to that of the first vehicle.

Drive train components: An internal inspection of the engine, transmission, and the differential revealed the same test results as the first vehicle. The rear differential housing leaf springs were found to have lost tension, thus causing the vehicle frame to rest on the axle and the ground.

Vehicle tires: The damage was similar to that observed for the first vehicle.

Fire cause and repairability: The extent of damage rendered the vehicle unrepairable. The fuel inlet fitting, which had been unscrewed at the carburetor to cause a fuel leak, was found to be still loose. Although it was evident that this fitting was loosened, it was not completely apparent that fuel had been leaking at that point prior to the fire. The cause of the ignition of the leaking fuel, being a partially exposed high tension spark plug wire, could not have been determined after the fire.

3. 1978 Chevrolet Impala

Following the first ignition attempt, it was found that the right rear axle bearing was destroyed. The heat generated caused the slave brake cylinder seal rubber to fail, resulting in the loss of brake fluid. A small fire in the immediate area was observed.

Fire cause and repairability: As only a small fire occurred during this attempt, the only damage was to the mechanical parts involved in the attempt.

After the fire due to the second attempt was over, an inspection of the vehicle showed the following: The extent and nature of the damage in the engine compartment, passenger compartment, trunk compartment, sheet metal body parts and the vehicle tires was found to be the same as in the previous two vehicles.

Drive train components: An internal inspection of the engine revealed baked oil deposits. With regard to the automatic transmission, the nature of the ignition attempt was to preload the automatic transmission and operate it beyond its designed capabilities. The after fire inspection found the automatic transmission with a large section of the upper portion of the torque converter bell housing melted and collapsed down upon the torque converter. With the removal of its pan, it was found to be devoid of oil and its internal components were severely damaged. This damage was not due to the heat of the fire but due to abuse as a result of the stall test. The rear differential was found to be unaffected by the preceding testing and the resulting fire.

Fire cause and repairability: The extent of the damage rendered the vehicle unrepairable. The severe damage evident in the transmission suggested that the transmission fluid had been expelled from the transmission and, once ignited, acted as an accelerant. Ignition of this fluid occurred in the engine compartment, but the source of this ignition could not be determined.

4. 1976 Ford Granada

After the fire due to the first ignition attempt was over, an inspection of the vehicle revealed the following:

The nature and extent of the damage in the engine compartment, in the passenger compartment, in the trunk compartment, to the sheet metal body parts and to the vehicle tires was found to be the same as that observed in the previous three vehicles.

Drive train components: No damage to the internal parts of the engine, transmission and differential occurred.

Fire cause and repairability: The extent of the damage rendered the vehicle unrepairable. No source of ignition could be determined from the physical condition of the vehicle.

5. 1975 Ford Country Squire

Only minor damage to the seats occurred from the smoldering cigarettes and cigars. The cause of this damage was obvious. The damage was repairable.

Discussion

The first ignition attempt on the GMC demonstrated that an engine fire is easily produced when a fuel leak in the engine compartment is combined with a fault or leak in the spark plug wiring. The resulting fire in this truck was confined to the engine area. This fire did not spread due to the lack of an adequate supply of fuel. In this truck, the engine compartment is large and the engine itself was relatively isolated in this area. The low amount of combustible material on the engine and the lack of material to conduct and transfer the heat and fire to the passenger compartment are probably reasons that this fire did not spread. The presence of leaking fuel on the engine, by itself, will not cause a fire. A spark or open flame is necessary for ignition. Previous tests have demonstrated that heat developed by the manifold is not sufficient to cause ignition.

The second ignition attempt on the GMC involved trying to produce a fire from an electrical short. Initially, all of these attempts were unsuccessful because the fuses and safety devices, that are part of a normal wiring system of a vehicle, all functioned correctly to prevent an electrical overload. A fire was finally produced only when a dead short to the brake pedal was created using a hot wire that was not protected by a fuse. Other attempts in previous tests also only succeeded after the fuses were bypassed.

The resulting fire in the GMC truck, caused by the electrical short, totally destroyed this vehicle. Two features of this fire were unusual. The first was that the entire truck, front to back, was destroyed. The spread of the fire to the rear of the truck was due to the failure of the fuel tank selector valve located under the driver's seat. This valve is connected to both fuel tanks on the truck. The burning away of material in the valve area allowed fuel to spread on the ground under the vehicle and eventually caused the auxiliary fuel tank to burn. If this truck had not had an auxiliary fuel tank and fuel tank selector valve, the fire produced by the electrical short would probably have been more confined to the cab. The second feature of this fire was that the gas cap and door covering the gas cap on the right fuel tank changed position. The gas cap on this tank was a threaded plastic type of cap. These plastic threads melted or contracted in the course of the fire. A build-up of pressure in the fuel tank caused this gas cap to be blown off against the door covering it. This door opened outwards, stayed open, and the gas cap fell to the

ground beneath this door. In contrast, the gas cap on the left fuel tank of the truck also burned away, however, this cap was found in place and the door covering it was closed. After the fire was out, the wires used to create a short to the brake pedal were evident.

The first ignition attempt on the Sunbird demonstrated that a hot catalytic converter and muffler can ignite combustible material (grass) that it comes in contact with under the vehicle. However, no fire was produced inside this vehicle itself and only cosmetic damage was evident. Damage to the vehicle might have been more extensive if there had been a significantly larger amount of dry grass or straw present.

The second ignition attempt on the Sunbird was an engine fire, similar to that induced in the GMC. Instead of producing a fuel leak from a cut in the fuel line, a threaded fitting on the carburetor was backed off in order to leak some gasoline. Again, ignition of this gasoline was due to a leak in a spark plug wire. This engine fire did spread to the passenger compartment and to the rest of the vehicle. The probable reason for this was that, in contrast to the GMC, there was more combustible material surrounding, and close to, the engine in this Sunbird. This more compact and crowded engine compartment allowed the fire to spread past the firewall.

Flames persisted in the engine compartment of the Sunbird even after most of the combustible material in that area had burned away. The lines used in the pollution control system and the fuel line from the fuel tank continued to vent gasoline vapors from the fuel tank through the burned ends of these lines in the engine compartment. These vapors burned like a blowtorch in the engine area and this burning continued as long as there was fuel in the tank. This blowtorch in the engine area would cause more damage to any metal components in its path because of its sustained burning. As the fire spread to the rear of the vehicle, the flexible connector between the gas cap filler neck and the fuel tank burned and eventually ruptured. This caused a sudden release of liquid gasoline onto the ground three to four feet away from the vehicle. This suddenly released fuel rapidly burned away and accelerated the destruction of the rear of this vehicle. After the fire was out, it was evident that the threaded fitting at the carburetor, that caused the initial fuel leak, had been loosened.

The first ignition attempt on the Impala, namely overheating the rear wheel bearing, was unsuccessful. With a different kind of vehicle that may contain more combustible material in the vicinity of the overheated bearing and with increased

load on the wheel, the heat and fire might be propagated to the rest of the vehicle and cause more extensive damage.

The second ignition attempt on the Impala simulated a situation where the vehicle is "stuck". An excessive load was put on the transmission and it was allowed to overheat to such an extent that oil vapors, emitted in the engine compartment through the transmission breather, ignited. This ignition, again, was likely due to a leak in the wiring system in the engine compartment. Alternatively, ignition might have been caused by contact between the hot vapors and the air (oxygen) or a hot manifold. The ignition temperature of the transmission oil (260-370°C) - (500- 698°F) is 100° to 130°C higher than its flash point (135-260°C)-(175-500°F). This fact combined with the difficulty in obtaining a suitable mixture of air and oil vapors in the engine compartment make ignition due to contact with the hot engine less likely. This fire in the engine compartment spread through the firewall and eventually destroyed the entire vehicle. In the course of this fire, all of the oil was expelled from the transmission and a portion of the transmission housing melted. The spread of the fire from the engine compartment to the passenger compartment is due to the presence of combustible material in the engine area, including the vaporized oil from the transmission.

Flames persisted in the engine compartment of the Impala, similar to that observed in the Sunbird, because of venting gasoline vapor through the fuel line in the engine compartment. Also, several electrical components in the Impala operated intermittently as the fire proceeded. This included turning on the windshield wipers, washers, horn and the front and rear lights, as well as engaging the starter motor.

After this fire was out, the overheating of the transmission, that caused the fire, was evident. Overheating of transmissions may be a more common problem in the future, with the general downsizing of vehicles compared to their towing loads.

The ignition attempt in the Granada involved the use of a flammable liquid. Simply pouring and igniting the fuel on the exterior of the vehicle was not sufficient. This caused the paint to checker but not blister. Pouring and igniting the fuel in the passenger compartment was successful and rapidly produced a fire that destroyed the entire vehicle. The fire was intense; it damaged the passenger compartment and sheet metal areas, caused the roof to sag and buckle, melted all the glass and removed the temper from metal, all in a manner similar to each of the other three vehicle fires. The drive train of this vehicle was undamaged by this

fire.

The liquid fuel (gasoline and fuel oil mixture) had been poured on the left side of the front and rear floor and seat areas of the Granada. In this vehicle, as in all the vehicles, a significant amount of floor material remained intact after the fire. There were no significant differences observed in the depth of charring in the floor areas, when comparing the left and right sides of the Granada, or when comparing it to the depth of charring observed in the other vehicles.

In the course of the fire in each of these vehicles, burning material in the passenger compartment produced large volumes of thick, black smoke. The

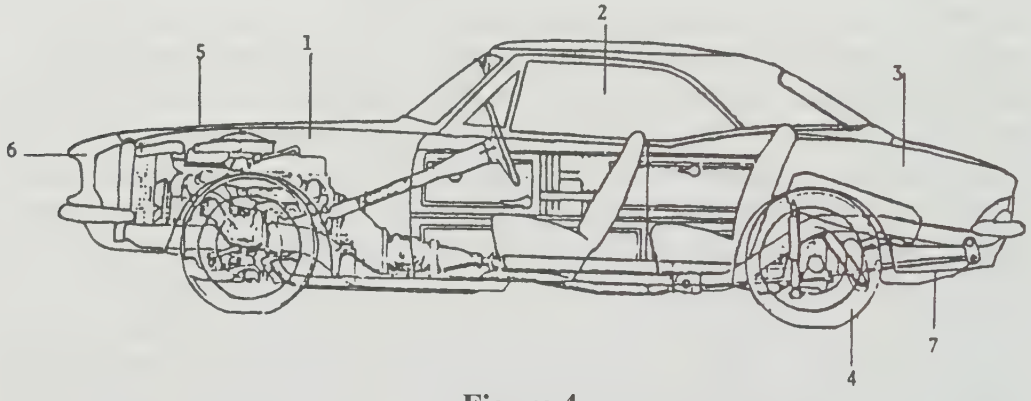


Figure 4

Estimated amounts in lbs. of combustible material in a car: (1) engine compartment, 50 lbs. (2) passenger compartment & dash area, 2-300 lbs. (3) trunk & spare tire, 15 lbs. (4) tires, 60 lbs. (5) paints, sealants & other coatings, 50 lbs. (6) fender extensions, bumper guards & other moldings, 50 lbs. (7) fuel tank with 10 gallon of fuel, 75 lbs. Total estimated weight of combustible materials: 5-600 lbs.

flame colors varied from orange (940°C),(1725°F) to orange-red (1100°C),(2012°F). Melted plastic materials from the engine and the passenger compartments would pool and burn on the ground at different points beneath the vehicle. This melted plastic would often cause a tire to start on fire. Almost without exception, each tire would burst approximately five to six minutes after they started to burn.

Hydrocarbon Levels

Great care must be taken in collecting samples from the scene. Gasoline was evidence in the soil samples collected near the GMC and Sunbird.This gasoline that was detected has nothing to do with the ignition source in these two vehicles, and is present only because of the manner in which these fires progressed. In both

of these fires, gasoline from the fuel tank was expelled onto the ground. The soil immediately under the fuel tank and soil two to three feet away from the vehicle, in line with the rupture, was sprayed with the gasoline. In the case of the Granada, the soil sample collected was from the ground where the gasoline and fuel oil mixture had been poured as a trailer. This trailer was evident in the soil after the fire, and both petroleum products were detected in this soil sample.

The debris samples collected from the interior of each vehicle all consisted of partially charred portions of the floor coverings. The floor areas of the vehicle are the last to be consumed in a fire and are often covered with glass, metal and other debris that falls from the upper areas of the passenger compartment. This tends to protect the combustible materials on the floor for a while, but, eventually, even this ;material is largely consumed in the fire. Residual fuel oil was detected in one of the samples of debris from the Granada. There was no evidence of the gasoline that was used.

No residual hydrocarbons from a flammable liquid were evident in the debris samples collected from the floor areas of the other three vehicles. None were expected since no flammable liquids were used in those vehicles.

The levels of lead, bromine and manganese detected indicate that there is no correlation between the presence of these elements on samples of fire debris and the use of gasoline as a method for starting a vehicle on fire.

Combustible materials in vehicles

The types of combustible materials that were found to be used in the passenger compartments of these vehicles consisted mostly of synthetic polymers. Some of these polymers or plastics can be made fire retardant, to varying degrees, by incorporating certain types of chemicals into their formulations. Some plastics can also be given a certain degree of fire resistance by the use of specialized types of fire retardants in their formulations. The ease with which these plastics burn, given a source of heat and a suitable supply of oxygen, is further influenced by factors such as their form (a form compared to a dense film or laminate) and their surroundings (the surface of the plastic may be covered by other plastics or materials). The flammabilities of different types of materials, including the wide variety of plastics, have been rated by applying various standardized tests. Plastics with higher flammability ratings, that is those that are easy to burn, are generally not used in vehicles. However, all the synthetic polymers found in a vehicle will burn, regardless of their fire retardant properties, given a sufficient heat source and oxygen supply. Therefore, since the term flammable infers that

a substance is easy to ignite, it is preferable to use the term combustible in describing the materials used in vehicle interiors. The amount of combustible material in a vehicle represents the potential fire load of that vehicle.

The trend in modern passenger vehicles is a reduction in overall vehicle size and weight for improved fuel economy. This has meant that more synthetic, light weight materials are incorporated wherever possible throughout the vehicle. Engine compartments are smaller and more crowded with components, especially in the sub-compact and compact vehicles. The passenger compartment is not generally reduced in size, but rather it is being fitted with more material for the comfort of the driver and passengers. The trunk area is variable and often it is not structurally separated from the passenger compartment. *Figure 4* is a skeletal side view of a typical, compact, North American passenger vehicle Conservative estimates of the amount (by weight) of the combustible material distributed throughout the vehicle are indicated. Assuming a gross vehicle weight of approximately 3,000 lbs., the combustible material, or fire load, in this vehicle would be about 600 lbs. or roughly 20% of the gross vehicle weight. If the weight of the drive train of this vehicle (roughly 1,000 lbs.) is subtracted, since the drive train is largely undamaged in a vehicle fire, the proportion of combustible material is about 30%. The amount of combustible polymeric material in a vehicle will continue to increase as some of the metal structural components are replaced with high strength polymers. Although all of these figures are approximate, they illustrate that a vehicle can contain several hundred pounds of combustible material and that this is a significant fire load relative to the overall weight of the vehicle. The physical damage that occurs to a vehicle in the course of a fire is due to the combustion of this fireload. This amount and type of combustible material, once ignited, will, logically, produce a very hot and sustained fire regardless of the method of ignition.

Conclusions

1. Vehicle Fires

The tests described in this work deal with vehicles that are totally consumed by fire and are noncollision related. These types of vehicle fires can be caused by accidental means, but some of these causes are less likely than others.

An electrical failure that causes wires to overheat can produce a vehicle fire. Electrical fires should not occur in the normal electrical system of vehicle that is protected by safety devices. Such electrical failures will occur in vehicles primarily under the dash or, where the normal wiring has been altered. Examples are the addition of stereo equipment and other accessories by unqualified workmen, or where the normal wiring has been otherwise tampered with. The

cause of such a fire, due to an electrical short, was evident in this test from an examination of the remains of the vehicle.

Leaking gasoline in the engine compartment can produce a fire only if there is a spark, such as from a faulty electrical system, or an open flame. Contact between liquid gasoline and a hot engine is not sufficient to ignite the gasoline. Such a fire, started in the engine compartment, can spread past the firewall throughout the vehicle. Depending on the nature of the fuel leak, the cause of such a fire may or may not be evident from an examination of the burned remains.

We could not produce a vehicle fire by overheating a catalytic converter. Combustible material, such as dry grass and straw, in direct contact with the converter or the muffler, however, will ignite. The amount of damage caused to the vehicle depends on the quantity of grass or straw present.

We could not produce a sustained vehicle fire by overheating a rear wheel bearing. Some heat damage and a small fire did occur in the immediate area of the wheel bearing, but it did not spread to the rest of the vehicle.

An overheated transmission can result in a vehicle fire. The fire is caused by the ignition of hot oil vapors, from the overheated transmission, in the engine compartment. The cause of such a vehicle fire should be evident from an examination of the vehicle.

The intentional use and ignition of flammable liquid did produce a vehicle fire. Once started, the intensity of the fire and the extent of damage it produced did not differ significantly from that observed in any of the other vehicle fires.

A vehicle fire, from its start to finish, progresses through three distinct phases. Temperatures in the range of 1000°C (1800°F), or more, are reached in a vehicle fire, regardless of its cause. These high temperatures can be sustained for about 20 to 30 minutes in the passenger compartment.

Modern motor vehicles contain a significant amount of combustible material (fire load). The increased use of synthetic material in the construction allows a fire to spread to all four corners of the vehicle.

A fire in a vehicle does not cause any wearing or mechanical damage to the working parts of the engine, transmission, or differential. Any damage in these existed before the fire. The routine analysis of oil samples from these components

is a useful aid in assessing the condition of the drive train prior to the fire.

Samples of debris from the fire scene, that are to be submitted to a laboratory for analysis, must be selected with care. Soil samples from the vicinity of the fuel tank and the motor should be avoided. In totally burned vehicles, the the only materials suitable for sampling are from the floor areas. The absence of any residual hydrocarbons from a flammable liquid on these materials does not mean that none were used to cause the vehicle fire. Similarly, the presence of any residual hydrocarbons from a flammable liquid in some soil samples does not necessarily mean that a flammable liquid was used to cause the vehicle fire.

These vehicle fire reconstructions demonstrate that the causes of vehicle fires are in fact limited, from a practical point of view, to only a few, definite possibilities. If it can be shown that a fire in a vehicle originated in the passenger compartment, then most of the accidental methods or ignition discussed here can be eliminated. The cause of such a fire is, therefore, probably the result of an intentional act. If it can be shown that a fire in a vehicle originated in the engine compartment, then some of the accidental methods discussed here are possible. A careful inspection of the vehicle and the circumstances may eliminate or indicate some of these methods."

Appendix

The following lists some of the observed events for each test vehicle. The times given are the actual time of day, using a 24 hour clock, in the form hours: minutes (i.e. hh:mm).

Vehicle #1: 1980 GMC Sierra Grande

Ignition Attempt #1

09:35 Engine started with fuel leak in gas line.
09:42 Engine restarted. Ignition.
09:44 Engine stops.
09:50 Hood opened. Air hoses and air cleaner burning.
10:10 Fire out.

Ignition Attempt #2

10:31 Attempt to short wires in engine compartment. Fusible link blew. Some smoke from insulation.
10:39 Second fusible link on starting motor blew.
11:28 Direct short under dash between hot wire from starter and ground. Vehicle was boosted by tow truck throughout. Smoke generated but

no ignition.
14:07 Repeat of above. Ignition occurs.
14:10 Smoke from dash.
14:16 Stopped boosting. Some open flame around dash. Windshield turning black.
14:20 Windshield cracks
14:22 Thermal fracture of left front window.
14:24 Windshield sags and falls in.
14:25 Intense fire in passenger compartment.
14:26 Roof sagged.
14:27 Plywood sheet in box starts to burn.
14:30 Left front window collapses. Engine compartment burning.
14:33 Door handle on right side falls off.
14:34 Fuel bypass valve to saddle tanks ruptures and fuel spills on ground.
14:37 Intense fire under right side of truck. Blowtorch effect from fuel tank. Gas pouring on ground and burning around right rear tire.
14:40 Left rear tire blew.
14:42 Entire rear of vehicle on fire. Fuel cap on ground and door to fuel cap compartment is open.
14:59 Fire dying down. Tires still burning.

Vehicle #2: 1980 Pontiac Sunbird

Ignition Attempt #1

10:48 Started engine. Converter pushed up against floor.
11:02 Timing advanced to engine and the carburetor to cause rich fuel mixture and heat converter.
11:48 Stopped and restarted engine with small leak in fuel line.
11:52 Placed some dry straw under convertor. Ignited and then went out.
11:59 Shut off gas leak. Inspected floor over convertor.
12:07 Placed more straw under convertor.
12:14 Straw smolders, burns and then goes out.
12:26 Some smoke in vehicle but no flames.

Ignition Attempt #2

12:38 Started engine with leaking gas fitting and a split in a high tension cable to spark plug.
12:40 Ignition occurs. Engine stops. Fire under hood.
12:45 Intense fire in engine compartment. Melted plastic and liquid dripping down from under vehicle.
12:46 Wind gusts.

12:48 Smoke from dash area.
12:49 Windshield cracks.
12:51 Flames in passenger compartment. Windshield falls in.
12:54 Right rear window collapses.
13:00 Right rear tire blew.
13:15 Interior fire dying down. Flames from trunk. Spare tire burning.
13:16 Fire around gas cap. Wind gusts and shift in direction.
13:30 Fuel tank venting and burning.
13:31 Left front shock blew.
13:39 Burst of gasoline on ground to the left rear of vehicle. Rubber connector between filler tube and tank collapses.
13:40 Left rear tire blew.
13:43 Intense venting of fuel tank.
14:01 Fire dying down. Tires still burning.

Vehicle #3: 1978 Chevrolet Impala

Ignition Attempt #1

13:20 Vehicle started.
13:30 Overheating right rear wheel bearing. Blue and white smoke coming from wheel.
13:33 Revved engine to about 90-100 kph. Screeching noise from steel bearing.
13:37 Engine stopped. Brakes failed. Some burning of brake line fluid in master cylinder.

Ignition Attempt #2

13:54 Restart engine.
13:55 Torque converter overload (stall test).
13:58 Smoke from transmission area.
14:02 Oil dripping on ground. Smoke from junction of bell housing and transmission.
14:03 Transmission fluid boiling out of air breather and out of filler spout.
14:06 Flames under hood. Engine stops. Windshield wipers come on.
14:07 Starter engages.
14:09 Vigorous flames under vehicle and in engine compartment.
14:12 Left front tire blew. Plastics dripping on ground.
14:14 Right front tire blew. Windshield separation. Some smoke from trunk.
14:16 Flames from engine and tires.
14:18 Windshield melting and sagging.
14:20 Windshield falls in.

14:21 Heavy fire inside vehicle.
14:23 Driver's side mirror fell off.
14:24 Right front window sags in.
14:31 Roof sags.
14:33 Driver's window sags in.
14:35 Intense fire in back seat.
14:38 Left rear window sags in.
14:40 Venting fuel line in engine area burning.
14:42 Back window sags.
14:44 Right rear tire burns.
14:49 Blowtorch effect from fuel line in engine compartment.
14:53 Left rear tire blew.
14:59 Fuel tank is venting and burning.
15:25 Fuel tank fire reducing in intensity. Trunk still burning.
15:41 Fire is dying.

Vehicle #4: 1976 Ford Granada

Ignition Attempt #1

10:22 Pour fuel on left side of hood. Ignite.
10:25 Fire out.
10:27 Pour fuel on trunk. Ignite. Dies.
10:33 Fuel on roof and door. Ignite.
10:35 Fire out.
10:37 Fuel on roof. Ignite.
10:39 Fire out. Soil sample collected.
11:01 Half gallon of fuel poured on left side of front and rear seat and floor. All windows down 2-3 inches. Ignite.
11:04 Flashover inside vehicle. Windows carboned up.
11:07 Windshield cracks. Dash on fire. Fan comes on.
11:08 Headlamps come on. Windshield washer comes on. Windshield separates.
11:09 Horn comes on and operates intermittently.
11:10 Park lights come on. Roof starts to sag.
11:14 Windshield falls in.
11:16 Driver's window falls in.
11:18 Fire in trunk.
11:19 Right front window falls in.
11:21 Right rear window cracks and shatters.
11:25 Left rear window falls in.
11:28 Fire in engine compartment. Spare tire blew.

11:32 Left rear tire blew.
11:34 Intense fire in engine compartment.
11:37 Left front tire blew.
11:39 Rear window sags.
11:45 Fuel tank bulges, fumes vent and burn.
11:48 Right rear tire blew.
11:51 All tires still burning.
12:20 Fire out.

Vehicle fire research in Germany has resulted in the publication of two very interesting reports. The first one deals with the subject *Fire propagation from carburetor Fires in passenger Vehicles* by Professor Mas Danner (Dr.-Ing.) President of the Allianz Center for Technology, Head of the Department of Vehicle Technology. Their findings were reported in the German periodical "Der Maschinenschaden" 50(1977) Number 2, pages 61-66.

The study was occasioned by a vehicle fire on the Autobahn that originated in the engine compartment. The reported spread of the fire was so rapid that there was some doubt expressed as to the driver's version of the fire.

Four vehicles were burned experimentally, three with front mounted engines and one with a rear mounted engine. In each vehicle 0.2 liters of fuel was used for ignition purposes at the carburetor with 1-2 liters used in a feeder container to simulate flow of fuel from the vehicles fuel lines. In each case the hood was raised after 0.8 minutes to simulate the actions of a driver on discovering smoke coming from under the hood, stopping and exiting from the vehicle and opening the hood.

The tests showed that in the case of a carburetor fire in the vehicles with front mounted engines flames spread from the engine compartment to the dashboard of the vehicle in 8-10 minutes. In one case, where the entire engine compartment was sprayed with about 2 liters of fuel this period was reduced to 4-5 minutes. In order for the fire to continue to spread and completely involve the vehicle's interior required a minimum of one minute and a maximum of three minutes was required. In all three cars with front mounted engines the fire reached the rear section of the car after an additional five minutes. In the case of the rear mounted engine car the elapsed time between ignition and total involvement of the interior was only 7.5 minutes. This was due to the rapid ignition and burning of the insulation in the engine compartment and rapid entry into the interior.

The vehicles used in these experiments were an Opel Kadett, a Volkswagen

type 1300 Beetle and two Fiat 124's. Notice the time lapse between ignition in these cases and entry into the vehicles interior. Compare this with our test conducted with the 1972 Ford LTD when 21 minutes elapsed between ignition in the engine compartment and entry into the vehicles interior. The difference, of course, is due to the size of the cars involved. Smaller cars will burn more rapidly because their combustible material is crammed into a smaller area. The fire load is concentrated and the flames do not have to travel a great distance. As the trend in down sizing of cars continues total burning time for vehicles will decrease.

A second series of tests conducted in Germany was reported on in a *Symposium on Fire/Explosions in Enclosed Spaces*, Stuttgart (1982) by H. Gossman and K. Maulhardt of the Hessisches Landeskriminialamt, Wiesbaden (Provincial Criminal Investigation Agency of Hessen, Wiesbaden) Dr. W. Lichtenber, Bundeskriminalamt, Wiesbaden (Federal Criminal Investigation Agency, Wiesbaden) and Dr. W. Schwanebeck. The tests were made with the cooperation of the Opel/Russelsheim Corporation and involved the burning of four Opel passenger cars. The main points to be determined by these tests were:

1. Possibilities of detecting residue of incendiary agents in cases of arson (Vapor-chamber sampling, isolation and identification of the residue of combustible liquids by high vacuum distillation and extraction).

2. Possibilities of detecting lead and bromide as "conducting elements" for the combustion of gasoline; establishing the value of detection of this kind of evidence in a case of arson in gasoline powered passenger vehicles.

3. The determination of electrical causes of fire.

In tests one and two fires were started in the passenger compartments with and without gasoline as an incendiary agent. Both vehicles were allowed to burn for 10 minutes before being extinguished. The driver's door was allowed to remain open and both fires were started in the area of the driver's seat. The fire without gasoline as an accelerant was started with rolled up newspapers. The overall destruction in both cases was similar but in the vehicle with gasoline as an accelerant an immediate steep temperature increase was noted at a measuring point in the front passenger floor while in the car burned with rolled up newspapers there was an "inductation period" of about 2 minutes before the temperature increase began. The intensity of the burning of the fire started with paper was no doubt increased by the fact that the windshield glass was not laminated and shattered quicker and more completely than the second vehicle

which did have a laminated windshield.

Attempts to detect gasoline as an identifiable accelerant were disappointing. High vacuum distillation of material from the mat underneath the driver's seat failed to identify the components as gasoline although the photo-ionization detector had indicated there was possibly some residual gasoline. Samples of smoke deposits failed to yield any significant traces of lead and bromine. (In this regard you will recall that the conclusions of the conductors of the Canadian experiments in 1982 were that finding these elements has no value in establishing the use of gasoline as a method for starting a vehicle fire).

An attempt was made to start one of the vehicles on fire by creating a short circuit in the engine compartment. The positive battery cable had a portion of its insulation peeled off and shorted against a section of the brake line. The insulation caught fire after a few seconds and produced a flame about 30 CM in height. The section of brake line became red hot and some of the escaping brake fluid caught fire but died of its own accord. The fire did not spread mainly because of lack of communication to other combustible materials.

The last test involved the pouring of gasoline onto the ground under the vehicle's engine (a front mounted engine) and then ignited. This fire was allowed to burn for 20 minutes with all doors and windows and the hood closed. A steep rise in temperature was noted in the engine compartment which levelled out after temperatures of about 700°C were recorded. The passenger compartment measuring points did not record temperatures of over 70°C. despite the fact the duration of the fire was twice as long as in the other fires. Damage was mainly confined to the engine compartment although the windshield, dashboard and firewall showed fire damage as well as some slight damage to the front seats. The rear of the interior was only smoked. The summary of the report advises that the test results should be regarded as preliminary experiments and that further tests should be made to improve and extend these findings.

In October of 1990 additional vehicle fire tests were held at Winnipeg, Manitoba, Canada. The vehicles involved were.

Vehicle #1 1985 GM Astro Van
V6 4.3L engine with throttle body injection

Vehicle #2 1987 Mercury Topaz
4 cylinder 2.3 L engine with electronic fuel injection

Transverse engine, front wheel drive

Vehicle #3 1986 Plymouth Caravelle
2.2L 4 cylinder engine with electronic fuel injection
Transverse engine, front wheel drive

Vehicle #4 1986 Buick Century
3.8L V6 engine with multiport fuel injection
Transverse engine, front wheel drive

Vehicle #5 1989 Oldsmobile Cutlass
3.1L V6 engine with fuel injection.
Transverse engine, front wheel drive.

All these vehicles were in operating condition at the time of the tests. As in the previous tests these experiments once again demonstrated that car fires will go through three distinct phases:

#1 Ignition — in this phase temperatures remain at or below 200°C (392°F)

#2 Consumption — in which $^{2}/_{3}$ of the combustibles in the vehicle are consumed.

#3 Cooling — in which the vehicle remains return to ambient temperature.

In the previous tests it was noted that maximum interior temperatures noted were 550°C (1022°F) in the engine compartment. The conclusions reached were that any damage to the vehicles engine, transmission and differential must have pre-existed the fire since these temperatures were not sufficient to cause massive damage.

In addition to the thermocouple temperature monitoring equipment the October 1990 tests also made use of radiometer readouts every 4 seconds that measured watts per square centimeter. This measures radiating heat in much the same manner that wind chill factor is determined. One other measurement taken that was not done in the previous tests was that of battery longevity. This is of importance since as long as a battery produces sufficient power an electric fuel pump can continue to operate and increase the intensity of a vehicle fire.

Vehicle #1 1985 GMC Astro Van

A leaking fuel line was created and ignition created with an electrical spark. Fire was instantaneous. Within 2 minutes there was smoke in the passenger compartment. Within 10 minutes the passenger compartment was fully engulfed in flames and shortly after the fire vented through the windshield and side windows. The fire communicated to the undercoating which aided in the fire spread. The fuel filler neck failed and a massive fire developed under the rear after 33 minutes. At 58 minutes the vehicle was totally destroyed. The battery maintained voltage for 18 minutes into the fire. Maximum temperatures recorded were:

Engine compartment -	900°C (1652°F)
Passenger Compartment -	950°C (1742°F)
Exhaust Manifold -	1050°C (1922°F)
Muffler -	850°C (1562°F)
Catalytic converter -	850°C (1562°F)

The extreme temperature in the engine compartment caused massive damage to the engine and transmission. The Radiometer readings were 3.8 watts per sq/cent at the side and 8.5 watts per sq/cent over the top.



A fuel leak in the engine compartment was the source of the fire in this Astro Van.



Left unchecked, the fire totally destroyed the van. In earlier days a complete burn out such as this was attributed to arson.

Vehicle #2 1987 Mercury Topaz

A car warmer was placed on the dash with the fan disabled and a loosened connection. Sufficient heat was developed to cause the plastic dash to melt and the experiment was discontinued. Had this not been done a fire would have developed. A transmission overload stall test was attempted and although there was a fluid overflow no fire developed. A fuel leak was created and ignition occurred through a disconnected spark plug wire. Within 5 minutes the fire entered the passenger compartment. It was then suppressed. An engine compartment temperature of 600°C (1112°F) was recorded.

Vehicle #3 1986 Plymouth Caravelle

After a transmission overload test in which the exhaust manifold temperature reached 500°C (932°F) an engine oil leak was created at the valve cover. Within 5 minutes the oil dripping onto the exhaust manifold ignited and a fire developed. This fire quickly burned into nearby fuel flex lines and involved the entire engine compartment. Maximum exhaust manifold temperature recorded was 650°C (1202°F).

Vehicle #4 1986 Buick Century

The drive wheels were spun in loose dirt in an attempt to start a tire fire. After 12 minutes the right tire blew out without a fire developing. Much heavy smoke was created, however, and it became necessary for the driver to wear a breathing apparatus. Weather and soil conditions may have prevented an outbreak of fire.

A transmission overload was created and after five minutes leaking transmission fluid ignited.The fire communicated to the power steering and fuel hoses, creating a fire of larger dimensions. The fire did not enter the passenger

compartment until 23 minutes after ignition when the windshield collapsed. Temperatures noted were:

Engine compartment -	650°C (1202°F)
Exhaust Manifold -	750°C (1382°F)
Passenger Compartment -	1200°C (2192°F)

The engine and transmission suffered severe damage and would not be salvageable. It was also noted that some of the spark plug wires had fallen away from their sockets. This could lead an unwary investigator to conclude that this had been done to furnish ignition for a deliberate fire.

Vehicle #5 1989 Oldsmobile Cutlass

Four attempts were made to cause a fire by creating electrical shorts. While smoke and fumes were caused no open fire resulted. The reduced voltage now in use in LED dash displays reduces the chance of fire. Electrical fires are still possibilities, however, in shorts in the primary system and in the case of added accessories. These tests are still being evaluated and as of the date of this writing the final report has not been prepared.

Some of the highlights noted are that:

1. Exhaust manifolds can now exceed 650°C (1202°F) and sometimes register higher than the exterior of the catalytic converter. The use of lighter metals in these components is believed to be the cause of these higher operating temperatures.

2. While engine blocks and pistons will survive most fires all other components in the engine compartment will likely be destroyed or severely damaged. Engine compartment temperatures of 550°C (1022°F) in the 1982 tests were noted while in the 1990 experiments 900°C (1652°F) was recorded in the engine compartment.

3. Engine oil and transmission fluid temperatures were recorded at 1200°C (2192°F) and 875°C (1607°F) respectively.

4. Passenger compartment temperatures remained consistent with the 1982 experiments.

5. In an average time of 10 minutes fire communicated from the engine

compartment into the passenger compartment. While this would allow occupants to escape under normal conditions it should be noted that within 3 minutes smoke and fumes can enter the interior and render injured or handicapped persons incapable of escaping.

6. Batteries maintained operating voltage until destroyed by fire. This can take as long as 20 minutes after the development of fire in the engine compartment.

7. There is no danger of a fuel tank explosion but there can be danger from fuel expulsion when the filler neck or cap fails due to build up of heat and pressure within the tank. There is also danger from exploding shock absorbers and lift assist struts when elevated temperature failures occur during the course of the fire.

8. While it cannot be eliminated as a fire cause, electrical failure or malfunction continues to be a less likely fire cause. When it does occur the resulting damage is not as great as from other sources such as fuels and lubricants.

9. Radiating heat from a burning vehicle can set fire to other combustibles either to the side or over the vehicle.

10. Important factors to consider in a vehicle fire are status of door and windows, location, wind direction and prevailing weather.

Fire Experiments 1988 Honda Prelude

One of the most difficult things encountered in vehicle fire research is the scarcity of equipment on which valid testing can be made. It is not too hard to come up with the upholstery or seat material to be used in test fires, nor is it difficult to obtain plastics and metal components to subject to high temperature reaction. When, however, it comes to a complete motor vehicle to be subjected to destructive fire testing, the best one can usually expect is a car or truck that is 15 years old and frequently in a wrecked condition. One can hardly expect to have donated a new vehicle in working order when the testing will result in its destruction. I realize that this is frequently done in collision testing but this is mandated by the Department of Transportation. Only the most elementary fire testing is required and this related only to upholstery fire spread and the amount of fuel leaked following a collision or roll-over. Both are important considerations but only relate to a small percentage of vehicle fires. I think the lack of fire experimentation done on present-day vehicles has led many fire investigators to being "stuck in the 60's" in that they have their fire origin and cause opinions based upon experi-

ments completed on vehicles 20 years of age or sometimes even older.

Bearing all this in mind, you can imagine my surprise and delight when in mid-1988 American Honda Motor Company offered to make available a 1988 Honda Prelude for fire cause and origin study. Since I was teaching a short course at San Jose State University sponsored by the Administration of Justice Bureau on Vehicle Fire Investigation, it was considered the best approach to have the vehicle donated to the school for this particular purpose.

This was eventually done and the 1988 Honda Prelude with only 4,947 miles on it, was delivered to San Jose State University at the Almaden Honda dealership in San Jose. It was then trailered to the California State Auto Association facility in Hayward for some preliminary testing and storage. Under the conditions of donation, the vehicle was not to be driven on any street or highway and after completion of this testing, was to be destroyed.

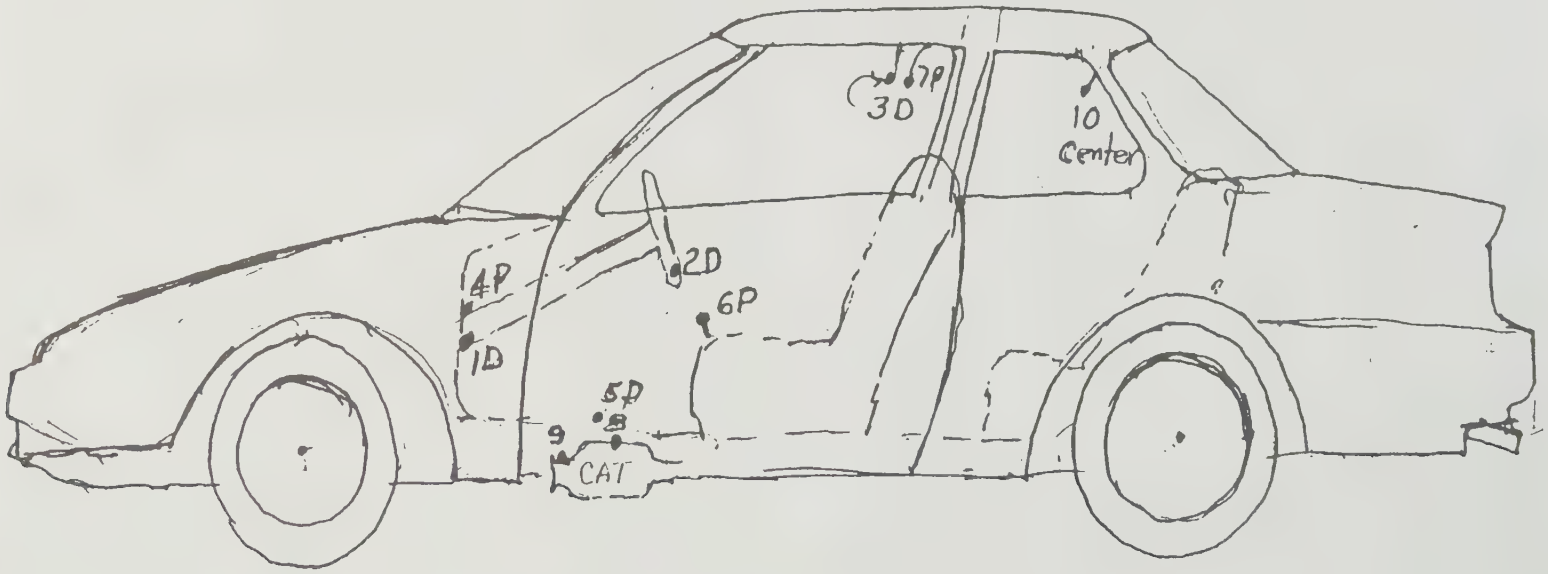
Some preliminary testing was done at the CSAA facility relating to operating temperatures of the vehicle's systems. After a 30-minute period of stationary operation in a high rpm idle condition, the following temperatures were recorded:

Intake manifold	160°F
Oxygen sensor in exhaust system	360°F
Exhaust manifold in engine compartment	345°F
Power steering fluid temperature	138°F
Engine oil temperature	208°F
Exhaust gas at tailpipe	130°F
Muffler Exterior temperature	224°F
Catalytic converter (measured at exhaust pipe entry into converter)	525°F

At the conclusion of our readings and after 40 minutes of operation, the exhaust manifold temperature inside the engine compartment had reached 407°F. After the engine had been shut down and the hood closed, this temperature had risen to 448°F within a five-minute period.

On April 5, 1989 the Honda Prelude was transported to a lot across the street from the Scotts Valley, California Fire Department. This fire department had volunteered their facilities and personnel in conducting these fire experiments on the vehicle. The car was equipped for normal operation. The fuel tank was 3/4's full and contained approximately 11.9 gallons of 86+ octane unleaded gasoline. The engine contained four quarts of SAE 10W-30 weight oil. Standard 5-speed

P=passenger
D=driver



1988 Honda Prelude

manual transmission contained two quarts of 20W-30 weight oil. The power steering system and reservoir held about a quart of Honda-approved power steering fluid. The radiator and coolant system contained 5.7 quarts of 50/50 mixture of water and ethylene glycol.

Three basic tests were envisioned for the experimental burning. These were:

1. Ignite four ounces of paper on the vehicle's back seat, close all doors and windows and observe:
   a. carbon deposits on window
   b. extent of fire damage
   c. length of time before oxygen supply is exhausted
   d. maximum temperature attained

2. Cause catalytic converter to overheat by disconnecting two spark plugs and inducing enriched fuel mixture. During this period the following would be poured on or applied to the exhaust manifold in the engine compartment:
   a. brake fluid
   b. automatic transmission fluid (Dexron II ATF)
   c. 50/50 mixture of water and ethylene glycol (coolant)
   d. dry rag inserted in crevice of exhaust manifold

3. Create a fuel leak in the line near the pressure regulator in the vicinity of the left side of the engine compartment just forward of the cowl. Ignite the vapors by either exhaust manifold heat or electrical spark. Observe progress of fire and temperatures attained.

Thermocouples with constant temperature readout capabilities were located at the following points in and under the vehicle:

1. Junction of steering column and in front of floorboard area, left side.
2. Lower side of steering wheel.
3. In headliner over driver's seat.
4. Front floorboard area, forward of passenger seat, right side.
5. Floorboard just forward of driver's seat, left side.
6. Forward edge of front passenger seat.
7. Headliner over passenger's set, right side, front.
8. Top of catalytic converter, underside of vehicle.
9. Forward section of catalytic converter at junction with the exhaust pipe.
10. Headliner over center of rear seat.

The experiments were begun at 11:15 a.m. The ambient temperature was 79°F

and the winds were calm at 2-1/2 to 5-1/2 mph out of the northwest.

Test #1:

4 oz. of loosely rolled yellow tablet paper was placed on the left rear seat and ignited with a match. Within two minutes the fire appeared to diminish but then began to climb the seat back. At three minutes and 15 seconds, thin wisps of smoke began to appear at the door lines. At three minutes and 55 seconds, the windows were completely covered with carbon, making it impossible to see inside the vehicle. At four minutes the temperature registered by thermocouple #10 had risen to 475°F. In less than a minute later the fire extinguished itself, having consumed all the available oxygen and the temperature began to drop. At six minutes it had descended to 250°F. At seven minutes it had descended to 217°F and after eight minutes and 47 seconds the headliner temperature had fallen to 195°F. After 13 minutes and 35 seconds the vehicle was opened. There was a dense, brownish colored smoke with a very acrid odor inside. The interior headliner temperature at the time the vehicle was opened was 157°F. The damage, apart from smoke and heat warping of some plastic items, was confined to a hole in the seat cushion about 10 inches in diameter and a corresponding burn on the seat back about 3/4ths of the distance to the top. It should be noted that during this time period temperatures at thermocouples location #1, #4, #5 and #6 did not exceed 156°F.

Test #2:

At 11:40 a.m. the engine of the 1988 Honda Prelude was started. At 11:44 a.m. catalytic converter temperature at thermocouple #9 was 167°F. Two minutes later one spark plug wire was disconnected and at that point temperature at thermocouple #9 was 242°F. At 11:48 a.m. the temperature at thermocouple #9 was 903°F. At 11:50 a.m. a second spark plug wire was disconnected. Within four minutes the temperature at thermo couple #9 was 940°F and one minute later the temperature had risen to 1115°F. At this point the catalytic converter was observed to be a glowing red color.

At 12:03 p.m. one ounce of brake fluid was poured on the exhaust manifold in the engine compartment. The result was steam and smoke but no flame was visible. At 12:05 p.m. one-half ounce of automatic transmission fluid (ATF Dexron II) was poured on the exhaust manifold in the engine compartment . The result was instantaneous fire. At 12:11 p.m. the temperature at thermocouple #9 which was located at the forward section of the catalytic converter at its junction with the exhaust pipe was 1102°F. At 12:12 p.m. a 50/50 mixture of water and glycol (coolant) was poured on the exhaust manifold in the engine compartment.

Steam and smoke followed but no outbreak of fire was observed. At 12:21 p.m. glycol coolant without water addition was poured on the exhaust manifold in the engine compartment. Steam and smoke followed but still no fire was in evidence. This should not be interpreted, however, to indicate that coolant either with or without water addition will not burn under any circumstances. Other experiments have shown that a fire can occur. At 12:27 p.m. a mechanic's shop rag was wedged into a crevice on the exhaust manifold in the engine compartment. One and one-half minutes later the rag began to smoke and at 12:31 p.m. it burned with glowing ashes.

Test #3:

For this test all spark plug wires were reconnected and the vehicle restarted. A small hole was cut in the fuel line near the pressure regulator as previously described. At 12:40 p.m. the experiment began. No fire developed during this fuel leak, no doubt because of the following:

a. The exhaust manifold is on the opposite side of the engine compartment from the point of leak, and

b. There had not been a sufficient accumulation of vapors to be ignited by the catalytic converter under the vehicle.

At 12:45 p.m. a spark plug wire was disconnected and the engine restarted. Instant flame developed at the leak source. At 12:46 p.m. the hood was closed. A large fan had been mounted in the front of the vehicle to simulate air movement over the car as if it were in motion. Since the engine continued to run, the fire rapidly increased in intensity due to the continued pumping of fuel into the engine compartment. Fire could be observed under the left front wheel after three minutes had elapsed. The exterior paint on the hood began to burn over the spot where the fire had first ignited. Fire could be observed under the cowl and the windshield began to crack. The following temperatures were recorded at the noted elapsed time from the start of the fire on test #3. Thermocouple #8 720°F at 7 minutes and 56 seconds. At thermocouple #8 1300°F at ten minutes and 21 seconds. At this point the horn began to blow because of fire generated short circuiting. At 11 minutes and 25 seconds after ignition fire has now entered the passenger compartment through the fire wall. Thermocouple #5 showed 650°F at 12 minutes and 37 seconds. At #6 560°F at 12 minutes and 42 seconds. At this point a tire blew out. At 15 minutes and 8 seconds thermocouple #3 showed

412°F, thermocouple #6 564°F at 15 minutes and 21 seconds. Thermocouple #7 showed 750°F at 15 minutes and 36 seconds and thermocouple #8 recorded 560°F at 15 minutes and 42 seconds. (The catalytic converter began to cool after reaching a high point of 1370°F at 13 minutes and 47 seconds into the burn. From this it can be assumed that the engine continued to run for over 12 minutes after the fire had ignited.) At 17 minutes after the ignition, thermocouple #4 showed a temperature of 135°F, #5 showed 1023°F and thermocouple #7 recorded 1150°F at 17 minutes and 18 seconds after ignition. #10 recorded 390°F at 17 minutes and 32 seconds. At 18 minutes and 45 seconds thermocouple #1 showed a temperature 980°F, #2 802°F and #3 1060°F at 18 minutes and 55 seconds after ignition. Thermocouple #5 recorded 800°F at 19 minutes and 5 seconds and #7 showed 1150°F at 19 minutes and 18 seconds. #10 now showed 500°F at an elapsed time of 19 minutes and 29 seconds after ignition. At this point the ambient temperature had climbed to 104°F. Vapors expelled by boiling fuel in the tank caught fire and burned in a "flare off" pattern above the fuel filler cap.

As the fire continued to burn the thermocouple readings continued to climb. At position #1 at 20 minutes and 55 seconds into the burn of the vehicle, a temperature of 1105°F was recorded. Thermocouple #2 showed 1230°F and #3 recorded 1115°F. At 21 minutes and 23 seconds #6 showed 1337°F, and #7 thermo couple was 1262°F. At 21 minutes and 37 seconds after the ignition thermocouple #10 now read 1103°F.

After 22 minutes and 7 seconds thermocouple #1 showed a temperature of 1200°F, #2 read 1070°F, and #3 showed a reading of 1229°F. At 22 minutes and 24 seconds #4 read 750°F, #5 450°F, and at 22 minutes and 40 seconds #6 showed a reading of 1221°F and #7 1284°F. #8 had dropped to 329°F and thermocouple #10 read 1215°F, exactly 23 minutes after the original ignition.

Within the next 10 minutes temperature at thermocouple #10 increased to 1285°F. The door glass melted and sagged inward on both sides. The fire vented itself through the collapsed windshield and moonroof and remaining combustibles in the interior burned furiously and a cloud of dark smoke with bright orange flames billowed up.

The connector between the filler neck and fuel tank failed and boiling gasoline was ejected onto the ground under the rear of the vehicle. Bright orange flames engulfed the vehicle's underside up to the front door hinges. The fire burned with great intensity for about three minutes and then began to diminish. In the

meantime plastic body panels at the vehicle's rear caught fire and burned, completing the fire cycle from front to rear.

After 45 minutes time had elapsed from ignition of the vapors from the fuel leak in the engine compartment, the fire was extinguished by members of the Scotts Valley Fire Department. The only suppressant used was water. By the time this was done the fire's intensity was greatly diminished and would have probably burned itself out within a half hour. After the fire was extinguished a heat pattern could be observed on the hood exterior at the point where the fire originated. It cannot be assumed that this would occur in every case, though.

Much valuable information is obtained from tests such as these. Some important points noted in these tests are:

1. 4 oz. of combustible material will exhaust the oxygen supply in an average size vehicle within five minutes burning time in a closed vehicle.

2. Prior to that time carbon deposits will have formed on the windows making visibility inside the vehicle almost impossible.

3. Automatic transmission or power steering fluid spilled on a hot exhaust manifold will ignite instantly.

4. With an overly rich fuel to air mixture a catalytic converter can exceed 1100°F exterior temperature.

5. A fuel leak in the presence of a spark will result in instant ignition of vapors and subsequent fire of great intensity.

6. Left to its own devices, an engine will continue to run after ignition of a fire in the fuel system in the engine compartment and pump additional fuel on the fire. The engine will eventually cease to operate due to insufficient oxygen or burning of the fuel line.

7. Vapors from boiling fuel will "flare off" above the fuel tank filler neck.

8. If allowed to burn, there is no difference in the appearance of a vehicle burned by accidental means than by a deliberate act. The only difference is the time factor involved.

Swedish Experiments

Experiments were conducted in Stockholm, Sweden on a 1980 Oldsmobile diesel 4 door sedan to simulate a fire of electrical origin. The fire was initiated by creating a short in the positive battery cable. Fire spread was enhanced by fuel and oil coated insulation and plastic components in the engine compartment. The vehicle had no prior damage and was in running condition. The wind was blowing 12-17 meters per second (about 25 MPH). Time figures are for elapsed time after the short produced a flame. This began at 00:01.

10:15 Open flames observed between hood, fender and firewall, left side
10:57 Flames seen in left front wheel well
12:30 More smoke and flames observed over left front hood and fender
13:00 Paint begins to burn over hood
13:20 Coolant begins to flow and run onto ground
14:00 Smoke visible at front grille
15:00 Flames visible at top of cowl, hood junction and begin to climb up windshield
16:00 Flames all way across windshield.
17:06 Heavy smoke from front of grille
17:20 Smoke visible in passenger compartment
17:30 Left front tire on fire
17:45 Flames across front of grille
18:38 Entire front end on fire
20:00 All windows carboned over
21:51 Fire is in passenger compartment. Tempered glass shatters.

Fire extinguished after 39 minutes. Maximum interior temperature: 850°C (1562°F)

July 1992 Vehicle Fire Experiments at Fresno, California

In early 1992 American Honda Motor Company contributed 44 late model vehicles to the California Conference of Arson Investigators (CCAI) for use in training fire investigators and suppression personnel. This event took place at the Fresno County Fairgrounds and was done in conjuction with the semi annual seminar of the CCAI. Over 300 people attended. Some of the salient points noted during these experiments follow:

1992 Honda four door sedan
A fuel leak was created in the engine compartment and a spark plug wire was damaged to provide an ignition source. An instant fire developed on engine start up. Gasoline could be observed leaking onto the ground under the engine compartment. Fire burned in and under the engine compartment and was extinguished after nine minutes. It did not enter the passenger compartment.

1992 Honda Accord four door sedan
An interior fire was set using two paper cups placed in cup holders between the two front seats. Newspaper was placed in the cups and then saturated with 12 ounces of denatured alcohol. Alcohol was also spread on the right front floor carpeting. Ignition was created by use of an electric "match". After 1:50 minutes the windshield began to exhibit a "crazed" pattern and at 2:52 after ignition the windshield failed and collapsed from the top center. After 6:37 minutes the fire was extinguished. The collapse of the windshield was directly over the point of the fire's origin.

1992 Honda two door sedan
A small quantity of pool chlorine was placed in a paper cup. This cup was placed in the forward console area under the center of the dash. About one ounce of brake fluid was poured into the cup over the dry chlorine. After 16 seconds had elapsed, a chemical reaction took place that resulted in a violent fire. After 20 seconds, flames were three feet high causing the dash, steering wheel and seat materials to melt and burn. This communicated to the headliner and after two minutes the forward area of the passenger compartment was involved. At three minutes and thirty seconds after the fire's ignition, it was extinguished.

1992 Acura two door sedan
This experiment was set up to simulate an engine compartment fire initiated by a stall test. The front wheels were blocked and the selector placed in "drive". The throttle was advanced to high RPMs. This condition was allowed to continue for twenty minutes and the transmission fluid boiled out of the dipstick tube, but did not ignite. While this was being observed a fire was noted at the rear of the vehicle. On checking this it was determined that the exhaust system became so overheated that the plastic covered rear bumper melted and ignited. The resulting fire was so intense that the entire rear section of the vehicle was heavily damaged before the fire was extinguished.

In addition to these experiments, numerous others were conducted. Power steering fluid ignited on an exhaust manifold after twenty minutes time. Engine

oil on the exhaust manifold ignited in just over thirty minutes. An electrical fire was created by shorting out an alarm system.

Vehicle Arson Demonstration
March 26-28, 1997 Monterey, California

At a seminar conducted by the Northern California Fraud Investigators Association, two vehicle arson demonstrations were conducted. The vehicle used was a Toyota Celica. Two methods were used, making use of materials easily attainable, but not leaving the trace of hydrocarbons normally associated with an accelerated fire.

Test #1
Place paper cup containing two ounces of dry pool chlorine in the center console under the dash. A loosely rolled page of newsprint was placed over this in the dash. A cap containing two ounces of brake fluid was poured over the dry pool chlorine in the paper cup. In 45 seconds a flame erupted from the cup, caused by the corrosion and heat produced by the chemical reaction. This fire quickly communicated to the paper and spread to the center of the dash. After about 2-1/2 minutes the fire was extinguished. If left unchecked it would have totally burned the vehicle.

Test #2
A one pound bag of potato chips was placed on the rear seat of the vehicle and a corner of the bag ignited. After about thirty seconds the oil in the potato chips began to vaporize and shortly after ignited. The fire quickly spread to the seat back and surrounding upholstery material. Within five minutes the fire had spread to the entire rear interior and was extinguished by fire fighters. This fire would have consumed the entire interior if left unchecked.

Experimental Vehicle/Garage Fire

Frequently the vehicle fire investigator is asked to determine the cause and origin of a fire involving an automobile in a garage. Almost immediately there is the speculation that the vehicle's system or systems was instrumental in the beginning of the fire. Such cases are very difficult to resolve since both the garage and vehicle contain fuels and systems that could create a fire. Most garages contain fuel, chemicals and electrical appliances. The presence of a motor vehicle adds to the potential for fire if there is a malfunction.

We have already enumerated the many causes of vehicle fires in earlier chapters of this book. The question arises often as to the behavior of a vehicle fire in a closed in garage situation. In order to attempt to ascertain better answers to this question, a 1989 Mercedes Benz 300E four door sedan was burned in a single car garage in Hartford, Connecticut in October 1997. The vehicle was in running condition with no prior damage. It was normally equipped. The garage rear door was left open in order to videotape the proceedings. The vehicle was driven into the garage under its own power.

At 12:00 PM three small fires were ignited in the engine compartment, one at the left firewall, one at the air cleaner and one at the right firewall. The hood was then closed.

12:01 Smoke observed at junction of hood and firewall. The smoke is white in color changing to gray. It is not dense.

12:03 Smoke continues. A burn pattern is seen at the hood and fender lines and right side forward of windshield.

12:09 Smoke increases, greater on right side than left. Smoke is now visible in garage.

12:12 More rapid and dense smoke developing in windshield/firewall area.

12:16 Fire visible at right center of windshield and firewall, begins spreading toward right.

12:18 Fire spreads across right side, about one foot up windshield.

12:20 Dense smoke in garage. Flames across windshield up to roof line, hood paint is burning, dense smoke in garage. Still no fire in vehicle interior.

12:25 Flames over one foot over roof line, at front.

12:26 Smoke in interior — windows smoked over.

12:29 Fire spreads across hood

12:30 Top of right front tire on fire

12:33 Smoke coming from under right rear wheel

12:34 Entire front end on fire, smoke coming from under right rear side. Smoke rolling out of garage door.

12:36 Flame over occurs at garage roof. Burning debris is falling on rear of vehicle.

12:37 Fire on both sides of car in garage. Back glass breaks out. Fire breaks through garage roof and rolls out from garage door.

12:39 Entire structure and car on fire. Large plume of flame and smoke from left rear side of vehicle as fuel tank releases.

12:43 Flames 20-30 feet above structure. Heavy black smoke.

12:46 Fire is diminishing, but apparent wind shift begins to intensify it.

12:50 More falling debris intensifies fire around car.

12:52 Remains of garage structure collapses, burning continues.

12:56 Final remains of garage fall in as fire diminishes — black smoke continues.

13:03 Fire department begins extinguishment activities.

13:12 Fire out.

1. A match is applied to the corner of this potato chip bag.



2. A short time later the oil in the chips vaporizes and the bag burns.



3. The fire spreads rapidly and the rear seat burns.



Motorhomes, Buses, Truck/Tractors Construction Vehicles & Recreational Boats

We will consider motorhomes first. There is some confusion in what a "motorhome" is. The term "mobile home" and "motorhome" are often used interchangeably. These vehicles are, in fact, completely different from each other as the following illustrates:

"Motor Homes" — A vehicular unit built on a self-propelled motor vehicle chassis, primarily designed to provide temporary living quarters for recreational, camping or travel use.

Please remember that a "Mobile Home" is indeed a home. It can be moved from site to site, but when it is placed there it has all the attributes of a house. It is, in fact, a house (or sometimes half of one) that can be firmly anchored to a foundation. In this way, it differs very much from a "Motor Home". (*RV Identification*, by D. Brickell & L. Cole, 1978, Ralph Egan MacDuff Publications)

In our book, "A Survey of Vehicle Fire Causes." we included 25 motorhome fires encountered during the survey period. The following was noted:

Cause
Splice in flex line leaked in engine compartment
Propane leak at refrigerator
Arson
Propane leak at stove inlet (cracked fitting)
Fuel pump sliced into fuel line in engine compartment
Power surge caused electrical system to burn out
Overheated engine caught oil, grease etc. on fire
Splice in fuel line leaked
Canister overflowed
Overheated engine
Arson
Undetermined (probable arson)
Battery cable shorted against air intake tube
Undetermined

Driver poured gasoline in carburetor - backfire
Electrical short in converter relay
Shorting in cable between starter and alternator
Fuel leak at carburetor inlet
Leak in refrigerator propane line
Propane leak from improper refrigerator installation
Leak in carburetor at accelerator pump
Electrical fuel pump installed leaked
Arson (attempted suicide)
Propane leak caused by cracked fitting
Fuel leak at carburetor inlet

This extensive list is an indication of the fire problems encountered in these vehicles. The main factors involved were:

1. Fuel line leaks
2. Propane leaks

These problems are frequently caused or enhanced by the "Coach Builder" who modifies the configuration of the original chassis builders fuel line design. As an example, some coach builders will splice in rubber composition flex line sections in place of the original steel alloy fuel line. These sections sometimes leak after age, heat and vibration have taken their toll.

Fires in motorhomes present a danger potential not found in regular passenger automobiles since, in addition to the regular fuels of propulsion, these vehicles carry propane for use in cooking, heating and refrigeration. This adds a double hazard heightened by the use of light construction material that is not flame resistant. The flame spread in these vehicles is often quite rapid and poorly located exits can cause a clear danger to occupants. One motorhome owner, who had escaped from his burning vehicle by squeezing through a window showed me the entrance/exit. It was located directly over the fuel tank and the filler neck was burning like a blow torch when the fire was discovered!

Cracked, loose or just plain unconnected propane lines led to five of the motorhome fires covered in the survey. One victim was blown through the door of his motorhome when he struck a match to light the stove. A repairing agency had forgotten to re-connect the propane line to the waterheater when some warranty work was done. When the victim turned on the gas from the outside, the motorhome interior filled with propane and the match set off a violent explosion and fire.

In another case someone spliced in an in-line electric fuel pump in the fuel line in the engine compartment. A connection failed and fuel vapors filled the engine compartment and quickly were ignited. The close confines of the cab over the engine arrangement of motorhomes causes poor air circulation in the engine compartment and heat build ups are common.

On some occasions spliced in flex line sections do not take into account a degree of off set in the fuel line to fuel inlet configuration. The result is strain placed on the spliced in section which can fail and allow the escape of fuel.

One practically new motorhome valued at over $80,000 caught fire in the vicinity of the refrigerator. Extensive damage was done before the fire was extinguished by the fire department. The following information is from our investigation report on the cause and origin of the fire:

"The propane supply line showed signs of bending and twisting at its junction with the inlet valve, indicating it had been forcibly pushed into the inlet and lightened. This caused the angle of junction to be about 60°. Removal of the inlet valve revealed the face of it to be worn and 'chewed', indicating the distinct possibility of cross threading. This condition was not concluded to be fire caused since there was no melting or dripping of the metal which appeared to be of zinc composition. The owner produced a document furnished to him when he purchased the motorhome regarding the refrigerator's operation. This document lists instructions and a diagram for installing this type refrigerator. The illustration shows the entry pipe coming from the floor behind the refrigerator in a long loop with a circular bend to absorb vibration. It states, 'The gas line should be routed in a manner to limit the possibility of vibration or abrasion'. This clearly was not done in this case.

Based on our observations, physical evidence found at the scene and information supplied by the owner, the following conclusions were made:

1. The installation of the refrigerator was not in accordance with the manufacturer's instructions.

2. The installation of the main gas supply line into the inlet valve was faulty, placing stress and strain on the inlet due to vibration.

3. Cracking of the inlet unit and wearing threads allowed propane gas to escape from the inlet valve, settling in the lower reaches of the vent space.

4. This flammable mixture was ignited by an electrical spark from the refrigerator's switch when it normally cycled during its operation on 110V current supplied by the outside source."

One of the major problems encountered by the motorhome manufacturer is the trade off of weight versus strength. Needless to say, no one would entertain the thought of buying a motorhome if its construction and weight rivaled an armored vehicle. While such a vehicle would offer strength and security it would be neither practical or affordable. The selection of light weight material by the manufacturer frequently includes wood, light plastics, cloth, foam insulation, paper and fiberglass. The flammability of these items is already clearly established. This increases the motorhome fire load made already heavy by the use of two separate fuel systems, i.e. gasoline or diesel fuel and propane.

The Recreational Vehicle Industry Association (RVIA) has in recent years promoted a set of voluntary standards relating to the design and installation of systems including heating, plumbing and electrical components. About 95% of the Recreational Vehicle manufacturers belong to RVIA. Industry standards used by RVIA were developed by the American National Standards Committee and are called "American National Standards for Recreational Vehicles". These standards include:

- Minimum flame-spread ratings for interior walls and ceilings.
- Alternate exits.
- Combustibility limits for fabrics and carpets
- Required installation of smoke detectors in larger RV's
- Grounding requirements for electrical systems.
- Circuit-breakers and fuse requirements
- Rules governing use of LP gas, including line installation, filling procedures and leakage testing.

These moves are certainly in the right direction, but still will be ineffective against situations such as the following segment from one of our investigation reports:

"The Insured reported that he was on his way home to Sacramento and did not stop en route. He was driving at about 55-60 mph when he heard a 'stutter noise' from the engine compartment and the engine began to miss. He pulled over into the center divider, but did not smell or see smoke at that time. When he came to a complete stop there was an explosion, shaking the motorhome. The noise was described as a 'thud'.

On the right the Insured saw fire rising by the passenger door window. The same could be observed by the left door window. The cab filled with smoke and he could not find the fire extinguisher. The Insured then made his exit from the vehicle through the right side entry door. He saw fire coming from under the front and engulfing the right front tire. He went around and observed the same on the opposite side. Within fifteen minutes fire had consumed the entire motorhome moving from front to rear."

This vehicle had only 2767 miles on it when it was destroyed. The value of the motorhome was $46,230. The cause of the fire was a leak at the carburetor inlet.

There have been attempts to have motorhomes included in FMVSS#302. Mr. Ed Horkey of Horkey & Associates, Tempe, Arizona filed a petition in 1990 protesting the inadequacy of FMVSS#302 and requesting that the standard be expanded and also cover motorhomes and similar recreational vehicle types. Their construction includes plywood, carpeting, plastics of all types and their fuels include both gasoline and propane. Mr. Horkey stated that FMVSS#302 is deficient in that the testing consists of:

"1. Horizontal mounting of samples. (not typical in the real world)

2. The test chamber is too small and provided without enough holes for venting. The profuse smoke put off by many samples is so great that oxygen is kept from the sample, again creating a non-real world situation."

Mr. Horkey continued his petition "HAI has conducted hundreds of burn tests on many sample materials...mounted vertically or at a 45° angle (found to be the most critical for many cases). A kitchen match is usually enough to start the ignition. The majority of materials (plastics used in cars, light trucks and van, motorhomes and RV's) burn vigorously; emit profuse quantities of smoke (probably toxic) frequently float black particle into the atmosphere and produce flaming droplets. We recently had some samples of a pick up truck dash material burn for a total of 14 minutes. If one multiplies the small sample weights (2"x2" squares weighing less that one ounce) it can be understood why the interiors of vehicles burn so vigorously and long, even without the addition of liquid vapors."

NHTSA, after some months of consideration, denied the Horkey petition on the grounds that "there is no reasonable possibility that a rule amending standard #302 would be issued at the conclusion of this rulemaking proceeding. Therefore

this proceeding is terminated."

There have been other critics of motorhome construction and materials and the application of FMVSS#302. In "Recreational Vehicles - Hazardous to Your Health and Pocketbook" by Laurie N. Solomon and Clarence M. Ditlow, published by The Center for Auto Safety, June 1982, the following appears on pages 39 & 40.

Up in Flames
Federal Standards Omit Too Much

Extensive use of flammable materials in RV's makes them fire hazards, particularly in view of the time spent by owners in their interiors. The problems of fire safety in RV's are compounded by federal standards that omit most parts of the RV, by weak industry standards, by restricted exits, and by volatile foamed plastics. Although Federal Motor Vehicle Safety Standard (FMVSS) 302 governs the flammability of interior materials supplied by the chassis manufacturer it does not cover the living quarters of all types of RV's. Motorhomes, mini-motorhomes, and truck campers are covered, but travel trailers, fifth wheel trailers, vans, and van conversions are excluded. FMVSS 302 is only applicable to cars, multipurpose passenger vehicles (MPV's), trucks and buses.

The National Bureau of Standards (NBS) criticized FMVSS 302 as being an ineffective standard. In a 1974 letter to NHTSA, James Winger, Head of Product Flammability Research in the Center for Fire Research at the NBS wrote:

'Our data shows that the major fire hazard of upholstered furniture and bedding in homes is smoldering ignition from cigarettes and the resultant smoke and lack of oxygen. Since campers and trailers are used as dwellings while camping, we would expect the fire hazard to be similar to that found in homes. If this supposition is correct, an appropriate standard must measure the resistance to smoldering rather than an open flame burn rate.'

Buses, like most commercial vehicles such as truck/tractors spend most of their time in actual operation. "Downtime" is kept to a minimum and sometimes maintenance is perfunctory. Deteriorating conditions can sometimes be overlooked and frayed or chafing electrical cables, hydraulic or fuel lines can be left for the next routine service and inspection. Sometimes this condition worsens and a fire will develop. A common cause of both truck/tractor and bus fires is one of

electrical origin where insulation on battery cables has worn against the frame or other metal component of the vehicle. Typical of such cases was a fire in the battery compartment of a 1981 GMC city transit bus.

The major fire damage was confined to the area of the battery compartment located forward of the rear axle on the left side at the lower level. The fire climbed the left side of the bus exterior, melting the windows and diminishing toward the top. The interior of the bus showed a corresponding but lesser pattern. The battery compartment and cover door showed a greater concentration of heat and fire damage at the rear than toward the front. The batteries were in place but had been heavily damaged by heat and fire. Again, the greater damage was to the rear.

These batteries were wired in series to produce the 24 volt system necessary for the operation of the bus. At the forward section of the compartment a converter-charger unit had fallen from its mounting and its cooling fins had melted in a face down position. Major connecting cables were still attached and while burned did not show signs of shorting. The battery hold down bracket showed evidence of heat and fire, but did not show evidence of any electrical contact.

In examining the battery compartment toward the rear it was noted that a main positive cable leading from the most rearward battery unit was missing. This cable leads to the rear of the compartment to a switch connection on the compartment wall. In examining similar unburned vehicles, it was noted that this cable should pass to the outside of the rear battery hold down bracket. One unburned bus examined revealed this cable had loosened at its wall connection and had rotated toward the center line of the bus, allowing contact with the hold down bracket. In examining the battery cable insulation at this contact point it was noted that the insulation was wearing and would soon allow contact to be made between the battery hold down bracket and the positive battery cable.

Examination of the battery hold down bracket revealed it to have melted at its inward side in a pattern resembling the circular shape of the battery cable. The outside of the bracket showed a copper colored sheen strongly suggesting that molten copper had flowed over its exterior. The entire length of the cable was missing and could not be found. It probably was left at the scene of the fire under the assumption that it had no value.

It should be noted that when copper and molten aluminum combine they form a low melting point alloy that melts at 1018°F. (548°C). This temperature is below

the melting point of aluminum 1220°F. (660°C) and far below copper 1981°F.(1083°C). Much of the copper strand positive battery cable would have been destroyed under these conditions.

It is my conclusion that this fire was caused by worn insulation allowing the positive battery cable to make direct contact with the aluminum alloy rear battery hold down bracket. The heat generated by this short circuit caused the insulation to burn and ignite other flammable material in the battery compartment.

In another bus fire case salt air, road dirt and water formed an electrolyte over the insulator on a solenoid, causing current to pass directly to it's metal body. This resulted in an engine compartment fire that did over $60,000 damage to the bus.

The expulsion of transmission fluid under pressure has been the cause of bus and truck/tractor fires as well as passenger cars. A case in point is the following account of an investigation into the cause of fire in a Flyer transit bus. The vehicle was powered by a 903 Cummins V8 diesel rear mounted transverse engine. The transmission was an Allison model #V730-D. The transmission had been installed two days prior to the fire.

The bus was severely damaged by fire, both internally and externally. The fire obviously originated in the rear engine compartment and diminished in intensity toward the front. The engine compartment showed more damage to the right side than the left, particularly at the lower right side. The left valve cover had melted. The left side of the transmission case had melted and ran down the side, exposing the torque converter housing. At the top of the transmission housing a sunken area was located, caused by melting. The alternator mounted above the transmission had melted. The transmission fluid filter mounted to the left and above the transmission had burned and fell from its mounting plate. Flex hoses carrying transmission fluid had burned away.An oil line leading from the engine oil pan to the block had burned away. Wiring was heat damaged with insulation burned away. A line leading from the starter solenoid to the alternator was burned in half a few inches from the solenoid. There was additional beading of copper on the wire toward the alternator. The battery box revealed extreme heat generated in the rear units. There was fire and heat damage to the underside of the bus extending beyond the rear axle. An examination of all fuel lines failed to disclose any signs of leaks or failures that might have caused the fire.

Examination of the transmission fluid dip stick revealed all the fluid had been expelled. The inside of the tube and dipstick top assembly showed signs of

oxidation and the spring clips on the dipstick tube had lost their tension due to heat, allowing the dipstick to move freely in and out the tube. The engine oil was checked and the dipstick showed a slight amount of oil remaining in the oil pan.

Cooling for the transmission fluid is provided by a series of copper pipes that direct a flow of coolant from the radiator to a small tank located in the right rear section of the bus. This tank's capacity would not exceed 2 gallons. Transmission fluid is circulated through this tank and returned to the transmission via flexible hoses. The copper tubing is joined together at various points by means of rubber hose sections secured to the tubing by clamps. This allows the system to have some flexibility instead of remaining rigid. In this manner the chances of cracking due to vibration are lessened. At the tank's extreme right side the coolant is expelled from the tank into a copper line which then returns the coolant to the engine block where it joins coolant circulating in the engine. The coolant is then extracted from the engine block, cooled by the radiator and the cycle repeats itself.

All the rubber hose connectors for this cooling system were burned away except the fitting at the return side of the transmission coolant tank. This hose showed evidence of tearing and disintegration not connected with the fire.

Based on examination of this vehicle and interpretation of the events prior to the fire it appeared the following occurred:

1. The rubber connecting hose between the transmission coolant tank ruptured, allowing the loss of fluid.

2. The transmission began to rapidly overheat, causing a build up of pressure inside the case.

3. This pressure caused the expulsion of superheated transmission fluid through the dipstick tube. Partially vaporized particles of this fluid were immediately ignited on contact with the hot surfaces of the exhaust manifold.

4. This fire spread rapidly throughout the engine compartment. During the course of the fire electrical shorting was caused by wiring contacting metal and resulted in the conditions observed.

This theory was substantiated by the reported statement of the driver that there was "smoke and liquid coming out from the rear of the bus" prior to the fire. This was probably steam and liquid coming from the ruptured coolant line.

Bus fires, while not as common as automobile fires, do occur with sufficient frequency to present the motor vehicle fire investigator with special problems.

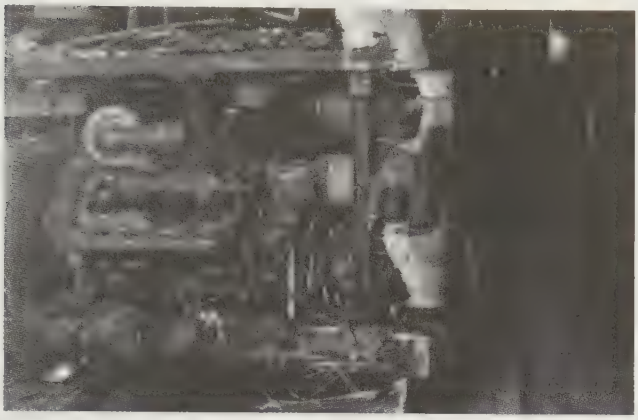
As observed by consulting mechanical engineer Joseph Carleton in his SAE Technical Paper #922456:

> Buses contain a number of unique hazards. Often, they have a people-density much greater than would be allowed in a building by fire codes. They may be moving at speeds of 70 MPH (or more!) The exits are often inadequate. There is the possibility of riders in wheelchairs, who would be nearest the front door. In the event of a fire, getting the wheelchair-bound passenger(s) off first would delay getting the other passengers off.

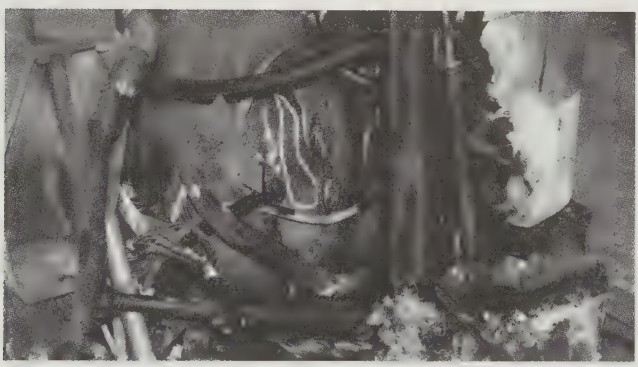
Of the seven bus fires reported in Mr. Carleton's paper, electrical short circuits were responsible for the cause and origin in six cases. This closely parallels our own experience in bus fire investigations.

In a case we handled in Honolulu,Hawaii we submitted the following report after our investigation into a fire occurring in a 1992 Van Hool T-800 Bus:

> According to information received, the fire occurred at about 10 AM on June 8th while the bus was in operation. The driver had reported no driveability or operational problems with the vehicle. He was signaled by a passing motorist who indicated to him that there was a fire in the rear of the bus. He pulled over, parking at 1932 Lustana Street, where thirty-five passengers were discharged. None were reported as injured.
>
> The Honolulu Fire Department was notified at 10:15 AM and arrived on scene at 10:20 AM.
>
> On July 31 examination of the vehicle began at the maintenance facility located at 435 Kalewa Street, Honolulu. It was identified and had a manufacture date of April 1992. The manufacturer isVan Hool NV, Lier 2500, Belgium. The recorded mileage is 85,945. The passenger seating capacity is 58. The overall length is approximately 50 feet with a width of 9 feet. The tire sizes are 12 R 22.5. The wheel totals are ten, two front, four drive and four "tag". The bus is powered by a 300 horse power Cummins diesel engine. The transmission is an automatic manufactured by Allison Corporation. Electrical power is supplied by a 24 volt system powered by two 12 volt direct



This devastating fire in the engine compartment of a transit bus...



was caused by expulsion of transmission fluid from the filler tube at arrow.

current batteries mounted at the right and left rear compartments of the vehicle. There are two 120 AMP alternators mounted in the engine compartment. These are air cooled units with internal voltage regulators.

The fire damage was at the rear of the bus and the rear wheel wells. There was greater fire and heat damage to the bus interior at the rear. Seats on the right side were burned more severely than the left. There are drain holes in the rear wheel wells and fire entered the interior through these holes. Entry was more easily accomplished on the right side than the left. The tires and wheels had been removed. Examination of the treads and side walls did not reveal any evidence of having been run flat causing heat to build up and cause ignition.

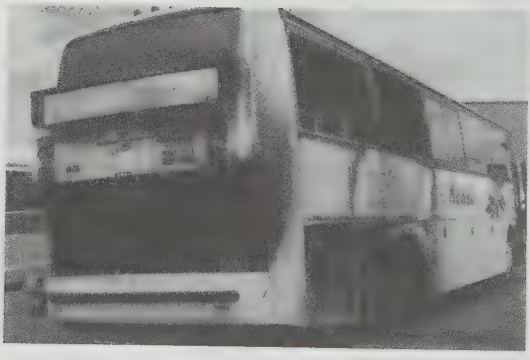
The vehicle's fuel system was examined to determine if any leak occurred that may have figured in the origin of the fire. Fires involving diesel fuel as the primary cause are rare since the fuel flash point is 100°Fahrenheit for flammable vapors. There was burning of the fuel lines in the engine



1) View of bus from left front.

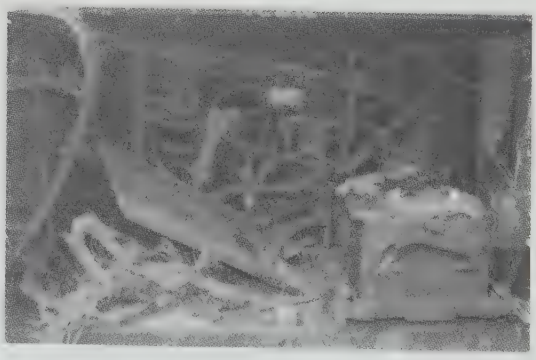
2) View of bus from right front.

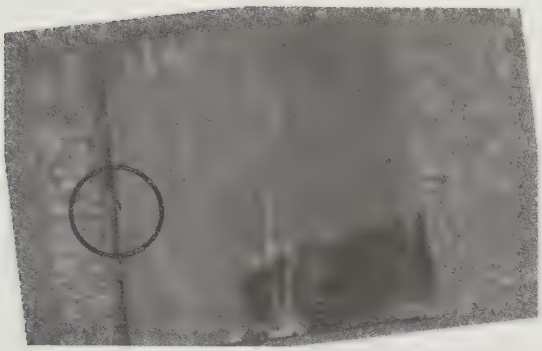




3) View of bus from right rear.

4) Battery compartment, right side.
Cover for relay panel is at arrow.

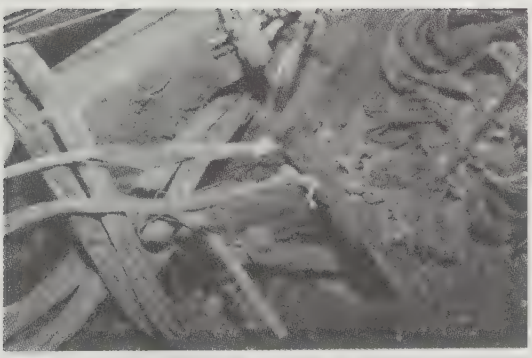




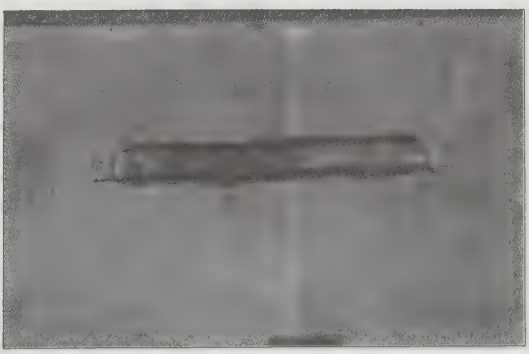
5) Relay cover removed. Note hole caused by electrical short.



6) Closer view, from inside panel cover.



7) Arrows show wiring that made contact with panel cover.



8) Battery hold down bracket removed. Note corrosion and molten copper.

compartment, but there was nothing to show this being the origin. The driver reported no problem that would indicate a fuel leak.

The engine oil level was full and was clean. There was no evidence of an oil leak being the cause of the fire. The underside of the engine was not exposed to fire and the oil filter case was not sooted. Transmission overloads have been determined to cause expulsion of transmission fluid on hot exhaust manifolds where ignition occurs. A check of the transmission fluid revealed the case to be full and the color good. There was no burned smell to the fluid and there were no burned particles suspended in the fluid as is found in cases where expulsion has occurred. There was no driveability problem reported by the driver that would indicate a transmission problem.

The braking system for this bus is assisted by an electromagenetic drive shaft retarder unit. Wiring to this unit had all the insulation burned away, but there was no evidence of arcing. Consultation with maintenance personnel revealed that there is no friction contact in this unit since it is controlled by an electromagnetic field.

The fire had burned into the air conditioning system, but there was no evidence that this was the cause of the fire. While the gas used in this system is considered not flammable, the light oil used to lubricate the seals occasionally does leak and burn. There was no report of any problem with the air conditioning system and it was eliminated as a possible cause.

The power steering reservoir is mounted on the right side of the engine compartment toward the rear. This reservoir is plastic and was totally burned away. The filter had dropped out and was lying on the metal shaft next to the engine. Power steering fluid will burn to the same degree as transmission fluid and can be a fire source when these vapors are exposed to high heat. This generally happens when the wheels are turned and pressure builds in the system. There is no evidence that there was excessive pressure in the system and no reports of any steering problems before the fire. This system was not involved in the origin or cause of this fire.

The cables from the right side battery system had been cut by the fire department personnel. A "hold down" bracket for the battery had been pushed askew and had been severely corroded at one end, possibly by the electrical contact. The other end. nearest the rear of the bus had a motlten copper deposit inside it. The other "hold down" post had its upper section and retaining nut completely melted away. Tracing of the battery cables revealed

them to curve inward toward the engine and then curve toward the front section of the engine compartment. Wiring from both alternators joined this wiring bundle and traveled toward the front. Wiring from this bundle turned downward and furnished power to the starter. As the bundle continued toward the front, there was a separation with wiring leading to the right underside and the other section curving upward and into a relay panel and junction box. The relay panel box is made of sheet steel about 1/16" in thickness. A cover piece composed of the same material was found in the right rear engine compartment. Examination of this piece revealed it to contain a melted hole about 3/4" long and 1/2" wide. There was molten copper on the edges of this hole, indicating contact with electrical wiring. Examination of the cover revealed the contact to have come from the outside. Two wires in the bundle leading from the battery showed evidence of extreme shorting. A cable and retaining bracket under these wires showed evidence of melting. There was what appeared to be secondary shorting in nearby wiring.

When the relay cover plate was placed in its original position, the two most severely shorted wires could be placed in the exact position where the hole was burned. Examination of a similar unburned bus showed that this wrapped wiring bundle was placed in such a position that it was tightly pushed against the relay box cover.

Copper melts at a temperature of 1981°Fahrenheit. One may readily see that the temperature required to melt copper and steel would be extremely high and would be rapidly communicated to other nearby material, such as wiring insulation, soft hoses, wrapping tape, road film, etc. Secondary shorting would occur when insulation burned away and wiring made contact with other metals.

It is my opinion that this electrical short and resulting arcing occurred when insulation on the wiring bundle lodged against the cover plate for the relay box wore away by friction and allowed contact between the wiring and cover. The resulting fire spread thoughout the rear of the bus.

Clever sabotage was concluded to be the cause of some truck fires in the San Francisco Bay Area.

This series of fires involving garbage trucks was first though to be caused by hydraulic hoses blowing off under pressure. At one of the scenes fire department

and company personnel were successful in obtaining a sample of the fluid from the roadway that had come from the rear of the cab just before the fire occurred. Samples were also obtained from the hydraulic oil tank of the truck for comparison. The laboratory reported back that the two oils were completely dissimilar. The oil from the tank was a hydrocarbon base while the one from the roadway was silicone base. This, of course, added to the confusion since silicone base oil has a flash point of over 600°F and is not considered readily combustible. A check of the garbage company's complete maintenance facility failed to reveal any such oil.

The fires, which numbered five in all, occurred within the first 2 $^1/_2$ hours of operation and were centered at the rear of the cab. The vehicles were all cab-over engine models and the drivers first noticed the fire as a flash with intense heat at their backs. It was finally concluded that some flammable substance, probably in a plastic bag, was being placed in the engine compartment on the exhaust system at the rear of the engine block. When exhaust heat melted the plastic bag the superheated vapors from the liquid caught fire.

We recommended a pre-operation inspection of all vehicles be instituted to include the cab, engine compartment and undercarriage. When this began to occur the fires ceased and did not resume. Our investigation revealed one employee who worked in the maintenance operation had previously worked for another company who had experienced fires in their trucks. These fires were immediately identified as arson and were not sophisticated like the series that we had under investigation. Perhaps he learned something in between times. At any rate, when investigation intensified and preventative methods were undertaken the fires stopped.

Some municipalities are introducing methanol powered buses into their fleets. An engine compartment fire developed in one of these buses in California and we were requested to make an inspection to determine if the methanol system was involved in the origin or cause of this fire. We did find a fuel hose that was leaking due to its rubbing against a radiator hose. This would allow fuel to run down the left side of the rear mounted transverse engine.

The fire and heat damage in the area of the leak and extending downward did not appear to be severe enough to confirm a fire of fuel origin. While the flash point of methanol is much higher than gasoline (-45°F), it is about one half that of diesel fuel (110°F). The combustion heat of methanol is 9600 BTU's per pound. Gasoline produces 20,800 BTU's per pound and diesel fuel 18,900 BTU's

per pound. From this it can be seen that a fire in the engine compartment of the bus resulting from a methanol fuel leak would not create as much heat, and therefore damage, as a diesel or gasoline generated fire. Since methanol vapor is heavier than air, it would sink to lower levels of the engine compartment where ignition could have been provided by exhaust manifold heat or a stray electrical spark. While there were other circumstances to consider in this case the methanol leak was deemed to be most significant and the cause of the fire.

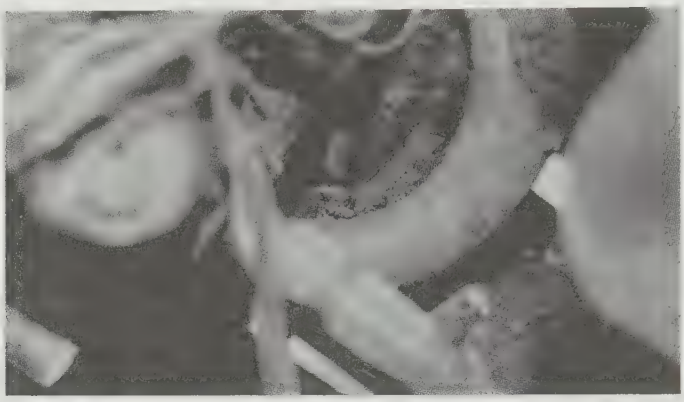
Construction vehicle fires are generally caused by electrical malfunctions. Other causes, such as fuel leaks, friction, overheating etc. should not be overlooked but the primary cause frequently turns out to be electrical.

A recent case involving an articulated lift truck was no exception to this rule. The operator started the vehicle and left it idling unattended for a warm up period of about ten minutes. A witness saw fire and smoke coming from the rear of the engine compartment forward of the radiator. Preliminary investigation suggested that a problem may have developed around the turbo charger, but once the top engine cover was removed we could observe that the heat concentration came from a lower area. Our examination continued and we quote from our investigation report:

"It was noted that at the point where the positive battery cable passes over a major hydraulic line leading from the forward pump to the rear drive unit there was a spot where the exterior of the line had burned away. This indicated that the battery cable insulation had become extremely hot and caused the hydraulic line exterior to deteriorate. In following the cable on its way to the engine starter motor mounted on the left lower side of the engine, it was noted that it passed under this major hydraulic line. The battery cable was pulled from under the hydraulic line and an area of electrical arcing was found. The hydraulic line was removed and on its underside, at a point where contact was made with the battery cable, there was evidence of electrical contact and a hole had been burned into the hydraulic line. The surrounding area showed evidence of fire and heat.

From our examination of this vehicle and its systems it was concluded that the positive battery cable, leading from the battery to the engine starter made contact with the hydraulic line leading from the pump to the rear of the vehicle. This contact was made after vibration and other motion caused the battery cable insulation to wear. This allowed direct contact between the bare cable and the metal braided cover of the hydraulic line and produced an electrical arc with a temperature in excess of 1900°F. This burned into the hydraulic line and fluid

under pressure was released. Resulting vapors were ignited. Hydraulic fluid is a light weight oil and has an ignition temperature of around 500°F. The heat produced by the electrical arc was more than sufficient to cause ignition. An intense fire followed."



A fire in a construction vehicle was caused when insulation wore away from an electrical cable and shorted against the main hydraulic line.

A rare occurrence in diesel engine operation is the "Runaway" caused by the ingestion of vapors from outside sources into the engine.

We were requested to investigate the cause and origin of an engine compartment fire in a 1991 Freightliner tractor that was attached to pressurized tank trailers used for the transportation and unloading of Liquified Petroleum Gas (LPG). The fire, had it not been contained, would have resulted in tremendous damage and danger to people in the area since it occurred in a propane storage yard. Our investigation report is quoted:

"The driver of the vehicle stated that he was making a delivery of butane. He hooked up his vapor hoses, then the liquid hoses and started the pumping operation by engaging the Power Take Off Unit. This is actuated by engaging the clutch since the unit is powered by the engine. He noticed that the lines were not extracting vapors from the customer's tank and were in a vacuum status. He disengaged the PTO and made sure the valves were all open on the vapor lines. He reversed the 4 way valve mounted on the compressor, restarted the truck and again engaged the PTO. The compressor "pop off" valve began to spin faster and to emit popping noises. The engine then began to increase its RPM's. The engine speed increased and all attempts by the driver to shut the engine down failed. He then attempted to put the vehicle in gear to cause the engine to stall, but before he could do this he noticed flames coming from the hood, the doors and the sides of

the vehicle. He was able to jump from the driver's side through the flames and exit the vehicle. The driver attempted to stop other vehicles in the area for safety reasons and requested that someone call the fire department. This was done and a short time later the fire department arrived and the fire was extinguished.

The vehicle was examined and identified by the VIN and license. There was some superficial burning of the interior door panels but little damage inside the cab. The hood exterior was more severely burned on the left than on the right side. The hood interior was burned in a similar pattern, almost through at a point just about over the ether cold start canister. The hood, of course, is of fiberglass material.

The burn pattern on the left side of the engine extended from the top down to the wiring for the starter motor. The burning on the right side did not extend as far forward as on the left side. A careful examination of the electrical connections and system from the battery box to the engine compartment was made. No evidence of malfunction or arcing was found. A trace was made of the fuel system from the tanks to the injectors. No evidence of any leaking was found. An extended fire pattern was observed in the area of the ether cold start canister mounted on the left side of the block at high center. The canister was pressure bulged in the direction of the firewall. The plastic line leading from the canister to the intake manifold had burned and melted away. Air conditioning lines in the vicinity had ruptured in the direction of the ether canister.

The function of the ether canister is to provide a shot of volatile vapors directly into the intake manifold at start up. Ether vapors have a flash point of -49°F and an ignition temperature of 356°F. Diesel fuel by comparison has a flash point of 100°F. The use of the ether enables the engine to be started and then the normal operating temperature raises the diesel fuel to the point where vapors can be ignited in the compression stage. A diesel engine does not require an electrical system to furnish an ignition spark as does a gasoline engine. The fuel/air mixture is ignited by the heat created in the compression stroke. When the engine began to race at excessive speed shutting off the switch would, of course, have no effect on the continuing operation of the engine.

"Runaway" diesel engines occur when airborne combustible vapors are taken into the intake system via the engine air supply. This has the same effect as continually depressing the accelerator and under extreme conditions the engine will literally tear itself apart and explode into a fireball. The only sure way to stop an engine under these circumstances is to shut off the air supply.

The driver reported difficulty with the vapor lines leading from the customer's tank. The "pop off" valve on the compressor confirmed this problem. While butane vapors are heavier than air and will sink, it should be remembered that the operation of a diesel engine requires large quantities of air. This would cause large amounts of air to be sucked into the intake system and with this would go a quantity of butane vapor. This would and did cause the engine to run at increased RPMs. The increase in RPMs would also cause a rapid rise in the turbo charger temperature, especially in a stationary operation. Exhaust surfaces under such conditions can exceed 900°F. This is more than sufficient to ignite butane vapors that have an ignition temperature of 761°F.

My conclusion was that the fire was the result of escaping butane vapors making their way into the air intake system of the vehicle. This caused a "runaway" condition in the engine and overheated exhaust surfaces ignited the butane vapors. This fire communicated to air conditioning hoses, wiring insulation and plastic lines on the firewall. The ether canister discharged parts of its contents under pressure and this increased the intensity of the fire." Protective devices can be installed on diesel engines to shut off the air supply and are commercially available.

The following list of causes for bus, truck/trailer and construction vehicles is recorded in our 1988 publication "A Survey of Vehicle Fire Causes". You will note that the electrical problems were a factor in almost one half of these cases.

Type	Cause
1974 International Conv.	Electrical short in radio
1977 Kenworth	Arson
1978 International 2 axle	Failed turbo bearing caused oil fire
1978 Cat Tractor	Hydraulic line failed
1978 Frieghtliner COE	Unknown
1979 Case Tractor	Electrical short in wiring under floorboard
1980 White Truck/Tractor	Short in radio-caused a fire
1982 Gillig Bus	Battery cable shorted against starter and block
1985 Van Pelt Fire Truck	Transmission overheated and expelled fluid
1985 Timberjack-Feller-Bucher	Electrical short in cables contacting hydraulic lines
1986 Volvo Garbage Truck	Arson-Sabotage

Recreational Boat Fires

The Boat — A vessel is any form of watercraft, other than a seaplane, capable of being used for transportation on the water. A boat is generally considered to be a small vessel. This book will only deal with boats under 65 feet in length and intended for recreational use. Recreational boats come in all sizes and shapes and would require a massive volume to adequately describe them in detail. We shall concentrate on boats utilizing machinery for propulsion or electrical generation. These boats range from small outboard powered open boats to large luxurious yachts. Even some sail boats have engines for propulsion in the event the wind does not cooperate.

Each recreational boat built or imported into the United States after August 1, 1972 is subject to the provisions of the Federal Boat Safety Act of 1971. There are two significant sections of this act that are of interest to the fire investigator. First is the requirement that all recreational boats are to have a 12-character hull identification number or "HIN", permanently attached to the hull. The composition of this number is as follows:

Calendar Year Method:	A B C	1 2 3 4 5	2 1 8 2
	Manufacturer Code	Hull ID	Month and Year Built
Model Year Method:	A B C	1 2 3 4 5	M 8 3 D

The choice of year method used is up to the manufacturer. When using the calendar year method, the 9th and 10th characters represent the month and the 11th and 12th characters represent the year. When the model year method is used, the 9th character is always the letter "M", the 10th and 11th characters are the model year, and the 12th character is the month in that model year. The boat model year runs from August 1 through July 31. As you can see from the example, the numbers appear to represent a different month and year of construction, but both numbers show that the boat was constructed in November of 1982. The following is a breakdown of the letter identifier for the model year months:

A=August
B=September
C=October
D=November
E=December
F=January
G=February
H=March
I=April
J=May
K=June
L=July

The significance of this number is that it tells the investigator what, if any, regulations apply to the fuel, electrical, and ventilation systems when the boat was built. In other words, it tells you what you can expect to find once you get into the system analysis.

The second significant section of the Federal Boat Safety Act of 1971 is the establishment of regulations governing the design and installation of the fuel, electrical and ventilation systems on recreational boats.

In addition to Federal regulation of recreational marine products, the marine industry itself has established standards for fuel, electrical, and ventilation systems. This is through the Boating Industry Association's (BIA) certification program.

The Inboard Marine Engine — An inboard marine engine is basically the same engine as those used in automobiles and trucks. In fact most modern marine inboard engines began life as an automobile engine block. The basic engine is then fitted with a special marine carburetor, alternator or generator, distributor, starter, fuel system, exhaust system, and cooling system. It is then connected to the marine transmission, which only contains a forward and reverse gear. Power is then transferred by way of an output shaft to a propeller or high speed water pump to drive the boat through the water.

There are two basic inboard engine drive systems commonly used today. One is the direct drive. This engine is mounted near the center of the boat with the drive shaft exiting the rear of the engine. It then passes through the bottom of the boat. The propeller is attached directly to this shaft. The other method is called a "V" drive. The engine is reversed and located in the rear of the boat with the drive shaft exiting the transmission toward the front of the boat. The drive shaft is connected to a second gear box which reverses the direction of the drive shaft, (but not the direction of rotation). The shaft then passes through the hull and is connected to the propeller. Both of these installations require that the boat be fitted with a rudder to control the direction of the vessel.

The Outboard Marine Engine — This is a special marine engine containing the engine, power train and transmission in one unit attached to the outside of the boat. All components except the fuel tank and the battery are located outside the hull. This unit usually drives a propeller, but can be fitted with a high speed water pump, (or jet pump), to provide the driving force. Unlike the inboard powered boat, this boat does not need a rudder. The entire unit is turned to control the

direction of thrust and thereby the direction of the boat. Outboard engines range in size and power from one cylinder and one horsepower to six cylinders and as much as three hundred horsepower. Outboard engines can be either two cycle (such as a lawn mower) or they can be four cycle (such as used in automobiles). Although these engines are especially designed and not modified automobile engines, the operating principles remain the same.

The Inboard/Outboard Marine Engine — (Commonly referred to as an I/O or inboard/outboard). This is the obvious next step in units designed for marine use. This engine is located inside the boat. The drive shaft exits the rear of the engine without using a transmission. The shaft then passes through the stern of the boat and is connected to the drive unit by a universal joint. The drive unit resembles the bottom half or lower unit of an outboard engine. It contains a right angle drive, the transmission and an output shaft. It also provides directional control in that the entire drive unit is turned to control the direction of thrust and direction of the boat.

A modification on this idea is the inboard/outdrive jet. This is also referred to as a jet drive or jet boat. The output of the engine is connected to a large high speed water pump located inside the hull. The output of the pump passes through the stern of the boat to a high pressure nozzle. Water is picked up by the pump and passed through the nozzle much like an aircraft jet engine. The high pressure output drives the boat. The nozzle is turned to change the direction of thrust and direction of the boat.

Electric Motors — Electric propulsion motors are gaining popularity as a replacement for outboard motors used for trolling. These units consist of a control head, a shaft extending below the water, and a sealed motor connected directly to a propeller. Power is supplied by a 12 volt battery.

The Fuel System — Since the engine is the heart of the boat, the fuel system is both its life blood and digestive system. Gasoline is stored in the fuel tank. It is drawn from the storage tank through a filter system and into the fuel pump and carburetor mounted on the engine. The carburetor mixes the gasoline with air and allows an atomized mixture to be drawn into the cylinder through the intake manifold. This is a simplified explanation of how any fuel system works. This is the type used in older boats and most outboard engine installations.

A long time ago, people involved in building and operating boats, and those who investigate accidents, began to realize that this simple system was not

adequate for boats. In earlier days, fuel tanks were made of any available material; tanks were not properly vented; the same hoses used for automobiles were being used in boats; no special provisions were made for engine parts; and nothing was done to insure system integrity. The end result of all this was a large number of boats and boaters lost as a result of explosions and gasoline fed fires. If you spring a leak in your automobile gas tank, where does the gas go? Under the car into an open and ventilated area. Not so in a boat. By the design of the boat, the fuel will run into the inner compartments, back to the engine, and "boom" we have another investigation. Prior to the Federal Boat Safety Act of 1971 and the regulations developed in the mid 70's as a result of the act, the fuel systems were whatever a manufacturer wanted to use. What he used depended on the quality of the manufacturer, whether he was concerned about product liability and whether he belonged to one of the industry trade organizations that had voluntary standards for their membership.

Beginning in the early 1970's, the recreational marine industry began to develop fuel system standards that were technically sound and tailored to boats. The standards were developed through cooperation of the National Fire Protection Association (NFPA), the American Boat and Yacht Council (ABYC), the Boating Industry Associations (BIA), and the U.S. Coast Guard. These standards became recognized as voluntary industry standards and were adopted by many manufacturers. These same standards became mandatory requirements under the Federal Boat Safety Act of 1971 with effective dates for specific requirements varying from August 1, 1977 to February 1, 1979 depending on the complexity of the requirement and the lead time necessary for compliance. These regulations affect all recreational boat manufacturers, dealers, engine installers, and boat owners. The system is designed for the sole purpose of prevention of fuel fires and explosions.

A brief summary of the major requirements of the fuel system regulations are listed below:

1. Permanently installed gasoline fuel tanks must meet impact, slosh, fire, and static pressure tests. The tanks must be made of specified non-corrosive material and must be adequately secured to prevent movement within the boat. The tank must be labeled to indicate compliance with the regulations: the manufacturer's name and address; the test pressure; the capacity; the material it is made of; and the month and year it was manufactured.

2. Each fuel tank vent must have a flame arrester that can be cleaned, unless the vent is itself a flame arrester. Vent flame arresters are made of layers of non-corrosive wire screens.

3. Diaphragm type fuel pumps must not leak from the pump if the primary diaphragm fails. This requirement calls for a means to prevent fuel from leaking back into the boat if the main diaphragm fails. This is normally accomplished by using a dual diaphragm pump with a sight glass connected between the diaphragms that will show a main diaphragm failure. The alternate method is to connect a "stripper" tube from the pump housing to the carburetor.

4. Electrically operated fuel pumps must not operate except when the engine is started or running.

5. Carburetors must be intended for marine use. Marine carburetors must meet certain leak tests before they are acceptable. If an engine uses an updraft or side draft carburetor, the carburetor must be fitted with a means to collect fuel from flooding and return it to the engine so it will be consumed. Most propulsion engines have downdraft carburetors, but auxiliary generators and some small engines have updraft or side draft types. The draft of the carburetors is determined by the direction of the airflow through the carburetor into the engine.

6. Except when used for a tank fill line, each tank fitting used with hose clamps must have a bead or flare to prevent the hose from sliding off.

7. Each metallic fuel line connecting the fuel tank with the fuel inlet connection on the engine must be made of seamless annealed copper, nickel-copper, or copper-nickel.

8. Each hose between the fuel pump and the carburetor must be "US Coast Guard Type A" hose. Each hose used for a vent line or fill line and each hose from the fuel tank to the fuel inlet connection on the engine must be "US Coast Guard Type A" or "US Coast Guard Type B" hose.

9. Each fuel line from the fuel tank to the fuel inlet connection on the carburetor must be above the level of the fuel tank top or have an anti-siphon device or an electrically operated fuel stop valve.

10. The entire fuel system must pass a static pressure test.

11. All engines must be equipped with an effective means of backfire flame control. Control can be obtained by fitting engine air intakes with Coast Guard approved flame arrestors or by ducting the air intakes outside the engine compartment to the open atmosphere. These open air intakes must be mounted so as to direct backfire flames away from the boat and its occupants. They must be vertical or directed over the stern of the boat.

As you can see, all of these requirements are intended to remove the fuel leg of the fire triangle. Once installed, this system must be properly maintained to be effective. Figure 1 is a simplified drawing of a typical gasoline fuel system, showing the major components and requirements. It is intended to be used in assisting you in tracing a boat fuel system.

Remember that these requirements only apply to gasoline engines, except outboard engines, that are used for electrical generation or mechanical power for propulsion. They do not apply to diesel engines.

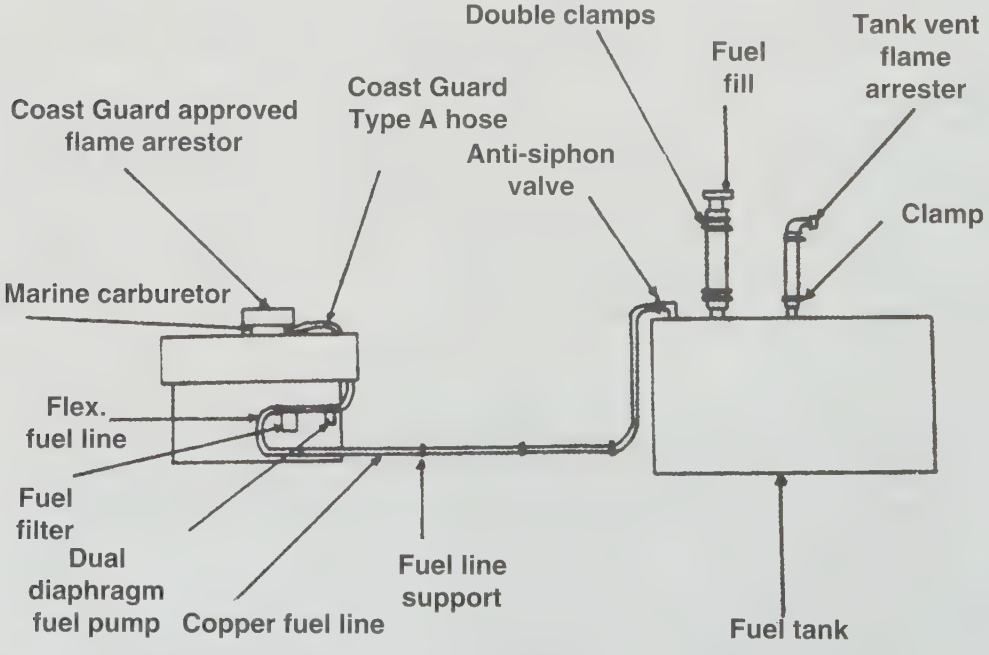


Figure 1
Gasoline Fuel System

The Direct Current (DC) Electrical System — The center of this system is the battery. The 12 volt battery has six compartments containing seven to fifteen plates. These plates are composed of lead and are submerged in a solution of sulfuric acid and water. This solution causes a flow of electrical current. This flow causes a difference in electrical potential at the battery terminals. The battery voltage, or potential, cause current to flow through the boat's electrical system. This flow is augmented and supplemented by the generator or alternator driven by a belt or gearing from the engine crankshaft and creates an electrical current by rotation of wire coils over an iron core. The current flow resulting is the same as previously described in this book.

The Alternating Current (AC) Electrical System — There is one other type of electrical system commonly found in recreational boats. This is the 120 volt or 240 volt alternating current system used to power such things as lights, refrigerators, air conditioners, or other major appliances. With the exception of the source this system is the same as a house electrical system. The source in this case is either from "dock side" with a heavy duty extension cord or from a self contained auxiliary generator. The power supplied from the dock is referred to as "shore power" and the heavy duty cord is referred to as "shore tie". The generator consists of either a gasoline or diesel engine, which is totally independent of the propulsion engine, driving an AC generator. The output of the generator, or shore power, is fed through a circuit breaker panel or distribution panel to the various items in the boat. This type of system will normally be found on larger boats which have space for large appliances and the generator unit. Parts of this system may be found to be intermixed with the DC system. For instance, light fixtures will contain a 12 volt DC circuit and bulb to be used when the generator is not running or the shore power is not connected. The same fixture will contain a 120 volt AC circuit and bulb which is used when the AC power is available.

Electrical System Requirements — As in the fuel system for boats, little attention was paid to the electrical systems until people began to realize that they were the ignition source in most boat fires and explosions. Again, the experts began to take a closer look and developed guidelines and standards to be followed in boat construction. These standards were developed parallel to, and in conjunction with, the fuel system standards and became regulations under the Federal Boat Safety Act of 1971. As in the fuel system regulations, the electrical regulations apply to all boats having gasoline engines, except outboards, that are used for electrical generation or mechanical power for propulsion. This set of regulations were phased in between August 1, 1977 and February 1, 1979.

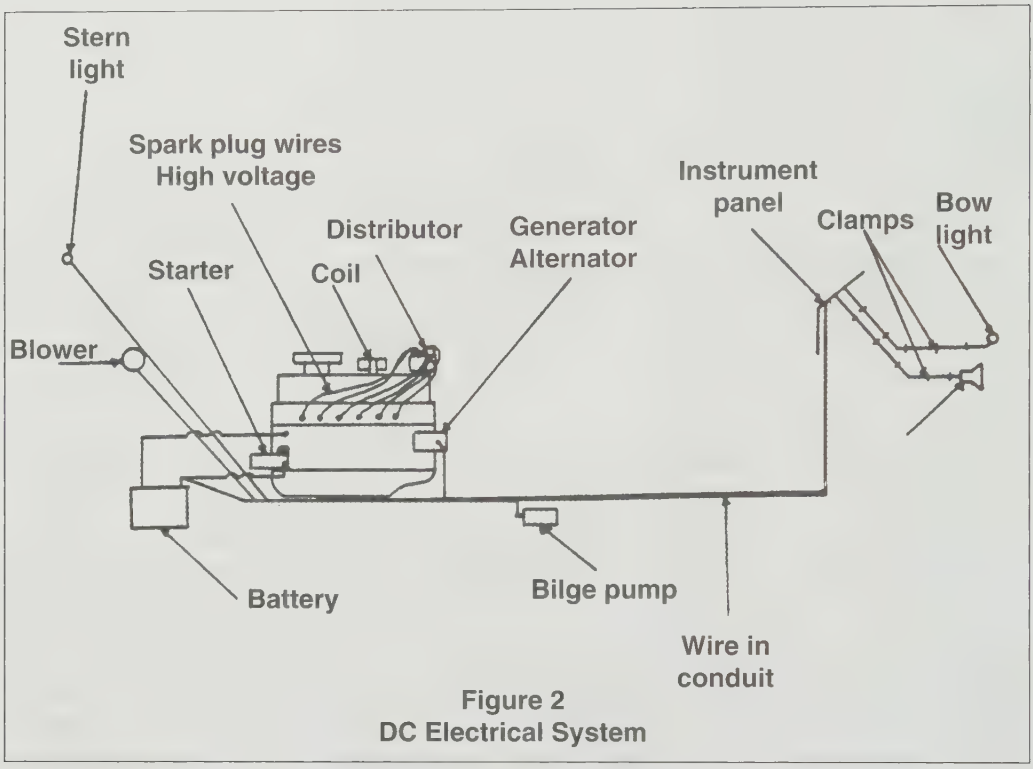


Figure 2
DC Electrical System

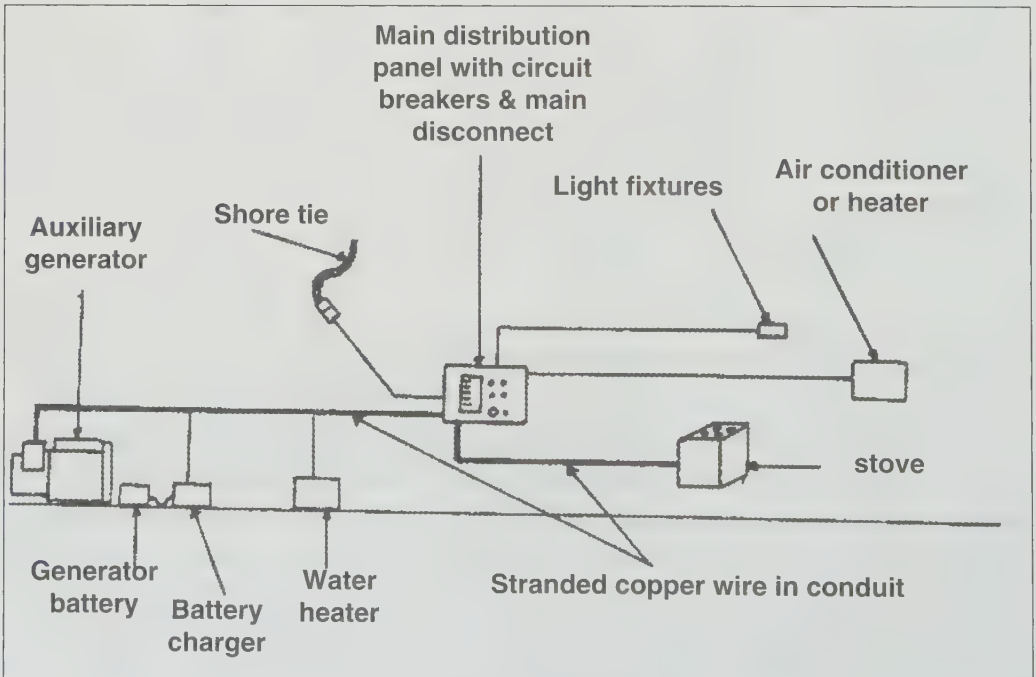


Figure 3
AC Electrical System

A brief summary of the major requirements of the electrical system are listed below:

1. Each battery must be installed so that metallic objects cannot come in contact with the ungrounded battery terminals. This is normally accomplished by housing the battery in a battery box or by covering the terminals with rubber "boots".

2. The battery cannot be installed above or below a fuel tank, fuel filter, or fitting in a fuel line. Additionally any fuel line running within 12 inches of the top of the battery must be covered with insulation.

3. As a battery is charged, hydrogen gas is generated inside. This gas escapes the battery through the caps and can form an explosive mixture. This gas must not be allowed to accumulate inside the boat.

4. Alternators, generators and starter motors must be intended for marine use. The items are specially designed for marine service and use seals, flame arresters and "potting" to make them capable of operation in an explosive atmosphere without igniting the environment.

5. Each conductor must be insulated stranded copper wire.

6. Each electrical component in the engine room must be fully enclosed or otherwise protected to prevent the ignition of gasoline vapors.

7. Each ungrounded current carrying conductor must be protected by a manually reset, trip free, circuit breaker or fuse.

8. Each conductor or group of conductors must be supported by clamps or straps not more than 18 inches apart unless the conductor or group of conductors is enclosed in a rigid duct or conduit.

9. A soldered connection must not be the sole means of connection between two or more conductors or between a conductor and connector. Wire nuts are not allowed. Crimped terminals and connectors are preferred.

As you can see with the electrical requirements, they are intended to remove the second leg of the fire triangle, the ignition. Figure 2 is a simplified drawing of a typical DC electrical system and Figure 3 is a simplified drawing of a typical AC system.

Remember that like the fuel system regulations, the electrical system regulations apply to gasoline engines. They are not required on diesel engines.

The Ventilation System — Ventilation systems are not new. They have been required since 1940. In those early days of powered boats it was determined that if fuel fumes were allowed to collect in the bottom of engine and fuel tank compartments an explosive situation could result. The Motor Boat Act of 1940 required boats to have some sort of system to remove fumes that might collect. This regulation required certain areas of a boat to be ventilated by the boat owner. In summary, the act stated:

All motorboats or motor vessels, except open boats, the construction or decking over of which is commenced after April 25, 1940 and which use fuel having a flash point of 110° F. or less, shall have at least two ventilator ducts, fitted with cowls or their equivalent, for the efficient removal of explosive or flammable gases from the bilges of every engine and fuel tank compartment. There shall be at least one exhaust duct installed so as to extend from the open atmosphere to the lower portion of the bilge and at least one intake duct installed so as to extend to a point at least midway to the bilge or at least below the level of the carburetor air intake. The cowls shall be located and trimmed for maximum effectiveness so as to prevent displaced fumes from being recirculated.

As you can see from the reference to a 110° F. flash point, this requirement was intended for only gasoline powered boats. The principle behind the ducting rests in the fact that gasoline fumes are heavier than air. The intake, or fresh air, duct extends to the area of the carburetor or the top of the fuel tank where most fuel fumes would originate. From there the "heavy" fumes will flow to the bottom of the compartment. The exhaust duct must be located in this lower area to collect and discharge these fumes.

When these regulations first came out, there was no requirement for any blowers or fans to be used in the system. Early attempts at ventilation provided "natural ventilation" to the compartments and relied on wind flow or movement of the boat to circulate fresh air into and out of these hazardous areas. The other problem with this early ventilation system was that it was the responsibility of the

owner to install and maintain it. This resulted in inadequate systems being installed and fires and explosions continued.

In order to provide better ventilation and more efficient removal of gas fumes, the mechanical blower was developed and gained widespread use throughout the boating industry. This is simply a small DC electrical powered fan installed in the exhaust duct. The intake to the fan comes from the duct leading to the bottom of the boat. The outlet of the fan is connected to the duct leading out of the boat. The fan then acts like a common vacuum cleaner and picks up the dirty air from the bottom of the boat and blows it out the duct. This system proved very effective in reducing fires and explosions and was adopted by the industry as a standard many years ago. Remember that under the Motor Boat Act of 1940, the ventilation was the owner's responsibility, so this was purely a voluntary standard.

The Coast Guard adopted the industry standards, with a few modifications, and published manufacturer requirements for ventilation effective August 1, 1980. A manufacturer is now required to provide a blower system in all boats with gas engines (either for propulsion or used as generators) in closed compartments. The engine compartment, and under certain circumstances, the fuel tank compartment must still have natural ventilation. The fuel tank compartment need not be ventilated if it contains a permanently installed tank that is vented to the outside of the boat and the compartment contains no spark producing devices. Boats having powered ventilation systems are required to have a statement near the ignition key which tells the operator:.

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WARNING: GASOLINE VAPORS CAN EXPLODE
Before starting engines operate blower for 4 minutes and check the engine compartment bilge for gasoline vapors.

Ventilation systems are not required by federal regulation for boats using diesel power. Ventilation is recommended by the boating industry as a voluntary standard and will be found on some of these boats.

The Exhaust System — This is the part of the system that removes the waste products from the boat's engine or auxiliary generator and carries them outside the boat. When the engine exhaust valves open, spent gases enter the exhaust manifold. From there the gases pass through a series of pipes and flexible couplings to the outside of the boat. On outboard and inboard/outdrive installations, the exhaust gases are fed through the drive unit to the propeller area where they exit underwater. This provides a silencer or muffler effect. There are two

types of exhaust systems used for marine installations. These are the wet exhaust and the dry exhaust.

The dry exhaust is similar to an automotive exhaust system where cooling is provided by air and it contains a muffler or silencer. The sections of a dry exhaust that are inside the boat are insulated and installed with maximum clearance for combustibles to prevent ignition.

A wet exhaust system uses water to cool the exhaust pipe and exhaust manifold on the engine. Engine cooling water is circulated through a water jacket around the exhaust manifold. Water is also injected directly into the exhaust pipe where it connects to the exhaust manifold.

Cooking and Heating Systems — There are many sources of fuel and a variety of appliances that are used for cooking and heating on boats. Due to size of the equipment and space limitations on boats, installed cooking and heating equipment will normally be found on boats longer that 20 feet.

The types of fuel ordinarily used for cooking are alcohol, liquefied petroleum gas (LPG) and electricity. Fuels such as wood, coal, kerosene, and solidified alcohol (canned heat) are rarely used these days. Electric stoves obviously are safe when properly used and maintained. These require no further discussion.

Alcohol is perhaps the most popular fuel for cooking on boats without AC electrical power. Alcohol is an expensive fuel but has an advantage on boats in that an alcohol fire can be extinguished by water. The principal of operation of an alcohol stove is the same as a gasoline camp stove. The alcohol is stored in a tank which has a hand pump installed, or in a gravity feed tank. The alcohol is pressurized (by pump or gravity) which forces raw fuel into a primer cup under the burner. The raw fuel is ignited with a match. The burning raw fuel heats the burner which causes the incoming alcohol to vaporize and burn with a small blue flame. It has a pattern similar to natural gas and is adjustable.

Liquefied petroleum gas is used for cooking and occasionally for heating. LPG (usually propane or butane) derives its name from the characteristics it exhibits. Under moderate pressure in a storage tank, the gas liquefies. When the pressure is released the liquid readily converts back to its gaseous state which burns like any natural gas. The advantage of the characteristics of LPG for use on boats is that it can be easily carried and stored in the liquid state. The primary hazard with LPG is that when it is in a gaseous state it is heavier than air. The

fumes will sink to the bottom of an enclosed compartment and when mixed with air, in proper proportions, will explode if ignited.

Although there are no federal regulations of cooking and heating equipment in boats, the National Fire Protection Association has developed guidelines and standards that should be followed in installation of this equipment. A summary of these guidelines follows:

1. All stoves shall be installed in an adequately ventilated area and be securely fastened. They must be manufactured, approved and labeled for marine use.

2. All combustible materials above and immediately surrounding stoves shall be effectively insulated.

3. Alcohol stove tanks must be made of corrosion resistant material with welded or brazed joints and fittings.

4. On LPG systems, the gas containers should be located on the deck or cabin top. Each gas container has to have a manual shutoff valve, a safety relief device and a reducing regulator.

5. All low pressure LPG lines must be made of copper with soldered or brazed connections.

Cabin heaters are fueled by electricity, coal, wood, LPG, alcohol, and kerosene. The same installation guidelines established by NFPA apply here and do not require elaboration.

There are, of course, many other elements and materials in the modern boat that can contribute to or be the cause of a fire. Plastics, rubber, paper, cloth and metals — all are used in the construction of a boat and their flammability is well established under the right conditions. Also remember that fiberglass reinforced plastics and wood are the two most common materials for building boat hulls.

Your first job is to determine what you are looking at. A boat that has burned on land or on a trailer will be almost totally consumed. You must talk with fire service personnel, the owner, witnesses and then dig through what's left to determine what the boat looked like prior to the fire. This is of absolute importance so that you will know what kind of systems you are going to be

examining. What kind of power did it have? Inboard, Outboard, Inboard/Outdrive? What kind of fuel? Was there an auxiliary generator? Did it have cooking equipment? What is the age of the boat? All of these questions must be answered prior to trying to determine the origin.

A boat burned in the water will be easier to identify because of the nature of fire. When the fire spreads to the hull, it will burn down slowly toward the water line. As the hull integrity is broken, the boat will start to sink, effectively raising the water line. As the fire reaches the water line, it will be cooled by the surrounding water and will run out of fuel. This, of course, will result in the fire being extinguished. In an outboard or inboard/outdrive, the boat will begin to sink at the stern because of the engine weight in that area. This will almost always leave the majority of the transom intact. The importance of this is that the hull identification number is on the transom. This will also reduce the damage to the engine as it will be underwater during the later stages of the fire. It is not unusual to find that the only part of the boat not involved in the fire is the outdrive unit. It is underwater from the time the fire starts and can reveal the manufacturer of the engine and a serial number for positive identification of the boat.

Once you have determined what you are looking at, you should begin to isolate the general fire cause. Try to determine whether the fire was caused by the fuel system, electrical system, ventilation system, cooking equipment or other sources such as the operator. A fire can be traced back to the general area of origin in a boat in much the same manner as in a structural or motor vehicle fire. Follow smoke patterns. Look for areas of charring. Most marine engines are mounted on wooden members. Examine them for charring. Look for evidence of arson, but be very careful. It is not unusual for a boat owner to carry a can of gas on his boat. There may also be large quantities of paint, paint thinner, linseed oil and oil soaked rags. These may all be very legitimate.

Now that you have things narrowed down a bit, let's look at some types of fires.

Hull Fires — A hull fire is usually a result of a fire originating at some other place and spreading by conduction or radiation to the hull. An aluminum hull will not burn, but will melt if the temperature is high enough. A fiberglass hull will burn similar to wood. If it appears that the hull damage is primarily exterior with little fire damage spreading into the interior of the boat, look for an external cause. If the major damage is confined to the interior of the hull, look for a source other than the hull itself.

Fuel System Fires — If your observations take you to the engine compartment, it is time to examine the fuel system which is involved in at least 80% of all boat fires. Let's begin at the fuel tank. Is it intact? Are the hoses and piping connected. Are the fittings tight? Are proper clamps installed? Are the hoses the right type? What type of material is the tank made of? A leak, loose connection or a disconnected line in this area will result in an accumulation of fuel and vapors within a closed compartment. This only needs a source of ignition to produce an explosion. On older boats, it is quite common to find vent lines loose or disconnected. This will allow fuel to spill into the boat each time it is fueled. A hot fire, given enough time, will burn even the best fuel line material leaving you wondering what it was made of. Look under the metal clamps. The hose immediately under the clamp will not be seriously damaged and will allow identification to be made. If a hose has been intentionally removed, the clamps will probably be found intact in the debris, having been pushed down the hose rather than removed.

Fuel tanks will rarely explode. What usually occurs is the tank will fail at a seam under extreme heat, or will melt, causing a large quantity of fuel to be dumped into the fire. We once investigated a fire that resulted from a fuel tank fill pipe failure. The pipe broke loose from the top of the tank allowing gasoline to run into the boat whenever it was filled or operated in rough water. The spilled fuel was ignited by an electrical short and an explosion occurred. The fire burned the area in and around the tank but never spread to the fuel in the tank, even though a large hole (between the pipe and tank) was present in the top of the tank.

Now let's work toward the engine. Check the condition of the fuel line, fittings and clamps. The most common failures in this area would be loose fittings, chaffing, disconnected lines or leaks.

At the engine check the fuel filter, fuel pump, fittings, and lines. Check the sight glass on a dual diaphragm pump. If fuel is in the bowl, a diaphragm failure has occurred. Is the pump a marine type or has it been replaced with an automotive type? Also check the flame arrestor. Is it an approved type for marine use? Did the owner take it off to make the engine run better? Has an automotive "paper" air cleaner been installed? While you are at the engine, check the area underneath. If fuel had been leaking from the engine components prior to the fire, it would have been pooled in this area. Fire damage would be apparent on the oil pan and underside of the engine components. This may not rule out the possibility that the fire originated there, but it certainly makes it unlikely. While in this area remember to check for charring of the engine mounting timbers. It can direct your

attention to one side of the boat or the other.

Electrical System Fires — A short in the wiring will result in sufficient heat to generate melting, separation and burning of the insulation. If this heated material comes in contact with other combustible material a fire is likely to result. You must also consider electrical components such as alternators, generators, starters, pumps, distributors and motors as sources of ignition for fuel fires. These are all items that create sparks in their normal operation and if they are not properly sealed, these sparks can reach spilled fuel or vapor concentrations. Have any items been replaced with automotive type? The fact that these items are "specialty" equipment makes them expensive. Owners are often tempted by the lower prices that can be obtained by going to an auto parts store.

Start at the battery and check the condition of the connections and the wiring. Are the connections tight? Have extra circuits been added? Is the battery secure? A battery in itself is usually not an ignition source because it produces no sparks. If a battery is neglected it can become lethal. Fires and explosions have occurred when batteries have broken loose and fallen against metal grounded objects. A loose battery connection will generate heat and sparks which can ignite fuel vapors.

Check for evidence of short circuits. A short will cause very high current flow which generates heat. The heat will be sufficient to melt and bubble the insulation from the point of the short to the source of the power. This high current flow will continue until the wire opens, breaks, or a fuse blows. The short can cause a fire in a very remote area away from the point where it occurs. A short in the instrument panel of a boat may not cause a fire at that location, but rather back in the engine room where the wire runs next to some wood or other combustible material. In examining wires be careful to separate the cause of the damaged insulation to either shorting or high current flow or external fire damage. As previously stated, a short can cause melting the entire length of a wire, including the area under clamps, where it passes through walls and inside wire bundles. External fire damage may only effect certain sections of a wire or only one side of a wire. If a conductor inside a wire bundle shorts, the exterior wires of the bundle will not be affected. External fire results in the opposite affect.

Don't forget to look at owner installed or modified equipment. Improper installations by the owner have resulted in electrical fires due to overloaded circuits with no circuit protection.

One last note about the battery. It was previously stated that normally the battery cannot be an ignition source. It can, however, be the source of fuel for a fire. While it is rare, battery explosions can occur. As we have previously discussed, hydrogen gas is generated in a battery when it is charging. If this gas is allowed to build up in a closed compartment or battery box an explosive mixture will result. Any spark introduced into the compartment can result in a violent explosion.

Ventilation System Fires — Fires do not occur as a result of the ventilation system itself. Fires that are associated with the ventilation system are in actuality either electrical system or fuel system fires. The ventilation system can, through improper use, lack of use, or failure, contribute to the conditions leading to a fire. The ventilation system is most often mentioned in fires and explosions occurring during or immediately after fueling. In checking the ventilation system, examine the blower, ducting and wiring. Examine the instrument panel to see if the ventilation system was turned on. Ventilation ducting most often used in boats is similar to household clothes dryer ducting — a plastic flexible tube with a wire insert. The plastic will burn away or melt, leaving the wire. This allows you to check the ducting runs for proper installation.

Other Causes — There are numerous additional items on a boat that could cause a fire. Cooking and heating equipment, cigarettes, linseed oil, oil soaked rags and stored gasoline. We will touch on each of these very briefly.

Cooking and heating equipment can be the cause of a fire if improperly maintained or used. Alcohol stoves will flare up with a rather large flame if the burner preheating process is not complete before the fuel is turned on. This has resulted in several serious fires when curtains and other combustibles have come in contact with the high flames. Failure in the fuel lines of an LPG system can result in fires. Malfunctioning safety valves that allow fuel to continue to flow after the unit is shut off, can also be a cause. The variety of this type of equipment in use today makes it difficult to cover all the possibilities. We recommend that if you suspect this equipment, contact the manufacturer for assistance in determining what may have occurred.

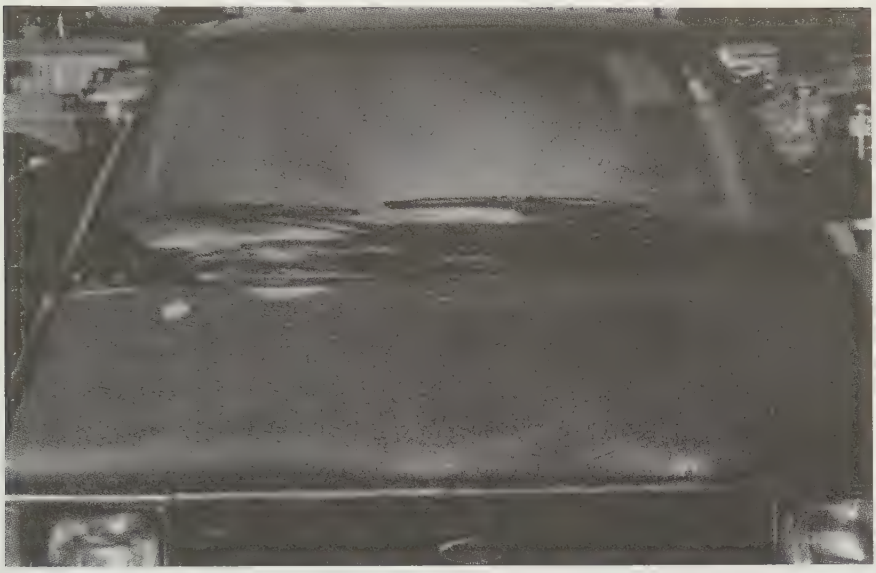
As in any other type vehicle or device used by, or lived in by man, the boat is subject to the careless smoker. Perhaps even more so than other vehicles because of the presence of large quantities of gasoline and other fuels in closed areas and the fact that the entire vessel may be made of combustibles. Check out the possibility with the owner and other witnesses.

In boats, especially large ones, it is not uncommon to find cans of paint, turpentine, paint thinner and linseed oil. These items are all used to keep the boat's appearance up and the owner likes to keep them handy. The inexperienced boat fire investigator gets excited when he finds a gallon can lying in the debris. Check with the owner, it may not be the container for an ignitable liquid. Linseed oil is commonly found on boats and is used to preserve unfinished wood trim. It is also prone to spontaneous combustion. A rag soaked in linseed oil can generate enough heat to self ignite in a very short period of time. The same thing can occur with other cleaning rags or oil soaked cloth though it may take a longer period of time.

The other hazard remaining to be discussed is stored gasoline. Often boaters are forced by necessity (lack of gas stations, high fuel prices, long trips, etc.) to carry extra gasoline while underway. This can be any amount ranging from a six gallon portable fuel tank to 55-gallon drums. If it is not properly secured with tight fittings and caps, it can, as a result of boat movement or thermal expansion from the heat of the sun, be a hazard just waiting for a spark.

(Source – The Investigation of Recreational Boat Fires – by Lee S. Cole & Robert C. Herold, 1983, Lee Books No longer in print.)

Ford Engine Compartment Fire

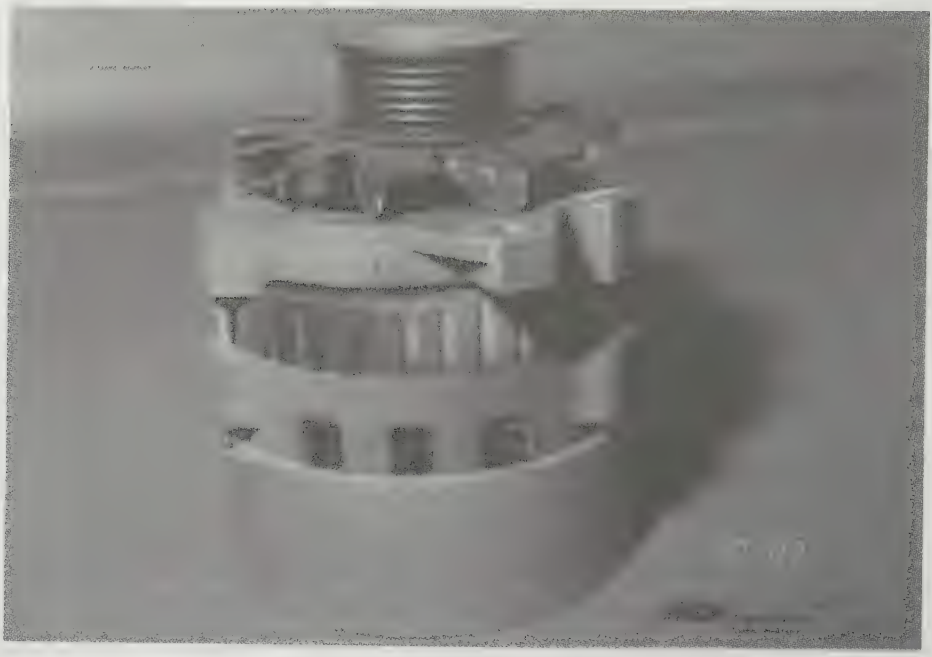


1. An engine compartment fire in this Ford Aerostar...



2. ...was caused by replacing a 65 amp alternator with a 75 amp unit. The additional output caused melting and failure at the B+ terminal.

GM Bearing Failure



1. A failed bearing in this General Motors alternator...

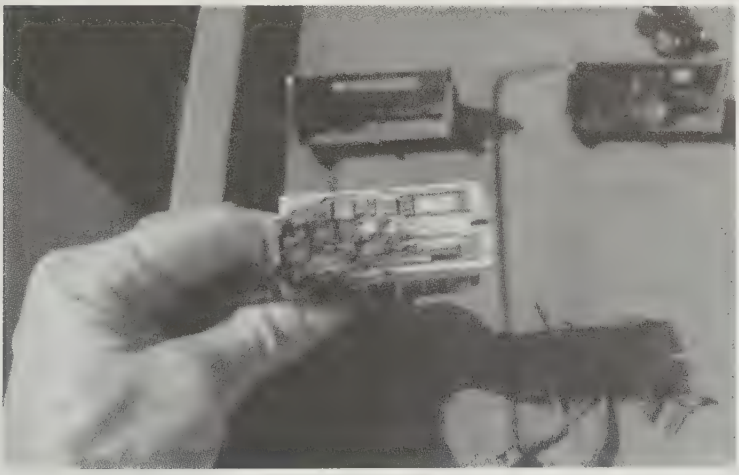


2. ...resulted in a fire when insulation melted and shorting occurred.

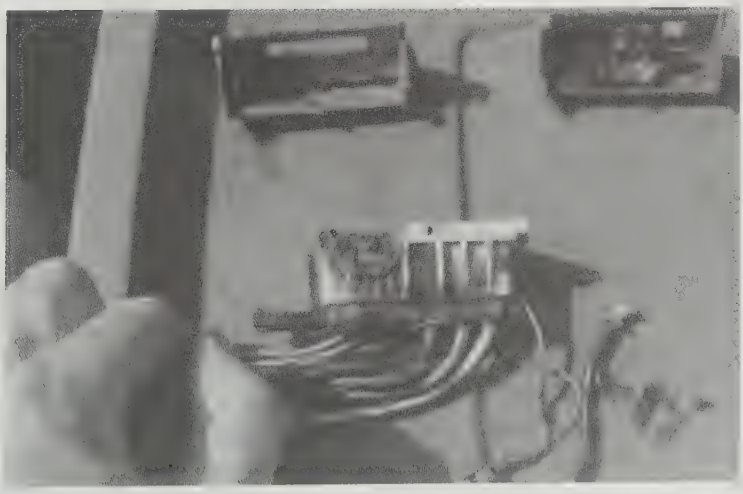
Ignition Switch Failures in Ford Products



1. Switch body is at center of photograph under key way.



2. Switch removed. Note heat concentration at end of battery terminal and ground. (Area above thumb of hand.)



3. Switch viewed from side.

Explosions

An explosion is a vigorous reaction with a sudden release of energy due to the rapid production of gases and liberation of heat. A "detonation" is a rapid and violent explosion with shock waves traveling faster than the speed of sound. High explosives such as dynamite have detonations in the hundreds of thousands of pounds per square inch. This produces the shattering and tearing effect observed in these cases.

The most common cause of explosions are:

1. Flammable liquids.
2. Gases — both natural and manufactured.
3. Explosives and blasting agents.
4. Steam, air, mechanical or electrical.
5. Unstable or explosive chemicals.
6. Dust, combustible metal particles and plastics.

In dealing with explosive and flammable material one frequently hears the term "lower and upper limits". This refers to the ratio between the amount of a potentially flammable vapor and the surrounding air. As a general rule the air contains about 21% oxygen.

"Lower Limits" are defined as the minimum concentration of vapor in the air below which flame does not occur on contact with an ignition source. This means that the mixture of air and fuel is too "lean" and will not burn.

"Upper Limits" refers to the maximum concentration above which ignition will not occur. This means that the air to fuel mixture is too "rich" to burn.

Since our study relates to motor vehicles we will consider only those explosions that are germane to this type of equipment. We have discussed in an earlier section the possibilities of battery explosion and the "blowing up" of the fuel tank during a fire. Let's examine some methods of deliberate vehicle destruction through explosives.

"Car Bomb Booby Trap" - This is a very simple procedure because the motor vehicle supplies the power source for igniting the explosive. Wires are run from an electrical blasting cap to a point along the ignition system and attached with alligator clips. When the ignition key is turned on the circuit is completed and the bomb detonated. Usually the explosive charge is centered at the lower portions of the firewall so that maximum force is directed at the individuals in the front seat area. Usually 3 sticks of dynamite are used along with two lengths of electrical wire with two clips for quick connections. One alligator clip is attached to the input side of the coil and the other fastened to the car's frame as a ground.

Other dynamite "Car Bomb Booby Traps" make use of a timing device such as a clock, a 9 Volt transistor battery and a triggering device for dynamite sticks. The unreliability of this is readily apparent since the precise actions of the intended victim cannot be exactly predicted.

Another method used — and a relatively simple one — is the use of heat generated by the exhaust system. Explosives are placed against the firewall and a fuse is wrapped around the exhaust manifold. After sufficient heat is developed the fuse is ignited and in turn sets off the explosive charge.

"The Mobile Bomb" — In the recent past political unrest in many countries has led to the introduction of a weapon of massive destructive power, the "Mobile Bomb". This is simply a car or truck loaded with explosives that is driven to its selected target and then detonated at a pre selected time. One needn't search far to be reminded of the effect of these weapons of terrorism. The bombing of the U.S. Marine Barracks and the Embassy buildings in Lebanon in 1983 and 1984 resulted in severe loss of life and property destruction. The same method of operation was used against the U.S. and French embassies in Kuwait on December 12, 1983. Conviction of those responsible for these acts later resulted in the highjacking of a Kuwait airliner in December 1984 and the death of two American passengers in an abortive attempt to secure the release of the convicted prisoners. They were subsequently released by Iraqi invaders of Kuwait in August 1990 and have since disappeared.

In December of 1983 an Austin 1300 GT loaded with explosives blew up just outside Harrods Department store in London, England at the very height of the Christmas shopping rush. Five floors were damaged and five persons were killed and 91 wounded. Two of the dead were police officers who had responded to a warning call that said "car bomb outside Harrods. Two bombs inside Harrods".

All these attacks, while separated by hundreds of miles, had two things in common. The act was politically motivated and the "container" for these destructive bombs had been a motor vehicle.

This technique seems to be growing in popularity as a device of terrorists. It should not be considered as a new departure since it has been used in the campaign against the French in Algeria. In Viet Nam it was employed by the Viet Cong and in August of 1970 was first used in the United States when a truck containing 1700 pounds of explosive processed ammonium nitrate fertilizer was exploded next to the Army Research Center at the University of Wisconsin in Madison. One person was killed and a number were injured. Property damage approached $7 million. Those responsible were making a "political statement". Most of these explosions have been triggered by either a timing or remote control device. The Mobile Bombs exploded in the Middle East were detonated by the driver who sacrificed his life in order to place the explosive laden vehicle in the most advantageous place. Fanaticism of this sort makes the job of prevention extremely difficult. It is best effected by an adequate and vigilant perimeter defense. It is not the function of this study to go into any great detail in this area, but certain considerations should be given when the defense of an installation is planned against a Mobile Bomb.

1. Perimeter fencing with controlled access. Entrances and exits should be kept to a minimum and manned 24 hours a day by adequately trained and equipped personnel who have a clear concept of their mission.

2. Zig zag obstructions should be placed inside the perimeter area to prevent a vehicle from making a suicidal dash toward the installation.

3. Parking should be restricted within close proximity to the installation to minimize any blast damage.

4. A careful check of any suspect vehicle should be made. In addition to the obvious search of the interior and engine compartment. The underside should be examined with the use of large mirrors.

The Federal Bureau of Investigation, United States Department of Justice has issued some excellent material on protective measures against suspected bomb devices. The following is from a bulletin issued on this subject:

"Do Not Disturb a Suspected Bomb

When a suspected bomb device has been located there are a number of actions which may be taken, by individuals not trained in bomb disposal, that do not involve moving the suspect item. Some of these measures, which may reduce the destructive potential of the device, are outlined below.

Immediate Steps

Evacuate: The distance and extent of personnel evacuation depends on the size, type, and location of the suspected device. The evacuees should be moved far enough from the item that they are protected from a bomb blast and flying debris. Evacuation must be directed calmly and logically in order to prevent a general panic, which can cause more injuries than a bomb explosion.

Secure the Area: Only authorized personnel should be allowed to enter the immediate vicinity of the suspected device. Securing the area will help prevent injury, and will keep the bomber from returning to rescue or dispose of his handiwork.

Notify:

(1) Bomb Disposal Personnel. Contact the appropriate bomb disposal personnel immediately and provide them with as much information as possible concerning the device itself, the bomb threat or warning, the possible perpetrators, and the surroundings.

(2) Other Emergency Personnel. Safety and security personnel on the premises, as well as police, fire, and emergency medical units, should be notified that a suspected bomb device has been found. Consideration should be given to notifying public utility companies, in case it becomes necessary for services to the building to be discontinued.

Steps to Reduce the Effects of an Explosion

Some of the effects of an explosion, and several protective measures which can be taken to reduce them, are listed below:

Blast: Blast is a positive pressure wave, emanating from an explosion, which is transmitted in all directions from the explosion. The wave, which is capable of producing a pressure of approximately 700 tons per square inch, tends to take the path of least resistance. Usually it is best not to attempt to confine the blast. Steps can be taken to provide a venting channel, or to open the area so that the blast will dissipate as rapidly and harmlessly as possible. Windows and doors may be

opened; objects near the device which could impede the blast wave or contribute to secondary fragmentation can be removed.

Fragmentation:
(1) Primary Fragmentation. Primary fragmentation consists of pieces of the bomb itself which are transmitted by the blast wave. Different types of devices have different fragmentation potentials. A pipe bomb containing a high explosive, for instance, would produce a large amount of fragmented material; a bomb which consists only of dynamite, fuse, and detonator would have virtually no primary fragmentation. An improvised barricade of materials that will not contribute to fragmentation, such as sandbags, mattresses, or a bomb blanket, can be erected around the suspected device. This may stop the fragments or reduce their speed. The device must not be disturbed, however, and enough standoff to avoid additional confinement should be maintained.



A pipe bomb placed between the hood and windshield of this Acura did heavy damage.

(2) Secondary Fragmentation. In many instances, much of the fragmentation will be secondary fragmentation. This consists of items near the bomb which are picked up and transmitted by the blast wave. If the effect of the blast is reduced, the amount of secondary fragmentation may also be reduced.

Heat: Heat is always a product of an explosion, and may range from 2000° to 4000° Centigrade (approximately 3630° to 7200° Fahrenheit). Although the thermal effect of an explosion is measured in fractions of a second, secondary fires frequently occur. Therefore, readily combustible materials should be removed from the vicinity of the device."

Most frequently the Fire/Arson Investigator enters the explosion scene after

the event has occurred. It is his, along with a staff of assistants, function to assemble the evidence necessary to answer the questions of who, what, where, when and why? In the "Handbook of Forensic Science" published by the Federal Bureau of Investigation, United States Department of Justice the following steps are recommended in making a bomb scene search:

"**A. Plan of action**: Formulate a plan adapted to the particulars of the bomb crime scene. This plan will include consideration of the creation of an on-scene command post; establishment of lines of supervision; assignment of various tasks such as photographing, fingerprint processing, crowd control, collecting evidence; protecting the crime scene; obtaining necessary equipment; periodically evaluating progress; providing pertinent information to the public; safety; etc.

B. Command post: Consider establishing an on-scene command post particularly at a large bombing which may require days or weeks to complete the crime scene search. The command post should coordinate efforts among investigative personnel and between representatives of other agencies and utilities, as well as handle inquiries from sightseers, persons associated with the scene, relatives of victims, and the press.

One person should be in overall charge of the bombing investigation, another over the actual crime scene search and another over the collection of the evidence. These three individuals must maintain close coordination and expeditiously exchange information on a continual basis. The evidence coordinator will report directly to the individual responsible for the overall bombing investigation.

C. Safety: Evaluate safety conditions at the outset of the crime scene search and on a continual basis throughout the search. Consider the possibility of a second bomb, a "jammed" bomb, or nearby residents, and personnel at the crime scene. Utilities, weakened walls, et cetera, which may create dangerous situations, should also be considered. Investigative personnel are not to assume the responsibility for handling a bomb.

D. Protection of crime scene: Take adequate safeguards to protect the crime scene from fire, law enforcement, utility, and rescue personnel as well as others such as sightseers, victims, and individuals with a personal interest in the property. Also, since most residues remaining after an initiation of an explosive are water soluble, the crime scene, as much as possible, should be protected against exposure to excessive moisture be it from rain, snow, broken water pipes, or any other source.

E. Photographs: Take appropriate photographs to give a photographic presentation of the crime scene. These photographs should be made immediately before, periodically during, and at the completion of the crime scene activity. Properly identify each photograph, coordinate the photographs with diagrams and/or blueprints or maps, and consider the advisability of aerial photographs.

F. Bomb scene specialists: If without a specialist(s) trained in handling and processing bomb scenes, make arrangements for obtaining such an individual(s). Although the basic principals of conducting a crime scene search apply in a bomb scene search, individuals with specialized knowledge of explosives, improvised explosive devices, damage produced by explosive charges, and facts associated with bomb scene searches, such as the search and collection of physical bombing evidence, are extremely valuable to the effective and efficient processing of a bomb scene. These specialists need not be qualified bomb disposal specialists. They should be the first persons, if possible, to be selected for the evidence and crime scene search coordinator positions.

G. Equipment: Promptly make arrangements to obtain the necessary equipment to move the debris and material at the scene. Although the equipment needed at the scene varies, the following have been used:

1. Hand tools: shovels, rakes, brooms, boltcutters, wire cutters, sledgehammers, hammers, screwdrivers, wrenches, chisels, hacksaws, magnets, flashlights, knives, 50' measuring tapes, and traffic wheel-measuring devices.

2. Other equipment: screens for sifting debris, wheelbarrows, metal trash cans, power saws, cutting torch equipment, ladders, portable lighting equipment, metal detectors, large plastic sheets, photographic equipment, and parachute harness with related rope and pulleys.

3. Heavy equipment: truck, front-end loader, bulldozer, crane, and shoring materials.

4. Personal equipment: hard hats, safety goggles, gloves (work and rubber types), foul weather clothing, coveralls, and work shoes.

5. Crime scene kit: usual equipment used for the collection, preservation, and identification of physical evidence.

6. Vehicle: If the bombed target was a vehicle, bring an identical vehicle, if possible, to the scene to assist in identifying fragmented and mutilated items.

H. Search for evidence: Bear in mind the search for evidence at a bombing crime scene is important because the crime scene may contain principal evidence which will lead to the identification of the bomber(s) and/or assist in the successful prosecution of the matter. The following guidelines are general in nature as the exact method of searching depends on various uncontrollable factors:

1. Place one person in overall charge of the collection of the evidence from the various collectors as valuable evidence may not be admissible in court if a proper 'chain of custody' cannot be established. Include the location where the items were found. A diagram of the crime scene is always useful.

2. Do not stop the search after a few items of evidence have been found.

3. Avoid the tendency to concentrate only on physical evidence, such as safety fuse, detonating cord, blasting caps, leg (electrical) wire, dynamite wrappers, batteries, clock and timing devices, electronic and electrical components, metal end cap from a TNT block, plastic end cap from a C4 block, explosive residues, and unconsumed explosives, which may represent a bomb as this can result in overlooking other valuable evidence, such as fingerprints, hair, fibers, soil, blood, paint, plastic, tape, tools, toolmarks, metals, writing paper, printing, cardboard, wood, leather, and tire tread-shoe print impressions.

4. Conduct a well organized, thorough, and careful search to prevent the necessity of a second search. However, have a secure 'dump' area for debris in the event a second search is necessary.

5. Normally, initiation of the search should start at the site of the explosion and work outward. If the bomb crater is in earth, obtain soil samples from the perimeter of the crater as well as from the sides and bottom making sure to dig into the substrata. If the crater is in another material, obtain similar samples.

6. Sift small debris through a $^1/_4$" wire screen onto an insect-type wire screen. Usually these screens are placed on 2 foot square wooden frames

constructed from 2" by 4" lumber.

7. X-ray the bodies of living and deceased victims who were in close proximity of the explosion site for possible physical evidence and if possible, have the evidence removed. (Their clothing should be retained as it may contain explosive residues.)

8. Search a sufficient distance from the site of the explosion as evidence has been found several blocks from the sites of large explosions.

9. Determine the possible flight paths of bomb components to prevent needless searching.

10. Search trees, shrubbery, telephone poles, and the roofs, ledges and gutter of buildings. Instances have occurred where physical evidence has been 'carried away' on the tires of fire and rescue vehicles.

11. Establish a search pattern for large areas. A line of searchers moving forward has been found to be a satisfactory method. A bomb scene specialist should follow the line of searchers to evaluate the items found, control the searchers, and furnish guidance. If a second search is desired, the positions of the searchers on the line should be rotated. Charting the area to be searched will insure a thorough search pattern.

12. Retain all items foreign to the scene and items which the searchers cannot identify after seeking the assistance of those familiar with the bombed target."

In many cases of Mobile Bomb, the Fire/Arson investigator's task is similar to the Auto Theft investigator's. The remains of the vehicle used must be located, re-assembled and properly identified. It is of primary importance to ascertain what type of vehicle was involved, who did it belong to and where did it come from. Bear in mind the fact that many vehicles used in these operations are stolen. Whatever remains at the scene must be collected and the identifying numbers ascertained. These numbers, referred to as Vehicle Identification Number or VIN, are placed on motor vehicles to distinguish them from other vehicles. Engines and transmissions also have distinguishing numbers and can be traced. Vehicle theft and identification specialists can ascertain this information and begin the tracing process in a short time. The circumstances of the vehicle's theft may lead to the identity of the persons responsible. Sometimes the Mobile Bomb will be a vehicle

registered to an individual or group responsible for its placement. As in the case of the intentional fire setter, many bombers are of the opinion that all evidence of their deed will be destroyed at the moment of explosion.This is seldom true and careful examination of the scene can lead to the identification of the suspect.

It is of great importance to locate parts of the vehicle bearing residue from the explosive. The blast pattern must be ascertained and searched for any remains of the bomb. The examination of residue from a bombing by a crime lab can result in the identification of the type of explosive used. Please remember that explosive residues must be protected against excessive moisture. The best container that can be used in the collection of this type of evidence is a new airtight paint can.

While not an explosive device, the "cigarette fuse" is an old and often used trick to set off other reactions. Often it is used in conjunction with a plastic bag or bottle containing gasoline. The fuse consists of: (1) match book and (2) lighted cigarette.

The cigarette is placed inside the match book parallel to the heads. The cover is closed, holding the cigarette in place against the match heads. When the cigarette burns down to the match heads, ignition takes place. The resulting fire is used to ignite other combustibles.

There are other methods of operation, of course, to destroy vehicles by explosives and we have only enumerated a few of them. In examining vehicles damaged by an explosion, caution should be exercised in order not to be misled. A few years ago some burned vehicles exhibited evidence of some intensive explosive force under the car. This explosion caused a jagged hole in the vehicle's underside and metal "shrapnel" bits inside the car.

Investigation disclosed that a ground fire under a vehicle could create heat intense enough to cause the hollow drive-shaft to explode. The explosion blasted debris from under the vehicle at about a 45° angle and the appearance of the shaft itself, after the blast, was similar to a scoop shovel. The sound of the explosion has been compared to a 20 MM shell and the damage caused in some cases has been equally as great. Recently we have observed exploded drive shafts in several American manufactured vehicles. Drive-shaft explosions may be uncommon, but they do occur. The same phenomena has been seen on the hollow shock absorbing extensions of bumpers and lift assist struts on vehicles. When these explosions occur these struts can be rapidly propelled over 100 feet from the vehicle and can be very hazardous to anyone nearby.



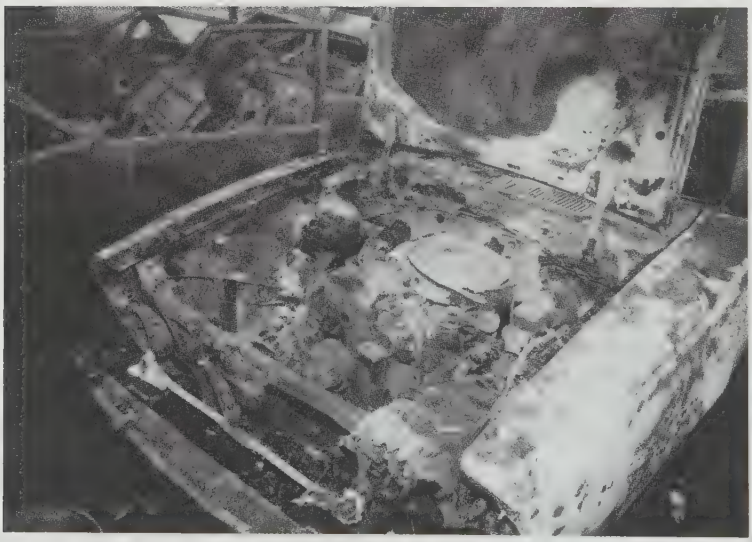
Establishing the origin and cause of vehicle fires also involving structures is very difficult.



Generally the overall burn is uniform with no apparent pattern.



Tops are frequently caved in from loss of tensile strength and falling debris.



Engine compartments are burned in a uniform pattern. Beware of designating exterior hood patterns as "hot spots", indicating possible areas of origin. These are frequently caused by falling debris.

Conclusion: Putting It All Together

No investigation should be accepted as merely routine. The case that seems to be lacking in investigative challenge can turn out to be anything but. As an illustration, consider the following from our book "Claims, Costs & Crimes: How Insurance Fraud Artists Steal Billions."

In a dedication page in his book "Homicide Investigation" Doctor Le Moyne Snyder writes "To those unknown and uncounted persons, prematurely deprived of life by malicious assault, and whose assassins have gone unpunished - this book is respectfully dedicated". I like to think there is a minimum number of cases where the "assassins have gone unpunished" but I realize that with the large number of violent deaths occurring daily it is highly probable that some of these are mistakenly labeled as "accidental" when in fact they are not.

Sometimes in a matter of days, weeks, months or even years, some event, incident, fact or rumor may surface and shine some light into what may be the dark cave of murder. When this does occur it behooves us to make certain that every effort is made to determine that the truth was not buried with the dead.

Tony Barrajas was a carpet layer. His work carried him all over the upper area of the San Francisco peninsula and he was frequently seen in his 1981 Ford Pickup, carpet rolls and tools in the bed and cab, on jobs in Daly City, Pacifica, South San Francisco, San Bruno, Millbrae and the other small towns and cities lying in the shadow of San Francisco. Sometimes he had to complete a job before leaving and would arrive late at his South San Francisco apartment. If he was going to be quite late he would call his wife, Marjorie, and let her know. Tony liked to drink. He might even be labelled a "heavy drinker" but he would not drink and drive. A conviction for that offense a few years back made a believer out of him and he waited until he was home before he began to consume his daily portion of his favorite Scotch. At age 59, Tony's life seemed reasonably happy. He had a good job and made decent wages. He and Marjorie, who was 36, had been married for over six years and had no debt problems. He was on reasonably good terms with his grown children from a previous marriage and was planning on visiting a new grandchild even though he voiced his disapproval of the living arrangements of the unmarried parents. The only real problem was Marjorie's son by a previous marriage. He was seventeen and said he hated Tony. His name was Joe and he and his fifteen year old sister lived with Tony and Marjorie. Joseph's animosity could be attributed to teenage rebellion against a stepfather. It wasn't

pleasant but no one made it a big deal.

At about 12:30 in the early afternoon on Friday, November 19, 1982 employees of a rifle range located in a remote area near a winding narrow road south of San Francisco called the police, advising them there was a Ford pickup at the bottom of a canyon near the range. The pickup was miraculously sitting upright on all four wheels after having plunged down the steep bank from the roadway high above. The first policeman to arrive on the scene took one look in the pickup cab and called for the coroner's office. Tony was on the floor of the pickup dead, his head smashed and bloody. He had not been wearing a seat belt and authorities concluded that he had suffered his injuries while being flung about the cab after running off the road. He had a blood alcohol reading of .11, more than enough to qualify as legally drunk.

Tony's movements for the preceding evening were traced. Marjorie said he came home at the usual time, didn't drink anything and between 8:30 and 9:00 p.m. left their apartment to visit his daughter and new grandchild who lived a few miles away. He also told her that he possibly would visit a friend in the same general area and that he might stay all night at his daughter's place. When he didn't return the following morning she became worried and tried to make a missing person report, but she was told by the Police that she would have to wait until 72 hours had elapsed. In the afternoon of November 19th she was notified by the police and coroner's office that Tony had been in an accident and was dead. Tony's daughter said that he never got to her house the night before. The answer was simple. Tony went to visit a friend, they got to drinking and by the time Tony decided he had better go home he was drunk. He ran off the narrow twisting road on the way home and the crash killed him. The death certificate gave the immediate cause as Subdural and Subarachnoid hemorrhage, fractured calvarium, extensive and trauma head. These injuries were sustained as the result of a highway accident. Case Closed. On November 22nd he was buried in the Italian cemetery in Colma.

On February 8, 1983 I received a request from a client insurance company to meet with a young Deputy Fire Marshal in South San Francisco and assist him in examining a 1980 Chevrolet Malibu sedan that he suspected had been deliberately burned. I contacted him by telephone and arranged a meeting for later that day. He told me that he had recently been assigned to the job and this was his first case involving a burned automobile. When I obtained the policy information from the insurance carrier I learned that the insured person was Marjorie Barrajas, a name that meant nothing to me at the time. There was no problem in establishing

that the fire destroying the car was arson. There was major damage in the passenger interior and very little in the engine compartment. There was no evidence of fuel leakage or electrical malfunction. Mrs. Barrajas had stated the vehicle was stolen from where she parked it on Mission Street in San Francisco. It was located in South San Francisco several hours later, burned. There was no evidence that the door or ignition locks had been forced. The location where the burning vehicle was found proved of interest. It was in the parking lot of a closed meat packing plant. Marjorie Barrajas had once been employed there. Why would a casual thief take the car there to burn it? Since nothing was stripped from it why burn it all? Was there some connection between the person who stole the car and Mrs. Barrajas? If the car had been locked as she claimed, why were there no signs of forced entry?

These questions and a few more needed answers. We contacted Mrs. Barrajas and found her quite vague and evasive. She admitted that she had filed no report with the San Francisco police and she couldn't furnish any descriptive information as to where and when the car had been parked on Mission Street. While we pondered the meaning of this turn of events we received a message that increased our suspicions. Mrs. Barrajas was dropping her claim against the insurance company for the theft and burning of her 1980 Chevrolet Malibu. When asked the reason for this action she replied that she didn't want to cause the insurance company any more trouble, "after all, they were so good to me when my husband was killed". An alarm went off in my brain. "Who was her husband? When and where was he killed?" Something was going "tilt" and we needed some answers right away.

At this point consider the position of the insurance company. They could take a release from Marjorie and allow the claim for the "stolen" and burned car to be dropped. No claim payment, no claim handling and no investigation expenses. An attractive offer with the savings of several thousand dollars. To their everlasting credit, the claims executives of the insurance company decided to push forward with the investigation and find out what was really going on.

We checked for other claims under Tony's auto insurance policy and found that on October 10, 1982, over one month before his fatal accident, Tony reported that someone had cut all the brake lines on his pickup truck while it was parked outside the apartment. A search of Department of Motor Vehicle records revealed an incredible event. On November 18, 1982 Marjorie Barrajas applied for a duplicate driver's license in the name of Marjorie Hernandez. This was on the very date that Tony went for his fatal ride. We learned that at Tony's funeral

Marjorie was accompanied by a young man named Jose Hernandez whom she introduced as her nephew. Court records showed that on December 1st, Marjorie gave a general power of attorney to Jose Hernandez to conduct all her affairs. His address was listed as the same as Marjorie's.

Jose was a small time criminal with an arrest record for auto theft, burglary, petty theft and grand theft. He was 23 years old and listed a prior address less than two blocks from Tony and Marjorie Barrajas. One of Tony's daughters told us that she was suspicious of her father's death and that Jose was Marjorie's boyfriend, not her nephew. She furthermore said that her father had not told her that he was coming over for a visit on the night of the 18th and that he had called at about 8:30 p.m. and made some general conversation, saying that he was "just laying on the waterbed, watching T.V." He sounded a little drunk, but there was nothing about the conversation that caused any alarm.

The insurance company's file on Tony's fatal accident held even more disturbing information. Marjorie told the adjuster that Tony didn't come home at all on the night of November 18th, but had called her at about 8:45 p.m. and said he would be stopping off at his daughter's house. At about 11:00 p.m. Marjorie called the daughter, asking for Tony, but was told that he had not been there. Tony's daughter told me and my associate, Dwight Bell, that this was a lie and Marjorie had made no such calls to her. The adjuster's photographs of the 1981 Ford pickup that Tony was driving showed no interior damage that would have likely occurred when his head struck with such force. A request was made for the report from the Coroner's office. Photographs showed Tony's body crammed into the floorboard of the pickup as if he had been shoved in from the right door side. His massive head wound clearly could not have been caused by anything inside the cab. So far Marjorie had been paid $56,000 in insurance settlement money and was due another $40,000. She and Jose bought a new Ford Van and, as neighbors reported, was having "one big party". Hardly a grieving widow!

Another interview with Tony's daughter produced even more suspicious circumstances. She pointed out that her father was lightly clothed and he would not have gone out that way on a rainy night. He was a heavy smoker and no cigarettes were found on the body. He always carried a pen and pencil in his shirt pocket but neither was found.

At this point a very serious problem presented itself. Unlike the tales of television and the movies, the private investigator does not welcome an opportunity to prove the authorities wrong or mistaken in a criminal case. We presented

the complete case to a Lieutenant of the California Highway Patrol responsible for the investigation of "staged" accidents. He agreed that the fatality was no accident and accompanied us to the Police Department responsible for the original investigation. We laid out the entire case, along with statements and evidence gathered, before the captain of the investigation division. The case was immediately reopened and two detectives assigned. The Coroner's office was also persuaded to re-examine their findings in light of new developments. They agreed and soon there was a new opinion rendered. Tony Barrajas was not killed in an accidental plunge over a cliff. He was murdered.

Marjorie's daughter had moved to Texas and was located there by police investigators. After a short interview she agreed to tell the truth. She didn't take part in her stepfather's murder, but she had been in the apartment when it happened. Marjorie had put some drugs in Tony's food. He had some drinks at dinner and then drank some more while he was in bed. He went into a deep sleep and Marjorie called Jose who came over with a friend. Jose and Marjorie's son Joe went into the bedroom where Tony was asleep and struck him on the head with a baseball bat. After doing this they came out into the living room, thinking they had killed him. A few minutes later they heard Tony groan and gasp for breath. They reentered the bedroom and struck him again with the bat. This time the job was done. He was dead.

They carried him out to his pickup truck, his shattered head wrapped in a piece of carpeting. With great effort he was stuffed into his pickup and the door closed. Joe drove the pickup, straddling the body on the floorboards while Jose and his friend followed in Marjorie's Malibu. She stayed behind to tidy up the apartment. They arrived at a point on the lonely road where the drop off would be most unobstructed. Lights were turned off. The Malibu was parked below the pickup which was then turned sharply toward the roadway edge. The accelerator was pushed and the engine raced. The drive selector was placed in "D" and the truck roared over the edge. Sounds of metal against earth and trees echoed back through the darkness. Satisfied, they ran to the Malibu and drove back to South San Francisco. The bloody bedding and the baseball bat were tossed into a convenient dumpster. Mission accomplished. And so to bed.

The skein now unraveled completely as more bits and pieces of evidence fell into place. On April 29, 1983 Marjorie was arrested and charged with first degree murder. A few weeks later Jose and Joe met the same fate as did the Good Samaritan who helped them dispose of the body.

The trial was held two years later and was highlighted by the defendant's bitter denunciations of each other and self serving statements, each claiming to be the dupe of the other. The jury promptly found them all guilty of murder. In June of 1985, Marjorie and Jose were given life terms of imprisonment. Joe was committed to the California Youth Authority. Jose's friend received lesser punishment because of his testimony.

What did all of this have to do with the 1980 Chevrolet Malibu Marjorie reported stolen and burned? Jose tracked bloody footprints on the carpeting after Tony was sent on his journey over the cliff. A few weeks later Jose was arrested for burglary and Marjorie had to bail him out. She was in between payments from Tony's insurance settlement so she put the car up for security. Jose jumped bail and she heard that the bondsman was looking for the car to seize on the bail default. If that happened she was afraid someone would notice the blood stains in the carpeting. The car was driven over to the old meat packing plant parking lot and burned. Then came the theft claim.

I think about Tony when I drive by the Italian cemetery in Colma. I never knew him in life, only in death, but I came to know him well. Rest in Peace, Tony. At least your assassins have not gone unpunished.

Source: Claims, Costs & Crimes: How Insurance Fraud Artists Steal Billions, by Lee S. Cole, Lee Books, San Anselmo, CA

Combustible Temperatures

The following tables may be useful to the fire investigator. Please remember this information is for general guidance only and is not absolute.

Temperature Colors of Flame in Degrees of Fahrenheit

900°	Faint red
1050°	Blood red
1175°	Dark Cherry
1250°	Medium Cherry
1375°	Cherry
1550°	Bright red
1650°	Salmon
1725°	Orange
1825°	Lemon
1975°	Light yellow
2200°	White

Color Temperatures of Steel and Iron in Degrees of Fahrenheit

800°	Purple blue
1000°	Dark red
1375°	Cherry red
1650°	Orange
1825°	Yellow
1975°	Light yellow

N.F.P.A. Melting Points (Approximately)

Aluminum	1150°-1500°F
Brass	1575°-1800°F
Copper	Average-1981°F
Glass	1400°-1600°F
Iron (pure)	2795°F

Liquid Hydrocarbon Fuels

a. Gasoline — Ignition temperature 536° to 800° degrees F depending on the octane rating.

Flash point minus 45° F.

Explosive limits: (percentage of vapor in air by volume)
Upper limits 7.6%
Lower limits 1.4%

b. Kerosene — Ignition temperature 480°F.
(No. 1 fuel oil) Flash point 100°F.

Explosive limits: (percentage of vapor in air by volume)
Upper limits 5%
Lower limits 0.7%

Gaseous Hydrocarbon Fuels

a. Natural Gas — Ignition temperature 1000° to 1200° F.

Explosive limits: (percentage of vapor in air by volume)
Upper limits 15.0%
Lower limits 4.5%

b. LP gases

1. Propane — Ignition temperature 871°F.

Explosive limits: (percentage of vapor in air by volume)
Upper limits 9.5%
Lower limits 2.2%

2. Butane -Ignition temperature 806°F.
Explosive limits: (percentage of vapor in air by volume)
Upper limits 8.5%
Lower limits 1.9%

Paints and Lacquers

Ignition temperature may be anywhere from 475° to 1000°F. Flash point usually 80°F or lower.

The flammable part of these materials is not the pigment, although certain metallic chromate pigments are, in themselves, flammable.

a. Amyl Acetate	Ignition temperature 715°F.
b. Acetone	Ignition temperature 1000°F.
c. Linseed Oil	Ignition temperature 650°F.
d. Mineral Spirits	Ignition temperature 473°F.
e. Petroleum Naptha	Ignition temperature 475°F.
f. Alcohols	Ignition temperature 750° to 900°F.
g. Turpentine	Ignition temperature 464°F.

Alkali Earths and Metals

a. Magnesium — Ignition temperature 1204°F, but may be as low as 900°F if the material is in finely divided powder form.

b. Sodium — Provides a reaction with water. Also reacts violently with inorganic acids.

c. Potassium — Undergoes the same reaction, only more violently than sodium.

Explosives

Choice of type will vary with the result intended. High detonation speed will give a shattering or a cutting action; low speed will result in a heaving action.

a. Gelatin Dynamites	Detonation speed up to 22300 ft. per second.
b. Straight Dynamites	Detonation speed up to 18200 ft. per second.
c. Blasting Dynamites	Detonation speed up to 3900 ft. per second.
d. Nitro-carbo-nitrate	Ammonium nitrate-fuel oil mixture. Produces a heaving action. Burns readily in a natural state, being composed of 60% oxygen. Because of this feature, cannot be extinguished by smothering.

Prevention of Vehicle Fires

Occasionally I am asked what can the owner/operator do to prevent a vehicle fire. On the spur of the moment this isn't easy to do without a lot heming and hawing around. I made the following list to be used on such occaions. No doubt you will think of some points that we didn't cover or some that you would want to change.

1. Since catalytic converters generate considerable operating heat, never park your vehicle over dry grass, leaves or paper debris.

2. If you work on your own car never splice in a piece of rubber hose in the fuel line.

3. If you change the fuel filter beware of overtightening or cross threading the fuel entry inlet nut.

4. Check the fuel pump for any sign of leaking fuel. If there is, the pump should be replaced.

5. Check the flex lines in the fuel line system. If they show signs of age or cracking they should be replaced.

6. Do not leave smoldering cigarettes in the ash tray. They can communicate to nearby paper and cause a major fire.

7. Oil leaking from the engine valve covers can drip down on the exhaust manifold and ignite. Check for leaks around gaskets.

8. Power steering hoses can rupture, allowing fluid to spray on a hot exhaust manifold. Check the hoses for signs of cracking and aging.
9. Check the fuses and circuit breakers. If any of these show signs of consistent failure or switching on and off you should have your car examined by an electrical specialist.

10. When installing stereos, air horns and similar "add on" systems, be careful not to create electrical shorting. This will not only cause a fire, but will also invalidate your new car warranty.

11. If you lose your fuel cap do not replace it with the first cap you find or stuff in a rag, etc. This can have very dangerous consequences.

12. Keep you engine compartment clean. A small fire will spread rapidly on accumulated oil and grease deposits.

Always remember that a modern motor vehicle is a very complicated piece of machinery and each of it's systems is interdependent on the other. A small problem can rapidly create a big one. As an example, I recently saw where a small gasket leak allowed oil to run down on the exhaust manifold. This created a small fire and burned two spark plug wires. This caused the engine to run extremely rough and unburned fuel ran into the exhaust system. This caused the catalytic converter to run extremely hot and the vehicle's interior ignited from radiant heat.



Oil leaking from a warped valve cover onto the exhaust manifold was the cause of this fire.

1. A stall test was conducted on this Acura with drive wheels blocked.





2-3. Before the transmission fluid could erupt in the engine compartment the super heated exhaust system ignited the plastic covered rear bumper.



Sources of Information

Advanced Technology Center
Calspan SRL Corp.
P.O. Box 400
4455 Genesee Street
Buffalo, NY 14225

American Automobile Manufacturers Association (AAMA)
7430 Second Avenue, Suite 300
Detroit, MI 48202
(313) 872-4311

American Trial Lawyers Association (ATLA)
1050 31st Street NW
Washington DC 20007
(202) 965-3500
(800) 424-2725

Association of International Automobile Manufacturers (AIAM)
1001 19th Street N., Suite 1200
Arlington, VA 22209
(703) 525-7788
E-mail: finnegan@aiam.org

Center for Auto Safety
2001 S Street NW, Suite 4 10
Washington DC 20009
202-328-7700

Class Action Reports, Inc.
4900 Massachusetts Avenue, NW, Suite 230
Washington, DC 20016
(202) 364-1031
FAX 363-6912
E-mail: bcmjr@aol.com

Consumer Product Safety Commission (CPSC)
Washington, DC 20007
Toll-Free HOTLINE:
(Within Maryland) (800) 492-2937
(Other Continental US) (800) 638-2772
Gopher address: cpse.gov

Dialog Information Services
2440 West El Camino Real
Mountain View, CA 94040-1400
(800) 334-2564

Environmental Protection Agency (EPA)
Manufacturers Operations Division
Mobile Sources
501 3rd Street, NW
Washington, DC 20560
(202) 233-9240 .
www.epa.gov

Federal Highway Administration (FHWA)
400 Seventh Street, SW (Nassif Building)
Washington, DC 20590
www.dot.gov

Office of Motor Carriers
Room 5103
Nassif Building
(202) 366-2519

Office of Safety & Traffic Operations Research & Development Turner Fairbanks Highway Research Center
6300 Georgetown Pike, Room T-306
McLean, VA 22 101
(703) 285-2104

Federal Trade Commission (FTC)
Bureau of Consumer Protection
6th and Pennsylvania Avenue, NW

Washington, DC 20580
(202) 326-3238
www.ftc.gov

General Accounting Office (GAO)
P.O. Box 6015
Gaithersburg, MD 20884-6015
(202) 258-4066 (Fax orders)
www.gao.gov

Gillis & Associates
1518 K Street, NW Suite 306
Washington, DC 20005
(202) 737-0766

Government Printing Office (GPO)
Superintendent of Documents
North Capitol & H Streets, NW
Washington, DC 20401
(202) 512-0571
www.access.gpo.gov/su-docs

Highway Loss Data Institute (HLDI)
1005 N. Glebe Road, Suite 800
Arlington, VA 22201
(703) 247-1600
www.hwy.safety.org

Highway Safety Research Center
730 Airport Road
University,of North Carolina
Chapel Hill, NC 27599
(919) 962-2202

House Committee on Commerce
2125 Rayburn House Office Building
Washington, DC 20515
(202) 225-2927
www.house.gov/commerce/welcome.html

House Committee on Transportation and Infrastructure
B370A Rayburn House Office Building
Washington, DC 20515
(202) 225-9446
www.house.gov/transportation

Insurance Institute for Highway Safety (IIHS)
1005 N. Glebe Road, Suite 800
Arlington, VA 22201
(703) 247-1500

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P.O. Box 446
512 W. Lancaster Avenue
Wayne, PA 19087-0446
(610) 688-6566
FAX: 688-7552
E-mail: news@mealeys.com

National Crash Analysis Center
George Washington University - Virginia Campus
20 101 Academic Way
Ashburn, VA 22011
(703) 729-8236
FAX: 729-8358

National Highway Traffic Safety Administration (NHTSA)
400 Seventh Street, SW
Washington, DC 20590

Toll-Free Auto Safety Hotline
(Within DC area)
(202) 366-0123
(Other Continental US)
(800) 424-9393
www.nhtsa.dot.gov

Crash Avoidance Research Database (CARDfile)
Center for Statistics and Analysis
(202) 366-1503

Crash Avoidance Rollover Study (CARS)
Office of Crash Avoidance Research
(202) 366-5662

Docket Room
Room 5109
Nassif Building
(202) 366-4949

Information Resource Management Division
Room 6130
Nassif Building
202-366-2590

National Center for Statistics and Analysis (NCSA)
Room 6125
Nassif Building
(202) 366-1503

Office of Crashworthiness Research Database
(202) 366-1862

Office of Defects Investigation (ODI)
Room 5326
Nassif Building
(202) 366-2768

Office of Public and Consumer Affairs
Room 5232
Nassif Building
202-366-9550

Technical Reference Division (TRD)
Room 5110
Nassif Building
(800) 445-0197

National Safety Council
1121 Spring Lake Drive
Itasca, IL 60143-3201
(708) 285-1121
www,nsc.org.nsc

National Technical Information Service (NTIS)
5285 Port Royal Road
Springfield, VA 22161
(703) 487-4600
Gopher address: marvel.loc.gov

National Transportation Safety Board (NTSB)
490 L'Enfant Plaza E., SW
Washington, DC 20594
(202) 382-6600

Recreational Vehicle Industry Association (RIVA)
1896 Preston White Drive
Reston, VA 22191-4363
(703) 620-6003

Senate Committee on Commerce, Science and Transportation
508 Dirksen Senate Office Building
Washington,DC 20510
(202) 224-5115
www.policydotnet/capweb/senate/senatecorn/com.htn-d

Society of Automotive Engineers (SAE)
400 Commonwealth Drive
Warrendale, PA 15096
(412) 776-4841
E-mail at sae@sae.org www.sae.org

Transportation Systems Center, DTS-32
Department of Transportation Kendall
Square Code 903
Cambridge MA 02142
(617) 494-3427

Department of Transportation
Bureau of Transportation Statistics (BTS)
400 7th Street, SW, Room 3430
Washington, DC 20590
(202) 366-3282
www.bts.gov
E-mail: info@bts.gov

Transportation Research Board Library
2101 Constitution Avenue
Washington, DC 20418
(202) 334-2933
www.nas.edu

Transportation Research Institute
University of Michigan
2901 Baxter Road
Ann Arbor, MI 48109-2150

Glossary of Terms

Following are a few terms one frequently hears in connection with both vehicle and other type fires.

Accelerant — A substance of any type used to enhance the burning qualities of a material and to increase the spread and intensity of fire.

Alternating Current (AC) — An electrical current that reverses direction in cycles or in polarity.

Ampere — A unit of electric current defined as the steady current that produces a force of 2 x 1-7 neutrons meter of length when flowing through parallel wires of small cross section and infinite length one meter apart in free space.

Combustion — This process is defined as rapid oxidation, accompanied by heat and light and is the actual operation of burning.

Conduction — This is transmission of heat from one object to another by contact. Fire may spread in this manner from the vehicle's interior to the engine compartment or vice versa.

Convection — This is the transmission of heat through a liquid or gas by the motion of its parts.

Direct Current (DC) — An electrical current flowing in one direction in a circuit.

Fire Point — This is the lowest temperature at which flammable liquid when exposed to heat and sufficient oxygen will give off sufficient vapors and will continue to burn. This is usually a few degrees above the flash point.

Flash Point — This is the lowest temperature at which a flammable liquid when heated in the presence of a source of ignition and sufficient oxygen will "flash" but not continue to burn.

Ignition Temperature — This is the lowest temperature required to initiate or cause self sustained combustion in the absence of a spark or flame for a flammable liquid or gas.

Ohm —The unit of electrical resistance, equal to the resistance of a conductor in which a current of one ampere is produced by a potential difference of one Volt.

Petroleum Products — These products are frequently involved in vehicle fires and are generally referred to as "hydro-carbons". These compounds range from gasoline as the lightest in weight and include kerosene, fuel oil, etc. to the heaviest, asphalt.

Radiation — This is the transmission of heat by waves. When we stand in front of a fire we become warm by this process.

Short or Short Circuit — This is defined as an electrical connection of low resistance or impedance, especially when accidental or unintended. An overloaded circuit will burn at its weakest point. A break in the wiring of an automobile will cause an incomplete circuit and consequently interrupt the operation of the vehicle. Heat is generated at the point of a "short".

Spontaneous Combustion — This is the act of burning from qualities inherent in the substance itself and not from any external cause. In order for this to take place the temperature must be raised to the degree at which the substance will ignite. This heat is generated by a slow oxidation process. It is most common in hay storage. Hay generates heat and when improperly stored can cause a fire.

Volt — A unit equal to the difference of electric potential which, when steadily applied to a conductor whose resistance is one ohm will produce a current of one ampere.

Watt — A unit of power of about 1/746 of a horsepower. Electrical power is generally expressed in watts and is called "wattage".

There are many other definitions, abbreviations, etc. that the investigator will be confronted with from time to time but space limits their listing. It is safe to say that in instances where an unknown term is encountered your best friend is your dictionary.

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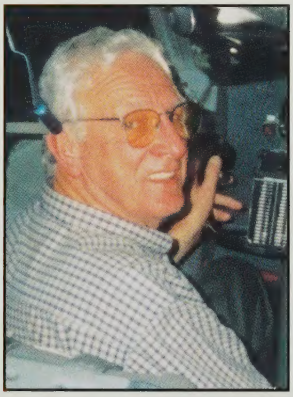
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